

2025 PM Guide Cards

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ARCS

Architectural

ARCS-BAR-01-06M

Security Bollards/Barricade

Rev-3 | 2022-01-01

This standard card applies to automatic hydraulic security Bollards/Barricades found at driveway entrances, and other areas where vehicular traffic must be controlled.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Review manufacturer's instructions.
3. Review Standard Operating Procedure for controlling hazardous energy sources.
4. De-energize, lock out, and tag electrical circuits.
5. This work should be scheduled at non-peak hours.
6. Notify affected personnel before performing PM (alarmed or security entrances).
7. Post "Out of Service" signs and/or barricades, as appropriate.

Checkpoint

1. Check that Bollards/Barricade extend to full height.
2. Check that Bollards/Barricade retract to full depth.
3. Check Bollards/Barricade for exterior damage.
4. Check and clean sump pits and pump.
5. Check that Bollards/Barricade fasteners are tight.
6. Check Bollard/Barricade safeties (proximity, Photo, infrared, pressure etc.).
7. Check Bollard/Barricade activation devices for proper operation (auto/manual/emergency)
8. Check Bollard operation under no power.
9. Inspect electrical power panel. Check electrical connections and wiring for loose or overheated conditions.
10. Inspect control panel for loose connections.
11. Check that traffic light functions, change lamps and clean lenses every 12 months.
12. Check hydraulic unit fluid level and pressures/ change fluid per manufacturer's recommendations.
13. Check hydraulic unit fluid connections, pump, piping and hoses for damage and leaks.
14. Clean motor and assembly.
15. Check the drive motor amperage against nameplate.
16. Clean drive mechanism compartment.
17. Inspect Bollard/Barricade coatings, decals and paint.
18. Check that all placards and warnings signs are posted.

Tools

TOOL_GROUP_A

This standard applies to davits, structures typically used to lower things over an edge. This standard is not intended to address all fall protection related equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. This section does not address powered man-lifts, harnesses, self retracting fall limiters, shock absorbers, and other related fall protection equipment that falls into the above requirements.
4. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.

Checkpoint

1. Visually Inspect for corrosion. Verify there is no evidence of anchor degradation and all components are securely mounted.
2. Check bolt torque.

Tools

TOOL_BASIC

This standard applies to Rope Descent System (RDS) anchorages which are devices installed specifically to support an RDS or fixed structural members of a building, such as posts, beams, girders, and columns that may also serve as RDS anchorages.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. This section does not address powered man-lifts, harnesses, self retracting fall limiters, shock absorbers, and other related fall protection equipment that falls into the above requirements.
4. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.

Checkpoint

1. Each anchorage is certified by a qualified person that it is capable of sustaining a minimum rated load of 5000 pounds (22.2kN) in any direction for each employee attached.
2. Provide a report of testing results, methods of analysis, and numbers of employees for which each anchor is tested.

Tools

None

This standard applies to warehouse or large overhead rolling doors.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Inspect general arrangement of door and mechanism, mountings, standards, wind locks, anchor bolts, counterbalances, weather stripping, door sweeps etc. Clean, tighten, and adjust and repair as required.
2. Operate with power from stop to stop and at intermediate positions. Observe performance of various components, such as brake, limit switches, door operating speed, motor, gear box, etc. Clean and adjust as needed.
3. Check operation of safety edges, stops, electric eye, treadle, or other operating devices. Clean and make required adjustments or repairs.
4. Check manual operation. Note brake release, motor disengagement, functioning or hand pulls, chains sprockets, clutch, etc.
5. Examine all wiring, motor, starter, push button, etc., blow out or vacuum if needed.
6. Inspect gear box, change, or add oil as required.
7. Perform required lubrication. Remove old or excess lubricant.
8. Clean unit and mechanism thoroughly. Touch up paint where required.
9. Clean up and remove all debris.
10. Check and replace worn/damaged signage or decals.

Tools

LUBRICANT-07 MAT_CLEANING-02 TOOL_BASIC

This standard applies to entrance doors operated by sensors or switches (i.e. handicap access doors).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Set suitable barriers at the entrance and exit of the door. Prevent obstructions from impeding pedestrian traffic around the work area.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check alignment of door and mechanism. Inspect mountings, hinges, mats, and trim, weather stripping, etc. Replace, tighten, and adjust as required.
2. Operate with power, observing operation of actuating and safety mats, door speed, and checking functions.
3. Check manual operation.
4. Inspect power unit, lubricate and tighten lines as required.
5. Check operation of control board relays, clean, replace, adjust contacts as required.
6. Inspect door operating unit, tighten lines, and adjust as required.
7. Clean and lubricate door pivot points.
8. On pneumatic or hydraulically operated door operators, check for correct operating pressures per manufacturer's instructions.
9. Clean up and remove all debris from work area.
10. Check and replace worn/damaged signage or decals.

Tools

TOOL_GROUP_B LUBRICANT-07

This standard applies to manually operated overhead doors. These doors normally range in size from 7 to 24 feet wide by 7 to 20 feet in height.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. If the door is equipped with intrusion alarms, notify the appropriate person before operating.
3. Schedule maintenance on these doors so that it does not interfere with loading dock operations.
4. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. Inspect general arrangement of door and mechanism, mountings, standards, wind locks, anchor bolts, counterbalance, weather-stripping, etc. Clean, tighten, and adjust as required. Make minor repairs as needed.
2. Inspect cables for frayed or broken strands or excessive rusting.
3. Inspect winding drum for tightness and proper tracking of cables.
4. Manually raise and lower door, noting that door tracks evenly and action of break release, functioning of hand pulls, chains, sprockets, clutch, etc.
5. If equipped, inspect gearbox; change or add oil as needed.
6. Perform required lubrication. Remove old or excess lubricant.
7. Clean unit and mechanism thoroughly. Touch up paint where needed.

Tools

GREASE_GUN-04 TOOL_BASIC RUST_INHIBITOR-02 LADDER-04

This standard applies to manually operated entrance doors, hinged and sliding type.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Set suitable barriers at the entrance and exit of the door. Prevent obstructions from impeding pedestrian traffic around the work area.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Hinged Doors:
 - a. Inspect the frame and supporting structures.
 - b. Inspect hardware; hinges, latch keeper, lock, etc. Apply graphite where needed, wipe off excess.
 - c. Inspect glass, putty, or retaining pieces. Correct any deficiencies.
 - d. Operate door to observe functioning of check. Adjust and service as needed.
 - e. Check opening/closing resistance pressure and speed in compliance with ADA recommended 8.5 to 10 Ft-lbf opening/ closing force
 - f. Touch up paint as needed.
 - g. Clean up and remove all debris from work area.
2. Sliding Doors:
 - a. Center door within track and remove, start by lifting up to remove from bottom track first.
 - b. Clean out track of any debris.
 - c. Inspect hardware; rollers, track etc. Correct any deficiencies.
 - d. Return door to track, start by inserting door into bottom track first.
 - e. Operate door to observe function.
 - f. Check opening/closing resistance pressure and speed.
 - g. Touch up paint as required.
 - h. Clean up and remove all debris from work area.
3. Revolving Doors:
 - a. Remove obstructions and clean out track.
 - b. Fold door. Note action and freedom of motion.
 - c. Inspect locking device, adjust as needed.
 - d. Clean pivot points, visible bearings and bushings and apply graphite.
 - e. Inspect felt or rubber seals.
 - f. Set the emergency fold pressure on the door to the manufacturer's specifications.
 - g. Touch up paint as required.
 - h. Clean up all debris from work area.

Tools

TOOL_BASIC BARRIERS-03 CLOTHS-01 GRAPHITE

The purpose of this standard is to perform a cleaning of the fire box and an inspection of the flue, chimney, fire damper, and smoke shelf. This function should be performed once a month, during the time of year that the fireplace is in use.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions.
- 3. Schedule shutdown with operating personnel.
- 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 5. Do not use any flammable lubricants or solvents. If lubrication is necessary, use only dry powdered graphite.

Checkpoint

- 1. Remove loose ash and vacuum inside of fire box (fireplace).
- 2. Inspect fire brick and mortar joints for cracks or breakage. Repair or replace as necessary.
- 3. Inspect fire damper, checking for proper operation.
- 4. Make a visual inspection of the flue and chimney, checking for obstructions.
- 5. Clean surrounding area.

Tools

VACUUM-01 SHOVEL_BROOM EYE_PROTECTION TOOL_BASIC

The work required by this standard applies to incinerators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Thoroughly clean furnace, ash pit, grates, etc.
2. Remove fly ash and soot from flue gas passages.
3. Examine furnace. Replace burned or damaged parts.
4. Inspect for loose, broken, or missing refractory or fire brick.
5. Examine all doors, inspect and/or clean out ports. Make them fit properly and stop any air leaks around them.
6. Check uptakes or connections to stack or chimney. Remove dirt, fly ash scale, etc.
7. Examine dampers for condition and freedom of motion.
8. Examine structure and supports. Look for warped or sagging members, cracks, or other indications of weakness.
9. Check charging chute, frame, cover, etc. Replace broken, missing, or defective parts.
10. Check all instruments, gauges, etc. Test for proper operation.
11. Repair any damaged or missing insulation. If the insulation contains asbestos, refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
12. Inspect stack or chimney for holes, cracks or deterioration.
13. Inspect spark arrestor or screen on top of stack. Repair holes as necessary or replace.
14. Clean up work area and remove all debris.

Tools

RESPIRATOR-01 GOOGLES TOOL_BASIC EQUIP_CLEANING-03

The work required by this standard applies to memorial or decorative fountains. These fountains normally operate during the summer season. They are usually drained and secured in the fall and are reactivated again in the spring.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Fall:
 - a. Drain fountain
 - b. Remove fountain head(s)
 - c. Clean scale from head(s) after removing and install cover plate.
 - d. Clean bolts and spacers. Replace defective bolts as needed.
 - e. Check and clean fill tube, remove spacer pipe and cap fill tube.
 - f. Clean overflows and strainers.
 - g. Check for structural damage. Repair as necessary.
 - h. Where applicable, add an approved safe winterizing/antifreeze agent to water to prevent freezing.
2. Spring:
 - a. Remove fill tube cap, install spacer pipe and install fill tube.
 - b. Remove fountain head cover(s) (if applicable) and install fountain head.
 - c. Clean basins.
 - d. Check for structural damage. Repair as necessary.
 - e. Remove tags from valves and fill fountain
 - f. Remove tags from pump and put into service.
 - g. Add an appropriate biocide.

Tools

TOOL_BASIC HIP_BOOTS BIOCIDES EQUIP_CLEANING-01

This standard applies to fences and gates found at driveway entrances, walkway entrances, and other areas where pedestrian or vehicular traffic must be controlled. This classification applies only to those fences and gates where a definite need exists for this type of maintenance to be performed.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. This work should be scheduled at non-peak hours.
6. Notify affected personnel before performing PM (alarmed or security entrances).
7. Post "Out of Service" signs and/or barricades, as appropriate.

Checkpoint

1. Gates:
 - a. Inspect all pivot points, hinges, latches, etc. Apply lubricant where needed, wiping off excess.
 - b. Check all locking devices. Lubricate as required.
 - c. Inspect center gate support rollers and lubricate as required.
 - d. Clean roller track of any debris.
 - e. Check bolts, fasteners, and mounting hardware. Tighten or adjust as necessary.
 - f. Check for any obstructions that retard full swing or movement of the gate.
 - g. Check that shrubs and trees are pruned clear of gate.
 - h. Check hold open devices for proper operation. Lubricate as required.
2. Fences:
 - a. Check posts and corner posts, support guys, and horizontal bars between each support post.
 - b. Check wire and anchor point; re-stretch and re-anchor if necessary.
 - c. Inspect fence anchors along the bottom of the fence and at the point where the fence is connected to the post.
 - d. Treat with galvanized protectant where rust has developed.
 - e. Apply weed control along entire base of fence. Consult the Safety Data Sheets (SDS) for hazardous ingredients and proper personal protective equipment (PPE).
 - f. Check that shrubs and trees are pruned clear of fencing.

Tools

RUST_INHIBITOR-07
LADDER-04

EQUIP_CLEANING-03

LUBRICANT-07

SIGN_OUT_OF_SERVICE-02

TOOL_BASIC

WIRE_STRETCHER

This standard applies to all installed sprinkler systems with nozzles.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. This work should be scheduled prior to flushing system.

Checkpoint

1. Check nozzle and remove corrosion/encrustation.
2. Check for proper spray pattern of nozzles, minimize over spray on surfaces or structures.
3. Replace washers and/or "O" rings if required.
4. Clean unit thoroughly and inspect for freedom of operation.
5. On rotating types:
 - a. Check throw regulator and spring assembly for tightness.
 - b. Lightly lubricate pivot arm and standard bushings.
 - c. Check adjustment screw and retaining spring.
6. Clean up work area on lawn, remove any debris around sprinkler heads.

Tools

TOOL_BASIC LUBRICANT-07 WASHERS

This standard applies to all installed flag poles with electric and manual cabling systems.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Electrical Flag Pole:
 - a. Adjust tension on cable.
 - b. Adjust spring adjustment cam.
 - c. Check stops on cable.
 - d. Inspect manual and automatic controls and adjust if necessary.
 - e. Lubricate all bearings.
 - f. Inspect cable, standards, hooks and clasps for wear and replace if necessary.
 - g. Check power and control wire and connections.
 - h. Check and test safeties and controls.
 - i. Inspect access doors and ports including gaskets.
 - j. Inspect anchor bolts.
 - k. Inspect pole surface condition.
2. Manual Flag Pole:
 - a. Adjust tension on cable.
 - b. Check stops on cable.
 - c. Lubricate all bearings.
 - d. Inspect cable, standards, hooks and clasps for wear and replace if necessary.
 - e. Inspect anchor bolts.
 - f. Inspect pole surface condition.

Tools

TOOL_BASIC MAT_CLEANING-02 LUBRICANT-07

This applies to gasoline powered, hand operated, rotary mowers and edgers. Maintenance should be scheduled after every 50 hours of operation or twice a season. Routine lubrication should be accomplished by operator.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Change engine oil. Note: Oil should be changed at end of season prior to laying unit up for winter.
2. Service air and fuel filters.
3. Sharpen or replace cutting blade.
4. Clean and gap or replace spark plug.
5. Inspect unit, clean debris from cooling air passages and make other needed adjustments.
6. Clean up work area and remove all debris.

Tools

TOOL_BASIC

This standard applies to manholes which contain electrical distribution cables or other electrical wiring and cabling.

Special Instructions

1. Review standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch. Complete GSA Form 3625 or COR approved equivalent.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review standard operating procedures for selection, care, and use of respiratory protection.
5. Wear appropriate protective clothing.
6. No open flames or smoking.
7. Use barricade around manhole.
8. Provide ventilation within manhole. Ensure that the exhaust from the gas or diesel-powered air compressor or blower is directed well away from the manhole.

Checkpoint

1. Test for gas in accordance with asset hazard assessment form and GSA form 3625.
2. Pump out water.
3. Clean out trash, debris, etc., and dispose of properly.
4. Inspect cable, racks, splices, etc.
5. Inspect structural features.
6. Inspect manhole seal, replace if necessary.

Tools

TOOL_BASIC RESPIRATOR-01 BARRICADES

This standard applies to manholes which contain sanitary sewer lines or other related piping systems.

Special Instructions

1. Review standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch. Complete GSA Form 3625 or COR approved equivalent.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review standard operating procedures for selection, care, and use of respiratory protection.
5. Wear appropriate protective clothing.
6. No open flames or smoking.
7. Use barricade around manhole.
8. Provide ventilation within manhole. Ensure that the exhaust from the gas or diesel-powered air compressor or blower is directed well away from the manhole.

Checkpoint

1. Test for gas in accordance with asset hazard assessment form and GSA form 3625.
2. Remove cover.
3. Observe flow.
4. Clean work area and remove all debris.
5. Treat for insect infestation as necessary.
6. Inspect manhole seal, replace if necessary.

Tools

TOOL_BASIC RESPIRATOR-01 BARRICADES

This standard applies to manholes which contain water, steam system, fuel oil or other systems.

Special Instructions

1. Review standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch. Complete GSA Form 3625 or COR approved equivalent.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review standard operating procedures for selection, care, and use of respiratory protection.
5. Wear appropriate protective clothing.
6. No open flames or smoking.
7. Use barricade around manhole.
8. Provide ventilation within manhole. Ensure that the exhaust from the gas or diesel-powered air compressor or blower is directed well away from the manhole.
9. Maintenance of any equipment located in the manhole will be performed under the appropriate standard card in conjunction with this maintenance activity.

Checkpoint

1. Test for gas in accordance with asset hazard assessment form and GSA form 3625.
2. Remove cover.
3. Observe flow.
4. Examine structural features of sewer line, interior of manhole, manhole frame and cover, etc.
5. If applicable, Check for water accumulation. Pump out water. If contaminated follow hazmat procedures for proper disposal.
6. Clean work area and remove all debris.
7. Treat for insect infestation as necessary.
8. Inspect manhole seal, replace if necessary.

Tools

TOOL_BASIC RESPIRATOR-01 BARRICADES

This standard applies to Built-Up roofing systems typically consisting of a waterproof membrane (bituminous) over a deck, an outer protective coating (rocks, gravel, roll roofing, cement shingles, metal, etc.) and metal flashing or coping at edges and seams.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's or installer's instructions (if available) and check whether roof is under warranty. If under warranty, refer all noted damage to the COR and contact issuer of the warranty so that repairs may be performed without voiding the warranty.
3. Avoid walking or standing near the roof edge unless there is a safe railing guard or parapet. Use fall prevention equipment, after receiving training, as needed to perform roof work safely.
4. Additional inspections should be performed after any severe winds or storms.
5. A maintenance check list and copy of the roof plan should be used at time of inspection and areas of concern should be clearly marked so they can be easily located by repair crews.
6. It is vitally important to identify and repair the root cause of any problem to prevent further damage to the roofing system.
7. Use care when working in high places, use safety belt if necessary.

Checkpoint

1. Check ceilings and the underside of the deck for signs of water entry such as stained ceiling tiles, dry rot in a wooden deck, or rust in a steel deck. Note all deteriorated areas on the roof plan for comparison later on the roof.
2. Walk around the perimeter of the building. Check for cracks and signs of water entry into the walls; examine exterior drainage accessories such as downspout, scupper heads and gutters for signs of leakage. Mark the deficiencies on the roof plan and proceed to the roof.
3. Check for ponding and plant growth (roof should drain within 48 hours of a rain fall). Accumulation of water (ponding) should be noted on the roof plan. Pay particular attention to areas near building air intakes.
4. Check for physical damage such as punctures, note location of patches, repairs and accumulation of debris especially near drains.
5. Check for wrinkles, buckles, bubbles and sponginess. Note exposure of bituminous coating due to loose or missing gravel.
6. Check all flashing for wind damage, missing caulking, curling, and exposed edges. Check flashing fasteners for looseness and deterioration. Check any fibrous material that might be asbestos for deterioration. If asbestos containing materials are present, check the building's asbestos management plan in the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
7. Check the condition of any joints, roof to wall joint in particular. Check the termination of roofing expansion joints at parapet walls. Be sure that water drains off the top of the coping and that it is well attached.
8. Check coping for deterioration, especially rusting, punctures, open seams and spalling or cracking in masonry coping.
9. Check whether walkways are in logical locations and in good condition with no blistering below them.
10. Check that any installed equipment such as antennas, HVAC equipment and flagpoles are properly installed according to building codes, including necessary I-beam supports for heavy A/C equipment and are flashed and secured to the building.
11. Cleanout and refill pitch pockets (penetration pockets) where filler is cracked or shrinking.
12. Check ballast (rock) for even spread; remove any with very sharp edges.
13. TPO Roofs (if applicable) must be kept clean for maximum reflectivity.
14. Check all roof hatches, doors and access/egress points. Inspect weather seals, replace if necessary. Check for proper latching/ opening pressure and swing. Inspect (locking mechanism where applicable)

15. Remove all trash, debris or unsecured material from roof and dispose of properly.

Tools

TOOL_BASIC LADDER-04 SAFETY_LINE

This standard applies to Shingle roofing systems typically consisting of a waterproof membrane (bituminous) over a deck, an outer protective coating of shingles, and metal flashing or coping at drip edges and valleys.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's or installer's instructions (if available) and check whether roof is under warranty. If under warranty, refer all noted damage to the facility manager or issuer of the warranty so that repairs may be performed without voiding the warranty.
3. Avoid walking or standing near the roof edge unless there is a safe railing, guard or parapet. Use fall prevention equipment, after receiving training, as needed to perform roof work safely.
4. Perform inspections twice annually in the spring and fall when the roof is driest and most accessible.
5. Additional inspections should be performed after any severe winds or storms.
6. A maintenance check list and copy of the roof plan should be used at time of inspection and areas of concern should be clearly marked so they can be easily located by repair crews.
7. It is vitally important to identify and repair the root cause of any problem to prevent further damage to the roofing system.
8. Use care when working in high places, use safety belt if necessary.

Checkpoint

1. Check ceilings and the underside of the roof deck for signs of water entry such as stained ceiling tiles, dry rot in a wooden deck. Note all deteriorated areas on the roof plan for comparison later on the roof.
2. Walk around the perimeter of the building. Check for cracks and signs of water entry into the walls; examine exterior drainage accessories such as downspout, scupper heads and gutters for signs of leakage. Mark the deficiencies on the roof plan and proceed to the roof.
3. Check for physical damage such as punctures, note location of patches, repairs and accumulation of debris especially near drains.
4. Check for wrinkles, buckles, and sponginess.
5. Check all flashing for wind damage, missing caulking, curling, and exposed edges. Check flashing fasteners for looseness and deterioration. Check any fibrous material that might be asbestos for deterioration. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
6. Check the condition of any joints, roof to wall joint in particular. Check the termination of roofing expansion joints at parapet walls. Be sure that water drains off the top of the coping and that it is well attached.
7. Check coping for deterioration, especially rusting, punctures, open seams and spalling or cracking in masonry coping.
8. Check that any installed equipment such as antennas, HVAC equipment and flagpoles are properly installed according to building codes, including necessary I-beam supports for heavy A/C equipment and are flashed and secured to the building.
9. Cleanout and refill pitch pockets (penetration pockets) where filler is cracked or shrinking.
10. Remove all trash, debris or unsecured material from roof and dispose of properly.

Tools

This standard applies to Extensive and Intensive Vegetative Roofs.

Special Instructions

1. Avoid walking or standing near the roof edge unless there is a safe railing, guard or parapet. Use fall prevention equipment, after receiving training, as needed to perform roof work safely.

Checkpoint

1. Check membrane, flashing and roof penetration condition for degradation and signs of leakage.
2. Clean drains and gutters.
3. Check irrigation system for uniform coverage, valve leakage, plugged heads, and leaking solenoids.
4. Check and clean cisterns.
5. Check operation of any pumps, floats, or irrigation controls. Irrigation controls tied to a weather service should have connection verified (verify appropriate weather station is used).
6. Check for invasive/damaging weeds and treat appropriately.
7. Assess any needs for fertilization.

Tools

TOOL_BASIC SAFETY_LINE LADDER-04

This standard applies to Key Card Access systems which control building access.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions for all system components.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

- 1. Disassemble card reader head and vacuum.
- 2. Check card printer unit and vacuum.
- 3. Check wiring for loose connections, discoloration, etc.
- 4. Change filter.
- 5. Change batteries.
- 6. Check cards to see if codes are in operation.
- 7. Reassemble unit.
- 8. Clean exterior and surrounding area of unit.

Tools

EQUIP_CLEANING-01 VACUUM_CLEANER-01 TOOL_GROUP_B SPARE_FILTER_BATTERIES

This standard applies to parking arm gates which control building parking access.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions for all system components.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

- 1. Lubricate mechanism with graphite.
- 2. Adjust linkage between motor and arm.
- 3. Check and adjust arm pressure.
- 4. Check and adjust sensitivity on magnetic coils embedded in asphalt.
- 5. Fill cracks in roadway with asphalt or appropriate filler where coils are imbedded.
- 6. Clean and adjust electric breakers.
- 7. Paint arm as required. Replace if damaged.

Tools

WRENCHES_TORQUE TOOL_GROUP_B LUBRICANTS EQUIP_CLEANING-03 ASPHALT_FILLER

This standard applies to Lightning Protection systems installed on the building to discharge to ground any electrical charges caused by lightning.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. During each inspection, check that:
 - a. All air terminals (lightning rods) are interconnected.
 - b. At least two down conductors are installed with their own ground connection.

Checkpoint

1. Inspect air terminals for corrosion and rigid attachment to structure.
2. Examine conductors and all connections for corrosion, strong mechanical joints which provide good electrical conductivity, and loose or broken fasteners.
3. Check for loops, sharp bends (less than 8" radius), (less than 90 degree bends), and frayed horizontal and vertical conductors.
4. Check for damaged guards and down conductors.
5. Inspect grounding attachment for permanency and corrosion (if practical).
6. Test resistance to ground for each down conductor.
7. Clean up work area and remove debris.

Tools

TOOL_GROUP_B

This standard is intended for Stormwater Management, Green Roofs.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, jurisdictional requirements shall be strictly adhered to.

Checkpoint

1. Check vegetation for coverage (50% after first year and 80% after second year for plugs or clippings/90% after first year for carpet or trays). Replace dying or dead vegetation or fill in bare spots as needed.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Green Roofs.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Weed to remove invasive weeds, tree seedlings, pests and thatch accumulation.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Green Roofs.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Check membrane, flashing and caulking for leaks and needed repairs.
2. Clean associated downspouts, gutters, drainage pipes, drain boxes, inlets, stone edge drains, scuppers and other structures draining to/from practice to make sure they are free of vegetation and debris.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. Contracted Certified Inspector: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Inspect water quality devices.
2. Provide water quality analysis to local jurisdiction as required (Post-construction, annually and after major maintenance or modifications to treatment system).
3. Inspect cistern for sediment buildup.
4. Check integrity of backflow preventer.
5. Inspect structural integrity of cistern, pump, pipe, electrical system and monitoring devices if applicable.
6. Replace damaged or defective system components.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Inspect and clean prescreening devices and first flush diverters.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Clear overhanging vegetation and trees over roof surface.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Inspect and clean storage cistern lids, paying special attention to vents and screens on inflow and outflow spigots. Check mosquito screens and patch holes or gaps immediately.
2. Inspect condition of overflow pipes, overflow filter path, and/or secondary stormwater treatment practices.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Keep gutters and downspouts free of leaves and other debris.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Rainwater Harvesting.

Special Instructions

1. Contracted Certified Inspector: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. For above-ground systems, winterize system in Fall and activate in Spring.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Permeable Pavement Systems.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.
2. Do not sand, re-seal, resurface, power wash, store snow piles containing sand, store mulch or soil material and stage construction activities on permeable surface.
3. Prevent trucks and other heavy vehicles from tracking, spilling, or grinding material onto permeable pavement.
4. Use only non-toxic and organic deicers in moderation and apply as either calcium magnesium acetate or pretreated salt.
5. Plow snow carefully with blades set 1" higher than normal.
6. Do not direct plowed snow piles or snowmelt to permeable pavement.
7. First 6 months - Pavement and contributing drainage area should be inspected at least twice after storm events that exceed 1/2" of rainfall and conduct any needed repairs or stabilization.

Checkpoint

1. Mow grass in grid paver applications.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Permeable Pavement Systems.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.
2. Do not sand, re-seal, resurface, power wash, store snow piles containing sand, store mulch or soil material and stage construction activities on permeable surface.
3. Prevent trucks and other heavy vehicles from tracking, spilling, or grinding material onto permeable pavement.
4. Use only non-toxic and organic deicers in moderation and apply as either calcium magnesium acetate or pretreated salt.
5. Plow snow carefully with blades set 1" higher than normal.
6. Do not direct plowed snow piles or snowmelt to permeable pavement.
7. First 6 months - Pavement and contributing drainage area should be inspected at least twice after storm events that exceed 1/2" of rainfall and conduct any needed repairs or stabilization.

Checkpoint

1. Remove any accumulated sediment in pretreatment cells and inflow points.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Permeable Pavement Systems.

Special Instructions

1. Contracted Service: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.
2. Do not sand, re-seal, resurface, power wash, store snow piles containing sand, store mulch or soil material and stage construction activities on permeable surface.
3. Prevent trucks and other heavy vehicles from tracking, spilling, or grinding material onto permeable pavement.
4. Use only non-toxic and organic deicers in moderation and apply as either calcium magnesium acetate or pretreated salt.
5. Plow snow carefully with blades set 1" higher than normal.
6. Do not direct plowed snow piles or snowmelt to permeable pavement.
7. First 6 months - Pavement and contributing drainage area should be inspected at least twice after storm events that exceed 1/2" of rainfall and conduct any needed repairs or stabilization.

Checkpoint

1. Mechanically sweep pavement with a regenerative sweeper or standard (dry) street sweeper to prevent clogging and replace joint material sucked up by vacuum if applicable.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Permeable Pavement Systems.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.
2. Do not sand, re-seal, resurface, power wash, store snow piles containing sand, store mulch or soil material and stage construction activities on permeable surface.
3. Prevent trucks and other heavy vehicles from tracking, spilling, or grinding material onto permeable pavement.
4. Use only non-toxic and organic deicers in moderation and apply as either calcium magnesium acetate or pretreated salt.
5. Plow snow carefully with blades set 1" higher than normal.
6. Do not direct plowed snow piles or snowmelt to permeable pavement.
7. First 6 months - Pavement and contributing drainage area should be inspected at least twice after storm events that exceed 1/2" of rainfall and conduct any needed repairs or stabilization.

Checkpoint

1. Stabilize the contributing drainage area to prevent erosion.
2. Remove any soil or sediment deposited on pavement.
3. Replace or repair any pavement surfaces that are degenerating or spalling.
4. Clean associated drainage pipes, inlets, stone edge drains, and other structures draining to/from practice.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Conduct a maintenance inspection.
2. Supplement mulch in devoid areas to maintain a 3" layer.
3. Prune trees and shrubs.
4. Add reinforcement plantings to maintain desired vegetation density.
5. Remove invasive plants using recommended control methods.
6. Remove dead or diseased plants.
7. Stabilize contributing drainage area to prevent erosion.
8. If water ponds for more than 72 hours, determine if removing and replacing the top few inches of planting soil will fix the problem or if a more serious issue is responsible.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Add reinforcement plantings to maintain desired vegetation density.
2. Remove invasive plants using recommended control methods.
3. Remove dead or diseased plants.
4. Stabilize contributing drainage area to prevent erosion.
5. If water ponds for more than 72 hours, determine if removing and replacing the top few inches of planting soil will fix the problem or if a more serious issue is responsible.
6. Conduct a maintenance inspection.
7. Supplement mulch in devoid areas to maintain a 3" layer.
8. Prune trees and shrubs.
9. Remove sediments in pre-treatment cells and inflow points.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Check and clear curb cuts, inlets, outlets and overflow drains for accumulated grit, leaves, and debris that may block inflow and outflow.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Remove and replace mulch layer.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Spot weed, remove trash and rake the mulch.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Spot weed, remove trash and rake the mulch.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Mow grass if applicable.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Remove and replace mulch layer (or establish a permanent plant-based ground cover in lieu of mulch).

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Bioretention.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Check curb cuts, inlets and overflow drains for accumulated grit, leaves, and debris that should be cleared so it doesn't block inflow and outflow.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Filtering Systems.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.
2. Pick up trash, debris, leaves and other materials around your property and in front of inlets.
3. Know the pollutant sources on property and try to eliminate them at the source.
4. Sweep paved areas to remove pollutants such as sediment and sand.
5. Store chemicals, used oil and pesticides in covered areas not exposed to rainfall.
6. Don't stockpile sand or salt unless in covered and contained area.
7. Don't wash vehicles or equipment on paved areas and prevent wash water from entering storm drain.
8. Mark underground sand inlets to be sure snow plows won't damage them.

Checkpoint

1. Check to see if sediment accumulation in the sedimentation chamber has exceeded six inches and if so, schedule a cleanout. Dispose of removed material in an acceptable manner (e.g., landfill).
2. Check to see if top three inches of sand are visibly discolored or water ponds on the filter bed for more than 72 hours with either indicating a need for replacement of the top layer. Dispose of removed material in an acceptable manner (e.g., landfill).
3. Check to see if inlets and flow splitters are clear of debris and are operating properly.
4. Check access points, concrete structures, pipes and outlets for any evidence of spalling, joining failure, leakage, corrosion, broken ladder steps, damaged manhole covers, etc. and repair as needed.
5. Ensure that the filter bed is level and remove trash and debris from the filter bed. Sand or gravel covers should be raked to depth of three inches.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Infiltration Best Management Practices.

Special Instructions

1. Landscape Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Clean out accumulated sediment from the pretreatment cell.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Infiltration Best Management Practices.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Ensure that the contributing drainage area, inlets, and facility surface are clear of debris and are stabilized and spot-reseeded where needed.
2. Remove sediment and oil/grease from inlets, pretreatment devices, flow diversion structures, and overflow structures.
3. Repair undercut and eroded areas at inflow and outflow structures.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Infiltration Best Management Practices.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Mow vegetated filter strips as necessary and remove the clippings.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Infiltration Best Management Practices.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Check for clogging of surface (ponding water) and replace pea gravel/topsoil and top surface geotextile fabric when clogged.
2. Inspect pretreatment devices and diversion structures for sediment build-up and structural damage.
3. Check observation wells 3 days after a storm event in excess of 1/2" in depth to make sure there is no standing water observed after three days which would indicate clogging.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Ponds - Wet Ponds.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

2. Cleanup shoreline to remove trash, debris and floatables.
3. Full maintenance inspection.
4. Open the riser to access and test the valves.
5. Repair broken mechanical components if needed.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Ponds - Wet Ponds.

Special Instructions

1. O&M Contractor: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Remove debris and blockages from spillway.
2. Repair undercut, eroded and bare soil areas.

Tools

STORMWATER_TOOLS

This standard is intended for Stormwater Management, Ponds - Wet Ponds.

Special Instructions

1. Contracted Landscaper: In addition to the procedure(s) outlined in this standard, local jurisdiction shall be strictly adhered.

Checkpoint

1. Mow the buffer and pond embankment if turf but native plantings are preferable.

Tools

STORMWATER_TOOLS

This standard applies to window washing scaffold that is deployed from the roof.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Set suitable barriers on the ground as necessary. Prevent obstructions from impeding pedestrian traffic around the work area.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Inspect structural features on roof. Remove obstructions from the track and from, on, or near the garage.
2. Inspect roof car, platform, steps, wire mesh panels, gate, hinges, hardware, etc.
3. Observe operation of drive motor and mechanism, brake, cable, reel, drive, wheels, guide rollers, etc. Adjust as necessary.
4. Examine telephone cable reel and make a test call.
5. Inspect operation of electric controller, direction switches, inching buttons, protective devices, limit switches, position interlocks, locking pins and sockets, etc. Adjust as necessary.
6. Check operation of manual and emergency controls, hand crank, motor disengagement, brakes, and other devices. Adjust as necessary.
7. Inspect fresh water and wash water tanks, pipelines, drains, inspect or access openings and covers, etc. Empty and wash out tanks after use.
8. Test operation of scaffold from low to high position and along track to assure safe operation. Test all control devices, interlocks, limits, etc.
9. Inspect hoist ropes for worn, frayed, or broken strands.
10. Perform any work prescribed by manufacturer that is not indicated here.
11. Lubricate all necessary moving parts per the manufacturer's recommendations.

Tools

LUBRICANT-02 TOOL_BASIC

CHLD-EQP-01-01M**Child Care Equipment**
Rev-3 | 2021-11-03

This standard is intended for Child Care equipment. This standard is intended to be used to ensure that the equipment and systems are maintained in a safe operating condition, there are no parts, pieces, conditions or situations that may cause injury or endanger the users or pedestrians when used in accordance with the manufacturers placards, recommendations, instructions or guidelines.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Obtain and understand the manufacturer's inspection and maintenance requirements for the equipment.
3. Read and understand manufacturer's instructions of each device before making adjustments to the equipment or to the system. Understand what effects making adjustments will have on the operation of the equipment PRIOR to making any changes.
4. Ensure that the equipment is not in use and that the area is free and clear of children and bystanders that may either be in the work area or may be in danger by in close proximity to tools and serviced equipment. This may require closing the area or constructing a safety barrier.
5. This PM will typically require one person.
6. If a condition is found to exist which is, unsafe, or may cause injury, immediate action must be taken to resolve the situation by securing the equipment or area from further use and notifying the building COR / Child Care operator.
7. Responsibility for correction of deficiencies may fall to others, such as the Child Care provider, cleaning and lawn maintenance programs, or field office manager/Child Care coordinator. Findings shall be reported to the appropriate manager.
8. Be particularly alert to potential entrapment hazards in playground equipment.

Checkpoint

1. Visually inspect equipment to ensure structural integrity, check all mechanical connections and fasteners for tightness.
2. Visually inspect equipment for body shearing or pinch points.
3. Visually inspect equipment for any parts or structure that could pierce or puncture.
4. Visually inspect all welded connections for signs of cracks or deterioration.
5. Visually inspect all rope and chain for proper coverings and for signs of wear.
6. Visually inspect any wheels and or bearings.
7. Lubricate moving parts as appropriate.
8. Visually inspect all areas and surfaces that are seating and sliding surfaces for sharp edges or indentations that may injure users.
9. Visually inspect all ground anchor points and supports.
10. Visually inspect any anti slip coatings or strips. Recoat or replace if necessary.
11. Visually inspect paint, decorative coatings or decals, repaint or replace.(No lead based paints shall be used). If lead paint is suspected, paint shall be tested. If positive for lead paint notify the COR immediately and follow proper procedure to mitigate.
12. Visually inspect safety notices and placards. Replace if necessary.
13. Check for any unsafe pinch points or conditions.
14. Visually inspect ground area beneath equipment to ensure no sharp objects and that the ground surface cushioning is adequate for the piece of equipment.
15. If equipment is found to be in an unsafe condition, the equipment shall be removed from service and placed in a safe condition until repairs can be made.

Tools

SOCKET_SET-01 BARRIER_KIT SCREW_DRIVER WRENCH_SET PLIERS

This standard is intended for Child Care secured perimeter systems. This standard is intended to be used to ensure that the areas designated for child care are secured and maintained in a safe condition, there are no, conditions or situations that may cause injury or endanger the users or occupants when the area is used in accordance with its designed intent and/or the occupants follow the specific instructions placards or guidelines set forth for the use of the space or area. Prior to conducting these tasks, review all specifications and requirements specific to Child Care areas.

Special Instructions

1. If a condition is found to exist which is, unsafe, or may cause injury, immediate action must be taken to resolve the situation by securing the equipment or area from further use and notifying the COR / Child Care Operator.
2. Responsibility for correction of deficiencies may fall to others, such as the Child Care Provider, cleaning and lawn maintenance programs, COR or Child Care Coordinator. Findings shall be reported to the COR.
3. Be particularly alert to potential entrapment hazards in playground equipment.

Checkpoint

1. Visually inspect the perimeter barrier, fence or walls for any breaches.
2. Visually inspect perimeter gates/doors or hatches to ensure they are secured from the outside.
3. Visually inspect that all means of egress from inside are fully functional and open under the required opening pressure.
4. Visually inspect that all Fire life safety devices/systems are functional.
5. Visually inspect ground area for any trip hazards.
6. Verify that electronic security alarms/CCTV systems are functional.
7. Visually inspect for trees or shrubs that may be obstructing or overhanging area.
8. Visually inspect that all warning and notice signs/ placards are in place.

Tools

SCREW_DRIVER WRENCHES-01 LADDER-01 FLASHLIGHT

This standard is intended for Child Care areas. This standard is intended to be used to ensure that the areas designated for child care use are maintained in a safe condition, there are no conditions or situations that may cause injury or endanger the users or occupants when the area is used in accordance with its design intent and/or the occupants follow the specific instructions placards or guidelines set forth for the use of the space or area. Prior to conducting these tasks, review all specifications and requirements specific to Child Care space for example: HVAC systems (temperature, air flow, outside air), Domestic water (temperatures/pressures).

Special Instructions

1. If a condition is found to exist which is, unsafe, or may cause injury, immediate action must be taken to resolve the situation by securing the equipment or area from further use and notifying the COR/Child Care Operator.
2. Responsibility for correction of deficiencies may fall to others, such as the Child Care Provider, cleaning and lawn maintenance programs, COR or Child Care Coordinator. Findings shall be reported to the COR.
3. Be particularly alert to potential entrapment hazards in playground equipment.

Checkpoint

1. Visually inspect fire life safety devices/equipment. Fire extinguishers for location, mounting and inspection date. Fire sprinkler heads not obstructed. Smoke detectors and pull stations unobstructed and functional. Emergency lighting is functional. Fire exits unobstructed and door opening pressure is correct.
2. Verify any emergency notification devices (Duress alarm/Phones) are functional.
3. Verify any security alarm/CCTV systems are functional.
4. Visually inspect any lifts/elevators for proper operation.
5. Visually inspect all electrical system components. Electrical rooms are locked; panels secured and locked Switches and switch plates in proper working condition. Outlets properly covered where required and cover plates fastened securely.
6. Visually inspect floor drains, covers secured and traps primed.
7. Check all service doors/ hatches to ensure they are secured and locked.
8. Visually inspect all windows for proper operation. Ensure max opening limits are correct and that locking mechanism is functional.
9. Visually inspect doors for proper operation. Door closer tension is correct and doors latch and lock properly.
10. Visually inspect all glass panes in windows /doors for cracks and loose panes.
11. Verify the space HVAC systems are functioning properly and the air flows and temperatures are within those specified.
12. Verify domestic hot water temperature is within range with no scalding hazards.
13. Verify drinking fountains are working properly.
14. Verify all hot water piping or hot surfaces are properly insulated or out of reach.
15. Inspect floor surfaces for tears, rips, bubbling, wrinkling and/or other trip hazards.
16. Inspect cabinetry for any loose attachments, faulty hinges, loose screws, sharp edges and that cabinet doors close and latch securely and properly.
17. Visually inspect lighting for failed lamps.
18. Visually inspect ceiling tile for water damage, damaged or loose tile.
19. Visually inspect wall surfaces for loose wall coverings and/or areas with paint that is chipped or peeling. If lead paint is suspected, paint shall be tested. If positive for lead paint notify COR immediately and follow proper procedure to mitigate.

Tools

CLNG-SCB-01-03M

Scrubbing Machine - Battery or Propane Powered
Rev-3 | 2021-11-03

This standard applies to battery or propane powered scrubbing machines used for scrubbing hard floor surfaces such as marble, ceramic tile, and terrazzo in lobbies, corridors, and toilet rooms. Daily equipment care should be accomplished by operator in accordance with established procedures.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's maintenance instructions.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

- 1. Battery Powered Scrubbing Machine:
 - a. Inspect all connections for tightness.
 - b. Vacuum or blow out dust from cooling air passages.
 - c. Lubricate according to manufacturer's instructions.
 - d. Check and adjust, as needed, belts and/or chain drive.
 - e. Check brakes, clutch, wheels, tires, steering, and frame for wear and damage. Replace or repair according to manufacturer instructions.
 - g. Inspect lights, buzzers, controls, flashers and wiring for tight connections and proper operation.
 - h. Check engine starter battery, charge if necessary.
 - i. Start machine and observe operation. Check suction on the dirty scrub water pickup.
- 2. Propane Powered Scrubbing Machine:
 - a. Inspect fuel tank, connections, and lines for leaks and deterioration. Tighten or replace as required.
 - b. Check for oil leaks and change oil, air, and fuel filters.
 - c. Change points, radiator, plugs, and rotor button if applicable.
 - d. Lubricate according to manufacturer's recommendations.
 - e. Check and adjust, as needed, belts and/or chain drive.
 - f. Check brakes, clutch, wheels, tires, steering, and frame for wear and damage. Replace or repair according to manufacturer instructions.
 - g. Inspect lights, buzzers, controls, flashers and wiring for tight connections and proper operation.
 - h. Check engine starter battery, charge if necessary.
 - i. Start machine and observe operation. Check suction on the dirty scrub water pickup.

Tools

BELTS GAUGE_GAP-02 GAUGE_TIRE-01 OIL-02 SPARK_PLUG TOOL_BASIC TOOL_BREAK_ADJUST VACUUM-01
CONSULT_MSDS

This applies to gasoline powered, hand operated, and reel type snow blowers.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's maintenance instructions.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. Routine cleaning, lubrication, and the application of rust preventive compound to bare metal parts of the equipment prior to storing shall be accomplished by the operator.

Checkpoint

- 1. Change engine oil (Note: Oil should be changed at end of season prior to storing for summer). Review the Safety Data Sheets (SDS) for proper disposal of used oil. If appropriate, recycle oil at an authorized station. Contact the COR if you have any questions.
- 2. Service air and fuel filters.
- 3. Clean and gap or replace spark plug.
- 4. Inspect blades.
- 5. Inspect tires.
- 6. Inspect belts or chains. Lubricate as necessary. Place a light coat of oil on chains prior to summer layup.
- 7. Inspect unit, remove any debris from cooling air passages, and make other needed adjustments.
- 8. Check engine starter battery, charge if necessary.
- 9. Start machine and observe operation.
- 10. Clean up work area, and remove all debris.

Tools

LUBRICANT-02 FILTERS_FUEL_AIR TOOL_BASIC CONSULT_MSDS GAUGE_GAP-02 SPARK_PLUG

This applies to gasoline or gas powered type sweepers used in driveways, parking lots, sidewalks, etc. Daily lubrication should be accomplished by the operator.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's maintenance instructions.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

- 1. Change oil and filter every fifty operating hours.
- 2. Service air and fuel filters.
- 3. Inspect engine, clean cooling air passages.
- 4. Clean and gap, or change spark plug.
- 5. Check oil level in gear boxes.
- 6. Adjust tension and/or replace V-belts.
- 7. Adjust brakes, brushes, and operating mechanisms as recommended by the manufacturer's instructions.
- 8. Clean entire unit.
- 9. Clean up work area and remove all debris.

Tools

GAUGE_GAP-02	TOOL_BREAK_ADJUST	TOOL_BASIC	SPARK_PLUG	OIL-01	BELT_DRIVE
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This applies to large trash compactors set an the site to handle building waste. Daily equipment care should be accomplished by the operator in accordance with established procedures.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's maintenance instructions.
3. Review standard operating procedures for selection, care and use of respiratory protection.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Extend ram, open door, check guide block retainers, and fit of guide blocks (Ram should not score wall).
2. Check operation of safety switches to insure they have not been by-passed. Check switches, contactors, disconnects. Tighten connections.
3. Grease couplings.
4. Check oil. Change if required.
5. Clean strainer in tank.
6. Clean pump exterior, motor exterior, line, etc. Remove trash and debris from mechanical areas.
7. Clean inside and outside of unit.
8. Lubricate according to manufacturer's instructions.
9. Start machine and observe operation.

Tools

LUBRICANT-07 TOOL_BASIC EQUIP_CLEANING-03

This applies to heavy duty, tank type cleaners used for vacuum cleaning carpet in office areas. Daily equipment care should be accomplished by the operator in accordance with established procedures.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's maintenance instructions.
3. Review standard operating procedures for selection, care, and use of respiratory protection.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check interior of tank for damage.
2. Inspect screens, bags, bag frames and exhaust parts.
3. Clean and wash all bags and bag frames.
4. Vacuum or blow out dust from cooling air passages.
5. Service filter.
6. Lubricate according to manufacturer's instructions.
7. Start machine and observe operation.
8. Operate unit and check for vacuum leaks.

Tools

RESPIRATOR-02

VACUUM_PORTABLE

VACUUM_BAG

TOOL_BASIC

SEALING_COMPOUND

This standard card applies to the vacuum producer on centralized, permanently installed vacuum cleaning systems. The motor or driver will be serviced using the standard for motors. Tubing and outlets will be serviced on a service call basis.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's maintenance instructions.
3. Review standard operating procedures for selection, care, and use of respiratory protection.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Lubricate as required.
2. Check vibration collars and seals for leaks.
3. Check interior of tank for damage.
4. Inspect screens, bags, bag frames and exhaust parts.
5. Clean and wash all bags and bag frames.
6. Remove tags and restore power.
7. Operate unit and check for vacuum leaks.
8. Check vacuum gauge and controls.

Tools

TOOL_BASIC

SEALING_COMPOUND

VACUUM_BAG

VACUUM_PORTABLE

RESPIRATOR-02

CTRL-DDC-01-01M**BAS Server**
Rev-1 | 2021-11-08

Building automation system (BAS) computers Front Ends act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers for a multiple workstation network. Some Front Ends are connected to the internet where they are controlled or accessed by different users.

Special Instructions

1. This maintenance should be completed at the same time as the weekly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel. Microsoft issues regular security patches the second Tuesday of every month.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
6. This procedure may require additional assistance from the IT department or the controls contractor.
7. It is recommended to have a backup of the system prior to completing this procedure.

Checkpoint

1. Perform a system back up.
2. Using the Windows Update utility, install any security patches issued by Microsoft. WARNING: Installing these patches may require a system reboot.
3. Back up monthly trends.

Tools

None

Building automation system (BAS) computers Front Ends act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers for a multiple workstation network. Some Front Ends are connected to the internet where they are controlled or accessed by different users.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Obtain and review manufacturer's information for servicing, testing and operating.
2. Schedule maintenance with operating personnel.
3. Obtain and review manufacturer's information for servicing, testing and operating.
4. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
5. This procedure may require additional assistance from the IT department or the controls contractor.

Checkpoint

1. Review Windows System, Security, and Application Event Logs. Correct any critical errors denoted by a red circle with a white "X". Make note of and/or correct any warning notices denoted by a yellow triangle with a black "!".
WARNING: Correcting these errors may require a system reboot.
2. If system data backups are completed automatically, verify the backup ran on schedule and without errors. Perform a monthly manual backup, if backups are run manually.
3. Back up monthly trends.

Tools

None

Building automation system (BAS) computers Front Ends act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers for a multiple workstation network. Some Front Ends are connected to the internet where they are controlled or accessed by different users.

Special Instructions

1. This maintenance should be completed at the same time as the weekly, monthly and quarterly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. Obtain As Built diagrams of installation.
6. Computers and the electronic components of which they are comprised are highly susceptible to electrostatic discharge. Ensure personnel adequately ground themselves prior to working on electronic equipment to avoid equipment damage and/or loss of data.

Checkpoint

1. Clean keyboard with clean, dry, compressed air and/or cleaning kit.
2. Clean mouse with a cleaning kit.
3. Shut down the computer, clean internal enclosure.
4. Check all internal cables are inserted fully into the motherboard and devices.
5. Back up system weekly.
6. Defragment hard drive using computer operating system or third party software.
7. Update Front End software, check with software provider.
8. Laptop Workstations:
 - a. Windows 7
 - i. Open the battery power meter dialog box by clicking on the battery icon, then click "more power options".
 - ii. Verify that the top of the window does NOT say "Consider Replacing Your Battery". If the checkbox "Warn Me If My Battery May Need To Be Replaced" is visible, make sure that it is checked. This checkbox may not be visible with a healthy battery.
 - iii. Replace battery if indicated.
 - b) Other operating systems
 - i. Charge battery to 100%
 - ii. Unplug charger for 10 minutes.
 - iii. If the battery has lost more than 10% of its charge (charge shows 90% or less), replace battery.

Tools

ANTI_STATIC_KIT TOOL_CLEANING-04

Building automation system (BAS) computers Front Ends act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers for a multiple workstation network. Some Front Ends are connected to the internet where they are controlled or accessed by different users.

Special Instructions

1. This maintenance should be completed at the same time as the weekly and monthly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel. Microsoft issues regular security patches the second Tuesday of every month.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
6. This procedure may require additional assistance from the IT department or contractor or the controls contractor.
7. It is recommended to have a backup of the system prior to completing this procedure.

Checkpoint

1. Defragment the hard drive using the Windows defragmentation utility or third party software on the workstation or server. This can take up to several hours to complete depending on the size of the hard drive.
2. Perform disk cleanup Windows disk cleanup utility or third party software on the workstation or server
3. Replace the backup media used for system backups with new media as recommended by the backup device's manufacturer.
4. Laptop Workstations:
 - a. Plug laptop into GSA network and get software updates.
 - b. Contact local IT to ensure laptop is updated correctly.

Tools

BACKUP_DRIVE

Building automation system (BAS) workstations act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers in a multiple workstation network.

Special Instructions

1. This maintenance should be completed at the same time as the weekly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel. Microsoft issues regular security patches the second Tuesday of every month.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
6. This procedure may require additional assistance from the IT department or the controls contractor.
7. It is recommended to have a backup of the system prior to completing this procedure.

Checkpoint

1. Perform a system back up.
2. Using the Windows Update utility, install any security patches issued by Microsoft. WARNING: Installing these patches may require a system reboot.

Tools

None

Building automation system (BAS) workstations act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers in a multiple workstation network.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule maintenance with operating personnel.
3. Obtain and review manufacturer's information for servicing, testing and operating.
4. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
5. This procedure may require additional assistance from the IT department or the controls contractor.

Checkpoint

1. Review Windows System, Security, and Application Event Logs. Correct any critical errors denoted by a red circle with a white "X". Make note of and/or correct any warning notices denoted by a yellow triangle with a black "!".
WARNING: Correcting these errors may require a system reboot.

Tools

None

Building automation system (BAS) workstations act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers in a multiple workstation network.

Special Instructions

1. This maintenance should be completed at the same time as the weekly, monthly and quarterly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. Obtain As Built diagrams of installation.
6. Computers and the electronic components of which they are comprised are highly susceptible to electrostatic discharge. Ensure personnel adequately ground themselves prior to working on electronic equipment to avoid equipment damage and/or loss of data.

Checkpoint

1. Clean keyboard with clean, dry, compressed air and/or cleaning kit.
2. Clean mouse with a cleaning kit.
3. Shut down the computer, clean internal enclosure.
4. Check all internal cables are inserted fully into the motherboard and devices.
5. Backup system weekly.
6. Defragment hard drive using computer operating system or third party software.
7. Update front end software, check with software provider.
8. Perform computer virus protection scans, weekly
9. Perform computer software error scans, weekly.
10. Perform maintenance as suggested by manufacturer's instruction book.

Tools

TOOL_CLEANING-04 ANTI_STATIC_KIT

Building automation system (BAS) workstations act as an application server and/or a database server for building mechanical, electrical and safety systems. These computers are generally standalone desktop computers used as servers in a multiple workstation network.

Special Instructions

1. This maintenance should be completed at the same time as the weekly and monthly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Schedule maintenance with operating personnel. Microsoft issues regular security patches the second Tuesday of every month.
4. Obtain and review manufacturer's information for servicing, testing and operating.
5. This procedure must be completed by a user with sufficient privileges to perform administrative and maintenance tasks on the computer.
6. This procedure may require additional assistance from the IT department or contractor or the controls contractor.
7. It is recommended to have a backup of the system prior to completing this procedure.

Checkpoint

1. Defragment the hard drive using the Windows defragmentation utility or third party software on the workstation or server. This can take up to several hours to complete depending on the size of the hard drive.
2. Replace the backup media used for system backups with new media as recommended by the backup device's manufacturer.

Tools

BACKUP_DRIVE

This standard is for Ethernet networks that are an integral part of a Direct Digital (DDC) Building Automation System (BAS). High performance and clean network construction are essential for the efficient and optimal operation of building mechanical systems. A facility should maintain a comprehensive list in the building equipment inventory of all networking devices and network details.

Special Instructions

1. This maintenance should be completed at the same time as the quarterly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. Schedule network outages for the affected segment(s) of network(s).
5. This procedure may require the assistance of the controls contractor.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check wiring at each connection for loose wires, broken or cracked insulation, and un-insulated wire length.
2. Check physical condition of networking device. Clean if needed.
3. Check electrical power connections.
4. Check that device is located appropriately on the network.
5. Clean and inspect all firewalls, routers, switches, hubs, modems, and gateways. The inspection must include a verification of proper settings per the manufacturer's and controls system requirements.

Tools

AIR_CANNED VACUUM_CLEANER-01 TOOL_GROUP_B

This standard is for Ethernet networks that are an integral part of a Direct Digital (DDC) Building Automation System (BAS). High performance and clean network construction is essential for the efficient and optimal operation of building mechanical system

Special Instructions

2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
4. Obtain and understand control system diagrams.
5. This procedure may require the assistance of the controls contractor.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Audit all Ethernet connections to the system. Verify all Ethernet connections are plugged into properly assigned BAS hub, router, and switch ports. Record and correct any discrepancies.

Tools

None

This standard is for non-Ethernet and field level networks that are an integral part of a Direct Digital (DDC) Building Automation System (BAS). High performance and clean network construction are essential for the efficient and optimal operation of building mechanical systems. A facility should maintain a comprehensive list in the building equipment inventory of all networking devices and network details.

Special Instructions

1. This maintenance should be completed at the same time as the quarterly maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
4. Obtain and understand control system diagrams.
5. Schedule network outages for the affected segment(s) of network(s).
6. This procedure may require the assistance of the controls contractor.
7. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check wiring at each connection for loose wires, broken or cracked insulation, proper shield landing, stray strands of wire, and un-insulated wire length. There should be no more than 1/8" wire showing at the connection point. It is important to have a small amount of wire showing to ensure there is no insulation disrupting the connection.
2. Check physical condition of networking device. Clean if needed.
3. Check electrical power connections.
4. Check that device is located appropriately on the network.
5. Check that trunk terminators are installed as required by the manufacturer's network setup instructions.
6. Check network repeaters, isolators, extenders, or other signal amplifying equipment is installed at the appropriate location and is configured per manufacturer's network setup instructions.
7. Check voltage and/or resistance readings are consistent with manufacturer's specifications for network communication. Verify the signal is a clean square wave without clipped or peaked edges and has no A/C interference present. Correct conditions on the network as indicated by waveform.
8. Conduct a communications speed test per manufacturer's instructions to verify current network configuration will support optimum network speed without data loss.

Tools

VACUUM_CLEANER-01 TEST_KIT AIR_CANNED METER_VOLT_OHM SPARES TOOL_GROUP_B OSCILLOSCOPE

This standard is for non-Ethernet and field level networks that are an integral part of a Direct Digital (DDC) Building Automation System (BAS). High performance and clean network construction are essential for the efficient and optimal operation of building mechanical systems. A facility should maintain a comprehensive list in the building equipment inventory of all networking devices and network details.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. This procedure may require the assistance of the controls contractor.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Run and review building level and floor level network diagnostic reports. Any errors or excessive changes of value may be an indication of impending network or device failure. Consult with the manufacturer's instructions or your controls contractor to resolve these issues.

Tools

None

This standard is for field level panels that are an integral part of a Direct Digital Control (DDC) Building Automation System (BAS). A facility should maintain a comprehensive list in the building equipment inventory of all field panels.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. Schedule service outages for the areas controlled by the field panel.
5. This procedure may require the assistance of the controls contractor.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check wiring at each connection for loose wires, broken or cracked insulation, stray strands of wire, and un-insulated wire length. There should be no more than 1/8" wire showing at the connection point. It is important to have a small amount of wire showing to ensure there is no insulation disrupting the connection. Tighten any loose connections. Ensure any required high voltage barriers are in place and installed correctly.
2. Check physical condition of the device.
 - a. Shut off power to the unit.
 - b. Blow out the panel with clean, dry air.
 - c. Vacuum any remaining dust
 - d. Turn power back on to the unit.
3. Check electrical power connections including incoming line voltage.
4. Check voltages on the main board or backplane of the panel as applicable per manufacturer's specifications.
5. Check all fuses for evidence of heating or weakening.
6. Check output voltages from each analog output are correct across the full specified voltage range.
7. Check all binary inputs and outputs open and close with the appropriate signal.
8. Check all analog inputs work across the full range of designed input.
9. Check all point labels are correct and up to date, if applicable.
10. Check all local man-machine interface ports work correctly.
11. Check all plug connections in the panel to ensure the plugs are fully seated.
12. Replace the panel back up battery with a manufacturer approved replacement battery here and in the power supply UPS if provided.

Tools

SPARES AIR_CANNED AC-DC_SIGNAL_GENERATOR BATTERIES-01 SCREW_DRIVERS TEST_KIT METER_VOLT_OHM

This standard is for field level controllers that are an integral part of a Direct Digital Control (DDC) Building Automation System (BAS). A facility should maintain a comprehensive list in the building equipment inventory of all field panels.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. Schedule service outages for the areas controlled by the field panel.
5. This procedure may require the assistance of the controls contractor.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check wiring at each connection for loose wires, broken or cracked insulation, stray strands of wire, and un-insulated wire length. There should be no more than 1/8" wire showing at the connection point. It is important to have a small amount of wire showing to ensure there is no insulation disrupting the connection. Tighten any loose connections. Ensure any required high voltage barriers are in place and installed correctly.
2. Check physical condition of the device.
 - a. Shut off power to the unit.
 - b. Blow out the panel with clean, dry air.
 - c. Vacuum any remaining dust
 - d. Turn power back on to the unit.
3. Check electrical power connections including incoming line voltage.
4. Check all fuses for evidence of heating or weakening.
5. Check output voltages from each analog output are correct across the full specified voltage range.
6. Check all binary inputs and outputs open and close with the appropriate signal.
7. Check all analog input work across the full range of designed input.
8. Check all point labels are correct and up to date, if applicable.
9. Check all local man-machine interface ports work correctly.
10. Check all plug connections in the panel to ensure the plugs are fully seated.
11. Verify the controller is on the logical floor level network and has the proper address.

Tools

AC-DC_SIGNAL_GENERATOR AIR_CANNED SCREW_DRIVERS SPARES TEST_KIT METER_VOLT_OHM

This standard is for electronic sensors that are an integral part of a Direct Digital (DDC) Building Automation System (BAS) and that measure variables such as temperature, humidity and pressure. Accurate measurement and reporting of control system variables are essential for the efficient and optimal operation of building mechanical systems. A facility should maintain a comprehensive list in the building equipment inventory of all electronic sensors. It is recommended that the inventory of sensors be recorded as equipment attributes of the field controller.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments or performing calibration. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. Ensure that the equipment associated with the sensors being checked is operational as needed to verify/calibrate sensor.
5. This PM will typically require two (2) people; one to measure and calibrate and the other to read the sensor value on the BAS.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Typical Sensors that require calibration include, but are not limited to:

- a. Outside air temperature
- b. Air temperature (e.g., air handler supply air, mixed air, return air)
- c. Water temperature (e.g., chilled water, condenser water, hot water)
- d. Relative Humidity
- e. Enthalpy
- f. Space temperature
- g. Static pressure (e.g., supply air static pressure, building static pressure)
- h. Differential pressure
- i. Air flow
- j. Water flow
- k. Carbon monoxide
- l. Carbon dioxide
- m. Oxygen
- n. Refrigerants

Checkpoint

1. Check that sensor is mounted securely and free of obstructions
2. Check physical condition of sensor.
3. Check that sensor is located appropriately.
4. Check the voltage, amperage, or resistance of the sensor in accordance with manufacturer's specifications. If the device falls out of this range, adjust or replace the device.
5. Check that physical location is consistent with graphical representation on BAS front end.
6. If the sensor has an associated setpoint, verify and confirm that the setpoint is reasonable and consistent with values specified in building documentation (e.g., building operating manual, controls as-built drawings, commissioning reports, re-commissioning reports)
7. Compare BAS value with field measurement using a calibrated handheld device at least as accurate as the desired BAS accuracy
8. For calibration of CO and CO, and CO₂ and other chemical sensors use a gas canister to create high readings for both the handheld device and the BAS sensor.

- 9. Perform sensor calibration using the standard gas kit and set the alarm or correct digital reading of the sensor to match the calibration.
- 9. Record at least five (5) separate readings from the BAS and handheld device.
- 10. Where BAS readings are outside accuracy specified for sensor, either calibrate or replace sensor.

Tools

CO2_CANISTERS HANDHELD_MEASURE_DEV METER_VOLT_OHM PRESSURE_GAUGE CAL-KIT SENSOR

This standard is for alarmable points that are an integral part of a Direct Digital (DDC) Building Automation System (BAS) and operate or control equipment in the building automation system. Accurate alarming of these points is essential for the efficient and optimal operation of building mechanical systems. A facility should maintain a comprehensive list of all points with the alarm property currently enabled.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams.
4. Ensure that the equipment associated with the alarm point being checked is operational as needed to verify operation of the control device.
5. This PM will typically require two (2) people; one to observe actual operation of the equipment the device controls and the other to monitor and control the BAS.
6. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. Obtain a list of alarm points from BAS.
2. Examine the list for any omissions due to renovation, remodel, program changes, and newly added equipment. Make points alarmable as required.
3. Examine the list for points deemed as no longer required to be alarmable due to renovations, remodels, program changes, or deleted equipment. Disable or remove alarm properties as required.
4. Examine the list for any points that are known to give nuisance alarms. If the point should be alarmable, correct the condition causing excessive alarming or adjust the alarm set point as required.
5. Examine the list of alarms for priority level. Adjust point alarms to the appropriate priority level.

Tools

None

This standard is for all Wireless pneumatic/ electronic controls that sense space temperature. This standard is intended to be used to ensure that the system is operating as intended, there are no air leaks in the control air system and to ensure that the devices are operating optimally for those devices associated with operating equipment controlled by the Building Automation System.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams and identify zone thermostat locations.
4. Ensure that the equipment associated with the end device being checked is operational as needed to verify operation of the control device.
5. This PM will typically require one person.

Checkpoint

1. From the server, run the manufacturers system diagnostic software for the wireless system. This diagnostic shall produce a report of all functional aspects of the wireless system indicating faults that should be addressed in this maintenance.
2. Review all zone set points at the server.
3. Inspect thermostat installation; ensure mounting is correct, fastened secure and that the thermostat is not blocked by equipment generating heat or furniture blocking air circulation.
4. Remove thermostat cover and lightly blow away any accumulated dust with canned low pressure air.
5. Listen for air leaks.
6. Replace battery.

Tools

None

This standard is for all Wireless pneumatic/ electronic controls that sense space temperature. This standard is intended to be used to ensure that the system is operating as intended, there are no air leaks in the control air system and to ensure that the devices are operating optimally for those devices associated with operating equipment controlled by the Building Automation System.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
3. Obtain and understand control system diagrams and identify zone thermostat locations.
4. Ensure that the equipment associated with the end device being checked is operational as needed to verify operation of the control device.
5. This PM will typically require one person.

Checkpoint

1. From the server, run the manufacturers system diagnostic software for the wireless system. This diagnostic shall produce a report of all functional aspects of the wireless system indicating faults that should be addressed in this maintenance.
2. Review all zone set points at the server.

Tools

None

This standard is for all pneumatic and electronic controls that sense temperature, pressure, differential pressure, humidity, and flow or operate devices such as valves and dampers. This standard is intended to be used to ensure that the system is operating as intended, there are no air leaks in the control air system and to ensure that the devices are operating optimally for those devices associated with operating equipment controlled by the Building Automation System.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Read and understand manufacturer's instructions of each device before making adjustments or performing calibration. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any changes.
2. Obtain and understand control system diagrams.
3. Before calibrating or adjusting pneumatic controls, adjust the pressure of the main control air supplying pneumatic sensors, thermostats, and controllers to manufacturer's specifications.
4. Servicing of pneumatic air compressors and control air reducing stations shall be performed at the same time to prevent duplication of some service check points.

Typical Sensors and Actuators to be checked or calibrated:

- a. Outside air temperature
- b. Mixed air temperature
- c. Return air temperature
- d. Discharge or supply air temperature
- e. Coil face discharge air temperatures
- f. Chilled water supply/return temperature
- g. Condenser water supply/return temperature
- h. Heating water supply/return temperature
- i. Wet bulb temperature or relative humidity (RH) sensors
- j. Enthalpy Sensors
- k. Space temperature sensors
- l. Economizer and related dampers
- m. Cooling and heating coil valves
- n. Static pressure transmitters
- o. Differential pressure transmitters
- p. Air or water flow
- q. Terminal unit dampers and flows
- r. General dampers
- s. Transducers and relays

Checkpoint

1. Pneumatic:
 - a. Listen for audible leaks and feel for any escaping air at tubing joints and terminations. Reed devices (i.e., thermostats) may produce audible air escape at setpoint, but any air noise should completely disappear when setpoint is significantly below (for direct acting) or above (for reverse acting) the signal level.
 - b. Check Control Air system pressure, ensure air is clean and dry. Blowdown all moisture separators in system.
 - c. Check physical condition of sensor or end device. Clean devices per manufacturer's recommendations.
 - d. Ensure sensor, actuator and controllers are mounted correctly, securely and are free of any obstructions. Check unit and linkage for proper opening, closing, binding, and loose connections. Adjust linkage as required for full range of operation.
 - e. Check that physical location is consistent system controls diagrams.
 - f. Check set point of controls (temperature, humidity, or pressure) against original sequence of operation and control diagrams.

- g. Check the unit over its range of control (minimum, midpoint, and maximum). Using a squeeze bulb and a pressure gage, impose simulated conditions to activate controls and check operation through full range or range of motion.
- h. Check for control point cycling.
- i. Check for correct pressure differential on all two position controllers (on-off-open-closed).
- j. Check condition and action of primary elements in the controllers and actuators. These can be bi-metallic strips, and or sealed bellows with capillary tubing for remote sensing, diaphragm operated pneumatic actuators.
- k. Note the action of the controlling device (thermostats, humidistats, and pressure stats) which changes the action of the controlled device (motors, valves, dampers, etc.).
- l. Check units for proper closing and loose connections.
- m. Check the condition and the ability of humidity sensing control elements (hair, wood, leather, or similar substances) to read the moisture changes and their action on the control mechanism.
- n. Check resulting action of the pressure sensing primary control elements such as diaphragms, bellows, inverted bells, and similar devices when activated by air, water, or similar pressure. Check operation of all relays, pilot valves, and pressure regulators.
- o. Check the resulting action of electrically controlled devices such as for valves, dampers, solenoids
- p. Replace air filters at sensors, controllers, and thermostats.
- q. Check all external restrictors for proper size and to verify the restrictor is installed per controls drawings. Verify the restrictor is not clogged.
- r. Use test kits and specific manufacturer's instructions whenever possible. If possible, replace rather than rebuild a control installed in the system. Take removed control device to shop for repair.
- s. If the sensor is out of calibration, follow manufacturer's instructions for calibration. NOTE: Some manufacturer's sensors cannot be field calibrated and must calibrated at the factory. Have a spare sensor for replacement if necessary.
- t. Any device that cannot be field calibrated should be replaced with equipment spares and turned in to shop for repair or to be sent out for factory calibration as required.

2. Electronic:

- a. Check that sensor is mounted securely and free of obstructions
- b. Check physical condition of device. Clean if needed.
- c. Check electrical connections.
- d. Check that device is located appropriately
- e. Check that physical location is consistent with graphical representation on BAS front end
- f. Check voltage and/or resistance readings are consistent with manufacturer's specifications for the device. Calibrate or replace if necessary.
- g. Check the device through its range and ensure that the controlled equipment responds correctly to the given command.
- h. Compare BAS value with field observation, i.e., open or closed, start or stop, to ensure device operation.
- i. Any device that cannot be field verified should be repaired or replaced with equipment spares and either thrown away, turned in to shop for repair or be sent out for factory service as required.

Tools

SQUEEZE_BULB	TEST_KIT	METER_VOLT_OHM	SPARES_CONTROL	RESTRICTOR	PRESSURE_GAUGE	HANDHELD_TEMP_DEV
TOOL_GROUP_B	AIR_FILTER					

This standard applies the control air distribution system for the HVAC system.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall system and the operation of the building PRIOR to making any changes.
2. Schedule shutdown with operating personnel, as needed.
3. Obtain and understand control system diagrams.
4. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform the following procedures concurrent with this procedure:
 - a. HVAC-AIR-01-06M Air Dryer, Refrigerated or Regenerative Desiccant Type
 - b. HVAC-AIR-02-06M Air Compressor
2. Verify correct main line pressure leaving the main line air regulator. If the pressure is not correct adjust or repair the regulator
3. Check all main line filters for evidence of clogging, leaks, oil, or water.
 - a. Replace the filter. If evidence of oil contamination is present, find the source of contamination and correct.
 - b. If there is evidence of water contamination, drain the device. Verify the air dryer is not bypassed.
 - c. If the device leaks, repair or replace per manufacturer's instructions.
 - d. If the device is clogged, unclog or replace device per manufacturer's instructions.
4. Check all pneumatic tubing for tightness and damage. Repair or replace lines as necessary.

Tools

FILTER_CARTRIDGE-03 TOOL_BASIC

This standard applies to all single and dual input receiver controllers in the HVAC pneumatic control system. These devices direct control air to a device to control or actuate it.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall system and the operation of the building PRIOR to making any changes.
2. Schedule shutdown with operating personnel, as needed.
4. Obtain and understand control system diagrams.
5. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. It is recommended that a spare calibrated receiver controller be taken into the field during this procedure to replace any non-functioning controller.

Checkpoint

1. Check inline filter for oil and moisture. Replace as required.
2. Check all external restrictors for proper size, correct installation, and clogs. Replace as required.
3. Verify correct main line pressure at the receiver controller. If the pressure is not correct the receiver controller will not function properly. Find the cause of incorrect pressure and correct.
4. Verify the spring range of all attached end devices.
5. Verify the range of all attached sensors.
6. Check all receiver controller modules, bellows, and bleed ports as applicable for evidence of clogging, leaks, oil, or water.
 - a. If evidence of oil contamination is present, replace the device.
 - b. If evidence of water contamination replace the device with a spare. Let the device dry out and recalibrate. If the device won't calibrate, discard the device.
 - c. If the device leaks, repair or replace per manufacturer's instructions.
 - d. If the device is clogged, unclog or replace device per manufacturer's instructions.
7. Follow the manufacturer's instructions for calibrating the device.
8. Check all pneumatic tubing for tightness. Repair or replace lines as necessary.
9. Change input and/or setpoint and verify the device stabilizes (doesn't hunt) and reaches setpoint.

Tools

SQUEEZE_BULB FILTER_AIR RESTRICTORS THERMOMETER_DIGITAL-02 TOOL_BASIC PNEUMATIC_FITTINGSOSCILLOSCOPE

This standard card applies to all control air filters installed in control air systems that are not part of the basic air compressor or air drier configurations.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
3. Schedule shutdown with operating personnel, as needed.
4. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall system and the operation of the building PRIOR to making any changes.
5. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.

Checkpoint

1. Valve filter out of service.
2. Remove and discard old cartridge.
3. Clean inside of housing.
4. Install new cartridge.
5. Open inlet valve and check for leaks.
6. Open outlet valve.

Tools

TOOL_BASIC FILTER_CARTRIDGE-03

CYBER-V1

Cyber Hygiene CheckList
Rev-0 | 2023-04-01



Uses SA Alternate developed by MHoffman.

1 PEOPLE

1.1 Account management

Task	Guidance/Requirement	Resource(s)
BSN Console Account Management Process Confirmation KeePass Password Mgmt Enterprise Rollout to PBS-OFM	Shared accounts will be removed by GSA IT. Users will each need to use individual credentials. Per Facility confirm accounts are updated/current.	
BSN Application Account Management Process Confirmation	Provide and/or confirm latest information to Regional KeePass Administrator for the local BMC applications.	KeePass Password Mgmt Enterprise Rollout to PBS-OFM
BMC Device Account Management Process Confirmation	Provide and/or confirm latest information to Regional KeePass Administrator for the BMC devices.	

1.2 ServiceNow

Task	Guidance/Requirement	Resource(s)
Are Points of Contacts updated per building?	Confirm and/or update the building point of contact listed in ServiceNow. Work with Regional ServiceNow leads for any required updates.	https://sites.google.com/a/gsa.gov/smart_buildings/resources Please Note: To access the list of FM Building contacts below, the user must sign into ServiceNow before they click on the links or they will get an error.

2 PORTFOLIO

2.1 BSN Inventory & Software Maintenance Agreements (SMA)

Task	Guidance/Requirement	Resource(s)
Confirm data elements are provided for BSN Inventory within NCMMS, including Firmware information.	Confirm required data elements are listed per BSN device. Follow NCMMS Desk Guide (Rev 2.), "Requirements for IT Assets" for WO process.	Alignment on Data Elements for BMC Devices National CMMS Policy Desk Guide - Rev. 2 : 25-July-2022 [DRAFT]
Confirm existing Software Maintenance Agreements (SMA) on system and provide update to Regional SB Program Manager.	Provide documentation for SMAs to Regional Smart Buildings Program Manager. Regional SB Program Manager will provide to GSA IT.	

2.2 Contracts

2.2 Contracts

Task	Guidance/Requirement	Resource(s)
CyberSecurity Contract updates		

3 PERFORMANCE

3.1 Building Recovery

Task	Guidance/Requirement	Resource(s)
Confirm last Building Recovery Test	[New/Needed] Store documents for BR in NCMMS, similar to certification process. FM to confirm at least bi-annual (at least every other year completion of BR documents.	
Process posted for reporting cybersecurity incidents		https://insite.gsa.gov/employee-resources/information-technology/security-and-privacy/it-security/report-it-security-incidents-and-suspicious-activity-immediately?term=

3.2 Incident Recovery

Task	Guidance/Requirement	Resource(s)
Confirm last Incident Recovery Test	If not conducted within the last X, then schedule with Y.	Incident Response Material

3.3 Alarms

Task	Guidance/Requirement	Resource(s)
Alarm Routing	Review alarms designated for text delivery. Only send critical alarms via text to avoid overburdening the network with cluttered alarms.	

~~Demarc and IT Closets, enclosures, etc. (TBTC~~

Cleaning of IT Closets; Demarc-room items- (including cleanup of cables, labeling of cables		Per telecomm guide
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4. Review of EOL Equipment & Existing Vulnerabilities

Task	Guidance/Requirement	Resource(s)
Review any pending moderate vulnerabilities requiring patches		

Special Instructions

- 1. Checklist must be filled out on the Details Tab (Review Actions Section) of the Work Order Tracking (HSE) application.

Checkpoint

Tools

None

DOM-WTR-FLUSH-01A

Annual Domestic Water Testing and Flushing

Rev-0 | 2023-06-10

Federally owned facilities under the jurisdiction, custody, and control of the U.S. General Services Administration (GSA), including facilities operating under a GSA delegation of authority, must implement this guidance if they meet the following criteria, regardless of occupancy. Applies to:

Buildings over 50,000 sqft. with water booster pumps (both conditions must apply); or Buildings over six stories; Buildings with a Child Care Center (CCC) or health unit, or Buildings that include multiple housing units with one or more centralized potable water heater systems that serve two or more housing units.

Special Instructions

Checkpoint

Tools

None

Federally owned facilities under the jurisdiction, custody, and control of the U.S. General Services Administration (GSA), including facilities operating under a GSA delegation of authority, must implement this guidance if they meet the following criteria, regardless of occupancy. Applies to: Buildings over 50,000 sqft. with water booster pumps (both conditions must apply); or Buildings over six stories; Buildings that include multiple housing units with one or more centralized potable water heater systems that serve two or more housing units.

Special Instructions

1. Breathing legionella bacteria infected mist may cause legionaire's disease. For flushing activities, wear the proper PPE identified in the workplace hazard assessment conducted per OSHA 1910.132(d). When flushing, flushing flow rates shall be as high as possible but not so high they create mist formation, splashing, or flooding.

Checkpoint

Tools

None

Federally owned facilities under the jurisdiction, custody, and control of the U.S. General Services Administration (GSA), including facilities operating under a GSA delegation of authority, must implement this guidance if they meet the following criteria, regardless of occupancy. Applies to Buildings with a Child Care Center (CCC) or health unit.

Special Instructions

1. Breathing legionella bacteria infected mist may cause legionaire's disease. For flushing activities, wear the proper PPE identified in the workplace hazard assessment conducted per OSHA 1910.132(d). When flushing, flushing flow rates shall be as high as possible but not so high they create mist formation, splashing, or flooding.
2. If there has been a prolonged shutdown or reduced operation it is recommended that hot water heater(s) be flushed at a minimum one time prior to initiating weekly building wide flushing. Ensure water temperature set point at the water heater is at a minimum of 140F. The Occupational Safety and Health Administration (OSHA) prescribes storing water at 140°F (60° C) and delivering it at a minimum of 122°F (50° C) to all outlets.

Checkpoint

Tools

None

DOME-WTR-FLUSH-01-W

Weekly Domestic Water System Flushing
Rev-0 | 2022-01-01

COVID19 - In accordance with recommendations and guidance from the Centers for Disease Control (CDC), perform a systematic flushing of the domestic hot and cold water system(s) of the building where there has been no to low water use for prolonged shutdown or reduced operation. This job plan applies to faucets, drinking fountains, toilets, urinals, decorative fountains, water/ice machines, showers, drain traps and any other unique water outlets, fixtures, or appliances that are within the O&M scope. This job plan does not apply to tenant owned appliances. Note - This job plan has specific record keeping requirements. Use attached flushing log template to document flushing activities.

Special Instructions

1. If there has been a prolonged shutdown or reduced operation it is recommended that hot water heater(s) be flushed at a minimum one time prior to initiating weekly building wide flush. Ensure water temperature set point at the water heater is at a minimum of 140F. The Occupational Safety and Health Administration (OSHA) prescribes storing water at 140°F (60° C) and delivering it at a minimum of 122°F (50° C) to all outlets.
2. Start at fixtures (e.g. bathroom sinks, breakroom/kitchenette sinks, water/ice machines, drinking fountains, etc.) closest to the water service entry into the building and progress to the furthest point from the service line, such as starting at lower levels of the building and moving upwards toward the top floors.
3. Flush the maximum amount of fixtures at a time while still maintaining adequate pressure. Flushing may need to occur in segments (e.g., floors, specific portions of the building) due to facility size and water pressure. The more fixtures being flushed simultaneously, the slower fresh water will flow inside the building due to pressure loss from multiple water fixtures being open.
4. Turn off or bypass the automatic motion fixtures where possible (e.g. putting painters tape over the sensor). These are the hands-free faucet functions. This will ensure the fixture continuously flushes and does not shut off every few seconds. Ensure to turn back on automatic motion fixtures after flushing is completed.
5. During flushing, operate all valves in the fully open position so that any particulate matter can be flushed through. Pay close attention to float-operated or other restrictive valves which need to be manually opened to clear particulates and prevent fouling of the valves.
6. Bypass (remove) any point-of-use filtration systems during flushings.
7. Recordkeeping - Document weekly flush/purge using the provided log template. Complete, date and re-attach log to work order.

Checkpoint

Tools

None

DWS-04-AQUALOGIX-01Y**Water Softener**
Rev-1 | 2019-01-01

This standard applies to the three most common types of serviceable water softeners: Lime, zeolite, and anthracite.

Special Instructions**Checkpoint**

Open and inspect main controls enclosure for any signs of water intrusion.

Check control panel power supplies for proper output voltage. The DC power supplies should have the output calibrated if offset is more than +/- 5% of rated output.

Inspect the FLECK motor and piston (behind black housing) to ensure no damage to the cam or the piston. Insect piston stem (where it enters the valve head) to make sure no sign of water.

Check the regeneration curves for each softener and ensure the proper results have been acquired. If not, close examination of the brine valve, piping, and injectors may be required. Brine injector and valve assembly (FLECK) may need to be disassembled and cleaned, possible integral screen filter replaced.

Inspect brine tank inlet using a flashlight, or remove the brine intake assembly, ensure free of debris and buildup.

Remove, clean and inspect Conductivity probes. Calibrate probes with 500uS and 1500uS standard solutions, using JUMO programing interface cable and software. Complete in process calibration if required (conductivity offset). Confirm temperature compensated conductivity output is correct using the JUMO programing interface cable and software. Complete in process calibration if required (Temperature compensation). Replace failed probes as required.

Verify System Operational

Side Stream Filter

Manually engage Soft Water Feed Valve to ensure it is free and clear. Move to a 50% position and visually ensure the actuator (motor) is moving back to its original home position. Immediately after the valve is manually engaged and is observed moving back to its home position, check main display to ensure the flow meter is registering a value, and flow meter is not obstructed.

If no value can be seen immediately after manually moving the valve, the meter may need to be removed and inspected for debris.

Manually engage Separate Source Backwash Valve to ensure it is free and clear. Move to a 50% position and visually ensure the actuator (motor) is moving back to its home (original) position. Immediately after the valve is manually engaged and is observed moving back to its home position, check main display to ensure the flow meter is registering a value, and flow meter is not obstructed.

If no value can be seen immediately after manually moving the valve, the meter may need to be removed and inspected for debris.

Inspect the FLECK motor and piston (behind black housing) to ensure no damage to the cam or the piston. Insect piston stem (where it enters the valve head) to make sure no sign of water.

Clean side stream pump cooling vents or cooling air intakes

Tools

GRAVEL_SAND_CHAR TOOL_GROUP_C RESPIRATOR-01

ELEC-ATS-01-01M**Automatic Transfer Switches**

Rev-1 | 2022-01-01

This standard card applies to Automatic Transfer Switches as part of the building electrical system and the emergency power system found in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform Monthly inspections following NFPA 110 (most recent edition at time of solicitation) procedures, outlined in Chapter 8 Operational Inspection and Testing.
2. All tested parameters and setpoints shall be within manufacturer's tolerances.

Tools

None

This standard card applies to Automatic Transfer Switches as part of the building electrical system and the emergency power system found in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform monthly visual inspections concurrently with annual maintenance and testing.
3. Schedule and coordinate work with operating personnel.
4. Obtain and review manufacturer's instructions.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Prior to annual maintenance testing perform thermographic survey, following procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
7. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Checkpoint

1. Perform electrical tests procedures for Emergency Systems, Automatic Transfer Switches outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to thermographic survey of the metal enclosed busways as part of the electrical system in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Checkpoint

Tools

None

This standard card applies to maintenance tests of metal enclosed busways as part of the electrical system in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing.
3. Schedule and coordinate work with operating personnel.
4. Obtain and review manufacturer's instructions.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Indoor Metal Enclosed Busways:
 - a. Perform electrical tests for Metal-Enclosed Busways following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
2. Outdoor bus structures:
 - a. Perform electrical tests for Outdoor Bus Structures following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
3. Test results shall conform to test values within the specified sections and or corresponding data tables within the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of capacitors as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.
3. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Checkpoint

Tools

None

This standard applies to maintenance tests of capacitors as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation
3. Schedule outage with operating personnel.
4. Obtain and review manufacturer's instruction (including time current characteristic curve).
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of capacitors as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of capacitors as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Schedule outage with operating personnel.
4. Obtain and review manufacturer's instruction (including time current characteristic curve).
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Capacitors and Reactors, Reactors Dry-Type, outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspection of liquid-filled reactors.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition.
2. Visually inspect anchorage, alignment, grounding, and required clearances.
3. Visually inspect for oil leaks.
4. All parameters shall be within manufacturer's tolerances.

Tools

None

This standard applies to oil-filled reactors.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Perform monthly visual inspections concurrently with annual mechanical maintenance testing.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
3. Perform visual and mechanical inspection procedures for Capacitors, Reactors, Reactors - Liquid Filled outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
4. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data..
5. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance tests of liquid-filled reactors as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Schedule outage with operating personnel.
4. Obtain and review manufacturer's instruction (including time current characteristic curve).
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Capacitors and Reactors, Reactors - Liquid Filled outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of low voltage cables as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of low voltage cables 600 volt maximum as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Schedule outage with operating personnel.
4. Obtain and review manufacturer's instruction (including time current characteristic curve).
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Cables, Low-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of medium voltage cables as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of medium voltage cables as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Schedule outage with operating personnel.
4. Obtain and review manufacturer's instruction (including time current characteristic curve).
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Cables, Medium and High Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to the thermographic survey of low voltage (1,000 Volts or less) Air Insulated-Case/Molded-Case circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of low voltage (1,000 Volts or less) Air Insulated-Case/Molded-Case circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Circuit Breakers, Air, Insulated-Case/Molded-Case outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of low voltage (1,000 Volts or less) power circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule thermographic survey with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of low voltage (1,000 Volts or less) power circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical tests procedures for Circuit Breakers, Air, Low-Voltage Power outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to thermographic survey of medium voltage (1,000 to 100,000 volts) Air Insulated circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of medium voltage (1,000 to 100,000 volts) Air Insulated circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical test procedures for Circuit Breakers, Air, Medium- Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspection of medium voltage (1,000 to 100,000 volts) Oil Insulated circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition
2. Visually inspect anchorage, alignment, grounding, and required clearances.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks and bushings.
5. Verify that breather vents are clear.
6. All parameters shall be within manufacturer's tolerances.

Tools

None

This standard applies to medium voltage (1,000 to 100,000 volts) Oil Insulated circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Perform monthly visual inspections concurrently with annual mechanical maintenance testing.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
3. Perform visual and mechanical inspections procedures for Circuit Breakers, Oil, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data.
5. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance tests of medium voltage (1,000 to 100,000 volts) Oil Insulated circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule outage with operating personnel.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Circuit Breakers, Oil, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of medium voltage (1,000 to 100,000 volts) Oil Insulated circuit breakers.

Special Instructions

1. Perform these tests only if equipment is safely accessible for the survey.
2. Schedule thermographic survey with operating personnel.
3. Obtain and review manufacturer's instruction (including time current characteristic curve).

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of medium voltage (1,000 to 100,000 volts) vacuum circuit breakers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Circuit Breakers, Vacuum, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

Clocks connected to central system are either digital, analog (pointer type), calendar clocks or special design clocks. These clocks times are synchronized by a central time controller module usually installed within the building. These controllers often incorporate a feature that will synchronize the clocks through a wireless or wired connection with a regional office.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This work should be done in spring and fall when time is changed from standard to daylight savings time in Spring and back one hour in Fall.

Checkpoint

1. Clean dirt and dust from interior and exterior of cabinet.
2. Adjust relays, check transmission of time signal.
3. Tighten contacts and terminal screws.
4. Burnish contacts if necessary.
5. Perform work suggested by manufacturer's instruction book.

Tools

TOOL_BASIC

This standard card applies to visual inspections for Direct Current Battery Systems using lead acid as the electrolyte. These can be flooded lead acid or valve-regulated lead acid types.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Wear personal protective equipment to avoid contact with acid droplets or fumes, including face shield, apron, gloves and respiratory protection.
5. Respiratory protection shall consist of an air purifying respirator fitted with cartridges for acid fumes, as minimum. Staff must be previously trained in respirator use and enrolled in respirator program before use.
6. Schedule and coordinate work with operating personnel.
7. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform visual and mechanical inspection procedures for Direct-Current Systems, Chargers, outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to visual inspections for Direct Current Battery Systems using lead acid as the electrolyte. These can be flooded lead acid or valve-regulated lead acid types.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Wear personal protective equipment to avoid contact with acid droplets or fumes, including face shield, apron, gloves and respiratory protection.
5. Respiratory protection shall consist of an air purifying respirator filter with cartridges for acid fumes, at minimum. Staff must be previously trained in respirator use and enrolled in respiratory program before use.
6. Perform monthly visual inspections concurrently with annual maintenance and testing.
7. Schedule and coordinate work with operating personnel.
8. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety for test procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
9. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform electrical tests following procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation. Compare and analyze inspection/test results in accordance with respective test values provided within the MTS.

Tools

TOOL_GROUP_B RESPIRATOR-07 GLOVES_SHIELD_APRON

This standard is for primary type (1.5-volt dry cell, one time full discharge, non-rechargeable) batteries used for switch gear, control circuits, fire alarm systems, sprinkler supervisory systems and transformer supervisory systems where the source of DC power must be reliable.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The maximum load for a No. 6 primary battery shall not be more than two amperes per cell.

Checkpoint

1. Conduct a load voltage test by measuring and recording the voltage when applying a one test ohm load on an individual (5-volt) cell.
2. Replace the battery when the voltage is reduced below one volt.
3. Replace a 6-volt battery assembly when a test load of four ohms reduces the voltage below four volts.
4. Clean terminals and tighten connections and check condition of wiring.

Tools

CLOTHS-02 TEST_LOAD BATTERIES-02 TOOL_GROUP_B

This standard applies to nickel cadmium type batteries that are used for switch gear, control circuits, fire alarm systems, sprinkler supervisory systems and transformer supervisory systems that require a reliable source of DC power.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Never smoke or carry an open flame in or near the battery room or area.
5. Use caution in handling the electrolyte, it is harmful to the skin and clothing. Wear personal protective equipment to avoid contact with acid droplets or fumes, including face shield, apron, gloves and respiratory protection.
6. Respiratory protection shall consist of an air purifying respirator fitted with cartridges for acid fumes, at minimum. Staff must be previously trained in respirator use and enrolled in respiratory program before use.
7. Never remove any connecting cables or straps while charger is on or there is a possibility of a load being on the batteries.
8. Wear acid resistant apron, gloves and plastic face shield when handling electrolyte.
9. Review standard operating procedures for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.

Checkpoint

1. With the battery charger disconnected, measure voltage for entire battery and record results.
2. With the battery charger disconnected, conduct a load voltage test by measuring and recording the voltage of the entire battery with the battery carrying the full connected load. An artificial load equal to the full load connected to the battery may be used for this test. Under load, the battery voltage shall not fall below 1.42-volts per cell nominal. Measure cells individually if possible.
3. With the battery disconnected, conduct a discharge test by following the manufacturer's instructions and measuring and recording the voltage of each cell while load testing the batteries for a minimum of 30 minutes. An artificial load equal to the full load connected to the battery may be used for this test. The voltage shall not fall below the levels specified.
4. With the battery charger connected and the batteries fully charged, conduct a charger test by placing an amp meter in series with the battery under charge. The charging current shall be the manufacturer's recommendation for the type of battery used.
5. Add distilled water to bring the electrolyte up to the proper level, and record the amount used. Abnormal use of water indicates overcharging.
6. Wipe salt and dirt accumulation from batteries with disposable cloth taking care not to drop foreign matter into batteries.
7. Clean terminals, tighten connections and check condition of wiring.
8. Check battery cable and battery supports for deterioration, paying special attention to wooden members.
9. Replace battery every three years. Dispose of old battery in accordance with all applicable Federal, state, and local regulations.

Tools

EYE_WASH APRON_GLOVE ELECTROLYTE TOOL_GROUP_B CLOTHS-02 RESPIRATOR-08

This standard card applies to visual inspections for Direct Current Battery System Chargers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule and coordinate work with operating personnel.
5. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform visual and mechanical inspection procedures for Direct-Current Systems, Chargers, outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to visual inspections for Direct Current Battery System Chargers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Perform monthly visual inspections concurrently with annual maintenance and testing.
5. Schedule and coordinate work with operating personnel.
6. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
7. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform electrical test procedures for Direct-Current Systems, Chargers, outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to emergency generators as part of the building electrical system and the emergency power system found in the building. This standard does not address the prime mover of the generator.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule and coordinate work with operating personnel.
5. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform visual and mechanical inspection procedures for Emergency Systems, Engine Generators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
2. Perform operational test in accordance with NFPA 110.

Tools

None

This standard card applies to emergency generators as part of the building electrical system and the emergency power system found in the building. This standard does not address the prime mover of the generator.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Perform monthly visual inspections concurrently with annual maintenance and testing.
5. Schedule and coordinate work with operating personnel.
6. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
7. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform electrical and mechanical test procedures for Emergency Systems, Engine Generators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to all Electric Vehicle Charging Stations consisting of a mounted 110-120VAC outlet where no cord or connector is provided for vehicle mating.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check that mounting bolts and fixtures are tight and unbroken. Remove corrosion and repaint as needed.
2. Clean and inspect receptacle for damage and replace broken receptacles. Inspect that receptacle hold inserted plugs firmly. Replace receptacle if accidental disengagement of the plug occurs.
3. Inspect that receptacle hold inserted plugs firmly. Replace receptacle if accidental disengagement of the plug occurs.
4. Verify Voltage at correct voltage is available against neutral and to ground. Verify neutral to ground values.
5. Test GFCI trip and reset functions to break and remake power to the receptacle.

Tools

None

This standard applies to all Electric Vehicle Charging Stations consisting of a mounted 220-240VAC power supply consisting of a cord and connector used to insert into the vehicle. Multiple cords may be built into a single mount.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Cords shall be cleaned and inspected for damage. Damaged chords shall be immediately replaced.
2. Charging connectors shall be cleaned and inspected. Damaged connectors shall be replaced. Blackened or pitted connectors are evidence of a loose fitting connector. These connectors shall be immediately replaced.
3. Charging system cord strain relief systems shall be inspected. Damages strain relief systems shall be immediately replaced, and electrical connections inspected for damage and tightness.
4. Check that mounting bolts and fixtures are tight and unbroken. Remove corrosion and repaint as needed.

Tools

None

This standard applies to all Electric Vehicle Charging Stations consisting of a mounted 220-240VAC power supply consisting of a cord and connector used to insert into the vehicle. Multiple cords may be built into a single mount.

Special Instructions

2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.

Checkpoint

4. Check that mounting bolts and fixtures are tight and unbroken. Remove corrosion and repaint as needed.
1. Cords shall be cleaned and inspected for damage. Damaged chords shall be immediately replaced.
2. Charging connectors shall be cleaned and inspected. Damaged connectors shall be replaced. Blackened or pitted connectors are evidence of a loose fitting connector. These connectors shall be immediately replaced.
3. Charging system cord strain relief systems shall be inspected. Damages strain relief systems shall be immediately replaced, and electrical connections inspected for damage and tightness.

Tools

None

This standard applies to Building Grounding Systems as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Schedule grounding system inspections with operating personnel.
2. Verify ground system is in compliance with ANSI/NFPA 70 National Electrical Code, Article 250
3. Inspect physical and mechanical condition.
4. Inspect bolted electrical connections using low resistance ohmmeter or calibrated torque wrench.
5. Inspect anchorage.
6. Bolt torque values for electrical connections shall correspond to OEM instructions or parameters outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Tools

None

This standard applies to Building Grounding Systems as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedures outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual visual inspections concurrently with 3-year electrical maintenance and testing.

Checkpoint

1. Perform electrical test procedures for Grounding Systems outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Instrument Transformers located within the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Perform these tests only if instrument transformers are easily accessible for survey.
4. Schedule thermographic survey with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of Instrument Transformers located within the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical test procedures for Instrument Transformers outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to dimmers and control panels, both at unit and remotely located, used to control light levels of general and special lights in auditoriums, large conference rooms, etc.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Dimmer Unit:
 - a. Remove necessary access covers and panels.
 - b. Check supply voltage.
 - c. Tighten all connections to main breaker, sub breakers, contactors, etc.
 - d. Check operation of contactors, clean and adjust as necessary.
 - e. Check operation of drive motor and drive mechanism. Lubricate as necessary.
 - f. Check for discoloration and overheating on rheostat and printed circuit board.
 - g. Check all wiring, including control wiring for deterioration, overheating, etc.
 - h. Clean interior of housing.
 - i. Clean rheostat contact surface.
 - j. Check dimmer rating against actual load.
2. Control Unit (Local and Remote):
 - a. Remove, disassemble and clean slide bars.
 - b. Check indicator lamps, replace as necessary.
 - c. Clean interior of unit.
 - d. Check all connection to slide bars, terminal strip switches, etc.
 - e. Check calibration of voltage meters.
 - f. Lubricate control buttons, slide bars and switches.
 - g. Reassemble unit.
3. Test operation of dimmer throughout full range noting light fluttering or level difference (required for each control unit).
4. If applicable, check time clock, programming and time schedules.
5. If applicable, check each relay form the BAS interface via BAS front end.
6. Replace covers on dimmer unit after making any adjustments.
7. Clean exterior of units.

Tools

MAT_CLEANING-05 MULTIMETER TOOL_BASIC AMP_METER LUBRICANT-07

This standard card applies to the group replacement of standard fluorescent lamps in office areas of a single building. Work performed by a two-person crew. Time is allowed for disassembly, washing, reassembly, and replacement of lamps in each fixture. For types of fixtures that cannot be disassembled, time is allowed to clean in place.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. This standard requires that the fluorescent light fixtures be washed every 5 years. However, since the expected life of the fluorescent light tubes are in excess of five years, the tubes should be replaced only every other washing. The existing tubes should be reused on the interim washing.
6. Turn off branch circuit at panel or individual light switch in room as appropriate. Do not attempt to wash energized lights.
7. Follow manufacturer's instructions on specular reflectors and specular parabolic lens (diffusers). Use gloves during handling because fingerprints are difficult to remove. Do not use industrial detergents as they will damage these items.
8. Ensure that the replacement lamps are of the same type and color temperature as removed from the fixture. T-12 and T-8 lamps require different ballasts. Using the wrong lamp in a fixture may cause the lighting system to not function properly and may lead to premature failure. Wherever possible, use energy efficient T-8 lamps and electronic ballasts.
9. If the ballast requires replacement, use only T-8 lamps and electronic ballasts as replacements. Replace ballasts in the entire fixture for uniform lighting appearance.
10. The disposal of old lamps may be regulated by state or local governments, depending on the number of lamps removed. Lamps should be disposed of as universal waste in accordance with local, state requirements. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. PCB fluorescent light ballasts must be removed and disposed of in accordance with 40 CFR 761, and state and local regulations.

Checkpoint

1. Disassemble fixture. Wash all removable parts with warm water and a mild detergent. Rinse and allow to drip dry.
2. Damp wipe remaining body of fixture in place.
3. Remove old fluorescent lamps and install new lamps.
4. Test light fixtures. Replace starters and ballasts where necessary. Note and report any needed electrical repairs.
5. Reassemble all removable parts to fixture.
6. Clean up area and remove any trash.

Tools

LADDER-04 TOOL_BASIC MAT_CLEANING-02

This standard card applies to the group replacement of standard fluorescent lamps in office areas of a single building. Work performed by a two-person crew. Fixture height is such that the work can be accomplished with the use of an eight foot stepladder. ♦Group relamping on a more frequent basis than once every five years must be justified and request for some submitted to Central Office through the regional office for approval.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Turn off branch circuit at panel or individual light switch in room as appropriate.
6. The disposal of old lamps may be regulated by state or local governments, depending on the number of lamps removed. Lamps should be disposed of as universal waste in accordance with local, state requirements. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
7. PCB fluorescent light ballasts must be removed and disposed of in accordance with 40 CFR 761, and state and local regulations.
8. Ensure that the replacement lamps are of the same type and color temperature as removed from the fixture. T-12 and T-8 lamps require different ballasts. Using the wrong lamp in a fixture may cause the lighting system to not function properly and may lead to premature failure. Wherever possible, use energy efficient T-8 lamps and electronic ballasts.
9. If the ballast requires replacement, use only T-8 lamps and electronic ballasts as replacements. Replace ballasts in the entire fixture for uniform lighting appearance.

Checkpoint

1. Remove louver or diffuser as necessary.
2. Remove all fluorescent lamps and install new lamps.
3. Test light fixture. Replace starters or ballasts where necessary.
4. Note and report any needed electrical repairs.
5. Reassemble louver or diffuser as required.
6. Clean up area and remove any trash.

Tools

TOOL_BASIC LADDER-04

This standard card applies to special lighting fixtures such as found in lobbies, porticos, court rooms, and auditoriums, and for fixtures above 12 feet.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Clean fixture thoroughly.
2. Check all sockets, replace as needed.
3. Inspect anchors or anchoring device, tighten as needed.
4. Examine fixture glass, side panels, diffusers, etc., for cracks, breaks, etc. Replace if necessary.
5. If group relamping is due, change all lamps; otherwise, replace only those that are burned out.
6. Check operation.
7. Clean up work area and remove all debris.

Tools

LADDER-04 TOOL_BASIC

This standard card applies to parking lot, street, loading dock, and perimeter lighting, and provides for group relamping and maintenance of such fixtures outside the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Open and tag switch.
2. Remove old lamp and clean fixture including reflector, refractor, and globes.
3. Inspect condition of wiring, contacts, terminals, and sockets. Look for evidence of overheating.
4. Install new lamp and assemble fixture checking gaskets for proper seat.
5. Test operation of automatic switches.
6. Inspect lamp standards and mounting devices.
7. Clean up work area and remove all trash.

Tools

LADDER-04 TOOL_BASIC

This standard card applies to parking lot, street, loading dock, and perimeter lighting, and provides for group relamping and maintenance of such fixtures outside the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Open and tag switch.
2. Remove old lamp and clean fixture including reflector, refractor, and globes.
3. Inspect condition of wiring, contacts, terminals, and sockets. Look for evidence of overheating.
4. Install new lamp and assemble checking gaskets for proper seat.
5. Test operation of automatic switches.
6. Inspect lamp standards and mounting devices.
7. Clean up work area and remove all trash.

Tools

TOOL_BASIC LADDER-04

This standard card applies to both fixed and portable spotlights used in the lighting and highlighting of performers and stage areas in auditoriums and large conference rooms.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Keep record of hour usage of spotlight bulb, noting its estimated life and date of installation.

Checkpoint

1. Check power supply cord and connection to plug.
2. Remove necessary access panels.
3. Check connection at, and operation of switch. Lubricate as needed.
4. Clean lens and reflector.
5. Lubricate and check operation of all slide rods, adjust rods and linkage.
6. Check bulb and socket, clean or change as required.
7. Clean housing interior.
8. Clean and check operation of color slides.
9. Replace removed access panels.
10. Lubricate wheels and pivot points as required.
11. Check mounting supports.
12. Clean exterior of unit.
13. Test.
14. Clean up work area.

Tools

TOOL_GROUP_A LADDER-04 LUBRICANT-07

This standard card applies to closed system emergency lights and Exit lights, inclusive of fluorescent ballasts that contain battery backup.

Special Instructions

- 1. In addition to the procedures outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instruction(s) shall be strictly adhered to.
- 2. Schedule and coordinate work with operating personnel.
- 3. Obtain and review manufacturer's instructions and NFPA 101 paragraph 7.9.3 and 7.10.9

Checkpoint

- 1. Inspect for structural defects and deposits.
- 2. Push test buttons for 30 seconds and observe light operation. (See manufacturer's instructions.) Illumination shall be undiminished for the duration of the test.
- 3. Check exit lights for proper arrow direction.

Tools

EQUIP_CLEANING-03	APRON_GLOVE	GREASE_SILICONE	HYDROMETER-01	TOOL_GROUP_B	VOLT_METER
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This standard card applies to closed system emergency lights and Exit lights, inclusive of fluorescent ballasts that contain battery backup.

Special Instructions

- 1. In addition to the procedures outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
- 2. Schedule and coordinate work with operating personnel.
- 3. Obtain and review manufacturer's instructions and NFPA 101 paragraph 7.9.3 and 7.10.9

Checkpoint

- 4. Inspect for structural defects and deposits.
- 5. Remove normal power for 90 minutes and observe light operation. (See manufacturer's instructions.) See NFPA 101 paragraph 7.9.2 Illumination shall be undiminished for the duration of the test.
- 6. Check exit lights for proper arrow direction.

Tools

APRON_GLOVE	EQUIP_CLEANING-03	GREASE_SILICONE	HYDROMETER-01	TOOL_GROUP_B	VOLT_METER
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This standard card applies to closed system emergency lights and exit lights, inclusive of fluorescent ballasts that contain battery backup.

Special Instructions

- 1. In addition to the procedures outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
- 2. Schedule and coordinate work with operating personnel.
- 3. Obtain and review manufacturer's instructions.
- 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 5. Use acid-resistant gloves, apron and plastic face shield.
- 6. Do not spark battery terminals or smoke while performing maintenance.

Checkpoint

- 1. Inspect for structural defects and deposits.
- 2. Clean off corrosion deposits and apply silicone grease to terminals.
- 3. Push test buttons and observe light operation. (See manufacturer's instructions.)
- 4. Check vent holes.
- 5. Clean exterior with dry cloth.
- 6. Disconnect power to check operation. The battery shall maintain the light output for 5 hours. If it does not, it needs to be replaced.
- 7. Record battery cell voltage. Replace cells below manufacturer's recommended low level.
- 8. If unit has to be discarded, review the Safety Data Sheet (SDS) for proper disposal of battery and electrolyte.
- 9. Clean up work area.
- 10. For exit lights check for proper arrow direction.

Tools

GREASE_SILICONE	EQUIP_CLEANING-03	HYDROMETER-01	TOOL_GROUP_B	VOLT_METER	APRON_GLOVE
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This standard card applies to wet cell emergency lights and Exit lights.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, NFPA 101.7.9 montly and annual tests shall be conducted and corrective actions taken. Additionally, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Do not spark battery terminals or smoke while performing maintenance.
6. Use acid-resistant gloves, apron and plastic face shield.
7. Consult the Safety Data Sheet (SDS) for the electrolyte and proper personal protective equipment (PPE).
8. Review standard operating procedures for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
9. When practicable, remove light units to shop area for inspection and maintenance.
10. Refer to the latest editions at time of contract solicitation of the American National Standards Institute/National Fire Protection Associate (ANSI/NFPA) publication 70B, Electrical Equipment Maintenance and the International National Electrical Testing Association publication, Maintenance Testing Specifications, as applicable.

Checkpoint

1. Inspect for structural defects and deposits.
2. Clean off corrosion deposits and apply silicone grease to terminals.
3. Inspect battery water level and take specific gravity readings. If reading is less than specified by battery manufacturer, the battery should be replaced with a freshly charged one. The old battery should be charged and tested before discarding. Review the MSDS for proper disposal of battery and electrolyte.
4. Add distilled water to raise electrolyte to proper level.
5. Push test buttons and observe light operation. (See manufacturer's instructions.)
6. Check vent holes.
7. Clean exterior with dry cloth, paint if necessary with an appropriate type and color paint. Caution: Do not paint over any indicator lights.
8. Unplug unit to check operation. The battery should maintain the light output for 90 minutes..
9. For exit lights check for proper arrow direction.
10. Clean up work area.

Tools

TOOL_GROUP_B	EYE_WASH	EQUIP_CLEANING-03	GREASE_SILICONE	HYDROMETER-01	WATER	APRON_GLOVE
BATTERY_DISPOSAL						

This standard applies to thermographic survey of Low Voltage (1,000 Volts or less) Motor Starters as part of the electrical system located in the building. These are typically located in a Motor Control Center in an equipment or electrical room.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instructions shall be strictly adhered to.
2. Schedule thermographic survey with operating personnel.
3. Obtain and review manufacturer's instructions

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Low Voltage (1,000 Volts or less) Motor Starters as part of the electrical system located in the building. These are typically located in a Motor Control Center in an equipment or electrical room.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instructions shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical test procedures for Motor Control, Motor Starters, Low-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of the Medium Voltage (1,000 to 100,000 volts) Motor Starters as part of the electrical located in the building. These are typically located in a Motor Control Center in an equipment or electrical room.

Special Instructions

1. In addition to the procedures outlined in this standard, the equipment manufacturer's recommended maintenance procedures and/or instructions shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule thermographic survey with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Medium Voltage (1,000 to 100,000 volts) Motor Starters as part of the electrical system located in the building. These are typically located in a Motor Control Center in an equipment or electrical room.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Motor Control, Motor Starters, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Low Voltage (1,000 Volts or less) Motor Control Centers as part of the electrical system located in the building. These are typically located in an equipment or electrical room.

Special Instructions

Refer to Standard ELEC-SWB-01-01Y for instructions, check points and safety requirements.

Checkpoint**Tools**

None

This standard applies to maintenance tests of Low Voltage (1,000 Volts or less) Motor Control Centers as part of the electrical system located in the building. These are typically located in an equipment or electrical room.

Special Instructions

Refer to Standard ELEC-SWB-01-03Y for instructions, check points and safety requirements.

Checkpoint**Tools**

None

This standard applies to thermographic survey of Medium Voltage (1,000 to 100,000 volts) Motor Control Centers as part of the electrical system located in the building. These are typically located in an equipment or electrical room.

Special Instructions

Refer to Standard ELEC-SWB-01-01Y for instructions, check points and safety requirements.

Checkpoint**Tools**

None

This standard applies to maintenance tests of Medium Voltage (1,000 to 100,000 volts) Motor Control Centers as part of the electrical system located in the building. These are typically located in an equipment or electrical room.

Special Instructions

Refer to Standard ELEC-SWB-01-03Y for instructions, check points and safety requirements.

Checkpoint

Tools

None

This standard is for induction, wound-rotor and synchronous motors in excess of 1 horsepower. The maintenance specified by this standard is not intended to require disassembly of the motor. This standard does not normally apply to motors rated less than 1 horsepower, for which maintenance is normally limited to cleaning and lubrication and is done with the maintenance of the driven machine (Parent).

Special Instructions

1. Schedule shutdown with operating personnel, as needed.
2. Review manufacturer's instructions.
3. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. In absence of manufacturer's instructions, use NETA Maintenance Testing Specifications as a guide.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform visual inspection procedures for Rotating Machinery, Induction or Synchronous Motors and Generators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

GLOVES-01 TOOL_GROUP_B

This standard is for induction, wound-rotor and synchronous motors in excess of 1 horsepower. The maintenance specified by this standard is not intended to require disassembly of the motor. This standard does not normally apply to motors rated less than 1 horsepower, for which maintenance is normally limited to cleaning and lubrication and is done with the maintenance of the driven machine (Parent).

Special Instructions

1. Perform monthly visual inspections concurrently with annual maintenance and testing.
2. Schedule shutdown with operating personnel, as needed.
3. Review manufacturer's instructions.
4. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. In absence of manufacturer's instructions, use NETA Maintenance Testing Specifications as a guide.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform mechanical inspection procedures for Rotating Machinery, Induction or Synchronous Motors and Generators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

TOOL_GROUP_B GLOVES-01

This standard is for induction, wound-rotor and synchronous motors in excess of 1 horsepower. The maintenance specified by this standard is not intended to require disassembly of the motor. This standard does not normally apply to motors rated less than 1 horsepower, for which maintenance is normally limited to cleaning and lubrication and is done with the maintenance of the driven machine (Parent).

Special Instructions

- 1. Perform monthly visual inspections and annual mechanical maintenance concurrently with 2-year electrical maintenance and testing.
- 2. Schedule shutdown with operating personnel, as needed.
- 3. Review manufacturer's instructions.
- 4. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
- 5. In absence of manufacturer's instructions, use NETA Maintenance Testing Specifications as a guide.
- 6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 7. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

- 1. Perform electrical testing procedures for Rotating Machinery, Induction or Synchronous Motors and Generators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

TACHOMETER-01	TOOL_GROUP_B	SAFETY_GOGGLES	GLOVES-01	EQUIP_CLEANING-03	VACUUM_CLEANER-06
LUBRICANT-07					

This standard applies to thermographic survey of all electric metering devices located in switchboards within the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of all electric metering devices located in switchboards within the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Obtain and review manufacturer's instruction (including time current characteristic curve).
3. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Schedule outage with operating personnel.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform electrical test procedures for Metering Devices, Electromechanical and Solid-State or Metering Devices, Microprocessor Based outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance testing of advanced metering devices.

Special Instructions

1. If meter is equipped with a service disconnect, remove power and check terminals and connectors for tightness.
2. Verify that the wire is properly stripped and that there is no excessive bare wire or pinching of the insulation in the terminals.
3. Install any firmware updates.
4. Inspect for signs of moisture, corrosion, dirt or contaminants, wear or cracking of insulation, presence of nameplate and that the meter is correct class for service.
5. Verify the control is firmly mounted, and that wiring is neat and orderly.
6. Obtain a current one-line drawing and verify meter is connected to the proper circuit. A copy of the one-line should be mounted or in a document sleeve in the vicinity of the meter. Verify wiring is to correct legs. Verify CTs are mounted to the correct legs and have the correct polarity.
7. Check the condition of any fuses.
8. Check for visible signs of overheating on wiring and terminals.
9. Perform thermal imaging.
10. Verify that the RMS voltage indicated by the meter agrees with the reading on a calibrated handheld meter between phases and phase to ground.

Checkpoint

See Special Instructions.

Tools

None

This standard applies to Network Protectors, 600V class.

Special Instructions

Refer to Standard ELEC-CKB-01-01Y and ELEC-CKB-01-03Y for instructions, check points and safety requirements.

Checkpoint

Tools

None

This job plan applies to photovoltaic solar panel systems and its components, including but not limited to the solar panel arrays, wiring, cabling, conduit, cabinets, combiners, meteorological sensors, mounting systems and fasteners. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturer's recommended maintenance and or the tasks and checks specified in this job plan.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Observe and record instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
2. Test metering connections for meter data flow and meteorological connections and signal strength.

Tools

None

This job plan applies to photovoltaic solar panel systems and its components, including but not limited to the solar panel arrays, wiring, cabling, conduit, cabinets, combiners, meteorological sensors, mounting systems and fasteners. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturer's recommended maintenance and or the tasks and checks specified in this job plan.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Visually inspect cabling for signs of cracks, defects, loose or pulled-out connections, overheating, arcing, short or open circuits, and ground faults. Maintain safe distance and use extreme caution near live electrical systems.
2. Use industry standard techniques to inspect panels (i.e. infrared imaging) for hot spots and panel failures.
3. Install any software updates, firmware updates, and security patches.
4. Inspect bolts, fasteners, other rack-mountings, and ballast devices (weighting). Tighten and grease per manufacturer's recommendations. Report failures of tightening devices to the Contracting Officer's Representative.
5. Observe and record instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
6. Test metering connections for meter data flow and meteorological connections and signal strength.
7. Remove any debris (tree branches, leaves, etc.). Clean PV panels strictly per manufacturer's recommendations. At a minimum, rinse with water to remove accumulated dust, pollen, etc.
8. Perform visual inspection of combiner boxes and control panels. Clean combiner boxes and control panels using compressed air and vacuum. Verify wiring is neat and orderly, Panduit and control covers are in place.
9. Test PV arrays voltage/current at the combiner boxes.
10. Inspect components for moisture, verify proper operation of any panel temperature controls.
11. Grease actuator gears and top off hydraulic fluid on tracker components (if applicable).
12. Test of SCADA and meteorological connections and signal strength.
13. Inspect cables for binding or wear (tracking systems).
14. Check and calibrate sensors, verify operation of wind sensors (tracking panels should 'flatten' during high wind conditions).

Tools

None

This job plan applies to photovoltaic solar panel systems and its components, including but not limited to the solar panel arrays, wiring, cabling, conduit, cabinets, combiners, meteorological sensors, mounting systems and fasteners. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturer's recommended maintenance and or the tasks and checks specified in this job plan.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Remove any debris (tree branches, leaves, etc.). Clean PV panels strictly per manufacturer's recommendations. At a minimum, rinse with water to remove accumulated dust, pollen, etc.
2. Observe and record instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
3. Test metering connections for meter data flow and meteorological connections and signal strength.

Tools

None

This job plan applies to photovoltaic solar panel systems and its components, including but not limited to the solar panel arrays, wiring, cabling.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform visual inspection of combiner boxes and control panels. Clean combiner boxes and control panels using compressed air and vacuum. Verify wiring is neat and orderly, Panduit and control covers are in place.
2. Test PV arrays voltage/current at the combiner boxes.
3. Inspect components for moisture, verify proper operation of any panel temperature controls.
4. Grease actuator gears and top off hydraulic fluid on tracker components (if applicable).
5. Test of SCADA and meteorological connections and signal strength.
6. Install any software, firmware updates.
7. Inspect cables for binding or wear (tracking systems).
8. Check and calibrate sensors, verify operation of wind sensors (tracking panels should 'flatten' during high wind conditions).
9. Observe and record instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
10. Test metering connections for meter data flow and meteorological connections and signal strength.
11. Remove any debris (tree branches, leaves, etc.). Clean PV panels strictly per manufacturer's recommendations. At a minimum, rinse with water to remove accumulated dust, pollen, etc

Tools

None

This job plan applies to photovoltaic system power inverters, and ancillary equipment including but not limited to, controllers, batteries, meters, and sensors. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturers recommended maintenance and/or the tasks and checks specified in this job plan. Meter readings for checkpoint #3 must be recorded on or about the last working day of the performance month.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Observe instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
2. Test metering connections for meter data flow and meteorological connections and signal strength.
3. Use the NCMMS Virtual Meter feature to record cumulative energy delivery (kWh) in accordance with training and guidance provided by GSA. Contact regional NCMMS POC for assistance, as needed. Monthly meter reading must be taken on or about the last work day of the month.

Tools

None

This job plan applies to photovoltaic system power inverters, and ancillary equipment including but not limited to, controllers, batteries, meters, and sensors. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturers recommended maintenance and/or the tasks and checks specified in this job plan. ♦ Meter readings for checkpoint #12 must be recorded on or about the last working day of the performance month.



Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Re-torque current-carrying conductor fasteners (screw lugs on terminal blocks)
2. Perform thermal imaging of electrical connections and components.
3. Clean or replace Inverter air filters.
4. Record measurements at the inverter to include inverter alarms, inverter control settings, input (DC) current and voltages and any other information available from the inverter data interface such as inverter temperature within the NCMMS Work Order Log. Compare with previous values and report any anomalies to COR.
5. If the inverter does not report temperature, add a temperature sensor to the inverter enclosure to identify overheating and record.
6. Install any recent software upgrades to inverter programming or DAS and monitoring systems.
7. Clean heat rejection fins.
8. Conduct performance testing: measure incident sunlight and simultaneously observe temperature and energy output; calculate PV module efficiency as a function of temperature and calculate the balance-of-system efficiency; compare readings with diagnostic benchmark (original efficiency of system.)
9. Perform visual inspection of inverters, and ancillary equipment. Clean vents, cooling fans, and filters to prevent overheating using compressed air and vacuum. Verify wiring is neat and orderly, and conduit and control covers are in place. Visually inspect components for signs of moisture, verify proper operation of any panel temperature controls.
10. Observe instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
11. Test metering connections for meter data flow and meteorological connections and signal strength.
12. Use the NCMMS Virtual Meter feature to record cumulative energy delivery (kWh) in accordance with training and guidance provided by GSA. Contact regional NCMMS POC for assistance, as needed. Monthly meter reading must be taken on or about the last work day of the month.

Tools

None

This job plan applies to photovoltaic system power inverters, and ancillary equipment including but not limited to, controllers, batteries, meters, and sensors. This job plan applies to all maintainable components, sub-components and parts in accordance with manufacturer's recommended maintenance and/or the tasks and checks specified in this job plan. Meter readings for checkpoint #4 must be recorded on or about the last working day of the performance month.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. All fall protection equipment is required by OSHA to be checked prior to each use.
3. Any and all personnel involved in the use or maintenance of fall protection or who may, in the course of their work, be present on any elevated surface without a guard rail are required to have regular training in fall protection safety. Conformance includes but is not limited to ASME, ASME A 120.1, IWCA I 14.1, OSHA 1910.66, CALOSHA, AISC, AWS and local code requirements.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform visual inspection of inverters, and ancillary equipment. Clean vents, cooling fans, and filters to prevent overheating using compressed air and vacuum. Verify wiring is neat and orderly, and conduit and control covers are in place. Visually inspect components for signs of moisture, verify proper operation of any panel temperature controls.
2. Observe instantaneous operational indicators on the faceplate of the inverter to ensure that the amount of power being generated is typical of the conditions; compare current readings with diagnostic benchmark; inspect inverter housing or shelter for physical maintenance required if present.
3. Test metering connections for meter data flow and meteorological connections and signal strength.
4. Use the NCMMS Virtual Meter feature to record cumulative energy delivery (kWh) in accordance with training and guidance provided by GSA. Contact regional NCMMS POC for assistance, as needed. Monthly meter reading must be taken on or about the last work day of the month.

Tools

None

This standard applies to visual inspections of Step Voltage Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition.
3. Visually inspect anchorage, alignment, and grounding.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks and bushings.
5. All parameters shall be within manufacturer's recommended tolerances.

Tools

None

This standard applies to Step Voltage Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
3. Perform visual and mechanical inspections procedures for Regulating Apparatus, Voltage, Step-Voltage Regulators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data..
5. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance tests of Step Voltage Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform monthly and annual inspections concurrently with 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Regulating Apparatus, Voltage, Step-Voltage Regulators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspections of Induction Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition.
2. Visually inspect anchorage, alignment, and grounding.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks and bushings.
5. All parameters shall be within manufacturer's recommended tolerances.

Tools

None

This standard applies to Induction Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.
5. Perform visual and mechanical inspections procedures for Regulating Apparatus, Voltage, Induction Regulators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. Sample insulating liquid from the main tank or common tank in accordance ASTM D923 as outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

4. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
 1. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
 2. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data.
 3. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance tests of Induction Regulators located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule outage with operating personnel.
5. Perform monthly and annual inspections concurrently with 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Regulating Apparatus, Voltage, Induction Regulators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
3. Perform visual and mechanical inspections procedures for Regulating Apparatus, Voltage, Induction Regulators outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Sample insulating liquid from the main tank or common tank in accordance ASTM D923 as outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data.
6. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspections of Load Tap-changers as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition.
2. Visually inspect anchorage, alignment, grounding, and required clearances.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks.
5. All parameters shall be in accordance with manufacturer's published data.

Tools

None

This standard applies to Load Tap-changers as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.
4. Perform monthly visual inspections concurrently with annual mechanical maintenance testing .

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Perform visual and mechanical inspections procedures for Regulating Apparatus, Load Tap-Changers outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Perform internal inspection procedures.
4. Visual and mechanical inspection test values shall be in accordance with information, data, and standards listed in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Results of insulating liquid tests shall conform to recommendations outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of Load Tap-changers as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule outage with operating personnel.

Checkpoint

1. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Perform monthly visual inspections and annual mechanical maintenance concurrently with 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
3. Perform electrical test procedures for Regulating Apparatus, Load Tap-Changers outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to all electrical system Protective Relays installed in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule and coordinate work with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Obtain and review manufacturer's instructions, circuit diagrams for relay and test set.

Checkpoint

1. Perform electrical test procedures for Protective Relays, Electromechanical and Solid-State or Protective Relays, Microprocessor-Based outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Low Voltage (1,000 Volts or less) Surge Arrester Devices as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction (including time current characteristic curve).

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of Low Voltage (1,000 Volts or less) Surge Arrester Devices as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction (including time current characteristic curve).
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing.

Checkpoint

1. Perform electrical test procedures Surge Arrester, Low-Voltage Surge Protection Devices outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Medium Voltage Surge Arrester Devices as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction (including time current characteristic curve).

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of Medium Voltage Surge Arrester Devices as part of the electrical system located in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instruction (including time current characteristic curve).
3. Schedule outage with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Surge Arresters, Medium- and High-Voltage Surge Protection Devices outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to thermographic survey of electrical switchgear and switchboard assemblies located throughout the building or site.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard card applies to maintenance tests of electrical switchgear and switchboard assemblies located throughout the building or site.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switchgear and Switchboard Assemblies outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Low-Voltage Air Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Low-Voltage Air Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Air, Low-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Medium-Voltage Metal-Enclosed Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Medium-Voltage Metal-Enclosed Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Air, Medium-Voltage, Metal-Enclosed outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Medium and High Voltage Open Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Medium and High Voltage Open Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Air, Medium-and High-Voltage, Open outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspection of Medium-Voltage Oil Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition
2. Visually inspect anchorage, alignment, grounding, and required clearances.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks and bushings.
5. Verify that breather vents are clear.
6. All parameters shall be in accordance with manufacturer's published data.

Tools

None

This standard applies to Medium-Voltage Oil Switches.

Special Instructions

1. Schedule work with operating personnel.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Review manufacturer's instructions for all system components.
4. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
5. Perform visual and mechanical inspections following the procedures and steps outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.
6. Remove oil sample of insulating liquid from the main tank or common tank following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Checkpoint

1. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
2. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data.
3. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to maintenance testing of Medium-Voltage Oil Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Oil, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Medium-Voltage Vacuum Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.
4. Perform these tests only if equipment is safely accessible for the survey.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance tests of Medium-Voltage Vacuum Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Vacuum, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to visual inspections of Medium-Voltage SF-6 Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition.
2. Visually inspect anchorage, alignment, grounding, and required clearances.
3. Inspect SF6 gas pressure alarms.
4. Test for SF6 gas leaks in accordance with manufacturer's published data.
5. All parameters shall be in accordance with manufacturer's published data.

Tools

None

This standard applies to thermographic survey of Medium-Voltage SF-6 Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.
4. Perform these tests only if equipment is safely accessible for the survey.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Medium-Voltage SF-6 Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, SF6, Medium-Voltage outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard applies to thermographic survey of Cutout Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard applies to maintenance testing of Cutout Switches.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Perform annual thermographic survey prior to 3-year electrical maintenance testing.
2. Review manufacturer's instructions for all system components.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Switches, Cutouts. following outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to annual thermographic survey of small dry transformers with windings rated 600 volts or less and sizes equal to or less than 167 kVA single-phase or 500 kVA three-phase.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Perform these tests only if equipment is safely accessible for the survey. Notify COR if unable or unsafe to complete.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard card applies to maintenance testing of small dry transformers with windings rated 600 volts or less and sizes equal to or less than 167 kVA single-phase or 500 kVA three-phase.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Transformers, Dry Type, Air-Cooled, Low-Voltage, Small outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to annual thermographic survey of large dry transformers with windings rated higher than 600 volts and low-voltage transformers larger than 167 kVA single-phase or 500 kVA three-phase.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Perform these tests only if equipment is safely accessible for the survey. Notify COR if unable to complete.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard card applies to maintenance testing of large dry transformers with windings rated higher than 600 volts and low-voltage transformers larger than 167 kVA single-phase or 500 kVA three-phase.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Schedule and coordinate work with operating personnel.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Transformers, Dry Type, Air-Cooled, Large outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to visual inspections for liquid filled transformers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.

Checkpoint

1. Visually inspect physical and mechanical condition
2. Visually inspect anchorage, alignment, and grounding.
3. Visually inspect for oil leaks.
4. Visually verify correct oil level in all tanks and bushings.
5. Verify that positive pressure is maintained on gas-blanketed transformers.
6. All parameters shall be in accordance with manufacturer's published data.

Tools

None

This standard card applies to the annual inspections and testing for liquid filled transformers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Schedule work with operating personnel.

Checkpoint

1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.
3. Perform visual and mechanical inspections procedures for Transformers, Liquid filled outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Visual and mechanical inspection parameters shall be in accordance with manufacturer's published data.
5. Results of insulating liquid tests shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to maintenance testing of liquid filled transformers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
4. Schedule and coordinate work with operating personnel.

Checkpoint

1. Perform electrical test procedures for Transformers, Liquid-Filled outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to Uninterruptible Power Systems as part of the building electrical system and the emergency power system found in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule and coordinate work with operating personnel.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform electrical test procedures for Emergency Systems, Uninterruptible Power Systems outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard card applies to Uninterruptible Power Systems as part of the building electrical system and the emergency power system found in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform monthly visual inspections concurrently with annual maintenance and testing.
3. Schedule and coordinate work with operating personnel.
4. Obtain and review manufacturer's instructions.
5. The InterNational Electrical Testing Association (NETA) Maintenance Testing Specifications are used as the guide for all Electrical Testing procedures. Follow qualifications, responsibilities and safety procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.

Checkpoint

1. Perform electrical tests procedures for Emergency Systems, Uninterruptible Power Systems outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

This standard is for thermographic survey of variable frequency drives (VFD) for motors rated at 5 horsepower or greater. This maintenance standard is typically used in conjunction with the maintenance of the driven machine (Parent).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule work with operating personnel, as needed.
4. Perform these tests only if equipment is safely accessible for the survey. Notify COR if unable to complete.
1. Perform thermographic survey following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

2. Review Thermographic Survey Suggested Actions Based on Temperature Rise information/data in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation. Perform recommended actions.

Tools

None

This standard is for maintenance testing of variable frequency drives (VFD) for motors rated at 5 horsepower or greater. This maintenance standard is typically used in conjunction with the maintenance of the driven machine (Parent).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel, as needed.
4. Perform annual thermographic survey prior to 3-year electrical maintenance testing following the procedures outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.

Checkpoint

1. Perform electrical test procedures for Adjustable Speed Drive Systems outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at the time of solicitation.
2. Test results shall conform to recommended Test Values outlined in the ANSI/NETA Maintenance Testing Specification (MTS), most recent edition at time of solicitation.

Tools

None

FIREDRILL-01-01Y

Annual Fire Drill Certification (WORK ORDER LOG NOTE CERTIFICATION REQUIRED.)

Rev-2 | 2024-10-25

This Job plan is for building to schedule, conduct and certify completion of their required annual fire drill. If the building's fire drill was conducted earlier this calender year, certify that completion using this work order.

Special Instructions

- 1. This work order is assigned to you as the ServiceNow Primary Point of Contact (POC) for the building listed on this work order and its purpose is to document the required annual fire drill performed at the facility.
- 2. If the last fire drill was conducted within this calendar year, the only action required for this work order is to complete the certification and change the work order status as described below.
- 3. If a fire drill has not been conducted prior to the issuance of this work order, schedule and conduct the annual fire drill for the facility within 30 days from the issue date of this work order.
- 4. After the fire drill is conducted, certify its completion by logging into NCMMS and copying the following message to the Work LOG tab of this work order: I certify that a Fire Drill was completed in the building assigned by this work order. The Fire Drill was completed on this date: _____
- 5. If you have questions about how to conduct a fire drill contact your regional fire protection engineer.

Checkpoint

Tools

None

FLSF-ALM-01-01M

Alarm Check Valves and Accessories
Rev-3 | 2022-01-01

This applies to all wet pipe sprinkler system's alarm check valves. When water flows in the sprinkler system due to the operation of one or more automatic fire sprinklers, the alarm check valve opens, allowing continuous flow of water into the system and transmittal of water pressure to electrical and/or mechanical water flow alarm devices.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
- 2. Review manufacturer's instructions.
- 3. Schedule shutdown with operating personnel, as needed.
- 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
- 6. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements.
- 7. Follow NFPA 25 (most recent version at the time of solicitation) for valve maintenance requirements.

Checkpoint

- 1. Refer to NFPA 72 and NFPA 25.

Tools

TOOL_BASIC

This standard applies to all supervisory signals of fire protection equipment and systems. The fire supervisory signal must be displayed at a constantly attended location on the site or at an approved remote monitoring location. These signals come from valves, air pressure devices, temperature sensors or water tank floats which are monitored at the fire panel or a remote location.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The COR and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. When feasible the position of valves, air pressure, temperature, or water level being monitored should be altered to actuate the signals.
6. Check all supervisory devices for damage, corrosion, and pitted electrical contacts.
7. Inspect conduit for loose joints, hangers, and clamps.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Refer to NFPA 72.

Tools

TOOL_BASIC

This standard applies to all smoke detectors, including ionization, photoelectric, combination ionization/photoelectric, photoelectric projected beam, and air sampling. These detectors may provide open area protection or HVAC duct protection and are initiated with the presence of smoke in the area or HVAC system.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The COR and the control center, central station or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. Check detectors for physical damage, obstruction, or corrosion. Replace damaged detectors.
6. Check for any changes in partitions or floor plan since detector installation may need to be rearranged.
7. Functional tests are to be made in place to assure smoke entry. Test duct detectors to assure that the device will sample the air stream.
8. When it is both feasible and safe, the environmental conditions being monitored should be altered to actuate the device.
9. Before each test, clean the detectors according to the manufacturer's instruction. In addition, the frequency of cleaning and tests must be based on results of regular tests and local conditions.
10. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
11. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Refer to NFPA 72.

Tools

TOOL_BASIC

This standard applies to all fire protection systems and equipment with waterflow alarms including mechanical water motor gongs, vane-type waterflow devices, and pressure switches that initiate audible or visual signals. These signals are initiated when water is flowing through the system indicating that a sprinkler or hose has been activated.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The COR and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Refer to NFPA 72.

Tools

TOOL_BASIC

This standard applies to all heat detectors, including: Spot-type - Sensitive element is confined to a small area. Fixed-temperature (non-restorable). Rate-of-rise (restorable). Combination fixed-temperature and rate-of-rise (has both non-restorable and restorable elements) Line-type - Sensitive element is in a continuous line. Cable type - fixed-temperature (non-restorable). Cable type - rate-of-rise (restorable). Sealed Pneumatic type - rate-of-rise (restorable).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. Check detectors for physical damage, obstruction, corrosion, or painting. Painting slows or inhibits detector response to a fire. Replace damaged or painted detectors and cables.
6. Check for any changes in partitions or floor plans since detector installation may need to be rearranged.
7. When it is both feasible and safe, the environmental conditions being monitored should be altered to actuate the device.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 72 National Fire Alarm Code (most recent edition at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to all building smoke control systems. Smoke control for the building may be divided into separate zones which will control ventilation systems based on activation of a fire detection device in the zone. This maintenance standard is to ensure that the operation in smoke mode functions correctly.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure will cause activation of an alarm and/or supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. Initiation of the smoke control system for each smoke control zone shall be accomplished by activation of a fire detection device that is programmed to initiate automatic smoke control within that smoke control zone.
6. Determine and list all programmed actions which should automatically occur upon activation of smoke control system for each smoke control zone. (Programmed actions are not standardized for the operation of air handling systems due to the various HVAC system design.)
7. The scheduling of the smoke control testing of the various smoke control zones should be evenly distributed during those months when outside temperatures are above freezing.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to all fire alarm panels and remote annunciators in both interior and exterior locations. These annunciators are for audio of voice notification during a fire event to the personnel in or around a building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure will cause the activation of the building fire alarm system and/or supervisory signals. The COR, control center, central station, and/or fire department must be notified prior to start and at completion of work.
5. When work requires the actuation of building evacuation systems, the work must be performed after hours and must be coordinated through the COR to prevent the unnecessary evacuation of building occupants.
6. Dispose of old batteries in accordance with all applicable Federal, state, and local regulations.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the most recent applicable NFPA code or standard at the time of solicitation.

Checkpoint

1. Refer to NFPA 72.

Tools

TOOL_BASIC

This standard applies to control panels for specialized fire suppression systems such as Halon 1301, carbon dioxide, preaction, deluge, water spray systems, and other systems with special shutdown features.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of the building fire alarm system and/or supervisory signals. The field office manager, control center, central station, and/or fire department must be notified prior to start and at completion of work.
5. When work requires the actuation of building evacuation systems, the work must be performed after hours and must be coordinated through the field office manager to prevent the unnecessary evacuation of building occupants.
6. If a test that includes actual discharge of the system is desired, special precautions must be taken with the using agency to prevent shutdown and discharge during routine work performance causing subsequent equipment or software damage. If an actual discharge test is not desired, be sure to prevent gas discharge, ventilation system, and computer system or other equipment shutdown.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to all fire alarm system central station transmitters that are not maintained by a central station supervisory service. These transmitters are used to send a fire event notification for the building to a central monitoring station. They are not maintained by the monitoring station.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. The field office manager, control center, central station, and/or the fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. When work requires the actuation of building evacuation systems, the work must be performed after hours and must be coordinated through the field office manager to prevent the unnecessary evacuation of building occupants.
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
7. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to equipment in Government-owned control centers providing proprietary supervisory station service. This standard ensures that the central monitoring station within the building sends and records the proper signals for a given fire event triggering the alarm(s).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work on these devices and systems must be coordinated with the control center's operators and the fire department that will receive the alarm.
5. The work required by this procedure should be performed only by employees knowledgeable of the equipment and the procedures involved.
6. Once daily operational tests are completed, all circuits are to be returned to normal as soon as possible.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to recorders in Government-owned buildings. Older fire systems use this paper spool type of device to record fire system events along with the associated date/time stamp.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This procedure is to be accomplished once weekly during the daily test and inspection required in the standard.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to fire alarm system event printers in Government owned buildings which are attached to the fire panel. These printers record all events which are reported into the fire panel and create a record for review.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions for all system components.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
- 5. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

- 1. Refer to NFPA 72.

Tools

VACUUM_CLEANER SPARE_RIBBON LUBRICANT-10 CLOTHS-02 BRUSH ALCOHOL TOOL_GROUP_B

This standard applies to fire alarm system audio control panels in Government owned buildings. These audio panels contain the audio/voice alerts used during fire events to broadcast instructions to building occupants.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The COR, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the most recent applicable NFPA code or standard at the time of contract award.
6. Follow NFPA 72 National Fire Alarm Code (most recent version at the time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Refer to NFPA 72.

Tools

TOOL_GROUP_B INDICATOR_SPARK CLOTHS-02

This standard applies to fire alarm system remote controllers in Government owned buildings.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions for all system components.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
- 5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
- 6. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

- 1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

AIR_COMPRESSED TOOL_GROUP_B SPARE_RIBBON CLOTHS-02 LUBRICANT-10

This standard applies to fire alarm remote amplifiers in Government owned buildings.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
5. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_GROUP_B AIR_COMPRESSED

This standard applies to all manual fire alarm pull stations in both interior and exterior locations. These devices provide a signal the building fire control system when activated initiating building fire alarms and signal to the central monitoring station.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. When alarm systems are connected to municipal systems, test signals to be transmitted to them will be limited to those acceptable to that authority.
6. Record all test results.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation) for fire alarm system operation and maintenance requirements. NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

MAT_CLEANING-02 TOOL_BASIC

This standard applies to Fire Alarms only and should follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation), defined by Chapter 10-Inspection, Testing and Maintenance.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Fire Life Safety (Fire Alarm) system testing shall be performed by a licensed and certified testing company as required to maintain the building UL certification.
3. Notify building occupants, alarm monitoring company, and/or local fire department prior to work.
4. Complete and file NFPA Inspection Testing and Maintenance Form (four pages).

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC

This standard applies to Fire Alarms only and should follow NFPA 72 National Fire Alarm Code (most recent edition at time of solicitation), defined by Chapter 10-Inspection, Testing and Maintenance.

Special Instructions

1. Fire Alarm System needs UL certification every five years.

Checkpoint

1. Refer to NFPA 72.

Tools

None

This standard applies to all swinging fire doors located in stairwells and exitways, sliding and vertical rolling doors located in fire walls in the building. These doors are required to shut during a fire event protecting against the spread of fire and/or smoke into other parts of the building or into vertical shafts such as elevators. Some doors will be alarmed, others will not. Proper maintenance will ensure that the stairwells and exitways remain safe for personnel egress during a fire event.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. The field office manager and the control center, central station or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the most recent applicable NFPA code or standards at the time of solicitation.
6. Follow NFPA 80 (Most recent version at the time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Refer to NFPA 80.

Tools

TOOL_BASIC LUBRICANT-07

This standard applies to all permanently installed electric generators used for providing emergency electrical power whose engines are powered by gasoline or natural gas. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review and follow manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. A written record of all inspections, service, tests, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identity of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
5. Have a properly serviced fire extinguisher in proper working order on hand.
6. Allow no open flames or smoking in the area.
7. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
8. Test medium and high voltage circuit breakers in accordance with circuit breaker maintenance standards.
9. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
10. Follow NFPA 110 (Most recent version at time of solicitation) for operation and maintenance requirements.

Checkpoint

Refer to NFPA 110

Tools							
ENGINE_TUNE_UP_KIT	ENGINE_OIL-02	GREASE	MAT_CLEANING-02	MEGGER	OIL-03	TOOL_BASIC	OIL_GOVERNOR
COOLANT	BELTS_FAN						

This standard applies to all permanently installed electric generators used for providing emergency electrical power whose prime movers are diesel engines. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review and follow manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. A written record of all inspections, service, tests, operation, and repairs to the emergency generator shall be maintained in NCMMS. This record shall include the date of maintenance, identity of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.

Checkpoint

1. Follow NFPA 110 (most recent edition at time of solicitation) Chapter 8 for monthly testing and maintenance requirements.

Tools

OIL-03 MAT_CLEANING-02 PROTECTION_HEARING TOOL_BASIC ENGINE_OIL-03

This standard applies to all permanently installed electric generators used for providing emergency electrical power whose prime movers are diesel engines. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
5. Have a properly serviced fire extinguisher in proper working order on hand.
6. Allow no open flames or smoking in the area.
7. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 110 and 111 (Most recent version at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Refer to NFPA 110.
2. Test glycol-water solutions regularly to determine the percentage glycol, pH, reserve alkalinity, inhibitor levels, and degree of contamination. The glycol solution must be checked at least once a year in accordance with the manufacturer's recommendations. Are batteries in compliance with NFPA 110 - A.5.6.4.5.1 ("It is recommended that lead-acid starting batteries be replaced every 24 to 30 months")?

Tools

COOLANT	FILTER_FUEL	GREASE	HOSE-02	MAT_CLEANING-02	OIL_GOVERNOR	PROTECTION_HEARING	TOOL_BASIC
BELTS_FAN	OIL-03	ENGINE_OIL-01	MEGGER	AIR_FILTER			

This standard applies to all permanently installed electric generators used for providing emergency electrical power whose prime movers are diesel engines. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review and follow manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
5. Have a properly serviced fire extinguisher in proper working order on hand.
6. Allow no open flames or smoking in the area.
7. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

OIL-03

MAT_CLEANING-02

PROTECTION_HEARING

TOOL_BASIC

ENGINE_OIL-03

This standard applies to all permanently installed electric generators used for providing emergency electrical power whose prime movers are diesel engines. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

- 1. In addition to the procedures outlined in this standard, the equipment manufacturer’s recommended maintenance procedures and instructions shall be strictly adhered to.
- 2. Review and follow manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
- 3. A written record of all inspections, service, tests, operation, and repairs to the emergency generator shall be maintained in NCMMS. This record shall include the date of maintenance, identity of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.

Checkpoint

- 1. Follow NFPA 110 (most recent edition at time of solicitation) Chapter 8 for 36 month testing and maintenance requirements.

Tools

ENGINE_OIL-03

TOOL_BASIC

PROTECTION_HEARING

MAT_CLEANING-02

OIL-03

This standard applies to all permanently installed emergency generators used for providing emergency electrical power. This standard applies primarily to the engine side of the generator. See electrical standards for the maintenance required on the electric generator side

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. The weekly maintenance checks shall be performed in conjunction with this monthly maintenance.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The COR and the control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
5. Qualified operating personnel shall be in attendance during the monthly generator operation.
6. Hearing protection is required whenever the engine is running.
7. Records of changes to the emergency electrical system should be maintained so that the demand likely to be produced by the connected load will be within the available capacity.
8. Review the Standard Operating Procedure for "Emergency Eyewash and Shower Equipment". Note the location of emergency eyewash and/or shower equipment.
9. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
10. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
11. Have a properly serviced fire extinguisher in proper working order on hand.
12. Allow no open flames or smoking in the area.
13. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
14. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the most recent applicable NFPA code or standards at the time of solicitation.
15. Follow NFPA 110 and 111 (Most recent version at time of solicitation) for operation and maintenance requirements.
16. If applicable, ensure the air quality/emissions permit is current and all requirements associated with the permit are followed.

Checkpoint

1. Refer to NFPA 110.

Tools

HYDROMETER-01	ENGINE_OIL-01	EYE_WASH	PROTECTION_HEARING	TACHOMETER-01	TOOL_GROUP_B	WATER
APRON_GLOVE	EQUIP_CLEANING-03					

This standard applies to all permanently installed emergency generators used for providing emergency electrical power. This standard applies primarily to the engine side of the generator. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
5. Have a properly serviced fire extinguisher in proper working order on hand.
6. Allow no open flames or smoking in the area.
7. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
8. Hearing protection is required whenever the engine is running.
9. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
10. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
11. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

TOOL_GROUP_B ENGINE_OIL-01 SPARE_AIR_FILTER BATTERY_TESTER EQUIP_CLEANING-02

This standard applies to all permanently installed emergency generators used for providing emergency electrical power. This standard applies primarily to the engine side of the generator. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Qualified operating personnel shall be in attendance during the monthly generator operation.
6. Hearing protection is required whenever the engine is running.
7. Records of changes to the emergency electrical system should be maintained so that the demand likely to be produced by the connected load will be within the available capacity.
8. Review standard operating procedure for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
9. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
10. Have a properly serviced fire extinguisher in proper working order on hand.
11. Allow no open flames or smoking in the area.
12. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
13. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
14. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

TOOL_GROUP_B SPARE_CRANKCASE_BREATHER EQUIP_CLEANING-02

This standard applies to all permanently installed electric generators used for providing emergency electrical power. See electrical standards for the maintenance required on the electric generator side.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Qualified operating personnel shall be in attendance during the monthly generator operation.
6. Hearing protection is required whenever the engine is running.
7. Records of changes to the emergency electrical system should be maintained so that the demand likely to be produced by the connected load will be within the available capacity.
8. Review standard operating procedure for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
9. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
10. Have a properly serviced fire extinguisher in proper working order on hand.
11. Allow no open flames or smoking in the area.
12. If the unit has a self-contained fuel tank, use only Underwriters Laboratories listed safety cans for fuel transfer.
13. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
14. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

SPARE_AIR_FILTER EQUIP_CLEANING-02 TESTER_COOLANT TOOL_GROUP_B COOLANT

This standard applies to portable emergency pumps and ventilating blowers and includes electric, LP gas, gasoline and pneumatic drive motors. This annual maintenance is to ensure that the equipment is in good operating order when needed in an emergency.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Have the proper fire extinguisher on hand. Ensure that the fire extinguisher is properly serviced and in proper working order.
4. Refer to appropriate standard and manufacturer's instructions for pump and motor maintenance.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

OIL-05 TOOL_BASIC GREASE_GUN-06

The purpose of this standard is to test the operation of emergency pumps and ventilating blowers. The standard applies to portable emergency pumps and ventilating blowers driven by electric, LP gas, gasoline and pneumatic drive motors.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Have the proper fire extinguisher on hand. Ensure that the fire extinguisher is properly serviced and in proper working order.
4. Notify all tenants before testing pumps and ventilators as the test may activate an alarm.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

TOOL_BASIC

This standard card applies to inline cartridge filters used on light fuel oils No. 1, 2, or 4, as well as the larger basket type used on heavier oils (#5 or #6).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Flammable liquids are being handled. Use all applicable safety precautions.
4. Check with operating personnel before starting work.
5. Secure and tag pumps, burners, and other necessary equipment.
6. Shut off and tag inlet and outlet valves.
7. Wear gloves while cleaning strainer. Consult the Safety Data Sheets (SDS) for proper personal protective equipment (PPE).
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Refer to NFPA 110.

Tools

TOOL_GROUP_C SOLVENTS GASKET-05

This standard applies to inline heaters as part of the fuel oil system supplying generator or boilers. These heaters are used to heat heavy oils as it passes from the tank and to the burners.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Schedule outage with operating personnel.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
- 5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

- 1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

LUBRICANT-07 TOOL_BASIC TUBE_CLEANING_MAT GASKET-02 VARSOL-02

This standard applies to all permanently installed electric generators that are driven by steam turbines and used for providing emergency electrical power for the steam plant. See electrical standards for the maintenance required on the electric generator equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's operating instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Check fire extinguishers for location and type. Ensure that the fire extinguishers are properly serviced and in proper working order.
4. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of
5. Check log to see if any abnormal condition has been noted.
6. Check with operating personnel before starting equipment.
7. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

TOOL_BASIC TACHOMETER-01 EQUIP_CLEANING-05

This standard applies to all permanently installed electric generators that are driven by steam turbines and used for providing emergency electrical power for the steam plant. See electrical standards for the maintenance required on the electric generator equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's operating instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Check fire extinguishers for location and type. Ensure that the fire extinguishers are properly serviced and in proper working order.
4. A written record of all inspections, service, tests, exercising, operation, and repairs to the emergency generator shall be maintained in an equipment log book and kept on the premises. This record shall include the date of maintenance, identification of service personnel, and notation of any unsatisfactory condition and the corrective action taken, including parts replaced.
5. Check logs to see if any abnormal condition has been noted.
6. Schedule PM on the steam turbine in conjunction with this PM.
7. Check with operating personnel before starting equipment.
8. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
9. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
10. Follow NFPA 110 and 111 (most recent edition at time of solicitation) for operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

EQUIP_CLEANING-05 TACHOMETER-01 TOOL_BASIC

This standard applies to load bank testing. Load banks shall be of the reactive type rather than the resistive only type.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Obtain and review manufacturer's operating instructions. One copy of the instruction manual(s) shall be kept in a secure, convenient location near the equipment and another kept in a different location.
3. Check fire extinguishers for location and type. Ensure that the fire extinguishers are properly serviced and in proper working order.
4. Load shall be at a minimum of 75% of rated capacity.
5. Where practical, load bank testing shall include transfer switches.
6. The recommended duration of the test is eight (8) hours; minimum of two (2) hours is required.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 110.

Tools

None

This standard applies to load bank testing where failure of EPSS to perform could result in loss of human life, health hazards or serious injuries. Health care facilities or assembly of occupancy greater than 1,000 people requires NFPA Level 1 testing.

Special Instructions

Level 1 systems are required to test generator under load for the duration of their class rating OR a minimum of four hours, whichever is greater, every three (3) years.

Checkpoint

Refer to NFPA 110.

Tools

None

This standard is for a monthly visual inspection of all fire extinguishers, and is in addition to the annual maintenance requirements.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Remove from service and discard all inverting-type fire extinguishers, including soda-acid, chemical foam, water cartridge-operated, and loaded stream cartridge-operated extinguishers.
5. Whenever an extinguisher is removed from service, immediately replace it with an extinguisher of a size and extinguishing agent appropriate for the hazard protected.
6. Each extinguisher shall have an inspection tag securely attached that indicates the month and year the inspection was performed and the initials of the person performing the inspection shall be recorded.
7. Records are to be kept of those extinguishers that were found to require correction, with an notation of when and how the corrections were made.
8. The requirement for an inspection tag and the recording of the inspector's initials does not preclude the use of electronic record keeping utilizing bar codes or some other method of insuring that the extinguishers were actually inspected.
9. In addition to the required tags and/or labels, a permanent file record is to be kept for each extinguisher. This file record should include the following information as applicable:
 - a) The date when each extinguisher was purchased and installed.
 - b) The maintenance date and the name of the person or agency performing the maintenance.
 - c) The date when last recharged and the name of the person or agency performing the maintenance.
 - d) The hydrostatic re test date and the name of the person or agency performing the maintenance.
 - e) Description of dents remaining after passing a hydrostatic test.
 - f) The date of the six-year maintenance for stored pressure dry chemical and Halon extinguishers.
 - g) The date when non rechargeable extinguishers are to be removed from service and discarded.
10. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform a visual inspection is a quick check to see that the fire extinguisher is in its proper location, that it is not blocked, is fully charged, and that it appears to be in good working order.
2. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10.

Tools

SEALS INSPECTION_TAG

This standard applies to the following types of extinguishers in service except for the non chargeable type: Stored Pressure Water, Loaded Stream, or Antifreeze AFFF (Aqueous Film Forming Foam)FFFP (Film Forming Fluoroprotein Foam) Dry Chemical with stainless steel shells Carbon Dioxide.

Special Instructions

1. NFPA reuired procedures and OEM instructions shall be strictly adhered to and take precedence over the procedures outlined in this standard.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Hydrostatic testing of extinguishers requires experienced personnel and suitable testing equipment. It is recommended that this service be performed by a fire equipment company specializing in this type of service.
5. If the hydrostatic testing is to be accomplished on the premises, it should be limited only to non compressed gas type extinguishers (water base, dry chemical, and dry powder). Compressed gas extinguishers (carbon dioxide) require special internal inspections and high test pressures.
6. Wheeled extinguisher testing is beyond the scope of this standard. Utilize a fire equipment company if wheeled extinguishers are on the premises.
7. Discharge and discard extinguishers when the cost of new extinguishers is cheaper than the cost of extinguishing agent and a hydrostatic test.
8. Refer to National Fire Protection Association Standard No. 10, Portable Fire Extinguishers for additional detailed information about hydrostatically testing extinguishers.
9. In addition, anytime an extinguisher shows evidence of physical injury or corrosion, and extinguishers with aluminum cylinders or shells are suspected of being exposed to temperatures in excess of 350 degrees F, they are to be immediately discharged and hydrostatically tested.
10. Extinguishers that are required to be returned to the manufacturer for recharging, are to be hydrostatically tested by the manufacturer.
11. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NPFA 25.

Tools

LABEL	TOOL_BASIC	TEST_EQUIP	SEALS	SCALE-02
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This standard applies to the following types of extinguishers in service except for the non chargeable type: Dry Chemical, Stored Pressure with mild steel shells, brazed brass shells, or aluminum shells Halon 1211 and 1301 Dry Powder, Stored Pressure, Cartridge-or Cylinder Operated, with mild steel shells.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Hydrostatic testing of extinguishers requires experienced personnel and suitable testing equipment. It is recommended that this service be performed by a fire equipment company specializing in this type of service.
5. If the hydrostatic testing is to be accomplished on the premises, it should be limited only to non compressed gas type extinguishers (water base, dry chemical, and dry powder). Compressed gas extinguishers (Halon 1211 and 1301) require special internal inspections and high test pressures.
6. Wheeled extinguisher testing is beyond the scope of this standard. Utilize a fire equipment company if wheeled extinguishers are on the premises.
7. Discharge and discard extinguishers when the cost of new extinguishers is cheaper than the cost of extinguishing agent and a hydrostatic test.
8. Refer to NFPA 10 (most recent edition at time of solicitation), Portable Fire Extinguishers, for additional detailed information about hydrostatically testing extinguishers.
9. In addition, anytime an extinguisher shows evidence of physical injury or corrosion, and extinguishers with aluminum cylinders or shells are suspected of being exposed to temperatures in excess of 350 degrees (F), they are to be immediately discharged and hydrostat.
10. Halon extinguishers requiring hydrostatic testing are to be returned to a fire equipment company so that the agent can be recovered.
11. Extinguishers that are required to be returned to the manufacturer for recharging, are to be hydrostatically tested by the manufacturer.
12. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TEST_EQUIP LABEL SCALE-02 SEALS TOOL_BASIC

This standard applies to fire extinguishers that use stored air or nitrogen pressure to expel the extinguishing agent. This includes extinguishers containing water, foam, dry chemical, dry powder, and Halon 1211. This standard does not apply to carbon dioxide extinguishers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This procedure is to be accomplished in conjunction with the monthly inspections required in this standard and the 5 year hydrostatic testing.
5. Every six years, stored pressure extinguishers with gauges and shells of non-stainless steel require additional maintenance procedures
6. Do not pressurize any extinguisher that shows signs of mechanical damage, corrosion, or burned in a fire. Relieve the pressure as indicated below and destroy the extinguisher.
7. Whenever an extinguisher is removed for servicing, a replacement extinguisher is to be provided of the same type and at least equal rating.
8. Each extinguisher shall have a tag or label securely attached that indicates the month and year the inspections, maintenance, and recharging were performed and shall identify the person and firm performing the service.
9. Always recharge an extinguisher with the same type of extinguishing agent as specified on the name plate.
10. No extinguisher shall be converted from one type to another nor be converted to use a different type of extinguishing agent.
11. Never mix multipurpose dry chemical agents with any other dry chemical agents.
12. Compressed air is satisfactory to pressurize water type extinguishers. Use standard industrial-grade nitrogen for pressurizing all other extinguishers.
13. When the discharging and recharging of Halon 1211 extinguishers is necessary as part of the required maintenance, an experienced fire equipment firm should perform the maintenance so that the Halon can be captured with a closed recovery system. Discharge Halon in accordance with Clean Air Act regulations.
14. Refer to Non rechargeable Fire Extinguishers Standard for maintenance on non rechargeable stored pressure extinguishers.
15. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

SEALS MAT_CLEANING-02 TOOL_BASIC INSPECTION_TAG SPARE_HOSE-01

This standard contains additional procedures to the annual maintenance required in Standard FLSF-FEX-02-01Y and applies to fire extinguishers that use stored air or nitrogen pressure to expel the extinguishing agent and have non-stainless steel shells of mild steel, brazed brass, or aluminum. This includes extinguishers containing water, foam, dry chemical, dry powder, and Halon 1211 and 1301. This standard does not apply to carbon dioxide extinguishers. This standard shall be used in conjunction with Standard FLSF-FEX-02-01Y whenever stored pressure fire extinguishers are discharged and have to be recharged.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This procedure is to be accomplished in conduction with the monthly extinguisher inspections and the 1year hydrostatic testing as required.
5. Review manufacturer's instructions, if available.
6. An experienced fire equipment firm should perform the maintenance of Halon 1211 and 1301 extinguishers so that the Halon can be captured with a closed recovery system.
7. Release pressure on extinguisher, without discharging extinguisher agent, and dismantle extinguisher.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TOOL_BASIC	INSPECTION_TAG	SPARE_HOSE-02	SEALS	PULL_PINS	O_RING	MAT_CLEANING-02	TAG_MAINT
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This standard applies to dry chemical and Halon extinguishers that cannot be recharged and are intended to be disposed of after use.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Non rechargeable extinguishers shall be removed from service and disposed of at a maximum interval of 1years from the date of manufacture.
5. When a non rechargeable dry chemical extinguisher is to be disposed of, it is to be discharged before discarding.
6. When a non rechargeable Halon extinguisher is to be disposed of, it is to be returned to the manufacturer, fire equipment dealer, or fire equipment distributor without being discharged so that the Halon can be recovered.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TOOL_BASIC

This standard applies to fire extinguishers that use a nitrogen cartridge to expel the extinguishing agent or the agent is self-expelled due to the agent being a compressed gas. This includes dry chemical, dry powder, Halon 1301, and carbon dioxide extinguishers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This procedure is to be accomplished in conjunction with the monthly inspection as required and required hydrostatic testing
5. Whenever an extinguisher is removed for servicing, a replacement extinguisher is to be provided of the same type and at least equal rating.
6. Each extinguisher shall have a tab or label securely attached that indicates the month and year the inspections, maintenance, and recharging were performed and shall identify the person and firm performing the service.
7. Always recharge an extinguisher with the same type of extinguishing agent as specified on the name plate.
8. No extinguisher shall be converted from one type to another nor be converted to use a different type of extinguishing agent.
9. Never mix multipurpose dry chemical agents with any other dry chemical agents.
10. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TOOL_BASIC	SPARE_HOSE-03	PULL_PINS	SEALS	O_RING	MAT_CLEANING-02	INSPECTION_TAG	SCALE-03
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This standard applies to water spray extinguishing systems. These systems are similar to automatic sprinkler systems except that they use open heads that are specifically designed to project a particular water pattern to a precisely defined area and are used to provide protection to special hazards such as cooling towers, electrical transformers, and cable trays.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. When extinguishing system service is required, use a fire service equipment company experienced in water spray extinguishing system service.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 25 (most recent at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. vPerform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TOOL_BASIC

This standard applies to water spray extinguishing systems. These systems are similar to automatic sprinkler systems except that they use open heads that are specifically designed to project a particular water pattern to a precisely defined area and are used to provide protection to special hazards such as cooling towers, electrical transformers, and cable trays.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Due to the special requirements for maintaining this type of extinguishing system, it is recommended that the servicing of the system be provided by a fire equipment company experienced in water spray extinguishing system service.
5. Review manufacturer's instructions.
6. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start as well as at the completion of work.
7. Annual preventive maintenance should be scheduled for the deluge valve, the detection system, the control panel, and the item being protected at this time.
8. Take necessary steps to prevent water damage during full flow drain test.
9. Prior to water flow test, install a pressure gauge at the hydraulically most remote spray nozzle. This can be done by removing the most remote nozzle, installing a tee with a pressure gauge attached and replacing the spray nozzle.
10. Perform these procedures in the spring after there is no danger of freezing.
11. Open and close all control valves slowly.
12. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
13. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

TOOL_BASIC

This standard applies to all fixed fire extinguishing systems utilizing carbon dioxide, Halon 1211 or 1301, or dry chemical and includes both total flooding and local application types.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. When extinguishing system service is required, use a fire equipment company experienced and certified in fire extinguishing system service.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 12, 13, 17,17A (most recent editions at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

SEALS **INSPECTION_TAG**

This standard applies to all fixed fire extinguishing systems utilizing carbon dioxide, Halon 1211 or 1301, or dry chemical and includes both total flooding and local application types.

Special Instructions

1. The procedures in this standard are in addition to the monthly inspection procedures.
2. Due to the special requirements for maintaining these extinguishing systems, it is recommended that the servicing of the systems be provided by a fire equipment company experienced and certified in fire extinguishing system service.
3. Obtain and review manufacturer's instructions.
4. Whenever heat or smoke detection systems are used to activate an extinguishing system, coordinate the procedures of this standard with maintenance of those sensing systems as appropriate.
5. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start of work.
6. When replacement cylinders are received from supplier, check that hydrostatic test date is current.
7. In addition to the required tags and/or labels, a permanent file record is to be kept for each extinguishing system. This file record should include the following information as applicable:
 - a. The date when each extinguishing system was purchased and installed.
 - b. The dates when maintenance was done and the name of the person or agency performing the maintenance.
 - c. The date when cylinders were last recharged and the name of the person or agency performing the maintenance.
 - d. The hydrostatic test date and the name of the person or agency performing the maintenance.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 12, 13, 17, 17A (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 10 and NFPA 25.

Tools

SCALE-01 SEALS TOOL_BASIC INSPECTION_TAG

This standard applies to all fire and smoke dampers in HVAC systems. These dampers are used to isolate the HVAC system from the fire event to preclude spreading smoke in the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Verify that dampers are not installed backwards. Air movement should always tend to close the damper.
5. Verify that dampers are not held open by the air stream.
6. NEVER replace the fusible link with a piece of wire.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 72.

Tools

TOOL_BASIC LUBRICANT-03

This standard applies to all of the various types of sprinkler heads installed in the building.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions for all system components.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. Inspect sprinkler heads immediately prior to the scheduled maintenance on dry pipe valves, deluge valves, or alarm check valves in wet pipe systems.
- 5. Replace any damaged sprinkler heads prior to performing maintenance on dry pipe, deluge, or alarm check valves.
- 6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
- 7. Follow NFPA 25 (most recent editions at time of solicitation) for system operation and maintenance requirements.

Checkpoint

- 1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

SPARE_SPRINKLER_HEAD TOOL_BASIC LADDER-02

This standard applies to those small portions of wet pipe sprinkler systems that are in unheated areas and are subject to freezing that use an antifreeze solution to protect the system from freezing.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start of work and at the completion of work.
5. Whenever the valve is closed, a clearly visible impairment tag is to be installed on the valve and the fire department connection. A typical impairment tag is illustrated in the NFPA Standard 25, Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems.
6. The use of antifreeze solutions in sprinkler systems is only intended for limited portions of wet pipe systems typically containing less than forty gallons of water in small unheated areas such as loading docks. Dry pipe sprinkler systems are to be installed to protect larger areas.
7. Prepare the antifreeze solution with a freezing point below the expected minimum temperature for the locality.
8. Sprinklers with antifreeze solution must be located below the interface between the water and antifreeze solutions so that the heavier antifreeze solution will prevent the lighter water from migrating into the unheated area.
9. Beyond certain limits, increasing the proportion of antifreeze does not lower the freezing point of the solution.
10. Always mix glycerin, diethylene glycol, ethylene glycol, and propylene glycol with water in the proper proportions before putting the solution into the piping because these materials tend to thicken near 32 degrees F.
11. Protect chlorinated polyvinyl chloride (CPVC) pipe from freezing only with glycerin.
12. A potable water system is one that is suitable for drinking.
13. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
14. Follow NFPA 25 for system operation and maintenance requirements.

Checkpoint

1. Refer to NFPA 25.

Tools

TOOL_BASIC HYDROMETER-03

This standard applies to all 1 in. fire hose in hose cabinets or racks used for interior firefighting.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. In general, GSA has eliminated racked fire hose in office type buildings. However, such hose may be maintained in storage facilities and facilities such as Federal Record Centers, where persons are trained in hose use. When replacement is made, the new hose shall be of woven synthetic fiber, single jacketed, with lining, equipped with a plastic combination straight stream-fog nozzle with shutoff. Hose and nozzle are available through the Federal Supply Schedule, Class 4210.
5. The following work is to be performed in conjunction with annual maintenance on Fire Department hose Connections. A flow test is to be conducted every five years by flowing the required volume of water at the design pressures to the hydraulically most remote, highest, or dead-end hose connection of each zone of the standpipe system. Contact the regional Safety and Environmental Management Branch for the correct procedures for the test.
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_BASIC

This standard applies to all 1" fire hose in hose cabinets or racks used for interior firefighting.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. In general, GSA has eliminated racked fire hose in office type buildings. However, such hose may be maintained in storage facilities and facilities such as Federal Record Centers, where persons are trained in hose use. When replacement is made, the new hose shall be of woven synthetic fiber, single jacketed, with lining, equipped with a plastic combination straight stream-fog nozzle with shutoff. Hose and nozzle are available through the Federal Supply Schedule, Class 4210.
5. The following work is to be performed in conjunction with annual maintenance on Fire Department hose Connections.
6. A flow test is to be conducted every five years by flowing the required volume of water at the design pressures to the hydraulically most remote, highest, or dead-end hose connection of each zone of the standpipe system. Contact the regional Safety & Environmental Management Branch for the correct procedures for the test.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

WATER_FLOW_TEST TOOL_BASIC TEST_HOSE

This standard applies to the standpipe system and hose connections in the building for use by the fire department. These components are typically located in stairwells and on the exterior of the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. Perform this work in conjunction with fire hose maintenance, where applicable.
6. When cracking the valve, do not stand directly in front of the opening.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_GROUP_C

This standard applies to the fire standpipe system and fire hose connections in the building for use by the fire department. These system components are typically located in stairwells and on the exterior of the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. The field office manager and the control center, central station or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. The following work is to be performed in conjunction with fire hose maintenance.
6. When opening the valve, do not stand directly in front of the opening.
7. Hydrostatic testing requires experienced personnel and proper water flow testing equipment. Contact the regional Safety and Environmental (S&EM) Branch for correct procedure and assistance. The S&EM Branch may decide to contract for this service.
8. Record the results of these checks in an appropriate format.
9. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

WATER_FLOW_TEST TOOL_GROUP_C

This standard applies to all fire department pumper connections to the building standpipe and sprinkler systems. Fire departments will connect a large hose to these connections and utilize their equipment to pressurize and deliver water for firefighting purposes.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Never stand directly in front of connection when removing cap.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

Refer to NFPA 25.

Tools

TOOL_BASIC

This standard applies to all fire hydrants, including wall and flush hydrants, in both interior and exterior locations.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. All hydrants are to be inspected after each operation.
5. Dry barrel (frost proof) hydrants are always to be checked in the fall before the first frost.
6. Wet barrel hydrants only have to be inspected annually.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_BASIC LUBRICANT-03 HYDRANT_WRENCH

This standard applies to all fire hydrants, including wall and flush hydrants, in both interior and exterior locations.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Flow test dry barrel (frost proof) hydrants in the fall before the first frost.
5. When flow testing hydrants in interior locations, use fire hose to direct the flow of water outside of the building.
6. Hydrant flow testing requires the use of at least two hydrants:
7. Pressure hydrant - The hydrant closest to the building.
8. Flow hydrant - The next closest hydrant from the pressure hydrant on the downstream side of the water supply.
9. Although hydrant flow testing is fairly simple and straight forward, guidance and assistance can be obtained from the Chief, Safety and Environmental Management Branch/Division if needed.
10. Coordinate flow testing with the regular semi-annual hydrant inspection of this standard.
11. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

HYDRANT_CAP TOOL_BASIC TAPE_MEASURING-01 PITOT_TUBE HYDRANT_WRENCH

This standard applies to all fire pumps powered by a diesel engine. The fire pumps are required to start automatically to provide water pressure into mid-rise and high-rise buildings.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
5. This standard establishes three extra check points to be performed once each month on all fire pumps that are equipped with automatic starting features.
6. Hearing protection is required whenever the diesel engine is running.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_GROUP_C PROTECTION_HEARING BATTERY_TESTER

This standard applies to all fire pumps powered by a diesel engine. The fire pumps are required to start automatically to provide water pressure into mid-rise and high-rise buildings.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
5. Qualified operating personnel shall be in attendance during the weekly pump operation.
6. Hearing protection is required whenever the diesel engine is running.
7. Tests of diesel engine driven pumps shall be conducted without flowing water.
8. First Test - On systems that do not have a manual start button, close starting circuit mechanically. Run long enough to ensure engine is running properly.
9. Second Test - Test shall be conducted by allowing automatic starting of the pump to occur. The diesel pump shall run a minimum of 30 minutes.
10. The following items shall be checked and tested in order to ensure that the pump assembly is in operating condition and is free from physical damage.
11. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_GROUP_B PROTECTION_HEARING BATTERY_TESTER

This standard applies to all fire pumps powered by a diesel engine. The fire pumps are required to start automatically to provide water pressure into mid-rise and high-rise buildings. These checks are in addition to the weekly and monthly checks required by this standard and NFPA.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. A yearly test shall be made at full pump capacity and over to make sure that neither pump nor suction pipe is obstructed. Guidance and assistance should be obtained from the Chief, Safety and Environmental Management Branch/Division.
6. Hearing protection is required whenever the diesel engine is running.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

COOLANT	TUNE_UP_KIT	ENGINE_OIL-02	EQUIP_CLEANING-03	OIL-04	PROTECTION_HEARING	TOOL_GROUP_C
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This standard applies to all fire pumps powered by a diesel engine. The fire pumps are required to start automatically to provide water pressure into mid-rise and high-rise buildings. These checks are in addition to the weekly and monthly checks required by this standard and NFPA.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. This standard establishes six extra check points to be performed once each quarter on all fire pumps that are powered by diesel engines.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

BATTERY_TESTER TOOL_GROUP_B

This standard applies to all fire pumps powered by an electric motor. This fire pump is required to start automatically when receiving a signal from the fire alarm panel or activated by fire department personnel. These checks are in addition to the weekly checks required by this standard and NFPA.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
5. This standard establishes two extra check points to be performed once each month on all fire pumps that are equipped with automatic starting features. (See Standard No. FLSF-PMP-02).
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_BASIC

This standard applies to all fire pumps powered by an electric motor. This fire pump is required to start automatically when receiving a signal from the fire alarm panel or activated by fire department personnel.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at the completion of work.
5. Qualified operating personnel shall be in attendance during the weekly pump operation.
6. Tests of electric motor driven pumps shall be conducted without flowing water.
7. First Test - On systems which do not have a manual start button, close starting circuit mechanically. Run long enough to ensure motor is running smoothly.
8. Second Test - Test shall be conducted by allowing automatic starting of the pump to occur. The electric pump shall run a minimum of 10 minutes.
9. The following items shall be checked and tested in order to ensure that the pump assembly is in operating condition and is free from physical damage.
10. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance with NFPA 25.

Tools

TOOL_GROUP_B BATTERY_TESTER

This standard applies to all fire pumps powered by an electric motor. This fire pump is required to start automatically when receiving a signal from the fire alarm panel or activated by fire department personnel. These checks are in addition to the weekly and monthly checks required by this standard and NFPA.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. A yearly test shall be made at full pump capacity and over to make sure that neither pump nor suction pipe is obstructed. Guidance and assistance should be obtained from the Chief, Safety and Environmental Management Branch/Division.
6. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_GROUP_C

This standard applies to all dry pipe, deluge, and preaction sprinkler valves. These valve operate to provide water into the piping when a fire event requires activation.

Special Instructions

1. Due to the special requirements for maintaining these types of valves, it is recommended that the servicing of these valves be provided by a fire equipment company experienced in this type of service.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Obtain and review manufacturer's instructions.
4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start of work and at the completion of work.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. If drains are not piped to the outside of the building, take necessary steps to prevent water damage during water flow alarm test. Rate of discharge may exceed capacity of floor drain.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_BASIC

This standard applies valves which are exposed to cold weather. These valve operate to provide water into the piping when a fire event requires activation.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
5. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_BASIC

This standard applies to all dry pipe, deluge, and preaction sprinkler valves. These valve operate to provide water into the piping when a fire event requires activation.

Special Instructions

1. Due to special requirements for maintaining these types of valves, it is recommended that the servicing of these valves be provided by a fire equipment company experienced in this type of service.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Obtain and review manufacturer's instructions.
4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start of work and at the completion of work.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. If drains are not piped to the outside of the building, take necessary steps to prevent water damage during water flow alarm test. Rate of discharge may exceed capacity of floor drain.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 25 (most recent version at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_BASIC

This standard applies to all strainers, filters and orifices. Dry pipe, deluge and preaction valves operate to provide water into the piping when a fire event requires activation.

Special Instructions

1. Due to special requirements for maintaining these types of valves, it is recommended that the servicing of these valves be provided by a fire equipment company experienced in this type of service.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Obtain and review manufacturer's instructions.
4. The work required by this procedure may cause activation of an alarm and/or a supervisory signal. The field office manager, control center, central station, and/or fire department that will receive the alarm and/or signal must be notified prior to start of work and at the completion of work.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. If drains are not piped to the outside of the building, take necessary steps to prevent water damage during water flow alarm test. Rate of discharge may exceed capacity of floor drain.
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_BASIC

This standard applies to all valves controlling water-based fire protection systems that have an indicating post that tells whether the valve is open or shut.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start of work and at the completion of work.
5. This work should be done when other scheduled maintenance is being performed that involves other water flow control valves. (See Standard Nos. FLSF-ALM-01, FLSF-VLV-01, and FLSF-VLV-03).
6. Whenever the valve is closed, a clearly visible impairment tag is to be installed on the valve and the fire department connection. A typical impairment tag is illustrated in the National Fire Protection Association Standard 25, Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems (most recent edition at time of solicitation).
7. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
8. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

MAT_CLEANING-02 TOOL_GROUP_C LUBRICANT-07

This standard card applies to all valves controlling water-based fire protection systems, including outside control valves and post indicating valves.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_GROUP_C

This standard card applies to all valves controlling water-based fire protection systems, including outside control valves and post indicating valves. This is for visual inspection only.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_GROUP_C

This standard card applies to all valves controlling water-based fire protection systems, including outside control valves and post indicating valves.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The work required by this procedure may cause the activation of an alarm and/or a supervisory signal. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
5. Whenever the valve is closed, a clearly visible impairment tag is to be installed on the valve and the fire department connection. A typical impairment tag is illustrated in the National Fire Protection Association Standard 25, Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems. Most fire system control valves are normally in the open position.
6. This work should be done when other annually scheduled maintenance is being performed that involves water flowing through valve(s).
7. See other portions of this standard for special annual maintenance requirements for post indicator valves.
8. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
9. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_GROUP_C

This standard card applies to all valves controlling water-based fire protection systems, including outside control valves and post indicating valves.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Obtain and review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
6. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

TOOL_GROUP_C

This applies to all pressure regulating type valves located in automatic sprinkler systems, fire department standpipe hose connections, and standpipe connections with 1.5 in. racked fire hose. These valves are found in combined automatic sprinkler and standpipe systems as well as in standpipe systems. Refer to an appropriate diagram of a typical system as necessary.

Special Instructions

1. Due to the special requirements for maintaining these types of valves, it is recommended that the servicing of these valves be provided by a fire equipment company experienced in this service.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. The purpose of this flow test is to determine if the system design water flow can be met with the current valve pressure settings.
6. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. Tamper switches may be provided. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
7. This work should be done when other annually scheduled maintenance is being performed involving water flowing through valve(s).
8. Check what the design pressure downstream from the pressure regulating valve is from the automatic sprinkler system design documents so that the pressure can be verified during the flow test. This design pressure will have been determined for the monthly inspections.
9. A pressure gage is required on both the upstream and downstream sides of the pressure reducing valve in order to accomplish this flow test.
10. Some valves can be adjusted in the field while others are preset and must be returned to the factory for adjustment. To establish the pressure setting, the following three variables must be known: (1) valve inlet pressure, (2) required outlet design pressure, and (3) required flow. Contact the Regional S&EM Office for assistance if you have any questions or if test procedures are not clear.
11. When valves are returned to the factory for testing or adjustment, care must be taken to reinstall them on the same floor from which they were removed. This is to ensure that valves with the proper pressure settings are reinstalled according to the system design. Design pressure settings will vary from floor to floor.
12. Waterflow testing requires experienced personnel and proper water flow testing equipment. Contact the regional Safety and Environmental S&EM Branch for correct procedure and assistance. The S&EM Branch may decide to contract out for this service.
13. Follow NFPA 72 National Fire Alarm Code (most recent version at time of contract award) for fire alarm system operation and maintenance requirements. Chapter 10 of NFPA 72 specifically covers the testing, maintenance and inspection of fire alarm systems.

Checkpoint

1. Refer to NFPA 25.

Tools

WATER_FLOW_TEST TOOL_GROUP_C

This applies to all pressure regulating type valves located in automatic sprinkler systems, fire department standpipe hose connections, and standpipe connections with 1.5 in. racked fire hose. These valves are found in combined automatic sprinkler and standpipe systems as well as in standpipe systems. Refer to an appropriate diagram of a typical system as necessary.

Special Instructions

1. Due to the special requirements for maintaining these types of valves, it is recommended that the servicing of these valves be provided by a fire equipment company experienced in this service.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Obtain and review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. The purpose of this flow test is to determine if the system design water flow can be met with the current valve pressure settings.
6. The work required by this procedure may cause the activation of an alarm and/or supervisory signal. Tamper switches may be provided. The field office manager and the control center, central station, or fire department that will receive the alarm and/or signal must be notified prior to start and at completion of work.
7. This work should be done when the standpipe systems are flow tested.
8. Check what the design pressure downstream from the pressure regulating valve is from the automatic sprinkler system design documents so that the pressure can be verified during the flow test. This design pressure will have been determined for the monthly inspections.
9. A pressure gage is required on both the upstream and downstream sides of the pressure reducing valve in order to accomplish this flow test.
10. Some valves can be adjusted in the field while others are preset and must be returned to the factory for adjustment. To establish the pressure setting, the following three variables must be known: (1) valve inlet pressure, (2) required outlet design pressure, and (3) required flow. Contact the Regional S&EM Office for assistance if you have any questions or if test procedures are not clear.
11. When valves are returned to the factory for testing or adjustment, care must be taken to reinstall them on the same floor from which they were removed. This is to ensure that valves with the proper pressure settings are reinstalled according to the system design. Design pressure settings will vary from floor to floor.
12. Waterflow testing requires experienced personnel and proper water flow testing equipment. Contact the regional Safety and Environmental S&EM Branch for correct procedure and assistance. The S&EM Branch may decide to contract out for this service.
13. The inspection, testing, repairs, and maintenance of all fire protection equipment and systems shall be in accordance with the requirements in the applicable NFPA code or standard.
14. Follow NFPA 25 (most recent edition at time of solicitation) for system operation and maintenance requirements.

Checkpoint

1. Perform inspection, testing, and maintenance tasks and activities in accordance NFPA 25.

Tools

WATER_FLOW_TEST TOOL_GROUP_C

HVAC-ACR-01-01M**Computer Room Air-Conditioning Unit, Package: or Special Systems**

Rev-2 | 2021-11-08

This PM standard applies to air conditioning units that may have the evaporator, compressor, fan unit components, and condenser within a single housing or may have the condenser separate from the housing. These units can be fan coils, heat pumps or spilt A/C units. This equipment is typically found in computer rooms, but may also be found in special purpose rooms such as CAD rooms or laboratories.

Special Instructions

1. Review manufacturer's instructions.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
3. Perform any required drycooler or air cooled condenser maintenance simultaneously with this PM.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the equipment.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet (SDS) for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Thoroughly inspect and clean interior and exterior of machine with wet/ dry vacuum, (remove panels).
2. Clean drain pan and note excessive corrosion, prepare and paint necessary. Consult the Safety Data Sheet to ensure that the paint is lead free.
3. Perform checks according to the type of unit.
 - a. Chilled water/Condenser Water units:
 - i. Check for water leaks on all lines, valves, strainers, coils, etc. If leaks are not able to be stopped or corrected, report leak status to supervisor.
 - ii. Clean strainer on chilled water unit.
 - b. Direct expansion units:
 - i. Check for refrigeration leaks on all lines, valves, fittings, coils, etc., using a halogen leak detector or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor.
4. Check refrigerant levels and recharge if needed. Consult the Safety Data Sheet for disposal requirements. Reclaimed and recycled CFCs are exempt from hazardous waste regulations (Consult 40 CFR Part 261).
5. Check condition of cooling and reheat coils. Use fin comb if needed to straighten fins.
6. A dirty coil surface can be cleaned using a coil cleaner solution and warm water.
7. Drain and clean humidifier drip pan, replace pan if applicable. Remove scale and paint if necessary.
8. Lubricate motor and fan bearings, if not sealed. Check alignment of motor and fan. Clean fan or blower.
9. Check belt tension and condition. Adjust or replace as required.
10. Replace prefilters if needed.
11. Replace final filters if needed.
12. Check compressor oil level if compressor has a sight glass.

13. Run machine, check action of controls, relays, switches, etc. to see that:
 - a. compressor(s) run at proper settings.
 - b. reheat coils activate properly.
 - c. humidistat activates humidifier.
 - d. suction and discharge pressures are proper.
 - e. discharge air temperature is set properly.
14. Check and adjust vibration eliminators. Replace if required.
15. Check and tighten all electrical terminals, connections, and disconnect switches.
16. Remove all trash or debris from work area. Consult the Safety Data Sheet for PPE.
17. If applicable, clean and test condensate pump and alarm.

Tools

CLAMP_METER	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02	PAINT-02	REFRIGERANT_RECOVERY-01
REFRIGERANT_TANK	SAFETY_GOGGLES	TOOL_CLEANING-01	TOOL_GROUP_A		

This standard card applies to those air conditioning machines that are designed to heat as well as cool a space or building with a water cooled condenser. Like package unit air conditioning machines, these have a evaporator, compressor, fan unit components, and a water cooled condenser. These will be split systems. For package type heat pumps, refer to the standard for package air handlers. Heat pumps are designed to be more efficient than resistance (emergency) heating at any temperature. Once the temperature falls below 32°F, a heat pump will need to utilize a supplemental heat source to meet the comfort needs. The other portion of a split system, i.e., a refrigeration condensing unit, is to be serviced in conjunction with this PM activity.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the equipment.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Inspect piping for evidence of leaks and vibration. If leaks are not able to be stopped or corrected, report leak status to supervisor.
2. Inspect all wiring for deterioration, and tighten electrical contacts. Check for corrosion, clean, prime, and paint as necessary.
3. Check mounting bolts and tighten if needed.
4. Check crankcase heater.
5. Check fan for vibration or excessive noise. Lubricate fan and motor if required.
6. Check refrigerant levels, recharge if necessary. Check for leaks if loss of refrigerant is detected, using halide leak detector and soap bubbles. If leaks are not able to be stopped or corrected, report leak status to supervisor.
7. Check temperature drop across condensing coil.
8. Check condensing coil and connected hoses for water leaks.
9. Clean air intake and screens; change filters as necessary.
10. Brush or pressure wash coil surfaces. Straighten fins with fin comb.
11. Check that reversing valve is energized in the "heat" mode and deenergized in the "cool" mode. Replace defective valves.
12. Check all electrical connections and fused disconnect switches.
13. Check electrical resistance heat if present for proper operation and cycling.
14. Check all controls, indoor and outdoor thermostats, timers, and control delays, especially for units with electric supplemental heaters. Repair or replace as necessary.
15. Check oil if compressor is equipped with a sight glass.

- 16. Clean up work area.
- 17. If applicable, clean and test condensate pump and alarm.

Tools

MAT_CLEANING-02	LUBRICANT-07	REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	LEAK_DETECTOR-02
REFRIGERANT	REFRIGERANT_TANK	SAFETY_GOGGLES	TOOL_GROUP_A	VACUUM_CLEANER-05	HOSE_FITTING-01

This standard card applies to air conditioning units and heat pumps that have remote condensers (split systems). The other portion of the split system, i.e., a refrigeration condensing unit, air or water cooled, or a packaged type chilled water unit, is to be serviced in conjunction with this PM activity.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. A/C machine maintenance should be scheduled to coincide with condensing unit or package chiller maintenance as noted above.
5. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
8. Recover, recycle, or reclaim the refrigerant as appropriate.
9. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal.
10. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
11. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
12. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Thoroughly inspect and clean interior and exterior of machine with vacuum (remove panels).
2. Clean drain pan and note excessive corrosion. Treat rusted areas with rust inhibitor. Ensure that the rust inhibitor chemical does not add volatile organic compounds or contaminants to the drain pan. If possible, rinse well after application or choose a less hazardous material. Consult the chemicals Safety Data Sheet (SDS) for this information.
3. Perform checks according to the type of unit.
 - a) Chilled water units:
 - i. 1). Check for chilled water leaks on all lines, valves, strainers, coils, etc. If leaks are not able to be stopped or corrected, report leak status to supervisor.
 - ii. 2). Clean strainer on chilled water unit.
 - b) Direct expansion units:
 - iii. Check for refrigeration leaks on all lines, valves, fittings, coils, etc., using a halogen leak detector or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor.
4. Check condition of cooling and reheat coils. Use fin comb if need to straighten fins.
5. Clean coils. Use detergent solution and warm water if coil is heavily soiled.
6. Drain and clean humidifier pan or pad, whichever applies. Replace pad if required. Remove corrosion as needed.
7. Clean and lubricate motor and squirrel cage fan(s). Check alignment of motor and fan. Check bearings for excessive wear.
8. Check belt tension and condition. Adjust or replace as required.
9. Replace pre-filters if needed.
10. Replace final filter if needed.
11. Run machine, check action of controls, relays, switches, etc., to see that:
 - a) Chilled water units:
 - i. Chilled water valve(s) are operating properly.

- ii. Reheat coils activate properly.
- iii. Humidistat activates humidifier.
- iv. Valves regulating water pressure are proper on cooling.
- v. discharge air temperature is set properly.
- vi. Check and record chilled water inlet and outlet temperatures.
- b) Direct expansion units:
 - i. Humidistat activates humidifier.
 - ii. Reheat coils activate properly.
 - iii. Discharge air temperature is set properly.
- 12. Check and adjust vibration eliminator mountings if equipped. Repair or replace if required.
- 13. If applicable, clean and test condensate pump and alarm

Tools

REFRIGERANT	FILTERS-02	GLOVES-01	LEAK_DETECTOR-02	REFRIGERANT_RECOVERY-02	REFRIGERANT_TANK	
SAFETY_GOGGLES	SPARE_BELT	TOOL_CLEANING-05	TOOL_GROUP_A	VACUUM-01	HOSE_FITTING-02	FIN_COMB

This standard card applies to special purpose or critical use, ceiling or wall mounted air conditioning units, i.e., mini-mates or ductless split type units. The unit may be either air cooled or water cooled. Humidifiers will be operated on those units serving computer space and will be inventoried and serviced under this standard card.

Special Instructions

1. Schedule outage with operating personnel.
2. Schedule PM on associated equipment in conjunction with this standard, i.e., air cooled condensers, glycol dry coolers, cooling tower, etc.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the equipment.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Thoroughly inspect and clean interior and exterior of machine with vacuum (remove panels).
2. Clean drain pan and note excessive corrosion. Correct as necessary.
3. Check for refrigerant leaks using a halogen leak detector, soap bubbles, or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor. Consult the Safety Data Sheet for disposal requirement. Reclaimed and recycled CFCs are exempt from hazardous waste regulations (Consult 40 CFR Part 261).
4. Check refrigerant levels and recharge if necessary.
5. Check condition of cooling and reheat coils. Use fin comb as needed.
6. Clean coils using detergent solution and warm water if coil is heavily soiled.
7. Drain and clean humidifier pan or pad, whichever applies. Replace pad if required. Remove corrosion, prime, and paint as needed.
8. Lubricate motor and fan bearings, if not sealed. Check alignment of motor and fan. Clean all fans or blowers.
9. Check belt tension and condition. Adjust or replace as required if belt driven.
10. On direct drive units, check set screws on fan shaft to make sure they are tight.
11. Replace filters as needed.
12. Check compressor oil level (not on hermetically sealed units) if compressor is equipped with a sight glass.
13. units) if compressor is equipped with a sight glass.
14. Run machine. Check action of controls, relays, switches, including fused disconnect type, etc., to see that:
 - a. compressor(s) run at proper setting.
 - b. reheat coils activate properly (if applicable).
 - c. humidistat activates humidifier (if applicable).
 - d. suction and discharge pressures are proper.
 - e. discharge temperature is set properly.
20. If applicable, clean and test condensate pump and alarm.
21. Clean up work area.

Tools

FIN_COMB	REFRIGERANT	GREASE_GUN-02	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02		
REFRIGERANT_RECOVERY-01		REFRIGERANT_TANK	SAFETY_GOGGLES	TOOL_CLEANING-03	TOOL_GROUP_A	LADDER-03	
VACUUM-01							

This PM standard applies to equipment which has the condenser, fan(s), and fan motor(s) enclosed within the same housing. The compressor and other components are at a separate location. PM of these other devices should be scheduled simultaneously with the units serviced by the condenser. If the condenser motor(s) is/are rated at 1 HP or higher, schedule PM of motor(s) with this PM.

Special Instructions

1. Review manufacturer's instructions.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
3. Review standard operating procedure for selection, care, and use of respiratory protection.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the equipment.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Remove debris from air screen and clean underneath unit.
2. Pressure wash coil with coil cleaning solution.
3. Straighten fin tubes with fin comb.
4. Check electrical connections for tightness.
5. Check mounting for tightness.
6. Check for and remove all corrosion or rust from unit and supporting steel, prime and paint as necessary. Consult the Safety Data Sheet to ensure that the paint is lead free. Consult the Safety Data Sheet for proper PPE.
7. Check fan blades and belts. Clean fan blades as necessary.
8. Check wires at condenser electrical fused safety switches for tightness and burned insulation. Repair as necessary.
9. Clean up work area.

Tools

REFRIGERANT_TANK	GLOVES-01	SAFETY_GOGGLES	FIN_COMB	HOSE_FITTING-01	MAT_CLEANING-02	PAINT-05
PRESSURE_WASHER	RESPIRATOR-01	TOOL_BASIC	REFRIGERANT_RECOVERY-01			

An evaporative condenser is similar to a cooling tower except the heated refrigerant is piped into the unit where water is sprayed over the coils and a fan blows air through the coils. The air flow increases the evaporation rate of the water thereby cooling the coils. The leftover water in the basin (sump) is recirculated through the unit. Makeup water is added to the system to counteract water lost due to evaporation.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Perform cleaning of the condenser in accordance with the PM standard for cooling tower cleaning before performing this PM activity.
4. Schedule performance of this PM activity prior to seasonal start-up.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations regarding refrigerant, as they apply to protection of stratospheric ozone.
7. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
8. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
9. Recover, recycle, or reclaim the refrigerant as appropriate.
10. If disposal of the appliance is required, follow regulations concerning removal of refrigerants and disposal of the appliance.
11. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
12. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
13. For refrigerant type units, closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.
14. Properly dispose of any debris, excess oil, and grease.
15. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management. Request copy of latest asbestos management plan from COR if necessary.

Checkpoint

1. Exterior Structural:
 - a. Inspect louvers for correct position and alignment, missing or defective items, and supports. Examine for ice damage.
 - b. Inspect casings and attaching hardware for leaks or defects.
 - c. Inspect condition of access doors and hinges. Repair as necessary.
 - d. Inspect the distribution system, including flange connectors and gaskets, caulking of headers, deterioration in distribution basins, splash guards, and associated piping.
 - e. Check screens.
 - f. Inspect stairways (if applicable), including handrails, knee rails, stringers, structure, and fasteners for rot, corrosion, security, and acid attack.
 - g. Shake ladders (if applicable) to verify security. Check all rungs. Verify compliance with Occupational Safety and Health (OSHA) regulations regarding height requirements.
 - h. Fan cylinders must be securely anchored. Check fastening devices. Note any damaged, corroded, or missing items. Verify proper tip clearance between the fan blade and interior of the cylinder. Verify compliance with OSHA requirements regarding height. Check its condition.
 - i. Apply protective coatings as needed on metal surfaces. Be sure rust and dirt have been removed first.

2. Interior Structural:
 - a. Inspect the distribution system piping for decay, rust, or acid attack. Check the condition and tightness of water and coil connections. Observe spray pattern of nozzles if possible and note missing and defective nozzles.
 - b. Inspect mechanical equipment supports and fasteners for corrosion. Check condition of springs or rubber vibration absorption pads, including adjusting bolts, ferrous members, and rubber pads.
 - c. Check valves, float valve, and continuous bleed line (should be open). Check operating condition of fire detection system, if installed. Check for corrosion of pipes and connectors. Check wiring of any thermocouple installed.
 - d. Inspect condenser fill for damage, ice breakage, deterioration, or misplaced, missing, or defective splash bars.
 - e. Examine interior structural supports. Look for iron rot of metal fasteners. Check condition of steel internals. Check condition and tightness of bolts.
 - f. Inspect the nuts and bolts in partitions for tightness and corrosion. Check condition of steel supports for rot and corrosion.
 - g. Check steel cold water basins for corrosion and general condition.
 - h. Check all sumps for debris, condition of screens, anti-turbular plates, and freely operating drain valves.
3. Mechanical:
 - a. Check alignment of gear, motor, and fan(s).
 - b. Check fan(s) and air inlet screens. Remove any dirt or debris.
 - i. Check hubs and hub covers for corrosion and condition of attaching hardware.
 - ii. Inspect blade clamping arrangement for tightness and corrosion.
4. Gear Box:
 - a. Clean out any sludge.
 - b. Change oil in gear reducer. Be sure gear box is full to avoid condensation.
 - c. Rotate input shaft manually back and forth to check for backlash.
 - d. Attempt to move the shaft radially to check for wear on the input pinion shaft bearing.
 - e. Look for excessive play of the fan shaft bearings by applying force up and down on the tip of a fan blade. (Note: Some output shafts have a running clearance built into them).
5. Power Transmission:
 - a. Check that the drive shaft and coupling guards are installed and that there are no signs of rubbing. Inspect the keys and set screws on the drive shaft and check the connecting hardware for tightness. Tighten or install as required.
 - b. Look for corrosion, wear, or missing elements on the drive shaft coupling.
 - c. Examine the exterior of the drive shaft for corrosion. Check the interior by tapping and listening for dead spots.
 - d. Observe flexible connectors of both ends of the shaft.
 - e. Inspect bearings, belts, and pulleys for excessive noise, wear or cracking, alignment, vibration, looseness, surface glazing, tension. Replace or repair as necessary.
 - f. Inspect condenser coil, fins, sprays, connections, etc. Clean if required.
 - g. Check water distribution. Adjust water level and flush out troughs if necessary. Check all piping, connections, and brackets for looseness. Tighten loose connections and mounting brackets. Replace bolts and braces as required.
 - h. Check nozzles for clogging and proper distribution.
 - i. Check pump. Lubricate as required.
 - j. Check water treatment equipment for proper operation and condition. Clean and paint as necessary.
6. Electrical:
 - a. Check electrical motor for excessive heat and vibration.
 - b. Inspect wiring, conduit, and electrical controls for loose connections, charred or broken insulation, or other defects. Tighten, repair, or replace as necessary.
 - c. Remove dust from air intakes and check for corrosion. Check TEFC motors for condition of air passages and fans.
 - d. If there is a drain moisture plug installed, see if it is operational.
 - e. Check motor and starters. Check amps and volts at operating loads.
 - f. Look for corrosion and security of mounting bolts and attachments.

Tools

SAFETY_GOGGLES TOOL_GROUP_A_AND_C PROTECTIVE_COAT-02 METER_VOLT_AMP MAT_CLEANING-02 LUBRICANT-07
 LADDER-06 GLOVES_WORK PRESSURE_WASHER

This standard applies to built-up air handlers over 5,000 CFM with chilled water and/or hot water and/or steam coils that are fed from central plant(s). O&M contractors may propose changes to this standard. Such changes are subject to GSA approval. Proposed changes to standards must be submitted showing markups of changes for GSA approval.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. Preventive maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts for cracks and excessive wear.
3. Check fan RPM against design specifications.
4. Check running motor amperage on all three phases. Compare with manufacturer's specifications.
5. Tighten all electrical connectors/lugs to proper torque.
6. If unit is a multi-zone air handler, then check each individual zone damper and associated controls.
7. Check bearing collar set screws on fan shaft to make sure they are tight.
8. Check dampers for dirt accumulations, clean as necessary. Check belt, repair or replace as necessary.
9. Check damper actuators and linkage for proper operation. Adjust linkage on dampers if out of alignment.
10. Lubricate mechanical connections of dampers sparingly.
11. Clean coils by brushing, blowing, vacuuming, or pressure washing.
12. Check coils for leaking, tightness of fittings.
13. Use fin comb to straighten coil fins.
14. Flush and clean condensate pans and drains, remove all rust prepare metal and paint. Consult the Safety Data Sheet to ensure that the paint is lead free. Hose down coils and drain pans and wash with an appropriate EPA approved solution approved solution.
15. Check belts for wear and cracks, adjust tension or alignment. Replace belts when necessary. Multi-belt drives shall only be replaced with matched sets.
16. Check rigid couplings for alignment on direct drives, and for tightness of assembly. Check flexible couplings for alignment and wear.
17. Before heating season (chilled water coils only): Drain cooling coils; blow down to remove moisture; refill with antifreeze and water solution; drain.
18. Check freezestat for proper temperature setting and operation.
19. Vacuum interior of unit.
20. Blow out motor air vents with compressed air only.
21. Check filter doors and access doors for proper gasketing and air leaks. Correct as necessary.
22. Lubricate fan shaft bearings while unit is running. Add grease slowly until slight bleeding is noted from the seals. Do not over lubricate. Remove old or excess lubricant.
23. Ensure that drains are clear and running. If applicable, test and clean condensate pump.
24. Clean up work area.
25. The maintenance contractor may propose a predictive maintenance program that would replace the above check points for preventative maintenance. The proposed predictive maintenance program would need to be approved by the contracting officer's representative (COR) and included in the Building Operating Plan.
26. Check Points for Equipment Functional Testing (Air Handlers controlled by Direct Digital Controls (DDC) Only).
27. Review sequence of operation for air handler.

28. Use government approved persistence commissioning tool (for example PACRAT) to verify that the air handler is operating according to the correct sequences of operation. Use approved persistence commissioning tool to also ensure that control loops are stable and sensors are reading correctly. Document anomaly reports from the persistence commissioning tool. Document resolution of anomaly reports through work orders on the Computerized Maintenance Management System (CMMS).

29. If an approved persistence commissioning tool is NOT installed then:

i. Set up trends on the building automation system for a minimum of two (2) weeks and trend the following points as appropriate for the subject air handler: supply air temperature, supply air enthalpy/ humidity, mixed air temperature, mixed air enthalpy/ humidity, return air temperature, return air enthalpy/ humidity, outside air temperature, outside air enthalpy/ humidity, supply air temperature setpoint, chilled water valve position, hot water valve position, economizer damper position, (cont)

mimum outside air damper position, outside air CFM, supply air CFM, supply and return fan VFD speed, supply and return fan kW, supply air static pressure, supply air static pressure setpoint, other DDC points critical to air handler operation.

ii. Review the trend data and check for stability of control loops.

iii. Review the trend data and confirm air handler is operating according to all the correct sequences. Typical air handler sequences that should be verified include: start/stop, optimum start, economizer operation, supply air temperature setpoint, supply air temperature re-set, static pressure re-set, demand controlled ventilation

iv. Document results of functional testing.

Tools

PRESSURE_WASHER-01	TOOL_GROUP_A	SAFETY_GOGGLES	TOOL_CLEANING-04	GREASE_GUN-02	GLOVES-01	FIN_COMB
TACHOMETER-01	SOFTWARE	VACUUM-01				

This standard applies to predicatively changing out filters for built-up air handlers over 5,000 CFM with chilled water and/or hot water and/or steam coils that are fed from central plant(s). This is accomplished by calculating the pressure drop across filters.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel as needed.
5. Preventative maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Differential pressure sensors should be installed across each filter bank and should report values to Building Automation System.
2. Determine the set point for each differential pressure sensor that should trigger a filter change. Some of the factors that should be considered include:
 - a. Manufacturer's recommendations for filter changes.
 - b. Operational issues (indoor air quality, number of service calls, etc.).
 - c. Cost of filter replacement.
 - d. Cost of increased energy usage due to filter loading.
3. Enter the determined operational set point into the system as a maintenance alarm or signal to the NCMMS system to generate a work order.

Tools

VACUUM-01 TOOL_BASIC RESPIRATOR-01 FILTER_REPLACEMENT

This standard applies to built-up air handlers 5,000 CFM and greater with chilled water and/or hot water and/or steam coils that are fed from central plant(s). This standard includes some predictive maintenance tasks and can be used as an alternative to the purely preventive maintenance standard. If the facility elects to use this standard as a substitute for the preventive maintenance air handler standard, then it is not necessary to follow the relevant air filter maintenance preventive standard. O&M contractors may propose changes to this standard. Such changes are subject to GSA approval. Proposed changes to standards must be submitted showing markups of changes for GSA approval. This standard may be used in lieu of standard HVAC-AHU-01-01Y.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. Preventive maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts for cracks and excessive wear.
3. Check fan RPM against design specifications.
4. Check running motor amps on all three phases. Compare with manufacturer's specifications.
5. Check belt, repair or replace as necessary.
6. Lubricate mechanical connections of dampers sparingly.
7. Clean coils by brushing, blowing, vacuuming, or pressure washing.
8. Check coils for leaking, tightness of fittings.
9. Use fin comb to straighten coil fins.
10. If unit is a multi-zone air handler, then check each individual zone damper and associated controls.
11. DDC:
 - a. Check to see that the operating control sensor activates the damper per design specifications. If not, recalibrate or replace the operating control sensor with the same temperature range sensor.
 - b. Check damper linkage for tightness or damage. Lightly oil moving parts using an approved lubricant.
 - c. Inspect damper(s) for free movement. Replace felt or other type seals as required.
 - d. Inspect connecting ductwork for air leaks. Correct leaks with approved duct tape or tighten connections, as required.
 - e. Tighten electrical connections to all servo-motor actuators, and test as applicable.
 - f. Check the heating or cooling valve (if present) for leakage around the stem or between the seat and disk. Repair or replace as needed.
 - g. Check velocity sensor tubing for cracks, tightness, or holes if applicable.
12. Pneumatic:
 - a. Check damper linkage for tightness or damage. Lightly oil moving parts using an approved lubricant for pneumatic systems.
 - b. Inspect damper(s) for free movement. Replace felt or other type seals as required.
 - c. Inspect mixing box and connecting ductwork for air leaks. Correct leaks with approved duct tape or tighten connections, as required.
 - d. Inspect damper actuator(s) for tightness to mounting brackets.
 - e. Inspect damper actuator diaphragm for leaks by performing a pressure test of the diaphragm.
 - f. Check the damper actuator spring range. Replace spring, adjust pilot positioner, or add pilot positioner as needed.

- g. If pneumatic actuator does not stroke properly, correct sticking valve stem or binding linkage. Replace or repair the diaphragm or actuator if necessary.
- h. Check the heating or cooling valve for leakage around the stem or between the seat and disk. Repair or replace as needed.
- i. Inspect the valve actuator for leaks by performing a pressure test of the diaphragm. Repair or replace as needed.
- j. Check the spring range of the valve actuator. Replace spring, adjust pilot positioner or add pilot positioner as needed.
- k. Inspect for air leaks around actuator and in the air line between controller and pneumatic actuator.
- l. Inspect operating control thermostat and/or pressure sensor for proper location and check main and branch air lines at thermostat for crimps, breaks, etc. Repair or replace if needed.
- m. Check all inline filters for oil and/or moisture. Replace as needed.
- n. Check all installed pressure gages for proper range and operability as applicable. Replace if needed.
- o. Perform a spring range check for all remaining end devices.
- p. Calibrate the operating control thermostat. Replace if it is defective with the same type action (direct or reverse action) and temperature range.
- q. Check to see that the operating control thermostat activates the damper per design specifications. If not, recalibrate the operating control thermostat.
- r. Flush and clean condensate pans and drains, remove all rust prepare metal and paint. Consult the Safety Data Sheet to ensure that the paint is lead free. Hose down coils and drain pans and wash with an appropriate EPA approved solution approved solution.
- s. Check rigid couplings for alignment on direct drives, and for tightness of assembly. Check flexible couplings for alignment and wear.
- t. Before heating season (chilled water coils only): Drain cooling coils; blow down to remove moisture; refill with antifreeze and water solution; drain.
- u. Check freezestat for proper temperature setting and operation.
- v. Vacuum interior of unit.
- w. Blow out motor air vents with compressed air only.
- x. Check filter doors and access doors for proper gasketing and air leaks. Correct as necessary.
- y. Lubricate fan shaft bearings while unit is running. Add grease slowly until slight bleeding is noted from the seals. Do not over lubricate. Remove old or excess lubricant.
- z. Ensure that drains are clear and running. Clean up work area.

Predictive Maintenance Checkpoints:

Vibration Analysis on Fans:

1. For each fan take baseline readings on all permanently installed vibration transducers.
2. Monitor permanently installed vibration transducers with BAS.
3. An on-staff or contracted Vibration Analysis Technician using specialized software will need to determine criteria for unacceptable vibration signals.
4. Set up maintenance alarms or signal CMMS to issue a work order when transducer output exceeds acceptable levels based on baseline readings.
5. Take corrective action by:
 - a. Having a full set of vibration readings taken by a qualified on-staff or contracted vibration analyst.
 - b. Following the analyst's diagnostic recommendations.
1. Differential pressure sensors should be installed across each filter bank and should report values to Building Automation System.
2. Determine the setpoint for each differential pressure sensor that should trigger a filter change. Some of the factors that should be considered include:
 - a. Manufacturer's recommendations for filter changes
 - b. Operational issues (indoor air quality, number of service calls, etc.)
 - c. Cost of filter replacement
 - d. Cost of increased energy usage due to filter loading
3. Enter the determined operational set point into the system as a maintenance alarm or signal to the CMMS system to generate a work order.

Check Points for Equipment Functional Testing (Air Handlers controlled by Direct Digital Controls (DDC) Only)

1. Review sequence of operation for air handler.
2. Use government approved persistence commissioning tool (for example PACRAT) to verify that the air handler is operating according to the correct sequences of operation. Use approved persistence commissioning tool to also ensure that control loops are stable and sensors are reading correctly. Document anomaly reports from the

persistence commissioning tool. Document resolution of anomaly reports through work orders on the Computerized Maintenance Management System (CMMS).

3. If an approved persistence commissioning tool is NOT installed then:

a. Set up trends on the building automation system for a minimum of two (2) weeks and trend the following points as appropriate for the subject air handler: supply air temperature, supply air enthalpy/ humidity, mixed air temperature, mixed air enthalpy/ humidity, return air temperature, return air enthalpy/ humidity, outside air temperature, outside air enthalpy/ humidity, supply air temperature setpoint, chilled water valve position, hot water valve position, economizer damper position...

(Continued) minimum outside air damper position, outside air CFM, supply air CFM, supply and return fan VFD speed, supply and return fan kW, supply air static pressure, supply air static pressure setpoint, other DDC points critical to air handler operation.

b. Review the trend data and check for stability of control loops.

c. Review the trend data and confirm air handler is operating according to all the correct sequences. Typical air handler sequences that should be verified include: start/stop, optimum start, economizer operation, supply air temperature setpoint, supply air temperature re-set, static pressure re-set, demand controlled ventilation.

d. Document results of functional testing.

Tools

TOOL_GROUP_A	VACUUM-01	SAFETY_GOGGLES	TOOL_CLEANING-04	TACHOMETER-01	SOFTWARE	PRESSURE_WASHER-01
GLOVES-01	FIN_COMB	FILTERS-02	GREASE_GUN-02			

This standard applies to packaged air handlers with A/C compressor(s) and/or a natural gas fired heating section. This includes A/C units with gas heat, A/C units only, Heat Pumps with air cooled condensers or any other type of packaged air handler with air conditioning. These units include all the components in one factory assembled unit, although may be comprised of different sections.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. Preventive maintenance on associated equipment should be accomplished at this time.
6. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
7. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
8. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
9. Recover, recycle, or reclaim the refrigerant as appropriate.
10. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal.
11. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
12. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
13. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts for cracks and excessive wear.
3. Check fan RPM against design specifications.
4. Check running motor amperage on all three phases. Compare with manufacturer's specifications.
5. Check dampers for dirt accumulations, clean as necessary. Check belt, repair or replace as necessary.
6. Check damper actuators and linkage for proper operation. Adjust linkage on dampers if out of alignment.
7. Lubricate mechanical connections of dampers sparingly.
8. If unit is a multi-zone air handler, then check each individual zone damper and associated controls.
9. Condenser:
 - a. Remove debris from air screen and clean underneath unit.
 - b. Pressure wash coil with proper cleaning solution.
 - c. Straighten fin tubes with fin comb.
 - d. Check electrical wiring and tighten loose connections. Check fused disconnect switches for condition and operation.
 - e. Check mounting for tightness.
 - f. Check for corrosion. Clean and treat with inhibitor as needed.
 - g. Check fan or blower for bent or damaged blades and imbalance.
 - h. Lubricate shaft and motor bearings on fans and remove old or excess lubricant.
 - i. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as necessary. Change badly worn belts. Multi-belt drives should be replaced with matched sets.
 - j. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an

adequate water treatment program. Consult water treatment standard for proper procedures.

- k. Clean coils by brushing, blowing, vacuuming, or pressure washing.

10. Evaporator:

a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.

- a. Inspect plumbing, valves and flanges for leaks and correct.

- b. Check coils for leaking, tightness of fittings:

i. Check for refrigerant leaks using a halogen detector or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor. Consult the Safety Data Sheet for disposal requirements. Reclaimed and recycled CFCs are

- ii. Check refrigerant levels and recharge if needed.

- c. Use fin comb to straighten coil fins.

- d. Remove debris from air screen and clean underneath unit.

- e. Inspect gaskets. Look for leaks between unit and structure, caulk as necessary.

- f. Clean condenser, cooling coil fins, and fans.

- g. Remove dirt or dust from all interior parts.

- h. Replace filter.

- i. Inspect and adjust damper.

- j. Lubricate motor and fan bearings.

- k. Check fan RPM to design specifications.

- l. Check bearing collar set screws on fan shaft to make sure they are tight.

- m. Check dampers for dirt accumulations. Check felt. Repair or replace as necessary.

- n. Check damper motors and linkage for proper operation.

- p. Lubricate mechanical connections of dampers sparingly.

11. Compressor(s):

- a. Lubricate drive coupling.

- b. Lubricate motor bearings (non-hermetic).

- c. Check and correct alignment of drive couplings.

d. Inspect evaporator tubes for scale. Clean if required. Leak test tubes using a halogen leak detector or suitable substitute.

- e. Add refrigerant per manufacturer's instructions if needed.

- f. Check compressor oil level.

12. Run machine; check action of controls, relays, switches, etc. to see that:

- a. Compressor(s) run at proper settings.

- b. Check compressor oil level.

- c. Run machine, check action of controls, relays, switches, etc., to see that:

- d. Compressor(s) run at proper settings.

- e. Reheat coils activate properly.

- f. Crankcase heater is operating properly. Gas and/or oil fired (if equipped).

- i. Check burner for flashback and tight shutoff of fuel.

- ii. Check operation of controls. Clean and adjust if necessary.

iii. Clean burner, chamber, thermocouple and control. (Use a high suction vacuum and/or brush.) Check combustion chamber for cracks, holes, or other defects.

- iv. Adjust pilot or electric ignition device.

- v. Inspect vent and damper operation.

- vi. Operate unit and adjust burner.

- vii. Check operation of safety pilot, gas shutoff valve, and other burner safety devices.

- viii. Check temperature differential and controls.

- ix. Check frame of unit with ohmmeter for proper electric ground.

- x. Replace covers (if any) and clean area.

- g. Electrical (if equipped).

i. Visually inspect for broken parts, contact arcing or any evidence of overheating. Inspect all wiring for deterioration.

ii. Check name plate for current rating and controller manufacturer's recommended heater size. (Heater size shall not be changed without the regional design engineer's approval.)

- iii. Check line and load connections and heater mounting screws for tightness.

- h. suction and discharge pressures are proper.

- i. discharge air temperature is set properly.

13. Check and adjust vibration eliminators. Replace as necessary.
14. Sample test the refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards. Based on the results, refrigerant may need to be replaced or recycled, and oil replaced.
15. Check and calibrate safety controls.
16. Gas Burner:
 - a. Check operation of all gas controls and valves including: manual gas shutoff; petal gas regulator; safety shutoff valve (solenoid); automatic gas valve; petal solenoid valve; butterfly gas valve, motor, and safety petal solenoid (if used.)
 - b. Check flue connections for tight joints and minimum resistance to air flow. (combustion chamber, flues, breaching, and chimney are clear before firing.)
 - c. Draft regulators require slightly negative pressure in the combustion chamber at maximum input.
 - d. On forced draft burners, gas manifold pressure requirements should correspond with modulating (butterfly) valve in full open position and stable at all other firing rates.
 - e. At high and low fire rates on burners equipped with OFF/LOW FIRE/HIGH FIRE control.
 - f. At single firing load point on boilers equipped with OFF/ON controls only.
 - g. Check burner for flashback and tight shutoff of fuel.
 - h. Check operation of automatic controls and combustion flame safeguards. Clean and adjust, if necessary.
 - i. Operation and adjustments should conform with manufacturer's instructions.
 - j. Flush and clean condensate pans and drains, remove all rust prepare metal and paint. Consult the Safety Data Sheet to ensure that the paint is lead free. Hose down coils and drain pans and wash with an appropriate EPA approved solution approved solution.
 - k. Check rigid couplings for alignment on direct drives, and for tightness of assembly. Check flexible couplings for alignment and wear.
 - l. Vacuum interior of unit.
 - m. Blow out motor air vents with compressed air only.
 - n. Check filter doors and access doors for proper gasketing and air leaks. Correct as necessary.
 - o. Lubricate fan shaft bearings while unit is running. Add grease slowly until slight bleeding is noted from the seals. Do not over lubricate. Remove old or excess lubricant.
 - p. Ensure that drains are clear and running.
 - q. Clean up work area.

Tools

TOOL_CLEANING-04	VACUUM-01	TOOL_GROUP_A	TACHOMETER-01	SAFETY_GOGGLES	PRESSURE_WASHER-01
GREASE_GUN-02	SOFTWARE	FIN_COMB	GLOVES-01		

This standard applies to packaged air handlers with compressors and/or gas burners. This standard includes some predictive maintenance Tasks and can be used as an alternative to the purely preventive maintenance standard. If the facility elects to use this standard as a substitute for the preventive maintenance air handler standard, then it is not necessary to follow the relevant air filter maintenance preventive standard. O&M contractors may propose changes to this standard. Such changes are subject to GSA approval. Proposed changes to standards must be submitted showing markups of changes for GSA approval.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule shutdown with operating personnel, as needed.
5. Preventive maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts for cracks and excessive wear.
3. Check fan RPM against design specifications.
4. Check running motor amperages on all three phases. Compare with manufacturer's specifications.
5. Check dampers for dirt accumulations, clean as necessary. Check belt, repair or replace as necessary.
6. Check damper actuators and linkage for proper operation. Adjust linkage on dampers if out of alignment.
7. Lubricate mechanical connections of dampers sparingly.
8. If unit is a multi-zone air handler, then check each individual zone damper and associated controls.
9. Condenser:
 - a. Remove debris from air screen and clean underneath unit.
 - b. Pressure wash coil with proper cleaning solution.
 - c. Straighten fin tubes with fin comb.
 - d. Check electrical wiring and tighten loose connections. Check fused disconnect switches for condition and operation.
 - e. Check mounting for tightness.
 - f. Check for corrosion. Clean and treat with inhibitor as needed.
 - g. Check fan or blower for bent or damaged blades and imbalance.
 - h. Lubricate shaft and motor bearings on fans and remove old or excess lubricant.
 - i. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as necessary. Change badly worn belts. Multi-belt drives should be replaced with matched sets.
 - j. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an adequate water treatment program. Consult water treatment standard for proper procedures.
 - k. Clean coils by brushing, blowing, vacuuming, or pressure washing.
10. Evaporator:
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Inspect plumbing, valves and flanges for leaks and correct.
11. Compressor(s)
 - a. Lubricate drive coupling.
 - b. Lubricate motor bearings (non-hermetic).
 - c. Check and correct alignment of drive couplings.

d. Inspect evaporator tubes for scale. Clean if required. Leak test tubes using a halogen leak detector or suitable substitute.

e. Add refrigerant per manufacturer's instructions if needed.

f. Check compressor oil level.

g. Run machine; check action of controls, relays, switches, etc. to see that:

h. Compressor(s) run at proper settings.

i. Suction and discharge pressures are proper.

j. Check and adjust vibration eliminators. Replace as necessary.

k. Sample test the refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards. Based on the results, refrigerant may need to be replaced or recycled, and oil replaced.

l. Check and calibrate safety controls.

Gas Burner:

a. Check operation of all gas controls and valves including: manual gas shutoff; petal gas regulator; safety shutoff valve (solenoid); automatic gas valve; petal solenoid valve; butterfly gas valve, motor, and safety petal solenoid (if used.)

b. Check flue connections for tight joints and minimum resistance to air flow. (combustion chamber, flues, breaching, and chimney are clear before firing.)

c. Draft regulators require slightly negative pressure in the combustion chamber at maximum input.

d. On forced draft burners, gas manifold pressure requirements should correspond with modulating (butterfly) valve in full open position and stable at all other firing rates.

e. At high and low fire rates on burners equipped with OFF/LOW FIRE/HIGH FIRE control.

f. At single firing load point on boilers equipped with OFF/ON controls only.

g. Check burner for flashback and tight shutoff of fuel.

h. Check operation of automatic controls and combustion flame safeguards. Clean and adjust, if necessary.

i. Operation and adjustments should conform with manufacturer's instructions.

j. Flush and clean condensate pans and drains, remove all rust prepare metal and paint. Consult the Safety Data Sheet to ensure that the paint is lead free. Hose down coils and drain pans and wash with an appropriate EPA approved solution approved solution.

k. Check rigid couplings for alignment on direct drives, and for tightness of assembly. Check flexible couplings for alignment and wear.

l. Vacuum interior of unit.

m. Blow out motor air vents with compressed air only.

n. Check filter doors and access doors for proper gasketing and air leaks. Correct as necessary.

15. o. Lubricate fan shaft bearings while unit is running. Add grease slowly until slight bleeding is noted from the seals. Do not over lubricate. Remove old or excess lubricant.

16. p. Ensure that drains are clear and running.

17. q. Clean up work area.

Tools

TACHOMETER-01	VACUUM-01	TOOL_CLEANING-04	SAFETY_GOGGLES	SOFTWARE	PRESSURE_WASHER-01	GLOVES-01
FIN_COMB	TOOL_GROUP_A	GREASE_GUN-02				

This standard card applies to those installations that utilize an air washer or a wet coil system on their air handler units. Air washers consist of a spray chamber, without coils, into which chilled water is sprayed for the purpose of cooling the air flowing through it. In a wet coil system, water is sprayed onto a chilled water coil for the purpose of providing better heat transfer between the coil and air. The air handler will be serviced using the standard for Air Handlers and the air washer or wet coil system should be scheduled for accomplishment at the same time. Direct or Indirect Evaporative Cooling Units shall be serviced under their specific Maintenance Standards.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule outage with operating personnel.
5. Open, lock, and tag out electric circuits serving motors for the air handler and spray pump.
6. Preventive maintenance on associated equipment should be accomplished at this time.
7. Secure and tag all water valves supplying the pan or sump.

Checkpoint

1. Examine and wire brush all structural elements including doors, chamber, piping, supports, pans, sumps, and framing.
2. Clean tank or sump and examine for leaks.
3. Treat with rust inhibitor and paint as required. Consult the Safety Data Sheet to ensure that the paint is lead free.
4. Clean suction screens.
5. Remove nozzles and clean. Replace as required.
6. Check piping for blockage or buildup. Clean or replace as required.
7. Check operation of float valve, mixing or automatic control valves and thermometers.
8. Check lights for water seal and operation. Replace as necessary.
9. Pumps and motors less than 1 HP will be serviced as part of this PM. Pumps and motors 1 HP and larger will be serviced using PM standard cards HVAC-PMP-01 and ELEC-MOT-01 respectively.
 - a. Clean and lubricate pump. Check and replace packing if applicable.
 - b. Blow out or vacuum motor windings and lubricate if required.
10. Remove tags and lockout from circuits for spray pump only.
11. Check with operating personnel before restoring circuits to the air handlers, to be certain personnel are not working on the unit.

Note: Pans and sumps should remain dry during winter operation. Tags should be removed from supply valves at the completion of this work, but the valves should be opened by operating personnel only when the unit is to be filled and placed in service.

Tools

TOOL_CLEANING-02 GASKET-02 GOOGLES TOOL_BASIC RUST_INHIBITOR-02

Hot air furnaces are used primarily to heat, but can be used with refrigerant coils to cool or as a backup for heat pump applications using multi-speed fan motors. These units can be horizontal or vertical, heating only or heating/cooling. \

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Review standard operating procedure for selection, care, and use of respiratory protection.
4. Schedule shutdown with operating personnel.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Coordinate other related preventive maintenance items, such as, motors, burners, fuel tank, humidifier, or other equipment.

Checkpoint

1. Remove furnace ends and access panels if applicable.
2. Check the fire box liner or refractory for cracks and leaks.
3. Check smoke stack for obstructions, leaks, etc.
4. Clean bottom of smoke stack (breaching).
5. Clean all fans and motors.
6. Check operation of controls and safeties.
7. Lubricate as required.
8. Check and clean plenum (clean cooling coils and check for leaks, if equipped.)
9. Replace furnace and access panels ends if removed.
10. Check all motors, belts, pulleys, shafts, etc. for alignment.
11. Treat all rusted areas with rust inhibitor and touch up paint.
12. Remove lock outs and tags. Restore fuel and power supply.

Tools

MAT_CLEAN_PATCH FLUE_BRUSH RESPIRATOR-05 RUST_INHIBITOR-05 TOOL_GROUP_A_AND_B VACUUM_CLEANER-02

Ultra-Violet (UV) Light systems are installed in the central air system where it suppresses mold grown on the cooling coils and surrounding areas to improve indoor air quality.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review manufacturer's instructions.
3. UV lights shall be replaced at a maximum of 10,000 run time or based on measuring intensity. Replace if below manufacturers recommended limit.
4. Avoid exposure to UV light, there is potential for eye or skin damage.
5. Schedule shutdown with operating personnel.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Coordinate other related preventive maintenance items, such as, motors, burners, fuel tank, humidifier, or other equipment.

Checkpoint

1. Check for evidence of loose connections.
 2. Replace all lamps (group relamping) if exceeding 10,000 hours use or below manufacturers minimum intensity level.
- Note: Even though a visible blue glow can be seen though the system viewpoint the glow is not UV light (which is invisible) but actually a by-product of the operation and lamps still require replacement.

Tools

TOOL_GROUP_A_AND_B PPE_UV_LIGHT

This PM applies to flues exhausting combustion gasses from fuel burning assets . These may be boilers, water heaters, combination air cooling/air heating units. This PM does not apply to Type B and Type BW gas venting systems defined in NFPA Standard 211. This PM does not apply to cooking hoods or rotating electricity generating equipment.

Special Instructions

1. Inspections to be performed by a state licensed inspector
2. The chimney exterior and surroundings shall be inspected at locations that can be accessed without removal of panels , doors or coverings.
3. Where panels, doors or coverings are opened as part of the performance of another task, such locations shall be examined as part of a level 1 inspection.
4. Readily accessible areas of the chimney that can be observed through openings such as clean outs or terminations shall be examined for the presence of a continuous flue liner, proper installation and no damage or deterioration.
5. When inspecting any flue in a chimney, all flues in that chimney will be inspected.
6. An inspection conducted during the cleaning or replacement of an appliance will include an inspection of the flue.
7. Inspections shall include verification that the flue is free of combustible material deposits or blockage or obstruction.
8. The connected appliances, their chimney connectors and surroundings shall be inspected for proper clearances, protection, damage, or deterioration and observable evidence of operating malfunction.
9. Chimney connectors shall be inspected for proper support, and fastening of joints, pitch, and securement to the chimney
10. Chimney connectors shall be inspected for damage, deterioration, internal blockage, or obstruction and freedom from combustible deposits.

Checkpoint

Tools

None

HVAC-AHU-INSP

Annual AHU Inspection to Verify Filtration (MUST USE WORKORDER TRACKING (HSE) TO PERFORM CHECKLIST)

Rev-1 | 2023-10-01

This annual inspection is to be carried out by the COR or COTR or their designee to verify O&M compliance with PWS mandated MERV ratings in AHUs. Also to verify both the quality of work in making the filter bank gap free and in ensuring the exterior of the AHU is free of gaps or openings that would allow air from outside the AHU to enter the AHU downstream of the filter bank. Additionally to verify that the area around the AHU is clear and free of obstructions that would impair performing the required maintenance. At the Review items below, check "yes" where items are satisfactory and "no" where items need work. Also create a supplemental work order for any review item needing work. This Inspection is to cover all AHU assets in the listed building. Ignore any references to a single floor or room in the Inspection LOCATION.

Special Instructions

1. Perform checklist from the Work Order Tracking (HSE) application.
2. After opening the Work Order, go to the Details tab and under Standard Actions fill out each Review Action with an entry to complete the checklist.
3. When the checklist is complete, change the Work Order status to Complete.

Checkpoint

Tools

None

This standard card applies to refrigerated or regenerative desiccant type air dryers with a capacity of 10 SCFM or greater. Those units with a capacity of less than 10 SCFM will be maintained in conjunction with the air compressor that they are associated with.

Special Instructions

1. Schedule this maintenance in conjunction with the maintenance on the associated air compressor.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the item.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. For refrigerant type units, closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Lubricate valves and replace packing, if necessary.
2. Check dryer operating cycle.
3. Inspect and clean heat exchanger.
4. Check outlet dew point.
5. Clean and lubricate blower.
6. Check automatic blow down devices.
7. Inspect and replace or reinstall inlet filters.
8. Refrigerated Type:
 - a. Check traps.
 - b. Check refrigerant level and moisture content. If low level or moisture is indicated, check for refrigerant leaks using a halogen leak detector or similar device. If leaks are not able to be stopped or corrected, report leak status to supervisor.
 - c. Clean and lubricate condenser fan motor.
9. Desiccant Type:
 - a. Replace filter cartridges, both prefilter and afterfilter.
 - b. Check the inlet flow pressure, temperature and purge rate.
 - c. Check the desiccant and replace if necessary.
 - d. Inspect and clean solenoids, purge valves, and strainers.

Tools

TOOL_GROUP_A EQUIP_CLEANING-04 FILTER_CARTRIDGE-04 FIN_COMB GASKET-01 HOSE_FITTING-01
REFRIGERANT_TANK REFRIGERANT_RECOVERY-01 REFRIGERANT

This standard card applies to air compressors which supply system air or Building Automation control air to a building. Compressors may come equipped with air dryers or be installed in multiples. Significant energy can be lost through leaks in the system.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions and equipment history record.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Coordinate motor PM on an annual basis.
5. Tank should be inspected and tested by qualified inspector.

Checkpoint

1. Perform normal tour checks and operations. Perform a visual inspection of the air system, noting any obvious leaks or portions of the air distribution network that may be subject to physical damage.
2. Change compressor crankcase oil.
3. Clean or replace air intake filter.
4. Check air dryer, automatic condensate drains, and air tank for proper operation. Clean condenser coils and cover grills.
5. Inspect oil separators for any sign of oil entering the system.
6. Inspect belt alignment and condition. Adjust or replace belts as required. Belts should be replaced in complete sets.
7. Check for corrosion and scale on water cooled units.
8. Clean heat exchange surfaces.
9. Check accuracy of gauges with calibrated test gauge.
10. On two stage compressor, check intermediate pressure.
11. Test relief valves, replace if leaking or the relief range is incorrect. Do not readjust safety relief valves in the field.
12. Check operation of compressor unloaders, repair or replace if not loading and unloading properly.
13. Check compressor suction and discharge valves for proper operation. Replace leaking valves.
14. Check cut in and cut out of compressor pressure controller, readjust if necessary for proper air pressure requirements. Do not exceed ASME maximum tank pressure.
15. Check to make sure belt guard is installed prior to putting air compressor back in service.
16. No pressure vessel is to have its hand hole or man hole covers removed unless the vessel is at atmospheric pressure.
17. Ensure the pressure vessel inspection certificate is posted at machine.
18. Check if air compressor is running excessively or frequently cycling on and off (possible leaks). Log hour meter readings.
19. Perform an air leak check of the compressor and air distribution network in the equipment room, using an appropriate ultrasonic scanning device. Check hoses, hose connections, hose fittings, quick couplers, filters, regulators and lubricators. Correct or schedule repair as a work item. Tag location and date of leaks. Repair air leaks as soon as possible.

Tools

TOOL_BASIC SCANNER_ULTRASONIC BELTS LUBRICANT-07 TEST_GAGE VACUUM_CLEANER-03 FIN_COMB

These units will be associated with special purpose or critical use packaged air conditioning units and refrigeration units, or would stand alone when utilized in free cooling.

Special Instructions

1. Schedule outage with operating personnel.
2. Obtain and review manufacturer's instructions for starter to be tested (including the time current characteristic curve).
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review standard operating procedures for selection, care, and use of respiratory protection.

Checkpoint

1. Dry Cooler Check Points:
 - a. Remove debris from air screen and clean underneath unit.
 - b. Pressure wash coil with coil cleaning solution. Check the Safety Data Sheet to ensure that the coil cleaner does not contain hydrofluoric acid or another irritating or hazardous compound.
 - c. Straighten fin tubes with fin comb.
 - d. Check electrical connections for tightness.
 - e. Check mounting for tightness.
 - f. Check for corrosion. Clean and treat with rust inhibitor and touch up paint as needed. Consult the Safety Data Sheet for hazardous ingredients and proper PPE.
2. Motors and Fans Check Points:
 - a. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as required. Change badly worn belts. Multi-belt drives should be replaced with matched sets.
 - b. Perform required lubrication and remove old or excess lubricant.
 - c. Clean motor with vacuum or low pressure air (less than 40 psi). Check for obstructions in motor cooling and air flow.
3. Expansion Tank Check Points:
 - a. Examine exterior of tank, including fittings, manholes, and handholes for leaks, signs of corrosion. Repair/paint as necessary.
 - b. Inspect structural supports and repair or replace damaged insulation or covering.
 - c. Clean, test, and inspect sight glasses, valves, fittings, drains, and controls.
 - d. Perform hydrostatic test if required.
 - e. Check antifreeze level with hydrometer and add glycol base antifreeze as required for protection to minus 40 degrees Fahrenheit.
4. Electrical Controls Check Points:
 - a. Visually inspect for broken parts, contact arcing, or any evidence of overheating.
 - b. Check motor name plate for current rating and controller manufacturer's recommended heater size. (Heater size shall not be changed without the COR's approval.)
 - c. Check line and load connections and heater mounting screws for tightness.

Tools

ANTIFREEZE	MAT_CLEANING-02	FIN_COMB	HYDROMETER-02	PAINT-04	PRESSURE_WASHER-01	RESPIRATOR-01
RUST_INHIBITOR-06	SAFETY_GOGGLES	TOOL_BASIC	VACUUM_CLEANER-01			

This standard card applies to the after-cooler/separator utilizing chilled water to condense moisture from large compressed air systems. These differ from mechanical/chemical type air dryers. Maintenance on this item should be performed with air compressors which supply the air through the after cooler.

Special Instructions

1. Schedule outage with operating personnel.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
3. Provide an alternate source of air, if necessary.
4. Schedule Preventive Maintenance (PM) on associated equipment if possible.
5. Secure air and chilled water valves and tag them.
6. Wear appropriate protective equipment.
7. Use caution when disassembling. Check for and relieve pressure where found.

Checkpoint

1. Unbolt and remove supply and discharge water lines.
2. Unbolt supply and discharge air flanges and lower assembly.
3. Remove tube bundle assembly.
4. Check tube bundle for deterioration or ruptured tubes.
5. Clean exterior of the tube of all scale buildup.
6. Flush out the tube bundle shell.
7. Visually inspect the shell, flanges, piping, etc. for deterioration, cracks, etc.
8. Clean the trap orifice of all carbon and heavy grease buildup.
9. Reassemble the tube bundle in the shell using new gaskets and seals.
10. Replace assembly and separator in pipe line, using new gaskets.
11. Reconnect the supply and discharge water lines.
12. Remove tags, open air and water valves, restore power, and start air compressor.
13. Check operation of unit. Check the air and water inlet and outlet temperatures.
14. Check all connections for leaks.
15. Wire brush, treat with rusticide and primer rusted areas. Consult the SDS for hazardous ingredients and proper PPE. Consult the Safety Data Sheet to ensure that the paint is lead free.

Tools

GASKET-03 HOSE-03 RUST_INHIBITOR-08 TOOL_GROUP_C VACUUM-02 HOIST-02 LADDER-04 BRUSH_WIRE

This standard card applies to both heating boilers and power boilers which use either natural gas or fuel oil for their fuel source. The standard card will identify which information applies to only one of these two boilers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform boiler external inspection with internal inspection and annual preventive maintenance.
3. Review manufacturer's instructions and ASME Boiler and Pressure Vessel Codes for boilers.
4. Review standard operating procedure for selection, care and use of respiratory protection.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. All automatically and manually operated control devices provided for controlling operation and safety of the vessel, steam or water pressure, hot water temperature, combustion, and boiler water level shall be inspected under operating conditions.
7. All associated valves and piping, pressure and temperature indicating devices, metering and recording devices, and all boiler auxiliaries shall be inspected under operating conditions.
8. All combustion controls attached to the boiler regardless of the fuel being fired must be in good working order or the inspection certificate shall be withheld.
9. Close and tag all hand or motorized valves required to isolate the boiler. Chain and lock all valves that are closed for safety. These valves shall not be unlocked and reopened unless authorized by the originator of the work authorization. Enter this information in the boiler log and sign.
10. Wear appropriate respirator, goggles, and gloves while in contact with hazardous materials.
11. Lighting: Flashlight should be used in preference to an extension light. When a portable extension light is used in a confined space, it shall not be operated at more than 12 volts.
12. Perform internal boiler inspection in accordance with requirements of American Society of Mechanical Engineers (ASME), Section VI, Recommended Rules for Care and Operation of Heating Boilers, and Section VII, Recommended Rules for Care of Power Boilers; and National Board Inspection Code Part 2 ANSI/NB-23, A Manual for Boiler and Pressure Vessel Inspectors.
13. Boilers shall be inspected and tested annually by RCI-1 commissioned inspector in accordance with NBIC Part 2 Inspection code.
14. Prepare boiler for internal inspection in the following manner:
 - a. Fuel supply and ignition system shall be locked out.
 - b. Water shall be drawn off and water side thoroughly washed out.
 - c. Manhole and handhole plates, washout plugs, inspection plugs in water column connections shall be removed.
 - d. The boiler shall be cooled and thoroughly cleaned.
 - e. All grates of internally fired boilers shall be removed.
 - f. Insulation and brickwork shall be removed as required to determine condition of boiler, headers, furnace, supports or other parts.
 - g. Pressure gage(s) shall be removed and tested.
 - h. Any leakage of steam or hot water into the boiler shall be prevented by disconnecting the pipe or valve at the most convenient point.
 - i. Before opening the manhole and entering any part of the boiler, the required steam or water system stop valves must be closed, tagged, and padlocked. All drain valves or cocks located between the two valves shall be opened.
15. Inspector will not enter boiler until satisfied that necessary safety precautions and pre inspection preparations have been made.
16. If a boiler has not been properly prepared for an internal inspection, the inspector should decline to make the inspection.
17. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the plan.

18. Account for all tools, materials, and equipment before closing boiler.

a. Visually inspect for corrosion and/or soot accumulation. Note: the temperature differential across the economizer can be taken to determine combustion efficiency.

Checkpoint

External Inspection Procedure:

1. Inspect overall cleanliness and accessibility of boiler and auxiliary equipment. Boiler fittings, valves and piping should be checked for compliance with ASME code and jurisdictional requirements.

2. Pressure Gages: Note the pressure reading indicated on pressure gage and compare it with another gage on the same system or with a standard test gage.

3. Water Level Gage Steam Boilers:

a. Observe the blowdown of the water gage in its normal manner and the promptness of the return of water in the gage should be noted.

b. During the test of the water level gage, water and steam connections should be blown separately to ensure both are clear.

4. Safety and Safety Relief Valves:

a. Safety valves should be tested by allowing the pressure in the boiler to rise to the popping pressure, and subsequent fall, to check the actual popping pressure and blowdown. If this is not practical, the valve should be tested by the boiler operator for free operation by use of the lifting lever, provided the boiler pressure is 75% or more of the set pressure.

b. Inspect valve discharge pipe to determine if discharge pipe is free and in accordance with ASME Code requirements.

c. When inspections reveal that a safety valve is not operating properly, the boiler shall be taken out of service and the valve shall be replaced or repaired.

5. Low Water Fuel Cutoff or Feed Controls: Observe the test of these controls after the drain has been opened. Close the drain and observe the promptness of the return to normal such as the silencing of an alarm or stopping of a feed pump should be noted. The float linkage and connections should be examined for wear and the float chamber should be free of sludge or other accumulation.

6. Blowoff Piping Power Boilers: Observe blowdown of the boiler in normal manner, check for freedom of piping to expand and contract and ensure there is no excessive vibration.

7. Flame Failure Cutoff and Alarm: Test flame failure shutdown and alarm on both pilot failure and main flame failure by securing fuel source. Ensure shutdown and alarm occurs in accordance manufacturers and ASME requirements.

8. Piping, Connections and Fittings:

a. Inspect piping to ensure there is provision for expansion and adequate support.

b. Piping and fittings should be examined for evidence of leakage and excessive vibration. Also closely examined to determine that they are properly rated for the service conditions to which they are subjected.

9. Scale, oil, etc.:

a. Examine all surfaces of exposed metal on waterside of boiler for deposits caused by water treatment, scale, oil, or other substances.

b. The smallest amount of oil is dangerous and immediate steps should be taken to clean affected surfaces and prevent further contamination, excess scale or other deposits should be removed by appropriate chemical or mechanical means.

10. Pressure controls (heating steam boilers): Verify that each automatically fired steam boiler is protected from over pressure by not less than two pressure operated controls one of which may be an operating control.

11. Determine if there is ample provision for expansion between the sections.

12. Check for excessive rust build-up between sections (cast iron type).

13. Examine all steam and water line to controls to determine that they are clear of scale and arranged to insure proper control operations.

14. Clean stems and shafts, and tighten packing nuts on valves and pumps.

15. Flush fuel oil strainer baskets and oil solenoid valve seats.

16. Dismantle low water fuel cut offs and water feeders to insure freedom from obstructions and proper functioning of the working parts. Always replace old gaskets before reassembling.

17. Inspect connecting lines to boiler for accumulations of mud, scale, etc., and clean as needed.

18. Examine all visible wiring for brittle or worn insulation, and make sure electrical contacts are clean, adequately tightened, and functioning properly.

19. On electrical type detection devices, replace vacuum tubes annually, and replace defective solid state devices.

20. Replace fusible plugs, if applicable.

21. Hydrostatic Test:

- a. The test pressure should not exceed 1 and 1/2 times the maximum allowable working pressure.
- b. The safety valve or valves should be removed or each disk shall be held down by means of a testing clamp.
- c. While at test pressure, time should be given to examine the complete pressure vessel for leaks prior releasing pressure.

22. Safety and Safety Relief Valves:

- a. Check valves for correct pressure setting and adequate discharge pipe supports.
- b. The safety or safety relief valve on a steam or hot water heating boiler should be tested manually once each month and pressure tested once each year.

23. Pressure Gages:

- a. When required, pressure gages shall be tested and calibrated.
- b. Location of steam pressure gage(s) should be noted to determine whether it is exposed to high temperature from an external source or to internal heat due to lack of protection by a proper siphon or trap.

24. Internal Inspection:

- a. When the boiler has cooled to the ambient boiler room temperature, wash out mud legs, and flush boiler.
- b. To protect the boiler from unnecessary stresses the boiler water temperature should be allowed to reach the boiler room temperature before draining the boiler. Drain boiler, tag valves and controls. NEVER attempt to remove a manhole or handhole cover without first properly venting the water or steam side of a boiler to the atmosphere. Prior to opening or entering a boiler it must be at atmospheric pressure.
- c. Inspect the boiler tube surfaces for rust and soot buildup. Rust is an indicator of condensation. The tubes should be brushed. Vacuum debris from all surfaces.
- d. Clean breaching, ducts, fireboxes, or main fire tube, and flue passages. Vacuum debris from all surfaces.
- e. Check and replace worn or damaged insulation. Repair the damage and remove the debris. If the insulation contains asbestos, refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
- f. Check refractory and brick work; if damaged notify supervisor and enter this data in boiler log.

25. Stays and stay bolts:

- a. Examine stays to determine whether or not they are in even tension. All fastened ends should be examined to determine whether cracks exist where stays are punched or drilled for rivets or bolts.
- b. Test firebox stay bolts by tapping one end of each bolt with a hammer and, where practical, a hammer or other heavy tool should be held on the opposite end to make the test more effective. An unbroken bolt should give a ringing sound while a broken bolt will give a hollow sound. Replace broken stay bolts.

26. Examine manholes, reinforcing plates and nozzles or other connections flange or screwed into the boiler for evidence of defects both internally and externally. If possible, observation should be made from the inside of the boiler as to whether connections are properly made to the boiler. Examine all openings leading to external attachments, such as water column connections, low water fuel cutoff devices, openings in dry pipes and openings to safety valves to ensure they are free from obstruction

27. Fire Surfaces:

- a. Examine for bulging and blistering:
- b. Inspect all plate or tube surfaces exposed to the fire also check whether any part of the boiler has become deformed by bulging and blistering.

28. Cracks:

- a. Examine vulnerable areas such as ligaments between the holes on watertube boiler drums, between tube holes on tube sheet of firetube boilers, at any flange where repeated flexing of the plate occurs during operation and around welded pipe and tube connections.
- b. Lap joint boilers are subject to cracking where plates lap in the longitudinal seam. If there is any evidence of leakage or other distress at this point, the inspector should thoroughly examine the area to determine whether cracks exist in seams. REPAIR OF LAP JOINT CRACKS ON LONGITUDINAL SEAMS IS PROHIBITED.

29. Corrosion:

- a. Inspect for corrosion.
- b. When active corrosion is found, provide advice to correct as necessary.

30. Grooving:

- a. Inspect for grooving. Examine as construction permits all flange surfaces, particularly the flanges of unstayed heads.
- b. Provide corrective advice for any defects found.

31. Firetubes: Examine closely for reduction in thickness near or at tube ends.

32. Watertubes:

- a. Inspect for corrosion, erosion, bulges, cracks, or any evidence of defective welds.

b. Examine short tubes and nipples used to join drums and headers, there is a tendency for fuel and ash to lodge in these areas and corrosion is likely in the presence of moisture.

33. Blowoff Piping:

- a. Inspect blowoff piping connections and fittings.
- b. Determine that blowout piping is properly secured and discharges at a safe point.

34. Brush all tubes clean of scale.

35. Brush plate surfaces clean, use vacuum cleaner.

36. Refill boiler with water and chemicals if boiler is to be put back in service or if wet-lay-ups are to be employed.

37. All safeties, interlocks, and alarms operate correctly in both automatic and manual operating modes. Verify proper sensor installation and calibration. The DDC control system relies on input from various sensors (such as temperature, pressure, and flow) in order to achieve the desired system operation. If sensors are not located correctly, or the measured value from any sensor to the control algorithm is incorrect, the system will not respond as intended.

38. Actuation and Sequencing:

- a. Verify that automatic isolation valves are installed and operate correctly, if applicable. (Automatic isolation valves are typically installed when multiple boilers are connected to a common supply header. Boilers with dedicated pumps generally do not have automatic isolation valves.) When an individual boiler is not operating, the isolation valve should be closed to prevent water from circulating through the unit.
- b. Verify proper boiler staging under normal operation, as well as under all failure and emergency operating modes, especially if multiple units are installed which are unequal in size.
- c. Verify that the boilers and primary/feed water pumps stage up and down per the sequence of operations under all operating modes.
- d. Verify that the time delay between boiler start/stop commands are per design.
- e. To remove residual heat from the boiler, the primary/feed water pump operation time delay, after the boiler is commanded OFF, is per design.

39. Setpoints and Reset Controls:

- a. Verify that the system operates and maintains hot water supply temperature setpoint under all operating modes, including automatic, manual, and failure/emergency modes.
- b. Verify proper coordination between individual setpoints and reset strategies.
- c. Verify that the control algorithms generate the proper water temperature setpoint based on the reset parameters specifies in the sequence of operations.
- d. Verify that the reset parameters are optimized for the system. In addition, ensure the reset control strategy does not result in a return water temperature from the building loads, which can cause the flue gasses to condense in non-condensing boiler systems.
- e. Verify that the O₂ trim controls, if applicable, operate to ensure that excess oxygen in flue gas is maintained at setpoint. If O₂ trim controls are not installed, review flue gas report and verify the boiler was tuned at high-fire and at least one intermediate part-load operating point.

40. Control Accuracy and Stability:

- a. Verify that all control loops stabilize within a reasonable amount of time (typically 2 to 5 minutes) after a significant load change such as start-up or automatic/manual recovery from shut down.

41. Document results of functional testing.

42. Reporting:

- a. During all tests the actual operating and maintenance practices should be noted by the Inspector and a determination made as to their acceptability.
- b. Record Review: Review the boiler log and records of maintenance and feed water treatment to insure that regular and adequate tests have been made on the boiler and controls.

43. Reports and Records:

- a. GSA Form 349, Inspection Report of Boiler, shall be prepared for each boiler when it is inspected.
- b. GSA Form 1034, Certification of Inspection, shall be issued when the boiler has been approved for operation. The original and one copy is required. The original is posted on or near the equipment and the copy is forwarded to the regional office, if required.

44. Conclusions: Any defects or deficiencies in condition, operating and maintenance practices of the boiler and auxiliary equipment should be discussed by the inspector with the owner or user at this time.

Tools

TEST_EQUIP_COMBUSTION	CHAIN_LOCKS	CHEMICALS_BOILER	HYDROSTATIC_PUMP	RESPIRATOR-01	SAFETY_GOGGLES
SCRAPER	TOOL_GROUP_C	VACUUM_CLEANER-04	SAFETY_SIGNS		

This standard applies to any boilers that use only electricity to heat the water for use. Generally, these units are smaller than the gas or oil fired boilers, unless there is no gas or oil available at the location.

Special Instructions

1. Perform boiler external inspection with internal inspection and annual preventive maintenance.
2. Review manufacturer's instructions and ASME Boiler and Pressure Vessel Codes for boilers.
3. Review standard operating procedure for selection, care and use of respiratory protection.
4. Review boiler wiring diagram.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. All automatically and manually operated control devices provided for controlling operation and safety of the vessel, steam or water pressure, hot water temperature, combustion, and boiler water level shall be inspected under operating conditions.
7. All associated valves and piping, pressure and temperature indicating devices, metering and recording devices, and all boiler auxiliaries shall be inspected under operating conditions.
8. Close and tag all hand or motorized valves required to isolate the boiler. Chain and lock all valves that are closed for safety. These valves shall not be unlocked and reopened unless authorized by the originator of the work authorization. Enter this information in the boiler log and sign.
9. Wear appropriate respirator, goggles, and gloves while in contact with hazardous materials.
10. Lighting: Flashlight should be used in preference to an extension light. When a portable extension light is used in a confined space, it shall not be operated at more than 12 volts.
11. Perform internal boiler inspection in accordance with requirements of American Society of Mechanical Engineers (ASME), Section VI, Recommended Rules for Care and Operation of Heating Boilers, and Section VII, Recommended Rules for Care of Power Boilers; and National Board Inspection Code Part 2 ANSI/NB-23, A Manual for Boiler and Pressure Vessel Inspectors.
12. Boilers shall be inspected and tested annually by RCI-1 commissioned inspector in accordance with NBIC Part 2 Inspection code.
13. Prepare boiler for internal inspection in the following manner:
14. Power supply shall be locked out.
15. Water shall be drawn off and water side thoroughly washed out.
16. Manhole and handhole plates, washout plugs, and inspection plugs in water column connections shall be removed.
17. The boiler shall be cooled and thoroughly cleaned.
18. All grates of the boilers shall be removed.
19. Insulation shall be removed as required to determine condition of boiler, headers, furnace, supports or other parts.
20. Pressure gauges shall be removed and tested.
21. Any leakage of steam or hot water into the boiler shall be prevented by disconnecting the pipe or valve at the most convenient point.
22. Before opening the manhole and entering any part of the boiler, The required steam or water system stop valves must be closed, tagged, and padlocked. All drain valves or cocks located between the two valves shall be opened.
23. Inspector will not enter boiler until satisfied that necessary safety precautions and pre inspection preparations have been made.
24. If a boiler has not been properly prepared for an internal inspection, the inspector should decline to make the inspection.
25. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management
26. Account for all tools, materials, and equipment before closing boiler.

Checkpoint

1. Inspect overall cleanliness and accessibility of boiler and auxiliary equipment. Boiler fittings, valves and piping should be checked for compliance with ASME code and jurisdictional requirements.
2. Pressure Gages: Note the pressure reading indicated on pressure gage and compare it with another gage on the same system or with a standard test gage.
3. Clean boiler by:
 - a. Placing boiler cleaning compound through appropriate opening in top of boiler. (Compound must not react against copper elements).
 - b. Operate boiler according to instructions.
 - c. Shut-off power by turning off pilot switch.
 - d. Open bottom blow-down valve to full open to let sludge and scale blow out of boiler.
 - e. Refill with water and close pilot switch.
4. Check nameplate for KW rating, voltage and amperage. Check electrical supply voltage to verify it conforms to boiler requirements.
5. Safety and Safety Relief Valves:
 - a. Safety valves should be tested by allowing the pressure in the boiler to rise to the popping pressure, and subsequent fall, to check the actual popping pressure and blowdown. If this is not practical, the valve should be tested by the boiler operator for free operation by use of the lifting lever, provided the boiler pressure is 75% or more of the set pressure.
 - b. Inspect valve discharge pipe to determine if discharge pipe is free and in accordance with ASME Code requirements.
 - c. When inspections reveal that a safety valve is not operating properly, the boiler shall be taken out of service and the valve shall be replaced or repaired.
6. Clean stems and shafts, and tighten packing nuts on valves and pumps.
7. Inspect connecting lines to boiler for accumulations of mud, scale, etc., and clean as needed.
8. Examine all visible wiring for brittle or worn insulation, and make sure electrical contacts are clean, adequately tightened, and functioning properly.
9. On electrical type detection devices, replace vacuum tubes annually, and replace defective solid state devices.
10. Replace fusible plugs, if applicable.
11. Reporting:
 - a. During all tests the actual operating and maintenance practices should be noted by the Inspector and a determination made as to their acceptability.
 - b. Record Review: Review the boiler log and records of maintenance and feed water treatment to ensure that regular and adequate tests have been made on the boiler and controls.
12. Reports and Records:
 - a. GSA Form 349, Inspection Report of Boiler, shall be prepared for each boiler when it is inspected.
 - b. GSA Form 1034, Certification of Inspection, shall be issued when the boiler has been approved for operation. The original and one copy is required. The original is posted on or near the equipment and the copy is forwarded to the regional office, if required.
13. Conclusions: Any defects or deficiencies in condition, operating and maintenance practices of the boiler and auxiliary equipment should be discussed by the inspector with the owner or user at this time.

Tools

SAFETY_GOGGLES	TOOL_GROUP_C	SAFETY_SIGNS	VACUUM_CLEANER-04	RESPIRATOR-01	HYDROSTATIC_PUMP
CHEMICALS_BOILER	CHAIN_LOCKS	SCRAPER	TEST_EQUIP_COMBUSTION		

This standard applies to any boilers with burners that use Natural Gas as their fuel source. These can be either natural draft or forced draft type of boilers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Verify that the annual inspections for the boiler have been satisfactorily performed.
4. As part of the combustion gas analysis, Nitrous Oxide (NOX) should be checked and compared with the local authority's requirements. Many areas now have maximum standards for NOX to assist in reducing pollution. Depending on the age of the boiler and burner, specific requirements may exist.
5. Some jurisdictions may require 3rd party (independent) combustion and operational testing to verify boiler and burner operation.

Checkpoint

1. Check boiler room for ventilation in accordance with the American Gas Association (AGA) burner requirements.
2. Check operation of all gas controls and valves including: manual gas shutoff; petal gas regulator; safety shutoff valve (solenoid); automatic gas valve; petal solenoid valve; butterfly gas valve, motor, and linkage to air louver; safety petal solenoid (if used.)
3. Check flue connections for tight joints and minimum resistance to air flow. (combustion chamber, flues, breaching, and chimney are clear before firing.)
4. Draft regulators require slightly negative pressure in the combustion chamber at maximum input.
5. On forced draft burners, gas manifold pressure requirements should correspond with modulating (butterfly) valve in full open position and stable at all other firing rates.
6. Take and record and record flue gas readings to determine boiler efficiency. Use manufacturer's instructions if available. If not, use attached table as a guide for performance criteria. If efficiency is low, check baffling and passes for short circuiting, and boiler for air infiltration. Adjust dampers and controls to optimize efficiency. Run test at following load points.
7. 100%, 70%, and 40% of rated full load for boilers having metering controls or modulation capacity at these load points.
8. At high and low fire rates on boilers equipped with OFF/LOW FIRE/HIGH FIRE control.
9. At single firing load point on boilers equipped with OFF/ON controls only.
10. Check burner for flashback and tight shutoff of fuel.
11. Check operation of automatic controls and combustion flame safeguards. Clean and adjust, if necessary.
12. Operation and adjustments should conform with manufacturer's instructions.

Tools

CLOTHS-01 TOOL_GROUP_C GAS_ANALYZER

This standard applies to any boilers that use Fuel Oil as their fuel source. These can be either natural draft or forced draft type of boilers. <

Special Instructions

1. Review manufacturer's instructions.
2. Verify that the annual inspections for the boiler have been satisfactorily performed.
3. As part of the combustion gas analysis, Nitrous Oxide (NOX) should be checked and compared with the local authority's requirements. Many areas now have maximum standards for NOX to assist in reducing pollution. Depending on the age of the boiler and burner, specific requirements may exist.
4. Some jurisdictions may require 3rd party (independent) combustion and operational testing to verify boiler and burner operation.

Checkpoint

1. Test and inspect burner (with or without firing) at rated pressure for leaks.
2. Perform timed trial for ignition on pilots and burners in accordance with instructions in the programmer timer.
3. Check automatic controls and combustion flame safeguards for normal operation. There should be no presence of oil discharge, ignition or flame.
4. Check pre-ignition purging capability of burner, combustion chamber, boiler passes, and breaching. Stack dampers should be fully open during purge and light off period.
5. Check delivery of fuel in relation to its response to the ignition system. Examine electrodes for carbon buildup, discoloration, distortion, and burning of parts, clean and adjust as necessary.
6. Check ignition transformer to supply dependable arc, adjust and regulate as required for clearance and air gap.
7. Clean and adjust draft regulator and air shutter on a natural draft burner to ensure excess air quantities are minimum for complete combustion. Test with flue gas analyzer.
8. On mechanical draft burners clean and check power driven fan blower.
9. Check out forced draft fan, clean fan and fan housing, check bearings, pulleys and or couplings and adjust belt tension if required replace worn belts and lubricate pivot points on linkages as necessary.
10. Check and replace filters. Check and clean water separators, primary and secondary filters.
11. Clean, check operation, and adjust controls and safeties.
12. Burners designed to change firing rates automatically should be checked for adequate proportioning changes in fuel and air rates.
13. Check oil level sight glass to see that burner maintains proper oil level (within 1/3") at rated output.
14. Check to ensure that power cannot feedback and energize ignition devices or feed valves after a control shuts off burner.
15. Clean or replace nozzles or cups, and check for tight shutoff of fuel.
16. Check stacks for smoke or haze and adjust burner accordingly.
17. Take and record flue gas readings to determine boiler efficiency. Use manufacturer's instructions if available. If not, use attached table as a guide for performance criteria. If efficiency is low, check baffling and passes for short circuiting, and boiler for air infiltration. Adjust dampers and controls to optimize efficiency. Tests should be run at the following load points:
 - a. 100%, 70%, and 40% of rated full load for boilers having metering controls or modulation capacity at these load points.
 - b. At the high and low fire rates on boilers equipped with OFF/LOW FIRE/HIGH FIRE control.
 - c. At the single firing load point on boilers equipped with OFF/ON controls only.

Tools

This standard card applies to large boiler room operations where the boiler PM standards do not sufficiently cover the maintenance required for the boiler room controls. These boiler controls include such items as boiler masters, draft and oil controllers, level meters, flow transmitters and recorders, ratio transmitters and totalizers, recorders and indicators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions before making any adjustments or calibrations.
3. Schedule work with operating personnel, as needed.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Review control wiring and piping diagrams.
6. Review and understand sequence of operation prior to testing.
7. Ensure appropriate site personnel are notified that alarms that may result from testing and to disregard them until testing is completed.

Checkpoint

1. Visually check equipment sensing lines, operating valves, and internal assembly for leaks.
2. Perform friction test.
3. Zero meter to scale, check pen(s) for zero indication, make necessary adjustments.
4. Simulate flow by use of check weights, provided water column, or other appropriate external simulation devices.
5. Check pen or indicator for proper chart or scale percentage at recommended check points; (or discretionary check points, if none are recommended) record actual indication; make minor adjustments as necessary.
6. Check control over full range. Set zero and full range stops as necessary.
7. Mark integrator or counter checks by simulating flow (usually 50% of full scale) and check number of counts over specific time period (minimum period of 30 minutes is recommended). Calculate correct number of counts for time period used and make necessary minor adjustments. Re-check after any adjustments, ascertain that the integrator or counter does not count at zero.
8. Repeat step #6
9. Open meter supply line to check for plugging. Put meter back in service.
10. Check electronic stations for proper electrical connections and pneumatic stations for any leaks by using a leak detector (soap solution).
11. Check operation of station panel switch and manual control for excessive friction and looseness of control knobs.
12. Check for correct supply pressure or voltage to the station.
13. Compare sender over it's range of control with an accurate external measuring device such as a U-tube, pneumatic calibrator, or electronic calibrator (all other measurements in the control loop should be made with the same device).
14. Check all indicators by simulating pneumatic or electronic signals to the appropriate port of terminal for proper indication to the related gauge or indicator (in most cases, mathematical calculations will be necessary to determine the proper indication at each test point).
15. Ascertain that the automatic signal is not restricted when the station is activated.
16. Record and report all deficiencies.

Tools

TOOL_BASIC SOAP_SOLUTION METER_DEVICE FLOW_METER

null

Special Instructions

10. Chimney connectors shall be inspected for damage, deterioration, internal blockage, or obstruction and freedom from combustible deposits.
9. Chimney connectors shall be inspected for proper support, and fastening of joints, pitch, and securment to the chimney
8. The connected appliances, their chimney connectors and surroundings shall be inspected for proper clearances, protection, damage, or deteriorationand observable evidence of operating malfunction.
7. Inspections shall include verification that the flue is free of combustible material deposits or blockage or obstruction.
6. An inspection conducted during the cleaning or replacement of an appliance will include an inspection of the flue.
5. When inspecting any flue in a chimney, all flues in that chimney will be inspected.
4. Readily accessible areas of the chimney that can be observed through openings such as clean outs or terminations shall be examined for the presence of a continuous flue liner, proper installation and nodamage or deterioration.
3. Where panels, doors or coverings are opened as part of the performance of another rask, such locations shall be examined as part of a level 1 inspection.
2. The chimney exterior and surroundings shall be inspected at locations that can be accessed without removal of panels , doors or coverings.
1. Inspections to be performed by a state licensed inspector

Checkpoint

Tools

None

This standard applies to baseboard radiator and convector heating units using hot water, steam, or electricity as the heating medium. These units are found in offices, entrance lobbies, vestibules, etc. They vary from simple finned tube baseboard units with no controls to zoned or individually controlled units. Valves can be on/off type or automatic temperature regulating valve.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management
4. Review building plans for special instructions where removal of wall panels is necessary.
5. Review standard operating procedures for selection, care, and use of respiratory protection.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Rope off or post signs at vestibules or entrances where work involves these areas.

Checkpoint

1. Check radiator valve for free turning and seating. Check packing.
2. Remove covers or wall panels. Note: Extreme care must be taken when removing marble or granite wall panels. These panels are extremely heavy and very fragile.
3. Check housing, braces, supports, hangers, and hardware for signs of deterioration or damage.
4. Check temperature or flow controls, shutoff valves, vents and traps for proper operation.
5. If radiator has automatic temperature regulating valve, remove valve cover and remove dirt by vacuuming.
6. For hot water radiators, check air bleed valve.
7. Remove and inspect seat of steam trap. Clean out trap. Follow other steam trap procedures as listed in trap maintenance standards.
8. Wire brush and treat with rust inhibitor all rusted areas.
9. Check coils, piping, and fin material for damage, leaks or looseness. Straighten finned material as necessary.
10. Vacuum or blow out finned tube area and interior housing.
11. Clean and replace covers or wall panels and caulk wall panels as required. Clean work area.

Tools

SIGNS_ROPE TOOL_BASIC VACUUM-03 GOOGLES_RESP_GLOVES RUST_INHIBITOR-01

This job plan is to be performed in each owned building cooled by at least one chiller. Chillers over twenty years old need not be overhauled at regional discretion. Chillers under warranty will follow warranty instructions.

Special Instructions

- 1. Enter or Verify chiller installation date is included in the chiller asset record.
- 2. Review overhaul status. Chillers in excess of 20 years of age need not be scheduled for overhaul and any scheduled overhauls may be recinded at COR discretion. This allowance does not apply to Mag Lev chillers whose multi year electrical maintenance must be completed as scheduled.

Checkpoint

Tools

None

This standard card applies to chilled water producing units that are self-contained, consisting of refrigeration compressors, condensers either air or water cooled, evaporator coils, refrigerant receivers, fan and motor, etc., contained in a common housing or assembly.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Schedule any OEM required, less than annual frequency, maintenance, with COR approval.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the item.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.
12. Remove access covers prior to accomplishing check points.

Checkpoint

1. Condenser:
 - a. Remove debris from air screen and clean underneath unit.
 - b. Pressure wash coil with proper cleaning solution.
 - c. Straighten fin tubes with fin comb.
 - d. Check electrical wiring and tighten loose connections. Check fused disconnect switches for condition and operation.
 - e. Check mounting for tightness.
 - f. Check for corrosion. Clean and treat with inhibitor as needed.
 - g. Check fan or blower for bent or damaged blades and imbalance.
 - h. Lubricate shaft and motor bearings on fans and remove old or excess lubricant.
 - i. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as necessary. Change badly worn belts. Multi-belt drives should be replaced with matched sets.
 - j. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an adequate water treatment program. Consult water treatment standard for proper procedures.
2. Evaporator:
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Inspect plumbing, valves and flanges for leaks and correct.
3. Compressor(s):
 - a. Lubricate drive coupling.
 - b. Lubricate motor bearings (non-hermetic).

- c. Check and correct alignment of drive couplings.
 - d. Inspect evaporator tubes for scale. Clean if required. Leak test tubes using a halogen leak detector or suitable substitute.
 - e. Add refrigerant per manufacturer's instructions if needed.
 - f. Check compressor oil level.
4. Run machine; check action of controls, relays, switches, etc. to see that:
 - a. Compressor(s) run at proper settings.
 - b. Suction and discharge pressures are proper.
 - c. Outlet water temperature is set properly.
 5. Check and adjust vibration eliminators. Replace as necessary.
 6. Sample test the refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards. Based on the results, refrigerant may need to be replaced or recycled, and oil replaced.
 7. Check and calibrate safety controls.
 8. Controls:
 - a. Check operation of all relays, pilot valves, and pressure regulators.
 - b. Check resulting action of pressure sensing primary control elements such as diaphragms, bellows, inverted bells, and similar devices when activated by air, water, or similar pressure.
 9. Motors:
 - a. Refer to the appropriate maintenance standard for motors.

Tools

FIN_COMB	SAFETY_GOGGLES	REFRIGERANT_RECOVERY-01	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02
MAT_CLEANING-03	PAINT-05	REFRIGERANT	REFRIGERANT_TANK	RESPIRATOR-01	TOOL_GROUP_A_AND_B
PRESSURE_WASHER-01					

This standard card applies to chilled water producing units that are self-contained, consisting of refrigeration compressors, condensers either air or water cooled, evaporator coils, refrigerant receivers, fan and motor, etc., contained in a common housing or assembly. This standard is for special purpose or critical use units.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
5. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
6. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
7. Recover, recycle, or reclaim the refrigerant as appropriate.
8. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the item.
9. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.
12. Remove access covers prior to accomplishing check points.

Checkpoint

1. Condenser:
 - a. Remove debris from air screen and clean underneath unit.
 - b. Straighten fin tubes with fin comb.
 - c. Check electrical wiring and tighten loose connections. Check fused disconnect switches for condition and operation.
 - d. Check mounting for tightness.
 - e. Check for corrosion. Clean and treat with inhibitor as needed.
 - f. Check fan or blower for bent or damaged blades and imbalance.
 - g. Lubricate shaft and motor bearings on fans and remove old or excess lubricant.
 - h. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as necessary. Change badly worn belts. Multi-belt drives should be replaced with matched sets.
2. Evaporator:
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Inspect plumbing, valves and flanges for leaks and correct.
3. Compressor(s)
 - a. Lubricate drive coupling.
 - b. Lubricate motor bearings (non-hermetic).
 - c. Check and correct alignment of drive couplings.
 - d. Inspect evaporator tubes for scale. Clean if required. Leak test tubes using a halogen leak detector or suitable substitute.
 - e. Add refrigerant per manufacturer's instructions if needed.
 - f. Check compressor oil level.

4. Run machine; check action of controls, relays, switches, etc. to see that:
 - a. Compressor(s) run at proper settings.
 - b. Suction and discharge pressures are proper.
 - c. Outlet water temperature is set properly.
5. Check and adjust vibration eliminators. Replace as necessary.
6. Sample test the refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards. Based on the results, refrigerant may need to be replaced or recycled, and oil replaced.
7. Check and calibrate safety controls.
8. Controls:
 - a. Check operation of all relays, pilot valves, and pressure regulators.
 - b. Check resulting action of pressure sensing primary control elements such as diaphragms, bellows, inverted bells, and similar devices when activated by air, water, or similar pressure.
9. Motors:
 - a. Refer to the appropriate Maintenance Standard for Motors.
 - b. Check status of water filter and clean filter or change as necessary.

Tools

MAT_CLEANING-03	TOOL_GROUP_A_AND_B	SAFETY_GOGGLES	RESPIRATOR-01	GLOVES-01	REFRIGERANT_RECOVERY-01
PAINT-05	REFRIGERANT_TANK	LEAK_DETECTOR-02	HOSE_FITTING-01	FIN_COMB	PRESSURE_WASHER-01
REFRIGERANT					

This standard card applies to chilled water producing units, consisting of refrigeration rotary screw or scroll compressor(s), water cooled condenser tube bundle, evaporator tube bundle, control panel, etc.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Schedule any OEM required, less than annual frequency, maintenance, with COR approval.
2. Review manufacturer's instructions.
3. Coordinate PM of refrigeration machine control panel with this PM activity.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.
6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lbs.) and leak rate (%) in NCMMS every time refrigerant is added to the equipment.
8. Prior to recycling, reclamation, or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.
9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Pump down system. Remove a small sampling of refrigerant using appropriate refrigerant recovery/recycling equipment, in accordance with manufacturer's instructions. Test the sample of refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards.
2. Perform spectro-chemical analysis of compressor oil annually to determine bearing conditions and replace as necessary.
3. As oil testing indicates drain and replace oil in compressor oil reservoir including filters, strainers and traps.
4. Review the Safety Data Sheet for proper disposal of used oil. If appropriate, recycle oil at an authorized station.
5. Change oil filter or element and clean housing.
6. Inspect filter for metal particles which could indicate bearing wear.
7. Clean strainer and nozzles of eductor.
8. Clean all water strainers in the system.
9. Inspect relief valves and piping. Check for valve for corrosion or foreign material and replace valves. Valves are safety devices for equipment and personnel, DO NOT attempt to repair or recondition. Increase valve inspection frequency if equipment installed in corrosive environment.
10. Inspect evaporator and condenser tubes.
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale.

- c. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an adequate water treatment program. Consult water treatment standard for proper procedures.
11. Test for leaks per manufacturer's instructions and EPA regulations (40 CFR Part 82). Refrigerant leak detection can be conducted with an electronic halide leak detectors, ultrasonic leak detectors or a soap bubble solution will be used along with dry nitrogen to pressurize chiller. If leak repair and verification tests cannot be completed within 30 days, notify the COR and supervisor. See Leak Repair Job Plan.
 12. Pull vacuum on refrigeration machine in accordance with manufacturer's instructions. Add refrigerant as required per specifications.
 13. Megger compressor and oil pump motors and record readings.
 14. Tighten all starter, control panel, motor terminals, overloads, and oil heater leads, etc. Loose connections can cause voltage spikes and overheating leading to malfunctions and failures.
 15. Check all electrical contacts for wear, pitting, etc.
 16. Check and calibrate overloads, record trip amps and trip times.
 17. Check and calibrate safety controls.
 18. Check external interlocks, flow switch, fans and pumps.
 19. Check dash pot oil in main starter.
 20. Check compressor(s) sequencing, if multiple stage system
 21. Check pressure and temperature transducers against gage on both the oil, refrigerant and water side systems.
 22. Clean up the work area. Properly recycle or dispose of materials in accordance with environmental regulations.

Tools

MEGGER	GAS_NITROGEN	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02	MAT_CLEANING-02	PAINT-01
REFRIGERANT	REFRIGERANT_RECOVERY-01	REFRIGERANT_TANK	SAFETY_GOGGLES	LUBRICANT-05	TUBE_CLEANING_WASH	
VARIAC	TOOL_GROUP_A_AND_C					

This standard card applies to chilled water producing units, consisting of refrigeration centrifugal compressor(s), water cooled condenser tube bundle, evaporator tube bundle, control panel, and associated equipment attached to the chiller.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. Schedule any OEM required, less than annual frequency, maintenance, with COR approval.
2. Coordinate PM of refrigeration machine control panel in conjunction with this PM activity.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. The replacement filter-drier cores for the high efficiency purge unit absorb water vapor from the ambient air. They are shipped in sealed containers and are not to be opened until they can be installed and sealed in the purge tank.
5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.
6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lb) and leak rate (%) in NCMMS every time refrigerant is added to the equipment in accordance with standard operating procedures, training, and guidance provided.
8. Prior to reclamation or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.
9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.
10. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
11. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
12. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Lubricate drive couplings.
2. Check and correct alignment of drive couplings.
3. Lubricate motor bearings (non-hermetic).
4. Lightly lubricate vane control linkage bearings, ball joints and pivot points. DO NOT LUBRICATE the shaft of the vane operator.
5. Remove refrigerant in accordance with manufacturer's instructions. Sample test the refrigerant and oil to verify compliance with the Air Conditioning and Refrigeration Institute standards.
6. Perform spectro-chemical analysis of compressor oil annually to determine bearing conditions and replace as necessary.
7. As oil testing indicates drain and replace oil in compressor oil reservoir including filters, strainers and traps. Manufacturers typically recommended that oil should be changed after the first year of operation and every five years thereafter.
8. Review the Safety Data Sheet for proper disposal of used oil. If appropriate, recycle oil at an authorized station.
9. Change oil filter, clean strainer.
10. Drain and replace oil in purge compressor.

11. Drain and replace oil in purge gearbox.
12. Change refrigerant filter/drier on cooling line to motor (hermetic). Check moisture indicator sight glass and if moisture present find source of water leak.
13. Clean all water strainers in the system.
14. Inspect relief valves and piping. Check for valve for corrosion or foreign material and replace valves. Valves are safety devices for equipment and personnel, DO NOT attempt to repair or recondition. Increase valve inspection frequency if equipment installed in corrosive environment.
15. Check pressure and temperature transducers against gage on both the oil, refrigerant and water side systems.
16. Inspect evaporator and condenser tubes.
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale.
 - c. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an adequate water treatment program. Consult water treatment standard for proper procedures.
17. Test for leaks per manufacturer's instructions and EPA regulations (40 CFR Part 82). Refrigerant leak detection can be conducted with an electronic halide leak detectors, ultrasonic leak detectors or a soap bubble solution will be used along with dry nitrogen to pressurize chiller. If leak repair and verification tests cannot be completed within 30 days, notify the COR and supervisor. See Leak Repair Job Plan.
18. Pull vacuum on refrigeration machine in accordance with manufacturer's instructions. Add refrigerant as required per specifications.
19. Megger compressor and oil pump motors and record readings.
20. Check dash pot oil in main starter.
21. Tighten all starter, control panel, motor terminals, overloads, and oil heater leads, etc. Loose connections can cause voltage spikes and overheating leading to malfunctions and failures.
22. Check all contacts for wear, pitting, etc.
23. Check and calibrate overloads, record trip amps and trip times.
24. Check and calibrate safety controls.
25. Clean up the work area. Properly recycle or dispose of materials in accordance with environmental regulations.

Tools

GLOVES-01	GAS_NITROGEN	HOSE_FITTING-01	LEAK_DETECTOR-02	LUBRICANT-05	MAT_CLEANING-02	MEGGER
PAINT-01	REFRIGERANT	REFRIGERANT_RECOVERY-01	REFRIGERANT_TANK	SAFETY_GOGGLES	TOOL_GROUP_A_AND_C	
TUBE_CLEANING_WASH	VARIAC					

This standard card applies to chilled water producing units using absorption refrigeration cycle. Components consist of absorber tube bundle, water cooled condenser tube bundle, evaporator tube bundle, system pumps, control panel, and other chiller equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Coordinate PM of refrigeration machine control panel and refrigeration machine controls in conjunction with this PM activity.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
6. Recover, recycle, or reclaim the refrigerant as appropriate.
7. If disposal of the equipment item is required, follow regulations concerning removal of refrigerants and disposal of the item.
8. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
9. Lithium bromide is a corrosive solution. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Clean solution strainer(s). Clean unit pump motor cooling circuit in accordance with manufacturer's instructions.
2. Check and clean all strainers and traps in steam or hot water supply, condensate return, and condensing water circuit.
3. Inspect relief valves and piping. Check for valve for corrosion or foreign material and replace valves. Valves are safety devices for equipment and personnel, DO NOT attempt to repair or recondition. Increase valve inspection frequency if equipment installed in corrosive environment.
4. Check pressure and temperature transducers against gage on all systems.
5. Inspect and clean evaporator and condenser tubes. If steam and hot water fired also include absorber coil.
 - a. Evaporator tubes should be inspected and cleaned of scale. Inspect and clean temperature sensors and flow switches.
 - b. Condenser tubes should be inspected and cleaned. Condenser tubes from open tower systems may have contamination or hard scale.
 - c. Excessive corrosion, scaling, erosion and algae typically indicate improper or lack of an adequate water treatment program. Consult water treatment standard for proper procedures.
6. For direct fired machines include inspection and cleaning of exhaust gas economizer.
7. For direct fired machines include gas burner PM per maintenance standard.
8. Inspect and clean temperature sensors and flow switches.
9. Use oil-dry nitrogen under positive pressure to check for air leaks around valves, gaskets, seals, etc.
10. Units with external purge pump system:
 - a. Check pulley alignment and V-belt extension.
 - b. Clean purge drive belts with safety solvent.
 - c. Change purge pump vacuum oil in accordance with manufacturer's instructions.
11. Review the Safety Data Sheet for proper disposal of used oil. If appropriate, recycle oil at an authorized station.
12. Lubricate purge pump motor with machine oil.
13. Check pumps, motors, controls, and lubricate as required (evaporator pumps, solution pump).
14. Add octyl alcohol to working fluids per manufacturer's charts and procedures.
15. Check and service system controls in accordance with manufacturer's instructions.
16. Clean up the work area. Properly recycle or dispose of materials in accordance with environmental regulations.

Tools

TOOL_GROUP_A_AND_C	LUBRICANT-05	SAFETY_GOGGLES	MAT_CLEANING-02	GLOVES-01
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This standard applies to control panels that are installed to protect and control the operation of central refrigeration machines (chillers). Typically the control panel is mounted on the chiller and controls chiller capacity, demand limit, temperatures, pressures as well as shut downs and safeties. The standard includes service to controllers, transmitters, sensors, relays, etc., associated with the central control panel.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Read and understand the manufacturer's instructions before making any adjustments or calibrations.
- 3. Obtain As Built diagrams of the control and safety systems.
- 4. Schedule work with operating personnel, as needed.
- 5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 6. Ensure appropriate site personnel are notified that alarms that may result from testing and to disregard them until testing is completed.
- 7. Replace defective control safeties found while performing preventive maintenance.

Checkpoint

- 1. Clean and calibrate all controlling instruments (temperature and pressure transducers, etc.) in accordance with manufacturer's instructions and maintenance standard.
- 2. Check and clean all electrical contacts and pneumatic orifices.
- 3. Check pneumatic tubing for leaks or damage. Repair or replace as required.
- 4. Check electrical wiring insulation and connections. Tighten or replace if necessary.
- 5. Replace recorder charts for temperatures, kilowatt, etc., add ink (if applicable).
- 6. Check for bad indicator lights and gauges and replace as necessary.
- 7. Test all controllers and set at proper set points.
- 8. Check operating data and analyze for proper operation. Note unusual conditions such as compressor surge on maintenance log.

Tools

GAUGE_TEMPERATURE	GAUGE_PNEUMATIC_CONTROL	MAT_CLEANING-03	METER_VOLT_OHM	PRESSURE_GAUGE-02
TOOL_GROUP_B	CONTROL_KIT			

This standard applies to refrigerant monitors or detectors used to detect refrigerant leaks, primarily in chiller rooms. All commonly used refrigerants except ammonia are heavier than air at atmospheric pressure and, if released, will displace oxygen in the room posing suffocation hazards to personnel. Refrigerants coming into contact with flames or hot surfaces can break down to acidic or toxic compounds. For these and other reasons refrigeration machinery rooms may be required to have a refrigerant detector to activate an audible and visual alarm and enable ventilation systems.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand the manufacturer's instructions before making any adjustments or calibrations.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Wipe down outside of unit.
2. Replace air filter on air sample intake to unit. When clogged air flow will be restricted and a filter or flow fault will occur. Samples can be taken from air surrounding the unit or through a sample tube from a remote location.
3. Calibrate unit with gas canisters that have specific levels of refrigerant. Two calibrations are required, Zero to calibrate when no refrigerant present and SPAN to calibrate level of refrigerant in parts per million (ppm) when unit should alarm. A ZERO scrubber (carbon filter) may also be used which will remove all impurities from air giving a zero reading.
4. The ppm level to alarm at varies based on refrigerant type; see International Machine code refrigerant classification. Name plate on unit should have levels listed.
5. Check that all relays sent to remote equipment close or open upon alarm.
6. Internal electronic components and sensors are typically not field serviceable and should be replaced either with new parts or sent to factory as applicable.

Tools

FILTERS-01 TOOL_BASIC MAT_CLEANING-01

This standard applies to high efficiency purge units for centrifugal chillers. The units purge the refrigerant of non-condensable gasses such as air or sometimes nitrogen which is used to lay-up chiller and the high efficiency denotes that a minimum of refrigerant is lost when purging. There are different units depending on whether the chiller is low pressure (R-11, R-123) or high pressure (R-134a, R-12, R-22). Units may be mounted on the chiller or remotely on a wall and connected with tubes.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
 2. Review manufacturer's instructions.
 3. Always purge chiller after it has been opened for servicing.
 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
 5. The replacement filter-drier cores absorb water vapor from ambient air, so they are shipped in sealed containers. Do not open them until the cores can be installed and sealed in the purge tank.
 6. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
 7. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
 8. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
 9. Recover, recycle, or reclaim the refrigerant as appropriate.
 10. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
 11. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
 12. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
 13. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.
1. Chiller or Refrigeration Systems

Checkpoint

1. The service points listed below cover the basic sequences and items to check out for all units. However due to the many models available some of the functions described below may be automatic such as liquid water drain out or purge unit isolation and do not require many steps to complete.
2. Check out and Water Removal
3. Perform the purge system control check as described in the manufacturer's instructions.
4. If applicable, check purge run time logs for excessive run times. This may indicate leaks in the system.
5. Purge Tank Service:
 - a. Isolate the purge tank by closing the valves on the purge tank inlet and liquid return lines.
 - b. With the purge condensing unit turned off, pressurize the purge tank through the purge tank drain valve to manufacturer's specifications and test the solenoid valves to insure they seal properly.
 - c. Check the purge tank sight glass to determine if there is water in the purge tank.
6. If the refrigerant level is visible in the sight glass but there is no water in the tank, then perform the following:
 - a. Connect a refrigerant hose from the purge tank Schrader valve to an access valve on the evaporator.
 - b. Open the vapor line and disconnect the refrigerant hose.
 - c. Proceed with the service procedure.
7. If, on the other hand, there is a layer of water on the refrigerant in the purge tank sight glass, then perform the following:
 - a. Connect a refrigerant hose from the purge tank Schrader valve to an approved containment vessel.
 - b. Disconnect the line at the pump-out compressor.

- c. Hold the service switch in the ON position. This will energize the solenoids and allow the purge tank to fill with air. The liquid refrigerant will drain into the container.
 - d. Pour or siphon the water from the top of the refrigerant in the container. Seal the refrigerant in the container for later use or other disposition.
8. Filter Drier Service:
- a. Empty the purge tank and relieve the test pressure.
 - b. Remove the filter-drier cores (typically two cores) from inside the purge tank.
 - c. Visually inspect the core and all internal parts. Clean where necessary.
 - d. Inspect the tank drain valve and line to insure it is free of any debris.
 - e. Replace the tank gasket.
 - f. Install new filter-drier cores and gaskets as necessary. Filter-drier cores are able to absorb moisture for one year assuming normal humid air infiltration.
 - g. Put the tank back in place using a new gasket.
 - h. Reconnect the pump-out line and solenoid valve.
 - i. Pressurize the purge tank to manufacturer's specifications through the isolation solenoids, with the solenoids energized. Check for leaks. Release the pressure.
 - j. Switch the purge system to the manual ON position and wait for the pump-out compressor to start.
 - k. If refrigerant needs to be returned to the system, remove the cap from the purge tank drain valve and connect a hose from the drain valve to the container. As the pump-out compressor operates, a vacuum is created in the purge tank, drawing the liquid refrigerant from the container into the purge tank. Disconnect the hose when completed and replace the valve cap.
 - l. Open the valves on the purge tank inlet and liquid return lines.
9. Purge pump-out and fault check:
- a. This procedure tests the ability of the purge to remove non-condensable from the unit. This is done by isolating the purge, adding air to the purge tank, and then using the purge compressor to remove the air again.
 - b. Note the timer setting and reset to the time recommended by the manufacturer.
 - c. Set purge control switch to turn on the purge condensing unit (see instruction manual).
 - d. Close the shutoff valves on the purge tank inlet and liquid return lines to isolate the purge tank.
 - e. Disconnect the line from the inlet of the pump-out compressor.
 - f. Energizing the pump-out solenoids and pump-out compressor. The purge will draw air into the purge tank back through the isolation solenoid valves.
 - h. Note: Repeat this procedure until sufficient air is drawn into the purge tank to initiate a pump-out sequence. Within approximately five minutes the pump-out compressor should start and run until the fault timer setting is exceeded. The pump-out compressor then shuts down.
 - i. De-energize the purge.
 - j. Reconnect the line at the inlet of the pump-out compressor.
 - k. Restore original fault timer setting, purge initiate pressure setting, purge cut-out pressure setting.
 - l. Restart the purge by turning the purge control switch to AUTO.
 - m. Open the shutoff valves on the purge tank inlet and liquid return lines.

Tools

TOOL_BASIC REFRIGERANT_RECOVERY-01 MAT_CLEANING-03

This standard applies to Chillers and HVAC appliances with a total refrigerant charge of 50 or more pounds of refrigerant (including HFCs), that have leaked 10% or more of their refrigerant charge. The leak rate must be calculated using the EPA's Annualizing Method. The SOP requires refrigerant leaks to be recorded in NCMMS and repaired by a Section 608 Certified Technician within 30 days of discovery.

Special Instructions

1. Reference the associated job plan to coordinate PM of the asset with this leak repair activity.
2. Reference PBS OFM's Refrigerant Management and Reporting Standard Operating Procedure (March 31, 2022) to calculate the leak rate based on EPA's Annualizing Method.
3. Attach a copy of the EPA Section 608 Technician Universal Certification within their respective Labor records as Qualifications in NCMMS.
4. Before opening appliances containing refrigerant, use certified refrigerant recovery or recycling equipment to evacuate the refrigerant to EPA-specified evacuation levels.
5. Attach supporting service documentation to the Work Order for the associated asset in NCMMS.

Checkpoint

1. For appliances containing CFC or HCFC refrigerants, perform a successful repair and initial verification test within 30 days of the appliance exceeding a 10% leak rate. The initial verification test must demonstrate leaks were repaired before refrigerant is added back into the appliance (recharged) and returned to normal operating conditions.
2. For appliances containing CFC or HCFC refrigerants, perform a follow-up verification test within 10 days of the successful initial verification test (and after the appliance has returned to normal operating characteristics and conditions to verify that the repairs were successful).
3. For HFC, HCFC, and CFC refrigerant, after the leak is repaired, update the Leak Rate percentage ('REF-Leak_Rate' Virtual Meter) in NCMMS.
4. For HFC, HCFC, and CFC refrigerant, after the leak is repaired update the asset's total refrigerant charge ('Total_Charge' attribute) in NCMMS if it changed due to the repair. If the asset has more than one refrigerant circuit, update the full refrigerant charge for each repaired refrigerant circuit (e.g., 'Full_Charge1') in NCMMS.
5. For HFC, HCFC, and CFC refrigerant, after the leak is repaired update the asset's total refrigerant charge ('Total_Charge' attribute) in NCMMS if it changed due to the repair. If the asset has more than one refrigerant circuit, update the full refrigerant charge for each repaired refrigerant circuit (e.g., 'Full_Charge1') in NCMMS.
6. For appliances containing CFC or HCFC refrigerants that are not continuously monitored by an automatic leak detection system, create a work order to conduct a leak inspection in the next calendar year (within 12 months of the leak repair).
7. For HFC, HCFC, and CFC refrigerant, notify the Facility Manager/COR if the appliance cannot be repaired within 30 days. Refer to PBS OFM's Refrigerant Management and Reporting Standard Operating Procedure (March 31, 2022).

Tools

None

Electronic vibration analysis can detect impending problems which can lead to failure. Manufacturers of centrifugal chillers, in particular, recommend vibration analysis be performed during routine maintenance to ensure safe operation of the chiller.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review manufacturer's instructions and guidelines for vibration analysis. Consult chiller manufacturer for pass/fail specifications as it pertains to vibration.
3. Always purge chiller after it has been opened for servicing.
4. Perform vibration analysis at various loads and compare against the previous vibration analysis.
5. Consult chiller manufacturer for baseline vibration data (for new chiller).

Checkpoint

1. Take baseline readings on all permanently installed vibration transducers.
2. Monitor permanently installed vibration transducers with BAS.
3. An on-staff or contracted Vibration Analysis Technician using specialized software will need to determine criteria for unacceptable vibration signals.
4. Set up maintenance alarms or signal CMMS to issue a work order when transducer output exceeds acceptable levels based on baseline readings.
5. Take corrective action by:
 - a. Having a full set of vibration readings taken by a qualified on-staff or contracted vibration analyst.
 - b. Following the analyst's diagnostic recommendations.

Tools

ACCELEROMETER

This standard card applies to all centrifugal and absorption type chillers.

Special Instructions

1. Coordinate performance of this PM activity with performance of annual PM on the central or packaged chilled water units. Schedule outage with operating personnel.
2. Complete an eddy current test of all heat exchanger tubes, both evaporator and condenser (plus concentrator and absorber in absorption units).
3. The test shall be performed in accordance with current requirements and procedures of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code Section V Nondestructive Examination, Article 8, Eddy Current Examination of Tubular Products and applicable recommended practice standards of the American Society for Testing and Materials for Eddy Current Testing.
4. A Certified Level II or higher technician or equivalent shall be used for this analysis in accordance with the American Society of Non-destructive Testing Recommended Practices, SNT-TC-1A, or current version.
5. The test is to be witnessed by the Contracting Officer's Representative or designated inspector.

Checkpoint

1. Prepare equipment for non-destructive testing (NDT). Remove heat exchanger heads, piping, clean tubes, and erect scaffolding as necessary.
 2. Test shall be recorded as required by the ASME code Section V (Article 8 - Appendix I, Article I-20).
 3. System calibration shall be confirmed hourly.
 4. The written procedure in paragraph I-23, Article 8 - Appendix I in the ASME code is required to be followed.
 5. Strip chart recordings shall be provided for:
 - a. Each calibration standard and artificial discontinuity comparator used. Annotate to identify each defect machined in the standard and calibration of each division on the chart.
 - b. Typical good tube in each bundle
 - c. For each defective tube, annotate to identify tube. Indicate nature and extent of defect.
 6. Test each tube to detect, as a minimum: leaks, saddle damage, pitting, interior erosion/corrosion, gasket condition, presence of 'tramp' metal, presence of tube sheet condition.
 7. Correct deficiencies as directed.
 8. Restore equipment to service.
 9. Reports and Records:
 - a. A copy of the magnetic tape record shall be maintained by the NDT contractor and furnished if requested by the Government.
 - b. A preliminary job site report shall be provided as soon as the test is completed.
- test reports include the following:
- i. Written test procedure
 - ii. Recommendations - List all tubes recommended for replacement or isolation.
 - iii. Make complete description of defects (location, depth, inside or outside surface).
 - iv. Map location - Show tube row, number, and support for each tube bundle.
 - v. Name of technician performing tests and evaluating data.
 - vi. Contractor's certification of technician qualifications.

Tools

None

This standard card applies to water cooled magnetic bearing chillers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.

2. Review manufacturer's instructions.

3. Coordinate PM of refrigeration machine control panel with this PM activity.

4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.

6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.

7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lbs.) and leak rate (%) in NCMMS every time refrigerant is added to the equipment.

8. Prior to recycling, reclamation, or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.

9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.

10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

Tools

SAFETY_GOGGLES	REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02
MAT_CLEANING-03	PAINT-05	REFRIGERANT	REFRIGERANT_TANK	RESPIRATOR-01	TOOL_GROUP_A_AND_B
PRESSURE_WASHER-01					

This standard card applies to water cooled magnetic bearing chillers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Coordinate PM of refrigeration machine control panel with this PM activity.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.
6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lbs.) and leak rate (%) in NCMMS every time refrigerant is added to the equipment.
8. Prior to recycling, reclamation, or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.
9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.
10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

Tools

SAFETY_GOGGLES	REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02
MAT_CLEANING-03	PAINT-05	REFRIGERANT	REFRIGERANT_TANK	RESPIRATOR-01	TOOL_GROUP_A_AND_B
PRESSURE_WASHER-01					

This standard card applies to water cooled magnetic bearing chillers.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions.
- 3. Coordinate PM of refrigeration machine control panel with this PM activity.
- 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.
- 6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
- 7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lbs.) and leak rate (%) in NCMMS every time refrigerant is added to the equipment.
- 8. Prior to recycling, reclamation, or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.
- 9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.
- 10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
- 11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

Tools

SAFETY_GOGGLES	REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02
MAT_CLEANING-03	PAINT-05	REFRIGERANT	REFRIGERANT_TANK	RESPIRATOR-01	TOOL_GROUP_A_AND_B
PRESSURE_WASHER-01					

This standard card applies to water cooled magnetic bearing chillers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.

2. Review manufacturer's instructions.

3. Coordinate PM of refrigeration machine control panel with this PM activity.

4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

5. Comply with U.S. EPA regulations 40 CFR Part 82 (CAA Section 608) for asset maintenance, service, repair, and disposal.

6. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.

7. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms. Record the total refrigerant charge (lbs.) and leak rate (%) in NCMMS every time refrigerant is added to the equipment.

8. Prior to recycling, reclamation, or destruction of recovered refrigerant, contact the COR. Provide GSA with refrigerant reclamation or destruction records.

9. If disposing of equipment, follow EPA regulations (40 CFR Part 82) for the evacuation and recovery of refrigerant and equipment disposal. The following disposal records must be provided to GSA: Type and quantity of refrigerant recovered; Date of refrigerant recovery; Type and quantity of refrigerant disposed of through recycling, reclamation or destruction.

10. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

11. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

Tools

SAFETY_GOGGLES	REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02
MAT_CLEANING-03	PAINT-05	REFRIGERANT	REFRIGERANT_TANK	RESPIRATOR-01	TOOL_GROUP_A_AND_B
PRESSURE_WASHER-01					

This standard applies to all heating and cooling coils that are in the building, including units that are part of an air washer or air handling unit. The coils will be water or steam to air heat transfer coils.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Schedule shutdown with operating personnel, as needed.
- 3. Reviews standard operating procedure for selection, care, and use of respiratory protection.
- 4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

- 1. Vacuum or blow out the fins, coils, etc.
- 2. Comb fins as required.
- 3. Remove obstructions to air flow.
- 4. Check coils for leaks. Correct or report any leaks.
- 5. Check drain pans for proper drainage, standing water, clogged drains and drain pipes. Clean drain pans as required. Check for rust on pans. Repair as necessary.
- 6. Test and inspect controls that protect coils against freezing.
- 7. Check for rust or corrosion around coil frame and coil mounting bracket. Clean, prepare for painting and coat with proper type paint as necessary.
- 8. Record differential air pressure across the coil.

Tools

COIL_COMB	MANOMETER	VACUUM_CLEANER-04	TOOL_BASIC	RESPIRATOR-01	BRUSH_RADIATOR	COIL_CLEANER-01
MATERIALS_PAINT	SAFETY_GOGGLES					

This standard applies to critical or special purpose units that have the compressors located inside the air cooled condensing units. These units are considered one-half of a split system. Condensing units are usually installed where compressor noise and/or vibration is undesirable and are connected with an air handler, packaged air conditioning (A/C) unit, modular A/C units (evaporators) that are ceiling or wall mounted, or a split system associated with walk-in freezers and/or refrigerators. If condensing unit is subject or exposed to salt water conditions see salt water location section.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's operation and maintenance instructions.
3. Should be serviced just prior to the cooling season
4. Schedule outage of unit with personnel in area the unit serves.
5. Review standard operating procedure for selection, care, and use of respiratory protection.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If disposal of the equipment is required, follow regulations concerning removal of refrigerants and disposal of the unit.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Checkpoint

1. Remove debris from air screen and clean underneath unit.
2. Pressure wash coil with coil cleaning solution. Rinse and neutralize (cleaning solution) in accordance with manufacturer's recommendations.
3. Straighten fins on tubes with a fin comb.
4. Check electrical connections for tightness.
5. Check unit mounts and tighten if necessary.
6. Check all refrigeration lines for support, and signs of wear
7. Thoroughly inspect and clean interior and exterior of machine.
8. Clean and treat all rusted areas. Touch up prime and paint as needed.
9. Check for refrigerant leaks using a halogen leak detector, soap bubbles, or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor.
10. Check refrigerant levels and recharge if needed.
11. Perform required lubrication and remove old or excess lubricant.
12. Clean motor with vacuum or low pressure air (less than 40 psi). Check for obstructions in motor cooling and air flow.
13. Visually inspect disconnect switches and contactors condition, proper operation, arcing or any evidence of overheating and fusing.
14. Check compressor crank case heater for current rating and controller manufacturer's recommended size before replacing heater. (Heater size shall not be changed without the Regional Design Engineer's approval.)
15. Check line and load connections and heater mounting screws for tightness.
16. Check condenser fan blades for dust buildup and clean if necessary.

17. Check condenser fan blades and moving parts for excessive wear.
18. Check fan motor RPM before replacing so the unit continues to operate at design specifications.
19. Check bearing collar set screws on fan shaft to make sure they are tight, if applicable.
20. Check dampers for dirt accumulation. Check felt; repair or replace as required.
21. Check damper motors and linkage for proper operation. Adjust linkage on vanes if out of alignment, if equipped.
22. Lubricate mechanical connections of dampers sparingly, if equipped.
23. Check compressor oil level (non-hermetically sealed units only) if compressor is equipped with a sight glass.
24. Run machine with service gauge manifold attached, checking action of controls, relays, switches, and other components to see that:
 - a. Compressor(s) run at proper settings.
 - b. controls activate properly.
 - c. controls activate unit.
 - d. refrigerant suction and discharge pressures are proper.
25. Check setting on controls and return and adjust back to normal operation if necessary.
26. Clean up the work area and properly dispose of debris and waste.

Note: Seal off all service ports with flare caps. Report any missing caps or dust covers.

Tools

REFRIGERANT_RECOVERY-01	FIN_COMB	GLOVES-01	HOSE_FITTING-01	LEAK_DETECTOR-02	MAT_CLEANING-04
PAINT-04	PRESSURE_WASHER-01	REFRIGERANT_MANIFOLD	REFRIGERANT_TANK	RESPIRATOR-01	RUST_INHIBITOR-04
SAFETY_GOGGLES	REFRIGERANT	TOOL_GROUP_A			

This standard applies to normal use units that have the compressors located inside the air cooled condensing units. These units are considered one-half of a split system. Condensing units are usually installed where compressor noise and/or vibration is undesirable and are connected with an air handler, packaged air conditioning (A/C) unit, modular A/C units (evaporators) that are ceiling or wall mounted, or a split system associated with walk-in freezers and/or refrigerators. If condensing unit is subject or exposed to salt water conditions see salt water location section.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Obtain and review manufacturer's operation and maintenance instructions.
3. Should be serviced just prior to the cooling season
4. Schedule outage of unit with personnel in area the unit serves.
5. Review standard operating procedures for selection, care, and use of respiratory protection.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If disposal of the equipment is required, follow regulations concerning removal of refrigerants and disposal of the unit.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
14. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and all labels on refrigerant containers.

Checkpoint

1. Remove debris from air screen and clean underneath unit.
2. Pressure wash coil with coil cleaning solution. Rinse and neutralize (cleaning solution) in accordance with manufacturer's recommendations.
3. Straighten fins on tubes with a fin comb.
4. Check electrical connections for tightness.
5. Check unit mounts and tighten if necessary.
6. Check all refrigeration lines for support, and signs of wear
7. Thoroughly inspect and clean interior and exterior of machine.
8. Clean and treat all rusted areas. Touch up prime and paint as needed.
9. Check for refrigerant leaks using a halogen leak detector, soap bubbles, or similar testing device. If leaks are not able to be stopped or corrected, report leak status to supervisor.
10. Check refrigerant levels and recharge if needed.
11. Perform required lubrication and remove old or excess lubricant.
12. Clean motor with vacuum or low pressure air (less than 40 psi). Check for obstructions in motor cooling and air flow.
13. Visually inspect disconnect switches and contactors condition, proper operation, arcing or any evidence of overheating and fusing.
14. Check compressor crank case heater for current rating and controller manufacturer's recommended size before replacing heater. (Heater size shall not be changed without the Regional Design Engineer's approval.)

15. Check line and load connections and heater mounting screws for tightness.
16. Check condenser fan blades for dust buildup and clean if necessary.
17. Check condenser fan blades and moving parts for excessive wear.
18. Check fan motor RPM before replacing so the unit continues to operate at design specifications.
19. Check bearing collar set screws on fan shaft to make sure they are tight, if applicable.
20. Check dampers for dirt accumulation. Check felt; repair or replace as required.
21. Check damper motors and linkage for proper operation. Adjust linkage on vanes if out of alignment, if equipped.
22. Lubricate mechanical connections of dampers sparingly, if equipped.
23. Check compressor oil level (non-hermetically sealed units only) if compressor is equipped with a sight glass.
24. Run machine with service gauge manifold attached, checking action of controls, relays, switches, and other components to see that:
 - a. Compressor(s) run at proper settings.
 - b. controls activate properly.
 - c. controls activate unit.
 - d. refrigerant suction and discharge pressures are proper.
25. Check setting on controls and return and adjust back to normal operation if necessary.
26. Clean up the work area and properly dispose of debris and waste.

Note: Seal off all service ports with flare caps. Report any missing caps or dust covers.
27. Remove debris from air screen and clean underneath unit.
28. Pressure wash coil with biodegradable coil cleaning solution. Apply solution and let it stand for approx. five (5) minutes. Rinse with low pressure water.
29. The most effective method of coil cleaning is to open the unit and clean coils from the inside out using a low pressure-type sprayer.
30. Straighten fin tubes with fin comb.
31. Check electrical connections for tightness.
32. Check mounting for tightness.
33. Check for and remove all corrosion or rust from unit and supporting steel, prime and paint as necessary. Consult the Safety Data Sheet to ensure that the paint is lead free. Consult the Safety Data Sheet for proper PPE.
34. Check fan blades and belts. Clean fan blades as necessary.
35. Check wires at condenser electrical fused safety switches for tightness and burned insulation. Repair as necessary.
36. Clean up work area.

Tools

SAFETY_GOGGLES	PRESSURE_WASHER-01	TOOL_GROUP_A	RUST_INHIBITOR-04	RESPIRATOR-01	REFRIGERANT_TANK
REFRIGERANT_RECOVERY-01	REFRIGERANT	LEAK_DETECTOR-02	PAINT-04	MAT_CLEANING-04	HOSE_FITTING-01
FIN_COMB	REFRIGERANT_MANIFOLD	GLOVES-01			

This standard applies to all dampers in the HVAC system including air handlers and remote dampers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule shutdown with operating personnel, as needed.
3. Review manufacturer's instructions.
4. During damper servicing, moveable surfaces should be cleaned and lubricated.
5. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.
6. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. Cleaning dampers annually with a power washer or with a bucket of soapy water and a brush. If dampers are on the intake of an air handler, ensure unit is shut down prior to cleaning.
2. Check damper for freedom of movement and proper operation. Drive actuator through full range.
3. Observe damper operation through full operating range, by activating controller.
4. Check damper surfaces for wear and clean vanes.
5. Check damper end bushings for excessive wear. Replace as needed.
6. Check actuator/damper linkage for proper operation. Adjust if needed. Tighten operator arm set screws.
7. Lubricate mechanical connections sparingly using an approved lubricant. Wipe off excess.
8. Check actuator for proper operation.
 - a. If it does not stroke properly, check for binding drive stem.
 - b. If actuator still does not operate properly, pressure test the diaphragm. If the diaphragm fails the pressure test, replace the diaphragm (pneumatic actuators).
 - c. If the actuator still does not operate properly, disassemble the actuator (pneumatic actuators only) and check the accumulator for excessive wear or binding. Repair or replace the actuator.
 - d. If the actuator has a pilot positioner (pneumatic only) calibrate the pilot positioner.
9. Check for proper operating voltage, signal voltage, and resistance as applicable on motorized actuators.
10. Check hydraulic fluid levels if applicable on electro-hydraulic motors. If low or discolored replace hydraulic medium or actuator per manufacturer's instructions.
11. If damper is used for outside air economizer, check economizer setpoints in the BAS.
12. For pneumatic actuators, check for air leaks around actuator and in air line between controller and actuator.
13. Lubricate actuator linkage sparingly. Wipe off excess lubricant. DO NOT LUBRICATE actuator/drive stem.
14. Clean off any corrosion or rust on damper frame and or damper blades and restore original finish.

Tools

EQUIP_CLEANING-03

TOOL_BASIC

LUBRICANT-02

This standard card applies to those installations that utilize a indirect evaporative cooler as a precoolers for an air handler system. Indirect evaporative cooling does not add moisture to the airstream to cool the air by means of contact with a cooling coil or other impermeable heat exchange surface. Typically, air or water is circulated through one side of the heat exchange surface while supply air is drawn through or past the other side, cooling the air. Heat exchange surfaces can be tubes or plates. These types of systems may have individual sumps or may also have a common sump. They may also be installed in conjunction with direct evaporative systems. If this equipment is part of an Air Handler, the Air Handler will be serviced using that standard card and the evaporative cooler should be scheduled for accomplishment at the same time.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times. Open, lock, and tag out electric circuits serving motors for the air handler, indirect evaporative cooling fan (if equipped) and circulating pump(s).
5. Preventive maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Examine and wire brush all structural elements including doors, chamber, piping, plates, supports, pans, sumps, and framing.
2. Clean tank or sump and examine for leaks.
3. Examine all heat transfer surfaces for leaks, cracks or evidence of scaling. Check piping and plates for blockage or buildup. Clean as required.
4. Treat with rust inhibitor and paint as required. Consult the Safety Data Sheet to ensure that the paint is lead free.
5. Clean pump suction screens.
6. Check operation of float valve, mixing or automatic control valves and thermometers.
7. Pumps and motors less than 1 HP will be serviced as part of this PM. Pumps and motors 1 HP and larger will be serviced using PM standard cards HVAC-PMP-01 and ELEC-MOT-01 respectively.
 - a. Clean and lubricate pump. Check and replace packing if applicable.
 - b. Blow out or vacuum motor windings and lubricate if required.
8. Check belts or drives on the indirect fan motors for wear, proper adjustment and proper alignment. Replace if necessary. Multiple belt installations require the belts to be replaced as sets.
9. Check outside (intake) air filters. Clean or replace as necessary.
10. If equipped, check exhaust filters. Clean or replace as necessary.
11. Remove tags and lockout from circuits for circulating pump only.
12. Check with operating personnel before restoring circuits to the air handlers, to be certain personnel are not working on the unit.

Note: Pans and sumps should remain dry during winter operation. Tags should be removed from supply valves at the completion of this work, but the valves should be opened by operating personnel only when the unit is to be filled and placed in service.

Tools

RUST_INHIBITOR-02 GOOGLES GASKET-02 TOOL_BASIC TOOL_CLEANING-02

This standard card applies to those installations that utilize a direct evaporative cooler as either a precoolers for an air handler system or as a stand alone system. Direct evaporative cooling adds moisture to the airstream to cool the air by means of a cooling pad or media. Typically, water is circulated via a small pump over the media while air is drawn through it cooling the air. These types of systems may have individual sumps or may also have a common sump. If this equipment is part of an Air Handler, the Air Handler will be serviced using that standard card and the evaporative cooler should be scheduled for accomplishment at the same time.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule outage with operating personnel.
3. Review manufacturer's instructions.
4. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times. Open, lock, and tag out electric circuits serving motors for the air handler, evaporative cooling fan (if equipped) and circulating pump.
5. Preventive maintenance on associated equipment should be accomplished at this time.

Checkpoint

1. Examine and wire brush all structural elements including doors, chamber, piping, supports, pans, sumps, and framing.
2. Clean tank or sump and examine for leaks.
3. Treat with rust inhibitor and paint as required. Consult the Safety Data Sheet to ensure that the paint is lead free.
4. Clean pump suction screens.
5. Check pad distribution piping and clean as necessary. Replace as required.
6. Check piping for blockage or buildup. Clean or replace as required.
7. Check operation of float valve, mixing or automatic control valves and thermometers.
8. Pumps and motors less than 1 HP will be serviced as part of this PM. Pumps and motors 1 HP and larger will be serviced using PM standard cards for the respective equipment.
 - a. Clean and lubricate pump. Check and replace packing if applicable.
 - b. Blow out or vacuum motor windings and lubricate if required.
9. Remove tags and lockout from circuits for circulating pump only.
10. Check with operating personnel before restoring circuits to the air handlers, to be certain personnel are not working on the unit.

Note: Pans and sumps should remain dry during winter operation. Tags should be removed from supply valves at the completion of this work, but the valves should be opened by operating personnel only when the unit is to be filled and placed in service.

Tools

TOOL_BASIC GOOGLES RUST_INHIBITOR-02 TOOL_CLEANING-02 GASKET-02

This standard card applies to self-contained humidification units containing a complete system or units which are part of a large air handler and controlled by a humidistat. These units are used to add or maintain the humidity at a prescribed level in a space in a given temperature range. These can be powered by a steam generator, supplied by site steam or electric.

Special Instructions

1. Review manufacturer's instructions.
2. Review standard operating procedures for selection, care, and use of respiratory protection.
3. Turn off water supply.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Use of work gloves may be necessary due to caustic residual mineral deposits.

Checkpoint

1. Operate humidistat through its throttling range to verify activation, or deactivation of humidifier.
2. Clean and flush condensate pans, drains, water pans, etc. Remove corrosion, and repaint, recoat as needed. If a corrosion preventive chemical is used, ensure that it does not become a part of the indoor air by creating large amounts of volatile organic compounds or irritants. Check the Safety Data Sheet (SDS) to see what hazardous products are present. If hazardous products are present rinse very well before the system is used. Ensure that the paint lead level is 0.06% or less.
3. Check condition of heating element. Clean steam coils.
4. Clean steam/water spray nozzles. Adjust/replace as needed.
5. Chemically clean exterior of coil to remove scale and encrustations present on the coil.
6. Inspect steam trap for proper operation.
7. Inspect pneumatic controller for air leaks.
8. Inspect water lines for leaks and corrosion. Tighten all connections and repair leaks.

Tools

TOOL_GROUP_A SAFETY_GOGGLES RESPIRATOR-01 PSYCHROMETER GLOVES_WORK COIL_CLEANER-02

This standard is for centrifugal fans typically found in air handling systems. Although they may be found in other systems such as exhaust systems This standard will typically be performed with the parent standard as noted below. Motor PM shall be performed at the same time as the fan PM.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule shutdown with operating personnel, as needed.
3. Review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. All tests shall conform to the appropriate NETA test procedure and the values used as standards shall conform to the manufacturer's, NETA, and ANSI Standards specifications.
6. Review standard operating procedure for selection, care and use of respiratory equipment.
7. Refer to appropriate standard cards and manufacturer's instructions for motor maintenance.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts excessive wear. Clean as needed.
3. Check fan RPM to design specifications.
4. Check bearing collar set screws on fan shaft to make sure they are tight.
5. Remove old or excess lubricant while fan is off.
6. Vacuum interior of unit if accessible. Clean exterior.
7. Lubricate fan shaft bearings while unit is running. Add grease slowly until slight bleeding is noted from the seals. Do not over lubricate.
8. Check belts for wear, adjust tension or alignment, and replace belts when necessary. Multiple belts should be replaced with matched sets.
9. Check pulley wheels and clean and lubricate as required.
10. Check structural members, vibration eliminators, and flexible connections. Check fan housing to ensure there is no damage and the housing is tight.
11. Remove all trash and clean area around fan and fan room.

Tools

GREASE_GUN-05 EQUIP_CLEANING-01 RESPIRATOR-01 TACHOMETER-01 TOOL_BASIC VACUUM-01

This standard is for propeller or axial fans typically found in air handling systems. Although they may be found in other systems such as exhaust systems. This standard will typically be performed with the parent standard as noted below. Motor PM shall be performed at the same time as the fan PM.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule maintenance with operating personnel.
3. Obtain and review manufacturer's information for servicing, testing and operating.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Clean unit, especially fan blades.
2. Inspect pulleys, belts, couplings, etc.; adjust tension and tighten mountings as necessary. Change badly worn belts. Multiple belts should be replaced with matched sets.
3. Perform required lubrication and remove old or excess lubricant.
4. Clean motor with vacuum or low pressure dry air (less than 40 psig). Check for obstructions in motor cooling and air flow.
5. Remove all trash and debris.
6. Start unit and check for vibration and noise.

Tools

VACUUM-01 TOOL_BASIC LUBRICANT-03 MAT_CLEANING-02

This standard applies to all fan coil units; with or without fan speed control in two or four pipe systems. These units may be found in various locations including wall mounted, under window mounted, or ceiling mounted.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule shutdown with operating personnel, as needed.
3. Review manufacturer's instructions.
4. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. Check fan blades for dust buildup and clean if necessary.
2. Check fan blades and moving parts for cracks and excessive wear.
3. Check running motor amperes. Compare with manufacturer's specifications.
4. Tighten all electrical connectors to proper torque
5. Check that the fan runs properly in all speeds as applicable.
6. Check dampers for dirt accumulations, clean as necessary. Check felt, repair or replace as necessary.
7. Check damper actuators and linkage for proper operation as applicable. Adjust linkage on dampers if out of alignment.
8. Lubricate mechanical connections of dampers sparingly as applicable.
9. Check the valve(s) for signs of leakage and proper operation. Repair or replace as necessary.
10. Clean coils by brushing, blowing, vacuuming, or pressure washing.
11. Check coils for leaking, tightness of fittings.
12. Use fin comb to straighten coil fins.
13. Check belts for wear and cracks, adjust tension or alignment as applicable. Replace belts when necessary.
14. Check rigid couplings for alignment on direct drives, and for tightness of assembly.
15. Before heating season on units with outside air (chilled water coils only):
 - a. Drain cooling coils.
 - b. Blow down to remove moisture.
16. Check freeze stat for proper temperature setting and operation on units with outside air.
17. Vacuum interior of unit.
18. Check filter door for proper gasketing and air leaks. Correct as necessary.
19. Change the filter with the correct size and type filter.
20. Ensure that drain(s) are clear and running.
21. Clean up work area.

Tools

FIN_COMB GLOVES-01 SAFETY_GOGGLES TOOL_CLEANING-04 VACUUM-01 TOOL_GROUP_A

This standard is normally used for replicable prefilters, throw away type pleated filters or similar filters in air handling systems. This standard will typically be performed with the parent standard as noted below.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions and become familiar with manufacturer recommended operating velocity. This information will assist in selecting proper filter efficiencies.
3. If asbestos or lead dust could be present on filters, wear respiratory protection after checking it for wear and cleanliness. Only wear clean respirators that are not broken.
4. Staff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with HEPA cartridges for potentially hazardous dusts and #M N95 disposal dust masks for other dusts.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. The efficiency of all filters used in GSA controlled space will conform to the requirements of ASHRAE test method 51-92.
7. Review equipment design and or filter design specifications where applicable for the equipment. The design filter MERV rating shall be determined and installed.
8. Record MERV Ratings for all filters replaced. Go to the Work Order Actuals Tab, then down to Materials at the bottom. Click the Select Materials button and pick the appropriate MERV rating. Update the Quantity if more than 1 filter was replaced.

Checkpoint

1. Replace filters when they lose their efficiency or when they are so clogged that they produce too much pressure drop across the filter. Check magnehelic gage to insure maximum pressure drop has not been reached.
2. If clogged filters could contain asbestos or lead dust, wet filters with soapy water first before removing and placing in plastic bag for disposal.
3. Replace filters if the frame is bent or warped, or if the filtering medium is punctured.
4. Replace filters with the arrows on the frame pointing in the direction of airflow.
5. Check filter size and ensure filters are installed to prevent leakage between the filter bed and its supporting frame.
6. Remove tags, restore to service, and check for proper operation.
7. Clean up work area and remove trash.

Tools

FILTER_REPLACEMENT RESPIRATOR-09 TOOL_BASIC VACUUM-04

This standard applies to changing disposable roll filter media as part of a larger air handler. This maintenance should be performed along with quarterly filter maintenance and any air handler maintenance as required.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. If asbestos or lead dust could be present on filters, wear respiratory protection after checking it for wear and cleanliness. Only wear clean respirators that are not broken.
5. Staff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with HEPA cartridges for potentially hazardous dusts, and 3M N95 disposal dust mask for other dusts.
6. Review manufacturer's instructions and become familiar with manufacturer recommended operating velocity. This information will assist in selecting proper filter efficiencies.
7. The efficiency of all filters used in GSA controlled space will conform to the requirements of ASHRAE test method 51-92.
8. Record MERV Ratings for all filters replaced. Go to the Work Order Actuals Tab, then down to Materials at the bottom. Click the Select Materials button and pick the appropriate MERV rating. Update the Quantity if more than 1 filter was replaced.

Checkpoint

1. Remove old filter media roll, HEPA vacuum heavy dust and remove debris. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
2. Inspect framework and structure. Look for loose or missing bolts, air leaks, condition of flashing or caulking.
3. Inspect all moving parts for proper alignment, freedom of motion, excessive clearance or play, Clean, adjust or tighten as necessary.
4. Install new media roll, inspect powered roll, and take-up roll for correct tracking of media. On manual operation, check wheel or hand crank.
5. On motor drives, check pressure sensing device(s) and pressure switches. Test settings for starting and stopping motor.
6. Inspect motor, starter, controls, selector switch for auto warning or indicator lights.
7. Check oil in gear case. Change or replenish as required. Perform required lubrication using graphite where it is suitable. Remove old or excess lubricant.
8. Remove all trash and debris to proper disposal area.

Tools

GOOGLES TOOL_BASIC LUBRICANTS RESPIRATOR-09 VACUUM-04

This standard applies to changing disposable roll filter media. This standard will typically be performed with any required air handler maintenance.

Special Instructions

1. Review manufacturer's instructions.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
3. De energize fan, media motor; tag and lock out circuit.
4. If asbestos or lead dust could be present on filters, wear respiratory protection after checking it for wear and cleanliness. Only wear clean respirators that are not broken.
5. Shaff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with HEPA cartridges for potentially hazardous dusts, and 3M N95 disposal dust mask for other dusts
6. Record MERV Ratings for all filters replaced. Go to the Work Order Actuals Tab, then down to Materials at the bottom. Click the Select Materials button and pick the appropriate MERV rating. Update the Quantity if more than 1 filter was replaced.

Checkpoint

1. Remove old filter media as required and install new roll. If clogged filters could contain asbestos or lead dust, wet filters with soapy water first before removing and placing in plastic bag for disposal.
2. HEPA Vacuum heavy accumulation of dust and remove debris.
3. Inspect for proper alignment and operation of automatic controls, adjust as necessary.
4. Clean and lubricate filter rollers and transport mechanisms as necessary for proper operation.
5. Remove all trash from area and room, put equipment back in operation.

Tools

GOOGLES RESPIRATOR-09 VACUUM-04 TOOL_BASIC

This standard applies to all electrostatic filters used in air handling and kitchen applications. This standard will typically be performed with the parent standard as noted below.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. If asbestos or lead dust could be present on filters, wear respiratory protection after checking it for wear and cleanliness. Only wear clean respirators that are not broken.
5. Staff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with HEPA cartridges for potentially hazardous dusts, and 3M N95 disposal dust mask for other dusts.
6. Ground bus trips, top to bottom.

Checkpoint

1. Before securing unit, check indicators for defective tubes or broken ionizing wires.
2. Secure filter unit and air handler.
3. Wash each manifold until totally clean. Units with water wash spray require approximately four (4) minutes with warm water or seven (7) minutes with cold water.
4. If dry filters are dirty, remove and clean or replace filter.
5. While cells are drying, look for defects, particularly broken wires or hum suppressor. Wipe insulators with soft dry cloth.
6. If unit requires disassembly, check it thoroughly, clean, and adjust as required.
7. Restore to service and check for evidence of shorts.
8. Clean work area and remove trash to proper disposal area.

Tools

RESPIRATOR-09 EQUIP_CLEANING-01 TOOL_BASIC VACUUM-04

This standard is for reusable filters and includes time for removing, cleaning, and replacing the filters. The throw-away filters are usually more economical than the viscous type. This standard will typically be performed with the parent standard as noted below.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. If asbestos or lead dust could be present on filters, wear respiratory protection after checking it for wear and cleanliness. Only wear clean respirators that are not broken.
2. Staff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with HEPA cartridges for potentially hazardous dusts, and 3M N95 disposal dust mask for other dusts.
3. The efficiency of all filters used in GSA controlled space will conform to the requirements of ASHRAE test method 51-92.
4. Filter efficiencies will be selected in accordance with the operating velocity recommended by the manufacture.
5. Liquid adhesive coatings used on air filters shall have a flash point no lower than 325 degrees F (163 degrees C) as determined by ASTM D93 and the filter will meet all NFPA and UL standards.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Remove filters and replace with filters that have been cleaned and recoated. Examine frame and clean it with a high suction vacuum. Repair or replace damaged frame.
2. Move dirty filters to cleaning station.
3. Clean, recoat, and store filters removed until next scheduled change.

Tools

GOOGLES TOOL_BASIC RESPIRATOR-09 VACUUM-04

This standard card applies to charcoal filters used in air distribution systems to clean air of undesirable or hazardous materials. These filters may be cleaned by group forces or require shipping to manufacturer for recycling.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Notify operating personnel before shut down.
3. Maintenance personnel shall wear air purifying respirators fitted with HEPA cartridges for hazardous dusts and the appropriate chemical cartridge for other chemicals and appropriate protective clothing and equipment when servicing filters involving hazardous materials, i.e., shooting ranges, research laboratories, chemical laboratories, photographic development labs, etc.
4. Filters involving hazardous materials, i.e., shooting ranges, research laboratories, chemical laboratories, photographic development labs, etc.
5. Follow manufacturer's recommendations for handling, transporting, cleaning and shipping filters.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Wearing gloves and respirator, remove old filters and place in plastic bags.
2. Clean filter holder, rack and surrounding area.
3. Install new filter.
4. Place cleaning materials in plastic bags for disposal if work involves hazardous materials. Consult the Safety Data Sheet for proper disposal.

Tools

VACUUM-04 TOOL_BASIC RESPIRATOR-10 FILTER_REPLACEMENT

This standard is used for all filter types that will require changing monthly due to special or unusual air quality conditions. Use the applicable quarterly standard that applies to the specific filter type used.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions and become familiar with manufacturer recommended operating velocity. This information will assist in selecting proper filter efficiencies.
3. For filters containing or trapping hazard chemicals or dusts, wear respiratory protection after checking it for war and cleanliness. Only wear clean respirators that are not broken.
4. Staff shall be trained and enrolled in a respiratory protection program before wearing respirators. Minimum respiratory protection shall be air purifying with cartridges suitable for the hazardous chemicals.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. The efficiency of all filters used in GSA controlled space will conform to the requirements of ASHRAE test method 51-92.
7. Review equipment design and or filter design specifications where applicable for the equipment. The design filter MERV rating shall be determined and installed.
8. Record MERV Ratings for all filters replaced. Go to the Work Order Actuals Tab, then down to Materials at the bottom. Click the Select Materials button and pick the appropriate MERV rating. Update the Quantity if more than 1 filter was replaced.

Checkpoint

1. Reference applicable Quarterly standard.
 - a. HVAC-FLT-01-03M Filter, Throw Away
 - b. HVAC-FLT-02-03M Filter, Roll Type, Disposable Media
 - c. HVAC-FLT-03-03M Filter, Electrostatic
 - d. HVAC-FLT-04-03M Filters, Viscous Type (Wire Mesh)
 - e. HVAC-FLT-05-03M Filter, Charcoal

Tools

RESPIRATOR-11 TOOL_BASIC FILTER_REPLACEMENT VACUUM-04

This standard applies to all permanently installed Cooling Tower Side Stream Filtration Sand Media Filter Systems

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check the unit for leaks, cracks, gage operation and sight glass clarity.
2. Check the unit for unusual vibration or noise.
3. Check filter inlet pressure gage reading is within unit specification. (See OEM literature for value.)
4. Verify operation of system valves.
5. Clean pre-strainer basket

Tools

None

null

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check and lubricate clamp on filter tank access port.
2. Check over drain assembly and media pack.
3. Remove any debris from the top of the filter.
4. Check the media for smooth soft texture free from clumps. Remove hardened sections.
5. Check media level. Fill as needed.

Tools

None

This standard applies to plate and frame heat exchangers. A plate and frame heat exchanger consist of plates separating the hot and cold fluids. Plate and frame heat exchangers have very high heat transfer surface area due to the large surface area of the plates in contact with the operating fluid. Plate and frame heat exchangers are compact compared to tube type heat exchanger of similar capacity. In commercial building application, plate and frame type heat exchangers are normally used in water side economizer systems.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions and drawings.
3. Open unit only when defective plates are suspected or leaks occur that cannot not be stopped by tightening.
4. If unit is opened follow manufacturer's instructions concerning disassembly, internal cleaning and reassemble.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Never open the unit when hot.
7. The pressure of both liquids should be simultaneously decreased gradually when shutting down a unit.
8. If leaks occur tighten to average dimension as shown on assembly drawing or unit data plate, follow manufacturer's suggested tightening sequence.
9. Back flush the unit when fibers or large particulate are present this can be accomplished by flushing the unit with clean water in reverse flow at 1 to 1 1/2 times product flow or by arranging piping and valves so the unit may be operated in reverse flow mode on the product side for certain periods of time.
10. Clean in Place (CIP) when back flushing does not provide desired results, The unit must be flushed clear to obtain a thorough cleaning and it is necessary to flow CIP solution bottom to top to ensure wetting of all surfaces the rate of flow should be greater than normal product flow rate.
11. Strainers should be installed in supply lines ahead of the exchanger when the streams contain significant solids or fibers.

Checkpoint

1. Check exposed bolt threads, upper guide bars, rollers in the moveable end frame and connecting frame. Clean and coat with a light grease.
2. Check unpainted carbon steel surfaces, clean and coat with light grease, SAE-30 oil or other rust inhibiting product.
3. Check pressure and temperature gauges, Back flush unit and clean in place annually or any time pressure/temperature profile exceeds set limits.
4. Check unit for leaks, follow manufacturer's trouble shooting guide to locate defective plates.
5. Check overall external appearance and condition of unit.

Tools

None

null

Special Instructions

1. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.

Checkpoint

- 1. Clean and wash permanent filters. Recharge wire mesh filters with oil.
- 2. Replace disposable filters.
- 3. Drain and clean condensate pan.
- 4. Clean coils by vacuuming or brushing.
- 5. Use fin comb to straighten coil fins.
- 6. Clean strainers for accumulation of dirt.
- 7. Check controls, trap, freeze-stat, and control stat for proper operation.
- 8. Check induction nozzles. Clean if required.
- 9. Clean interior unit surfaces to remove soil.
- 10. Damp wipe exterior surfaces.
- 11. Clean surrounding floor area, and dispose of dirt and debris properly.

Tools

FILTERS-02 FIN_COMB MAT_CLEANING-03 TOOL_GROUP_A VACUUM-01

This procedure applies to all types of centrifugal pumps found in various mechanical systems throughout the building.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system PRIOR to making any changes.
3. Schedule service outages for the areas controlled by the field panel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. For replacement of mechanical seals, see the addendum at the end of this standard. It is generally not a good idea to tamper with pumps using mechanical seals if they are otherwise performing properly. Since mechanical seals can cost as much as the pump, it is usually not cost effective to risk damaging the seal by performing an annual internal inspection of the pump.

Checkpoint

1. Visually inspect the energized pump and motor. Look for fluid leaks originating from the area of the mechanical seal. Replace seal even if leaking only slightly. If seal is leaking, perform the following:
 - a. Disconnect and remove pump/motor coupling from pump shaft.
 - b. Clean external pump housing and shaft.
 - c. Remove flange on pump housing to remove pump mechanical seal assembly while noting how the seal is assembled on the shaft and in the pump housing.
 - d. After old seal has been removed, clean surfaces of shaft sleeve.
 - e. If this sleeve is galled or damaged, a new seal will not hold for long. It is best to replace the shaft sleeve at this time.
 - f. If the shaft sleeve is not damaged, follow seal manufacturer's instructions and install new mechanical seal.
 - g. Install new pump housing "O" ring or gasket, reinstall pump housing.
 - h. Slowly open discharge and suction valve, then rotate pump shaft to assist in setting mechanical seal.
2. Look for fluid leaks from other areas of the pump, flanges and gaskets, pressure gauges and fittings, bypass tubes or piping, and air release valves.
3. Look for bearing oil or grease leaks.
4. Observe coupling for excessive noise or obvious misalignment.
5. Observe fluid temperatures, pressures, and bearing temperatures if thermometers and pressure gauges are available.
6. De-energize pump and motor.
7. Observe pump and motor during shutdown. Often times, vibration and bearing problems become more noticeable during a coasting slowdown of the pump and motor.
8. Close suction and any bypass valves. Bleed off pressure but do not drain pump.
9. Observe pressure gauges for signs that the discharge check valve is not closing tightly.
10. Close discharge valve. Bleed off pressure but do not drain pump. Observe pressure gauges for signs that the discharge and suction valves are not closing tightly.
11. Lubricate pump and motor bearings. DO NOT OVER LUBRICATE
12. Use a torque wrench to check the tightness of all bolts, especially the bolts of cast iron pumps.
13. Closely inspect the pump-motor coupling.
14. Check motor and pump alignment.
15. Inspect and clean suction strainers and blowdown valves.
16. Clean pump and casing surfaces.
17. Other pumps:
 - a. Check that base bolts are securely fastened.

- b. After shut-down, drain pump housing, check suction, discharge and check valve for holding.
- c. Remove cover gland and packing.
- d. Remove corrosion from impeller shaft and housing cover.
- e. On pumps with oil ring lubricated bearings, drain oil, flush, and fill to proper oil level with new approved type oil.
- f. Inspect water rings, seals, and impeller.
- g. Clean pump suction strainers, and pump packing water seal filter/strainer.
- h. Replace packing and reassemble.
- i. Start and stop pump, noting vibration, pressure, and action of check valve.
- j. Adjust packing.
- k. Lubricate impeller shaft bearings. Do not over lubricate.
- l. Check motor and pump alignment.
- m. Check drive shaft coupling.
- n. Clean up work area and remove all debris.

Tools

GREASE_GUN-01 HOIST TOOL_GROUP_C EQUIP_CLEANING-03 ALIGNMENT_INDICATOR

This standard card applies to vacuum pumps used in printing plants, photographic plants, and other operation that requires a vacuum to operate. These units vary in size and appearance. They may resemble a small tank-less air compressor or a larger tank mounted and belt driven air compressor.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system PRIOR to making any changes.
3. Schedule service outages for the areas controlled by the field panel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Drain oil from reservoir if applicable. Place oil in a proper receptacle for oil disposal.
2. Remove oil filter and replace, if applicable.
3. Remove oil sight glass and clean as needed.
4. Fill with the appropriate type and amount of oil if applicable. Refer to manufacturer's specifications for the equipment.
5. Remove the exhaust filter and replace.
6. Check the exhaust chamber and oil return line for clogging. Clean if needed.
7. Inspect the exhaust chamber for excess oil. Wipe out if needed.
8. Inspect the vacuum inlet filter.
9. Wipe out the interior of the filter housing.
10. Wipe exterior surfaces of the pump free of debris and oil.
11. Remove and clean the vacuum diverter valve. Replace with new if needed.
12. If applicable, check the drive belt for cracks fraying and stretching. Replace if worn.

Tools

GLOVES-01 TOOL_GROUP_A TOOL_CLEANING-04 SAFETY_GOGGLES

These pumps are generally present in steam systems, when condensate is returned to the steam plant to be reused or is expelled from the system. When steam is supplied by a local utility the condensate is not usually recirculated, since it is usually monitored for billing purposes. In some cases condensate pumps are present at coiling coils with dehumidification loads.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. This maintenance should only be performed when it will not prevent operation of the steam boiler.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Operate unit to check for steam binding.
2. Check condensate temperature. It should be approximately 30 degrees (F) / 17 degrees (C) below steam temperature if traps are not leaking.
3. Examine flanges for steam leaks, replace gaskets as necessary.
4. Pump receiver down.
5. Turn condensate to sewer.
6. Shut down unit.
7. Clean and examine receiver, vent pipe, inlet and discharge openings for corrosion; remove all external corrosion, and paint with appropriate type paint.
8. Clean and adjust motor float switch and float operation on high low water level. Inspect float rods and pressure switches.
9. Make any adjustments as necessary.
10. Check alignment of motor and pump coupling with straight edge. Align if necessary.
11. Lubricate pump and motor.
12. Adjust packing glands and change packing when necessary.
13. Examine vacuum breaker operation.
14. Inspect ball floats, rods, and other linkage, adjust as necessary.
15. Properly dispose of all debris.

Tools

LUBRICANT-07 LEVEL_6INCH METAL_CLEANING TOOL_GROUP_C CLOTHS-01 GASKET-04

These standard applies to Test-And-Balance.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
2. Review most recent certified TAB report.
3. Verify that any subsequent remodeling, tenant improvements, or installation of major electrical devices have been reviewed by a mechanical engineer for impacts on heating and cooling loads.

Checkpoint

1. Using flow hoods, traverse pitot or anemometer to verify minimum and maximum air flows against the report.
2. Use pressure gauges to verify water flow rates against the report.
3. Verify readings agree with BAS readings, to within 5-10%.

Tools

PRESSURE_GAUGE-03 FLOW_HOOD ANEMOMETER

This maintenance standard applies to most terminal units (mixing boxes) which are found in most types of air handling systems. The terminal unit may be VAV, CAV, dual duct, with or without heating/re-heating or cooling, and be pressure dependant or pressure independent. These mixing boxes will be controlled by either pneumatics or Direct Digital Controls (DDC). The re-heat coil maintenance shall be combined with this standard.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's specifications.
3. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any control changes.
4. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.
5. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. DDC:
 - a. Check to see that the operating control sensor activates the damper per design specifications. If not, recalibrate or replace the operating control sensor with the same temperature range sensor.
 - b. Check damper linkage for tightness or damage. Lightly oil moving parts using an approved lubricant.
 - c. Inspect damper(s) for free movement in mixing box. Replace felt or other type seals as required.
 - d. Inspect mixing box and connecting ductwork for air leaks. Correct leaks with approved duct tape or tighten connections, as required.
 - e. Tighten electrical connections to all servo-motor actuators, and test as applicable.
 - f. Check the heating or cooling valve (if present) for leakage around the stem or between the seat and disk. Repair or replace as needed.
 - g. Check velocity sensor tubing for cracks, tightness, or holes if applicable.
2. Re-heat (if applicable):
 - i. Check electrical connections and insulators on re-heat coils.
 - ii. Vacuum or blow out the fins, coils, etc.
 - iii. Comb fins as required.
 - iv. Remove obstructions to air flow.
 - v. Check coils for leaks. Correct or report any leaks to COR.
 - vi. Test and inspect controls that protect coils against freezing.
 - vii. Check for rust or corrosion around coil frame and coil mounting bracket.
 - viii. Clean, prepare for painting and coat with proper type paint as necessary.
 - ix. Record differential air pressure across the coil.
3. Pneumatic:
 - a. Check damper linkage for tightness or damage. Lightly oil moving parts using an approved lubricant for pneumatic systems.
 - b. Inspect damper(s) for free movement in mixing box. Replace felt or other type seals as required.
 - c. Inspect mixing box and connecting ductwork for air leaks. Correct leaks with approved duct tape or tighten connections, as required.
 - d. Inspect damper actuator(s) for tightness to mounting brackets.
 - e. Inspect damper actuator diaphragm for leaks by performing a pressure test of the diaphragm.
 - f. Check the damper actuator spring range. Replace spring, adjust pilot positioner, or add pilot positioner as needed.

- g. If pneumatic actuator does not stroke properly, correct sticking valve stem or binding linkage. Replace or repair the diaphragm or actuator if necessary.
- h. Check the heating or cooling valve for leakage around the stem or between the seat and disk. Repair or replace as needed.
- i. Inspect the valve actuator for leaks by performing a pressure test of the diaphragm. Repair or replace as needed.
- j. Check the spring range of the valve actuator. Replace spring, adjust pilot positioner or add pilot positioner as needed.
- k. Inspect for air leaks around actuator and in the air line between controller and pneumatic actuator.
- l. Inspect operating control thermostat and/or pressure sensor for proper location and check main and branch air lines at thermostat for crimps, breaks, etc. Repair or replace if needed.
- m. Check all inline filters for oil and/or moisture. Replace as needed.
- n. Check all installed pressure gages for proper range and operability as applicable. Replace if needed.
- o. Perform a spring range check for all remaining end devices.
- p. Calibrate the operating control thermostat. Replace if it is defective with the same type action (direct or reverse action) and temperature range.
- q. Check to see that the operating control thermostat activates the damper per design specifications. If not, recalibrate the operating control thermostat.
 - i. Re-heat (if applicable)
 - ii. Check electrical connections and insulators on re-heat coils.
 - iii. Vacuum or blow out the fins, coils, etc.
 - iv. Comb fins as required.
 - v. Remove obstructions to air flow.
 - vi. Check coils for leaks. Correct or report any leaks.
 - vii. Check for rust or corrosion around coil frame and coil mounting bracket.
 - viii. Clean, prepare for painting and coat with proper type paint as necessary.
 - ix. Record differential air pressure across the coil.

Tools

TAPE_DUCT	DRAWINGS	LUBRICANT-07	SAFETY_GOGGLES	TOOL_CALIBRATION	TOOL_GROUP_B	MAT_CLEANING-03
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This maintenance standard applies to most terminal units (mixing boxes) which are found in most types of air handling systems. The terminal unit may be VAV, CAV, dual duct, with or without heating or cooling, and be pressure dependant or pressure independent. These terminal units will be controlled by Direct Digital Controls (DDC) ONLY. This standard includes some predictive maintenance tasks and can be used as an alternative to the purely preventive maintenance standard for DDC controlled units only. The use of a fault detection diagnostic tool, such as VAV Box Performance Assessment Control Charts (VPACC, form NIST) which uses a small number of control charts to assess the performance of VAV boxes is the primary means for this predictive maintenance task. This standard assumes that the diagnostic tools is installed and functioning properly and that staff has been trained on its use. O&M contractors may propose changes to this standard. Such changes are subject to GSA approval. Proposed changes to standards must be submitted showing markups of changes for GSA approval. This standard may be used in lieu of standard HVAC-TMU-01-01Y. However, this standard is performed quarterly.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's specifications.
3. Read and understand manufacturer's instructions of each device before making adjustments to the device or to the system. Understand what effects making adjustments will have on the overall Building Automation System and the operation of the building PRIOR to making any control changes.
4. For pneumatic systems, verify proper main line pressure per manufacturer's specifications.
5. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work. Follow lock out/tag out procedures at all times.

Checkpoint

1. DDC:
 - a. Check to see that the operating control sensor activates the damper per design specifications. If not, recalibrate or replace the operating control sensor with the same temperature range sensor.
 - b. Check damper linkage for tightness or damage. Lightly oil moving parts using an approved lubricant.
 - c. Inspect damper(s) for free movement in mixing box. Replace felt or other type seals as required.
 - d. Inspect mixing box and connecting ductwork for air leaks. Correct leaks with approved duct tape or tighten connections, as required.
 - e. Tighten electrical connections to all servo-motor actuators, and test as applicable.
 - f. Check the heating or cooling valve (if present) for leakage around the stem or between the seat and disk. Repair or replace as needed.
 - g. Check velocity sensor tubing for cracks, tightness, or holes if applicable.

Predictive Maintenance Checkpoints:

- a. Ensure that diagnostic tool has been installed properly.
- b. Review all alarms associated with the diagnostic tool and any corrective actions that have been taken as a result. All alarming functions that were originally set up in the diagnostic tool should be responded to at the time of the alarm.
- c. Review all trend data generated by the diagnostic tool where alarms are present.
- d. Based on the results, repair or correct the terminal unit mechanical or control related deficiency as required by qualified staff.
- e. Ensure that the diagnostic tool remains in operation and that operators are trained on its use.

Tools

Cooling towers are used in an open loop cooling water system. The most common use of a cooling tower is to cool the condenser water of a chiller. A cooling tower is composed of a fan, spray nozzles, heat transfer media and a water basin (sump). Within the cooling tower the water is sprayed over a heat transfer media called fill, as the first stage of cooling. The second stage of cooling occurs when the cooling tower fan is turned on. This fan is either a constant speed fan, a two speed fan or a variable speed fan.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule performance of this PM activity prior to seasonal start-up. Consider the time needed to effect any required repairs.
5. Perform cleaning of the tower in accordance with PM standard
6. Wear air purifying respirator fitted with HEPA cartridges, or 3M N95 disposable mask, after being enrolled in a respiratory protection program and receiving training on respirator use.
7. 3M mask shall only be worn if it is sufficiently appropriate for the potential airborne hazards.
8. Properly dispose of any debris, excess oil, and grease.
9. If materials to be worked on, such as the wet deck panels (tower fill media), are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Exterior Structural:
 - a. Inspect louvers for correct position and alignment, missing or defective items, and supports.
 - b. Inspect casings and attaching hardware for leaks or defects. Check the integrity and secure attachment of the corner rolls.
 - c. Inspect for loose or rotten boards on wood casings. Examine from the interior. Extensive damage may require replacement with fiberglass sheathing.
 - d. Inspect condition of access doors and hinges. Repair as necessary.
 - e. Inspect the distribution system including flange connectors and gaskets, caulking of headers on counterflow towers, deterioration in distribution basins, splash guards, and associated piping on crossflow towers. If configured with water troughs check boards for warpage, splitting, and gaps.
 - f. Examine the drain boards for damage and proper drainage. Check the fasteners also.
 - g. Inspect stairways including handrails, knee rails, stringers, structure and fasteners for rot, corrosion, security and acid attack.
 - h. Shake ladders to verify security, and check all rungs.
 - i. Check the security, rot, and corrosion on walkway treads. Check treads, walkways, and platforms for loose, broken, or missing parts. Tighten or replace as necessary.
 - j. Ladders must be checked for corrosion, rot, etc. Verify compliance with Occupational Safety and Health regulations regarding height requirements. Check ladder security.
 - k. Check fan decks and supports for decay, missing and broken parts, and gaps. Check the security.
 - l. Fan cylinders must be securely anchored. Check fastening devices. Note any damaged, missing, or corroded items. Watch for wood rot and corrosion of steel. Verify proper tip clearance between the fan blade and interior of cylinder. Verify compliance with OSHA requirements regarding height. Check its condition.
 - m. Apply protective coatings as needed on exterior surfaces. Be sure rust and dirt have been removed first.
2. Interior Structural:

a. Inspect the distribution system piping for decay, rust, or acid attack. Check the condition and tightness of connections and branch arms. Observe spray pattern of nozzles if possible and note missing and defective nozzles. Note condition of the redistribution system under the hot water system.

b. Inspect mechanical equipment supports and fasteners for corrosion. Wood structural members in contact with steel should be checked for evidence of weakness. Check condition of springs or rubber vibration absorption pads, including adjusting bolts, ferrous members, and rubber pads.

c. Check valves and operating condition of fire detection system. Check for corrosion of pipes and connectors. Check wiring of any thermocouple installed.

d. Check drift eliminators and supports. Remove any clogging debris. Replace missing blades.

e. Inspect tower fill for damage, ice breakage, deterioration, and misplaced, missing, or defective splash bars.

f. Examine interior structural supports. Test columns, girts, and diagonal wood members for soundness by striking with a rubber mallet. A high pitched, sharp sound indicates good wood, whereas a dull sound indicates soft wood. Probe rotted areas with a screw driver to determine extent of rot. Look for iron rot of metal fasteners in contact with wood. Check condition of steel internals. Check condition and tightness of bolts.

g. Inspect the nuts and bolts in partitions for tightness and corrosion. Look for loose or deteriorated partition boards. Note if partitions are installed so as to prevent wind milling of idle fans. Make sure wind walls parallel to intake louvers are in position. Boards or transite members should be securely fastened. Check condition of wood or steel supports for rot and corrosion.

h. Check wooden cold water basins for deterioration, warps, splits, open joints, and sound of wood. Inspect steel basins for corrosion and general condition. Inspect concrete basins for cracks, breaking joints, and acid attack.

i. Check all sumps for debris, condition of screens, antitubercular plates, and freely operating drain valves.

3. Mechanical:

a. Check alignment of gear, motor, and fan.

b. Inspect fans and air inlet screens and remove any dirt or debris.

4. Check hubs and hub covers for corrosion, and condition of attaching hardware.

5. Inspect blade clamping arrangement for tightness and corrosion.

6. Gear Box:

a. Clean out any sludge.

b. Change oil. Be sure gear box is full to avoid condensation.

c. Rotate input shaft manually back and forth to check for backlash.

d. Attempt to move the shaft radially to check for wear on the input pinion shaft bearing.

e. Look for excessive play of the fan shaft bearings by applying a force up and down on the tip of a fan blade. Note: Some output shafts have a running clearance built into them.

7. Power Transmission:

a. Check that the drive shaft and coupling guards are installed and that there are no signs of rubbing. Inspect the keys and set screws on the drive shaft, and check the connecting hardware for tightness. Tighten or install as required.

b. Look for corrosion, wear, or missing elements on the drive shaft couplings.

c. Examine the exterior of the drive shaft for corrosion, and check the interior by tapping and listening for dead spots.

d. Observe flexible connectors of both ends of the shaft.

e. Inspect bearings, belts, and pulleys for excessive noise, wear or cracking, alignment, vibration, looseness, surface glazing, tension. Replace or repair as required.

f. Check water distribution. Adjust water level and flush out troughs if necessary. Check all piping, connections, and brackets for looseness. Tighten loose connections and mounting brackets. Replace bolts and braces as required.

g. Check nozzles for clogging and proper distribution.

h. Inspect keys and keyways in motor and drive shaft.

4. Electrical:

a. Check electric motor for excessive heat and vibration. Lubricate all motor bearings as applicable. Remove excess lubricant.

b. Inspect fused disconnect switches, wiring, conduit, and electrical controls for loose connections, charred or broken insulation, or other defects. Tighten, repair, or replace as required.

c. Remove dust from air intakes, and check for corrosion. Check TEFC motors for conditions of air passages and fans.

d. If there is a drain moisture plug installed, see if it is operational.

e. Check amps and volts at operating loads, recommend pitching of fan blades to compensate.

f. Look for corrosion and security of mounting bolts and attachments.

5. Winterizing: (if applicable) If Cooling Tower is seasonal.

- a. Check that all electrical energy switches condenser pumps, fans, sump heaters have been de-energized and tagged out.
- b. Check that tower is drained, water make-up has been secured at an interior building location tagged off and exterior line is drained and left open to prevent freezing.
- c. Check that tower drains are locked in the open position and tagged.
- d. If there are piping heaters, check to make sure they are energized and working.
- e. Check that tower supply and return lines have been drained down to a point inside the building where water in them will not freeze and that the water has been pre treated to prevent corrosion and bacteria growth.
- f. Check that any water treatment pumps that serve the cooling tower system have been secured.
- g. Drain and secure cooling tower filtering systems that are exposed to freezing conditions.

Tools

TOOL_CLEANING-04	TOOL_GROUP_C	SAFETY_GOGGLES	RESPIRATOR-10	PRESSURE_WASHER	LUBRICANT-09	LADDER-06
GLOVES_WORK	AMP_PROBE	PROTECTIVE_COAT-02				

This applies to all cooling towers and evaporative condensers. Those located on the mezzanine or lower levels and near fresh air intakes are particularly important. Since cooling tower and evaporative condensers can become contaminated with microorganisms which are harmful.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform work before seasonal start-up (unless system has remained clean and free of bio-deposits), before seasonal shutdown, and quarterly during the cooling season.
3. If unit is not used year round drain and clean at least twice per year.
4. Review manufacturer's instructions.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work times.
6. Wear air purifying respirator fitted with HEPA cartridge, or 3M N95 disposable mask, after being enrolled in a respiratory protection program and receiving training on respirator use. 3M mask shall only be worn if it is sufficiently appropriate for the potential airborne hazards.
7. Ensure that there are safe and sturdy ladders and platforms to perform the lifting and cleaning required.
8. 3M mask shall only be worn if it is sufficiently appropriate for the potential airborne hazards.
9. If biological growth is excessive, have the water tested for legionella bacteria and if positive, clean and disinfect using the Wisconsin protocol. Then have a qualified water treatment specialist review your treatment program.
10. Refer to Table of Values for information on chlorine use.
11. If materials to be worked on, such as the wet deck panels (fill), are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Close building air intake vents within the vicinity of the cooling tower until the cleaning procedure is complete.
2. Shut down, drain, and flush the cooling tower with water (check with state to determine if there are any restrictions on discharging the water). Isolate the cooling tower from the rest of the condenser water system where applicable.
3. Clean the wet deck, remove all debris, and dispose of properly. If the wet deck panels contains asbestos, it is considered hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
4. Inspect the tower, the tower basin and holding tank for sediment and sludge, and any biological growth.
5. Using low pressure water hose or brushes, clean the tower, floor, sump, fill, spray pans and nozzles and removable components such as access hatches, ball float, and other fittings until all surfaces are clean and free of loose material. Porous surfaces such as wooden and ceramic tile towers will require additional cleaning and brushing. Clean cracks and crevices where buildup is not reached by water treatment.
5. Use caution when cleaning and handling sludge or water with biological growth, to make sure droplets or spray is not inhaled. Clean off skin with soap and water after the cleaning.
6. Clean all system strainers and strainer housings.
7. Remove drift eliminators and clean thoroughly using a hose, steam, or chemical cleanser.
8. Check fan and air inlet screens and remove any dirt or debris.
9. Reassemble components, and fill tower and cooling system with water.
10. Monitor the water pH and maintain pH within a range of 7.5 to 9.5. The pH can be monitored with litmus paper or a pH meter.
11. Add chemicals need to maintain water chemistry properties recommended by a water treatment consultant.
12. Perform the following if a more thorough disinfectant cleaning is needed:
 - a. Add a silicate-based low or non-foaming detergent as a dispersant at a dosage of 10-25 pounds per thousand gallons of water in the system.

- i. If the total volume of water in the system is not known, it can be estimated to be ten (10) times the recirculating rate (gallons per minute) or 30 gallons per ton of refrigeration capacity.
- ii. The dispersant is best added by first dissolving it in water and adding the solution to a turbulent zone in the water system, such as the cooling tower basin near the pump suction.
- iii. Contact a professional water treatment specialist for a dispersant which may be safely used without interfering with the operation of the system.
- b. Add chlorine disinfectant to achieve 25 parts per million (ppm) of free residual chlorine.
 - i. Maintain 10 ppm of free residual chlorine in water returning to the cooling tower for 24 hours.
 - ii. A swimming pool test kit may be used to monitor the chlorine. Follow the manufacturer's instructions. Test papers such as those used to monitor restaurant sanitizing tanks may also be used.
 - iii. Monitor every 15 minutes for two hours to maintain the 10 ppm level. Add chlorine as needed to maintain this level.
 - iv. Two hours after the slug dose or after three measurements are stable at 10 ppm of free residual chlorine, monitor at two hour intervals to maintain the 10 ppm of free residual chlorine.
 - v. Some kits cannot measure 10 ppm. In this case dilute the test sample with distilled water to bring it within the test set range.
- c. After 24 hours, drain the system (check with state to determine if there are any restrictions on discharging the water).
- d. Adjust bleed, float, central valve for desired water level.
- e. Open any building air vents that were closed prior to the cleaning of the cooling tower.
- f. Implement an effective routine treatment program for microbial control.
- g. Document all maintenance and cleaning procedures by date and time. Record the brand name and the volume or weight of chemicals used.

Tools

TOOL_CLEANING-04	BOOTS_RUBBER	CHEMICALS	CLOTHING_WATERPROOF	GLOVES-02	LITMUS_PAPER
PRESSURE_WASHER-02	RESPIRATOR-04	TEST_KIT_POOL	TOOL_GROUP_C	SAFETY_GOGGLES	

Unit heaters are usually used to heat otherwise unconditioned spaces. These units can be horizontal or vertical and are usually heating only.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check valve for full stroke operation in both directions.
2. Check valve for signs of abnormal wear and leaks. Replace packing if needed.
3. Clean the coil with vacuum cleaner.
4. Comb the fins as needed.
5. Clean all fans and motors.
6. Check operation of controls and safeties.
7. Lubricate as required.
8. Check all motors, belts, pulleys, shafts, etc. for alignment.
9. Treat all rusted areas with rust inhibitor and touch up paint.

Tools

VACUUM_CLEANER-02 TOOL_GROUP_A_AND_B RUST_INHIBITOR-05 MAT_CLEAN_PATCH

Unit heaters are usually used to heat otherwise unconditioned spaces. These units can be horizontal or vertical and are usually heating only.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Review standard operating procedure for selection, care, and use of respiratory protection.
4. Schedule shutdown with operating personnel.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Coordinate other related preventive maintenance items, such as, burners, fuel tank, humidifier, or other equipment.

Checkpoint

1. Remove access panels if applicable.
2. Check the fire box liner or refractory for cracks and leaks.
3. Check oil pump for leaks. Repair or replace as needed (oil units only).
4. Change the oil filter and oil spray nozzles (oil units only).
5. Check all gas lines for leaks. Repair as needed.
6. Check smoke stack for obstructions, leaks, etc.
7. Clean bottom of smoke stack (breaching).
8. Clean all fans and motors.
9. Check operation of controls and safeties.
10. Lubricate as required.
11. Replace access panels if removed.
12. Check all motors, belts, pulleys, shafts, etc. for alignment.
13. Treat all rusted areas with rust inhibitor and touch up paint.

Tools

RUST_INHIBITOR-05 FLUE_BRUSH RESPIRATOR-05 TOOL_GROUP_A_AND_B VACUUM_CLEANER-02 MAT_CLEAN_PATCH

This standard applies to all valves which are controlled through the BAS system and operated with an electric motor, pneumatic actuator or electric device such as a solenoid.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan.
5. For electronic motors complete steps 6a to 6e. For pneumatic actuators perform steps 7a to 7i.

Checkpoint

1. Clean unit and make visual examination of all parts.
2. Operate from limit to limit. Observe operation, look for binding, sluggishness, action of limits, etc.
3. Determine if valve seats and holds properly.
4. Check condition of packing.
5. Apply graphite to moving parts of valve.
6. DDC:
 - a. Check to see that the operating control sensor activates the valve per design specifications. If not, recalibrate or replace the operating control sensor with the same temperature range sensor.
 - b. Inspect actuator for free movement.
 - c. Tighten electrical connections to all servo-motor actuators, and test as applicable.
 - d. Check all voltages or resistance at the actuator. Refer to manufacturer's specifications for values.
 - e. Skip to step 1.
7. Pneumatic:
 1. a. Inspect actuator diaphragm for leaks by performing a pressure test of the diaphragm.
 2. b. Check the actuator spring range. Replace spring, adjust pilot positioner, or add pilot positioner as needed.
 3. c. If pneumatic actuator does not stroke properly, correct sticking valve stem or binding linkage. Replace or repair the diaphragm or actuator if necessary.
 4. d. Inspect for air leaks around actuator and in the air line between controller and pneumatic actuator.
 5. e. Inspect operating control thermostat and/or pressure sensor for proper location and check main and branch air lines at thermostat for crimps, breaks, etc. Repair or replace if needed.
 6. f. Check all inline filters for oil and/or moisture. Replace as needed.
 7. g. Check all installed pressure gages for proper range and operability as applicable. Replace if needed.
 8. h. Check to see that the operating control thermostat activates the actuator per design specifications. If not, recalibrate the operating control thermostat.
 9. i. Clean up work site.

Tools

This is applicable to both condenser water and chilled water systems. Included are open recirculating systems, closed systems, air washers, and sprayed coil unit and evaporative condensers. Once through systems are not applicable. Note: This maintenance standard does not eliminate daily testing where it is an established policy.

Special Instructions

1. Chemicals must comply with the Environmental Protection Agency (EPA) regulations and handled in accordance with occupational safety requirements. Employ personal protection against corrosive or hazardous treatment chemicals as appropriate.
2. Be familiar with the Safety Data Sheets of any chemicals used in the water treatment program.
3. Water treatment specialists must be properly trained.
4. Water treatment must be based on proven standard engineering practices. Treatment methods that claim mysterious magnetic or catalytic forces shall not be used.
5. Follow treatment as directed by manufacturer or water treatment company.
6. Maintenance includes chemicals, chemical feeding, maintaining proper water conditions, controlling bleed off, protecting idle equipment, and recordkeeping.
7. Select water treatment methods which protect the life of equipment, maximize heat transfer, and minimize scale, corrosion, solid matter buildup, algae, fungi, biological growth, and water usage. Alternate biocides. Changes to higher dosage levels may be required to control growth.
8. Ensure chemicals are properly stored, test equipment clean, and that chemicals have not passed expiration date.
9. Special attention must be paid to wooden towers and solar heating systems which can be more difficult to properly maintain.
10. Physical inspections of towers and piping systems are to be made to determine water treatment effectiveness.
11. Maintain records and test results.

Checkpoint

Tools

TOOL_BASIC GLOVES_SHIELD_APRON

This is applicable to both condenser water and chilled water systems. Included are open recirculating systems, closed systems, air washers, and sprayed coil unit and evaporative condensers. Once through systems are not applicable. Note: This maintenance standard does not eliminate daily testing where it is an established policy.

Special Instructions

1. Chemicals must comply with the Environmental Protection Agency (EPA) regulations and handled in accordance with occupational safety requirements. Employ personal protection against corrosive or hazardous treatment chemicals as appropriate.
2. Be familiar with the Safety Data Sheets of any chemicals used in the water treatment program.
3. Water treatment specialists must be properly trained.
4. Water treatment must be based on proven standard engineering practices. Treatment methods that claim mysterious magnetic or catalytic forces shall not be used.
5. Follow treatment as directed by manufacturer or water treatment company.
6. Maintenance includes chemicals, chemical feeding, maintaining proper water conditions, controlling bleed off, protecting idle equipment, and recordkeeping.
7. Select water treatment methods which protect the life of equipment, maximize heat transfer, and minimize scale, corrosion, solid matter buildup, algae, fungi, biological growth, and water usage. Alternate biocides. Changes to higher dosage levels may be required to control growth.
8. Ensure chemicals are properly stored, test equipment clean, and that chemicals have not passed expiration date.
9. Special attention must be paid to wooden towers and solar heating systems which can be more difficult to properly maintain.
10. Physical inspections of towers and piping systems are to be made to determine water treatment effectiveness.
11. Maintain records and test results.

Checkpoint

Tools

TOOL_BASIC GLOVES_SHIELD_APRON

This standard is intended for maintaining chemical free water treatment systems such as eH2O and Flozone. Each manufacturer has specific requirements; manufacturer recommendations shall be followed. This standard summarizes the typical tasks found in manuals for chemical free water treatment systems.

Special Instructions

1. Manufacturer specific guidelines shall be followed. This standard only summarizes typical annual tasks and shall not be considered inclusive of all necessary activities.

Checkpoint

1. Per manufacturer recommendations, check hours of operation of the devices (i.e. purifier, oxygen concentrator). If run hours exceed 8,000 hours, devices shall either be re-built or replaced. Consult manufacturer recommendations.

Tools

CONSULT_MANUF

This standard is intended for maintaining chemical free water treatment systems such as eH2O and Flozone. Each manufacturer has specific requirements; manufacturer recommendations shall be followed. This standard summarizes the typical tasks found in manuals for chemical free water treatment systems.

Special Instructions

1. Manufacturer specific guidelines shall be followed. This standard only summarizes typical quarterly tasks and shall not be considered inclusive of all necessary activities.

Checkpoint

1. Per manufacturer recommendations, all sensors and probes shall be cleaned monthly. Follow manufacturer specific procedures for cleaning each probe. Typical probes include pH probes and conductivity probes; typical sensors include Oxygen percentage.
2. Per manufacturer recommendations, all filters shall be inspected, cleaned and replaced if necessary (if filter feels wet to the touch). Follow manufacturer specific procedures for inspecting, cleaning and replacing filters. Note: it may be necessary to perform a manual backwash, consult manufacturer specifications and recommendations.
3. Per manufacturer recommendations, desiccant indicators shall be inspected. Follow manufacturer specific procedures for verifying moisture content: the manufacturer shall be notified if high moisture content is present.
4. Per manufacturer recommendations, a visual inspection of the equipment shall be performed, with specific focus on loose fittings or other notable damage.
5. Per manufacturer recommendations, sensors and probes shall be calibrated. Follow manufacturer specific procedures for calibrating sensors.
6. Per manufacturer recommendations, coupons shall be changed out (this is manufacturer specific). Follow manufacturer specific procedures for replacing coupons.

Tools

CONSULT_MANUF

Applies to: Buildings with Cooling Towers and Evaporative Condensers and Buildings with Humidifiers, Misters, Decorative Fountains or Other Unique Water Features that Release Water Aerosols

Special Instructions

8. The CDC Legionella Sampling Procedure for Decorative Fountains Samples will consist of () 1L bulk water and a biofilm swab per basin (number of swabs dependent on size and complexity of the fixture)
3. Ensure that the sampling subcontractor has the capabilities to collect, transport, and analyze samples including "chain of custody" documentation and final reporting.
2. Ensure the sample collector is qualified and experienced in microbiological sampling methods for Legionella.
 1. Breathing legionella bacteria infected mist may cause legionaire's disease. For flushing activities, wear the proper PPE identified in the workplace hazard assessment conducted per OSHA 1910.132(d). When flushing, flushing flow rates shall be as high as possible but not so high they create mist formation, splashing, or flooding.
4. Ensure the sampling subcontractor, as needed, has contacted the testing laboratory before sampling and has received sample collection materials, information on sampling methodology, paperwork, and transport. Subcontractor follows all instructions from the testing laboratory closely, including the completion of the chain-of-custody.
5. Ensure the laboratory that the sampling subcontractor is using is accredited for Legionella analysis according to the international laboratory quality standard ISO 17025:2017 by an accreditation body that is itself accredited according to ISO 17011:2017 (e.g., AIHA LAP), certified by the CDC Environmental Legionella Isolation Techniques Evaluation (ELITE) program, or as required by specific state requirements (e.g. the New York State Environmental Laboratory Approval Program for New York State).
6. The CDC Legionella Sampling Procedure for HVAC Cooling Towers Samples will consist of () Make-up water (water added to replace water loss because of evaporation, drift, or leakage) 1 Sample 1L bulk water, () Collection basin (an area below the tower where cooled water is collected and directed to the sump) 1L bulk water and a biofilm swab at the water line ()
9. The CDC Legionella Sampling Procedure for Humidifiers, Misters and unique water features Samples will consist of () Bulk water (as close to 1L as possible) and at least one biofilm swab of a moist surface
7. Continued. Sump (a depressed chamber contiguous to the basin, where water flows to facilitate pump suction; may also be used as collection point for silt and sludge) 1L bulk water and a biofilm swab at the water line () Storage tank or reservoir in the system 1L bulk water () Drift eliminators or other surfaces that remain moist 1 biofilm swab () Heat sources (e.g., chillers) 1L bulk water.

Checkpoint

Tools

None

KTCH-DSH-01-03M**Dishwashing Machine**
Rev-1 | 2021-11-08

This maintenance standard applies to electric and steam operated cafeteria kitchen dishwashing equipment.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Notify cafeteria operator and get permission prior to performing all maintenance.
3. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
4. Review manufacturer's instructions.
5. Review standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits and fuel service.
7. Follow all instructions of Material Safety Data Sheets for lubricants and cleaners.

Checkpoint

1. Check with operator or manager for any deficiencies, verify cleaning program.
2. Check motor and bearings for excessive noise, vibration, and overheating. Clean motor ventilator openings
3. Check electric insulators, connection and wiring, including inside access panels and junction boxes, and final connections. Tighten loose connections.
4. Test electrical controls, signal lights, timer, and OFF/ON switches. Test timer and switches.
5. Examine all pump suction and discharge connections for leakage, adjust packing nuts as required.
6. Check temperature regulator and adjust or calibrate as required.
7. Check thermostatic control solenoid valve for a minimum of 100 degrees prewash, 140 degrees for wash, and 140 degrees or 180 degrees F for final rinse. (Low temp machines at 140 degrees F.)
8. Check operation of wash and rinse spray mechanism for spray coverage and drainage.
9. Inspect soap and spray solution feeder lines; clean as necessary.
10. Inspect water/steam lines and fittings for leaks; tighten fittings as necessary.
11. Check packing glands on wash, rinse, and drain valves; add or replace packing as required. Tighten nuts, bolts, and screws.
12. Check lubricant in gear case; add manufacturer's recommended oil if required.
13. Inspect splash curtain for tears, clearance, and water tightness; adjust if required.
14. Check proper operation of solenoid valve and float in fill tank; adjust as required. Check and repair insulation as needed.
15. Check proper operation of micro-switch.
16. Check doors for operations of chains and counterweights, warping, alignment and water tightness. Replace door gaskets if needed
17. Clean lime off thermostatic probe and heating elements. Drain booster heater to remove scale from the bottom.

Tools

TOOL_GROUP_A

This maintenance standard applies to electric or gas fired cafeteria kitchen fryer equipment, including deep fat, counter-top, drop-in, and free standing varieties.

Special Instructions

1. Notify cafeteria operator and get permission prior to performing all maintenance.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. If any safety deficiencies are found which could cause injury or damage, including tank leaks, notify the cafeteria operator immediately and secure the equipment from further operations.
4. Review manufacturer's instructions.
5. Review standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits and fuel service.

Checkpoint

1. Check pilot and flame on gas operated unit; adjust as necessary.
2. Check all gas connections for tightness and for leaks.
3. Check all electrical connections and wiring for tightness and signs of overheating.
4. Check thermostat; calibrate, if necessary.
5. Check basket raising mechanism.
6. Check basket/rack for bends, breaks, or defects; straighten bends or repair as necessary.
7. Check operation of unit.
8. Check flue for proper draft or obstructions.
9. If electric model, check heating elements by amperage draw and check against nameplate.
10. Check nuts, bolts, and screws for tightness; tighten or replace as necessary.
11. Verify that drop-in units must have flange to counter seal.
12. Clean interior walls and elements to obtain maximum heat transfer.
13. Inspect grease compartment.
14. Lubricate gas valves.

Tools

TOOL_GROUP_A

This maintenance standard applies to electric operated or gas fired cafeteria kitchen grill equipment, including flat-top, open grill, drop-in, or free standing varieties.

Special Instructions

1. Notify cafeteria operator and get permission prior to performing all maintenance.
2. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
3. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
4. Review manufacturer's instructions.
5. Review standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits and fuel service.

Checkpoint

1. Check with operator to verify cleaning program.
2. Check with operating or area personnel for any deficiencies.
3. Check nuts, bolts, and screws for tightness; tighten or replace as required.
4. Inspect grease trough, drip tray, splash guard, and surface condition.
5. Examine gas utility supply line, valve packing, tighten fittings as required.
6. Examine/clean burners and elements.
7. On gas operated units, check pilot and gas burners for uniform flame; adjust as required.
8. On electrically operated units, check switches, connections, and wiring for loose or overheated conditions.
9. Check calibration of thermostats; calibrate if required.
10. Check flue for proper draft or obstructions.
11. Lubricate gas valves.
12. Check elements to obtain maximum heat transfer take amperage measurement and check against nameplate.
13. Examine burner guards, grates, covers or cook top surfaces for cracks or damage.
14. Check all controls, mechanisms for proper operation; adjust as required.
15. Check electric power line condition, switch, disconnect, etc.; or check condition of gas supply, valves, regulators, and inspect pilot, check for gas leaks.
16. Examine control knobs and indicating lights; adjust/replace as required.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ice makers, including storage and dispenser types.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
6. Review standard operating procedure for controlling hazardous energy sources.
7. De-energize, lock out, and tag electrical circuits.
8. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
9. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
10. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
11. Recover, recycle, or reclaim the refrigerant as appropriate.
12. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
13. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
14. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
15. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.
16. Only approved cleaning chemicals shall be used.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Visually check for refrigerant, oil and water leaks.
3. Inspect ice condition/size.
4. Check and tighten any loose screw-type electrical connections.
5. Check all controls; adjust if necessary.
6. Examine water connection; open and close water valve; test ice dispensing valve and (door) metering adjustment.
7. Check and clear ice machine draining system (drain vent, strainer and trap).
8. Examine condition of bin doors-closure, hinges, gaskets, handles and ease of slide; lubricate as required. Check storage bin condition.
9. Clean motor, compressor, and condenser coil.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ice cream and shake maker equipment.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
6. Review standard operating procedure for controlling hazardous energy sources.
7. De-energize, lock out, and tag electrical circuits.
8. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
9. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
10. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
11. Recover, recycle, or reclaim the refrigerant as appropriate.
12. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
13. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
14. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
15. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Examine the differential pressure pump. Check the "O" ring seal and replace if defective.
3. Test for proper soft serve ice cream texture.
4. Inspect wash kit faucet (single lever swing spout).
5. Check beater motor, clean and grease assembly.
6. Check refrigerant charge.
7. Check machine amperage draw against nameplate.
8. Inspect gear reduce oil level and lubricate fan motor.
9. Examine pulley alignment, belt condition, and belt tension.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen kettles, including steam, electric, gas, and tilting, (braising pan).

Special Instructions

1. Preventive maintenance services, should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Obtain and review manufacturer's instructions.
4. Notify cafeteria operator and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify the cleaning program.
2. Examine utility supply line, valve packing, specialties, and insulation.
3. Inspect safety pressure valve.
4. Lubricate tilting gear mechanism and trunnion bearings, if applicable.
5. Calibrate thermostats as required.
6. Check and replace leaking packing washers.
7. Examine water supply control and fill valve.
8. Inspect temperature gauge, lid hinge and condition. Lubricate hinge.
9. Test timer, switches, pilot light.
10. Check for adequate steam pressure to unit.
11. Examine condensate trap, thermostatic trap, and regulator.
12. Lubricate valves.
13. Tighten all screws in electrical wiring connections, i.e., panels, junction boxes, final connections, etc.
14. Check elements to obtain maximum heat transfer.
15. Examine handles, knobs and controls for tightness and safe condition.
16. NOTE: Remaining checks pertain to gas units.
17. Check for gas leaks.
18. Check and clean burner orifices.
19. Clean and adjust pilot light assembly.
20. Check air shutters to make sure air/gas mixture is correct.
21. Check flue for obstructions and proper draft.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ovens, including wet/dry steam, char broiler, convection or baking.

Special Instructions

1. Preventive maintenance services, should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Check all controls, mechanisms for proper operation; adjust as required.
3. Examine utility supply line, piping, valve packing, specialties, and insulation; look for leaks.
4. Check electric power line condition, switch, disconnect, etc.; or check condition of gas supply, valves, regulators, and inspect pilot, check for gas leaks.
5. Check to ensure ovens and oven racks are level.
6. Check gaskets and seals; check doors for tightness and warping; lubricate hinges and repair as necessary.
7. Examine handles, knobs and controls for tightness and safe condition.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ovens, including wet/dry steam, char broiler, convection or baking.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Check all controls, mechanisms for proper operation; adjust as required.
3. Examine utility supply line, piping, valve packing, specialties, and insulation; look for leaks.
4. Check electric power line condition, switch, disconnect, etc.; or check condition of gas supply, valves, regulators, and inspect pilot, check for gas leaks.
5. Check and clean fan blades for convection ovens.
6. Check to ensure ovens and oven racks are level.
7. Check the operation of thermostats; calibrate if required
8. Clean and adjust gas burners.
9. Check safety pilot and solenoid.
10. Clean and adjust pilot light assembly.
11. Check flue for proper draft or obstructions.
12. Lubricate gas valves.
13. Tighten all electrical connections in panels, junction boxes, final connections, etc.
14. Clean interior walls and elements to obtain maximum heat transfer.
15. Check gaskets and seals; check doors for tightness and warping; lubricate hinges and repair as necessary.
16. Examine handles, knobs and controls for tightness and safe condition.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen refrigerators/freezers (walk-in units).

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Review manufacturer's instructions.
3. Notify cafeteria operator and get permission prior to performing all maintenance.
4. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
5. Review standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
14. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Verify indicator light on; check compartment temperature.
3. Examine evaporator for proper clearances/slope and air flow.
4. Examine condensate drain and drain heating.
5. Examine handles, hinges and tightness of door closure.
6. Examine safety door release and fan shut down safety switch.
7. Inspect lighting for burnt out lamps.
8. Inspect door gaskets for damage and proper fit; adjust gaskets as required and lubricate hinges with food grade oil.
9. Check door gasket heater.
10. Check box floor for water or ice accumulation.
11. Check box for excessive ice build- up and open seams.
12. Verify defrost cycle and timer operation.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen refrigerators/freezers (walk-in units).

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Review manufacturer's instructions.
3. Notify cafeteria operator and get permission prior to performing all maintenance.
4. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
5. Review the standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
14. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Verify indicator light on; check compartment temperature.
3. Examine evaporator for proper clearances/slope and air flow.
4. Examine handles, hinges and tightness of door closure.
5. Examine safety door release and fan shut down safety switch.
6. Inspect lighting for burnt out lamps.
7. Check starter panels and controls for proper operation, burned or loose contacts, and loose connections.
8. Clean evaporator coil, evaporator drain pan, blowers, fans, motors, and drain piping as required; lubricate motor(s).
9. Inspect defrost systems for proper operation, including timer; adjust as required. Have automatic defrosters adjusted as required so freezer will defrost during off peak hours.
10. Check operation of thermostats; calibrated as required.
11. Check coil superheat and adjust to manufacturers recommendations.
12. Inspect and service all electric motors.
13. Inspect door gaskets for damage and proper fit; adjust gaskets as required and lubricate hinges with food grade oil.
14. Check door gasket heater.
15. Check box floor for water or ice accumulation.
16. Check box for excessive ice build-up and open seams.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen refrigerators/freezer (Reach-In, Pass-Thru, Display case) Types.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Review manufacturer's instructions.
3. Notify cafeteria operator and get permission prior to performing all maintenance.
4. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
5. Review the standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
14. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Verify indicator light on; check compartment temperature.
3. Examine evaporator for proper clearances/slope and air flow.
4. Examine condensate drain and drain heating.
5. Examine handles, hinges and tightness of door closure.
6. Examine safety door release and fan shut down safety switch.
7. Inspect lighting for burnt out lamps.
8. Inspect door gaskets/doors for damage and proper fit; adjust gaskets as required and lubricate hinges with food grade oil.
9. Check door gasket heater.
10. Check box floor for water or ice accumulation.
11. Check box for excessive ice build- up and open seams.
12. Verify defrost cycle and timer operation, if applicable.
13. Clean condenser coil.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen refrigerators/freezer (Reach-In, Pass-Thru, Display case) Types.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. Review manufacturer's instructions.
3. Notify cafeteria operator and get permission prior to performing all maintenance.
4. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.
5. Review standard operating procedure for controlling hazardous energy sources.
6. De-energize, lock out, and tag electrical circuits.
7. Comply with the latest provisions of the Clean Air Act and Environmental Protection Agency (EPA) regulations as they apply to protection of stratospheric ozone.
8. No intentional venting of refrigerants is permitted. During the servicing, maintenance, and repair of refrigeration equipment, the refrigerant must be recovered.
9. Whenever refrigerant is added or removed from equipment, record the quantities on the appropriate forms.
10. Recover, recycle, or reclaim the refrigerant as appropriate.
11. If appliance is disposed, follow regulations concerning removal of refrigerants and disposal of the appliance.
12. If materials containing refrigerants are discarded, comply with EPA regulations as applicable.
13. Refrigerant oils to be treated as used oil or hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
14. Closely follow all safety procedures described in the Safety Data Sheet for the refrigerant and to all labels on refrigerant containers.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Verify indicator light on; check compartment temperature.
3. Examine evaporator for proper clearances/slope and air flow.
4. Examine handles, hinges and tightness of door closure.
5. Examine safety door release and fan shut down safety switch.
6. Inspect lighting for burnt out lamps.
7. Check starter panels and controls for proper operation, burned or loose contacts, and loose connections.
8. Clean evaporator coil, evaporator drain pan, blowers, fans, motors, and drain piping as required; lubricate motor(s).
9. Clean condenser coil and condensing unit section.
10. Clean and inspect defrost evaporation trays/pans.
11. Inspect defrost systems for proper operation, including timer; adjust as required. Have automatic defrosters adjusted as required so freezer will defrost during "Off Peak" hours.
12. Check operation of thermostats; calibrated as required.
13. Check coil superheat and adjust to manufacturers recommendations.
14. Inspect and service all electric motors.
15. Inspect door gaskets for damage and proper fit; adjust gaskets as required and lubricate hinges with food grade oil.
16. Check door gasket heater.
17. Check box floor for water or ice accumulation.
18. Check box for excessive ice build- up and open seams.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ranges, including electric, or gas; open burner, hot plate, and griddle top. For oven base, see oven guide.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and COR and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Check all controls, mechanisms for proper operation; adjust as required.
3. Examine utility supply line, piping, valve packing, specialties, and insulation; look for leaks.
4. Check electric power line condition, switch, disconnect, etc.; or check condition of gas supply, valves, regulators, and inspect pilot, check for gas leaks.
5. Examine burner guards, covers or cook top surfaces for cracks or damage.
6. Examine handles, knobs and controls for tightness and safe condition.

Tools

TOOL_GROUP_A

This maintenance standard applies to cafeteria kitchen ranges, including electric, or gas; open burner, hot plate, and griddle top. For oven base, see oven guide.

Special Instructions

1. Preventive maintenance services should be performed by a qualified manufacturer's service representative.
2. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
3. Review manufacturer's instructions.
4. Notify cafeteria operator and COR and get permission prior to performing all maintenance.
5. If any safety deficiencies are found which could cause injury or damage, notify the cafeteria operator immediately and secure the equipment from further operations.

Checkpoint

1. Check with operating or area personnel for any deficiencies; verify cleaning program.
2. Check all controls, mechanisms for proper operation; adjust as required.
3. Examine utility supply line, piping, valve packing, specialties, and insulation; look for leaks.
4. Check electric power line condition, switch, disconnect, etc.; or check condition of gas supply, valves, regulators, and inspect pilot, check for gas leaks.
5. Examine burner guards, covers or cook top surfaces for cracks or damage.
6. Clean and adjust gas burners and pilot light assembly.
7. Clean electric heating elements and check amperage against nameplate.
8. Examine handles, knobs and controls for tightness and safe condition.
9. Check automatic burner lighters and safety controls.
10. Lubricate gas valves.
11. Check electric power line condition (switch, disconnect, etc.), or check condition of gas supply, valves, regulators.

Tools

TOOL_GROUP_A

MHDL-CRN-01-03M**Crane - Electric**
Rev-2 | 2021-11-08

This standard card applies to electric driven cranes mounted on overhead tracks, mounted on walls or mounted on the floor. It is not to be applied to hydraulic lifts.

Special Instructions

1. Coordinate related PM items at this time, i.e. ELEC-MOT-01, ELEC-MCC-03 Motor Control Center, etc.
2. Review manufacturer's instructions for operation and maintenance.
3. Review standard operating procedure for controlling hazardous energy sources.
4. Shut off electrical power, tag and lockout of service.
5. The height of the crane may require the use of scaffolding or hydraulic lifts to accomplish this PM.

Checkpoint

1. Inspect trolley wheels and track for condition.
2. Check braking system, including any hydraulic reservoirs.
3. Inspect all structural features including framing, supports, bracing, anchors, etc.
4. Scrape, wire brush, treat with rust inhibitor and paint all rusted or bare metal areas.
5. Inspect cables for worn, frayed or broken strands. Adjust tension if necessary.
6. Inspect pulleys, sheaves, chains, etc. for alignment and wear.
7. Check operation of controls, including limit switches.
8. Lubricate in accordance with manufacturer's recommendations.
9. Check oil in gear case, add or replace in accordance with manufacturer's instructions.
10. Remove tags and lockout, restore power.
11. Test operation.

Tools

TOOL_GROUP_B

This standard card applies to small freight elevators or lifts intended to carry objects rather than people. Dumbwaiters found within modern structures, including both commercial, public and private buildings, are often connected between multiple floors.

Special Instructions

- 1. Complete all maintenance requirements included in the applicable ASME A17 Elevator / Escalator Safety Codes.
- 2. Utilize the Maintenance Control Program (MCP) that is required to be developed and provided by the contractor for each piece of equipment as required in section 8.6 of ASME A17.1. MCP shall include all maintenance tasks and their frequencies with month dependent upon each unit's characteristics (equipment age, condition, accumulated wear, usage, improved technology, manufacturer's recommendations, etc.)
- 3. Ensure MCP is available for each unit. Ensure documentation of tasks performed under the MCP requirements are available and complete for review.
- 4. Follow lock out/tag out procedures at all times.
- 5. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

Tools

SAFETY_SIGNS TOOL_GROUP_B LUBRICANT-07 MAT_CLEANING-05

This standard card applies to small freight elevators or lifts intended to carry objects rather than people. Dumbwaiters found within modern structures, including both commercial, public and private buildings, are often connected between multiple floors.

Special Instructions

1. Review manufacturer's instructions.

Checkpoint

1. Inspect and clean hoistway. Remove trash from pit.
2. Examine sheaves, cable, counterweight, etc. Look for loose bolts, misalignment, weak or improper cable fasteners, etc. Make safety or reliability tests if anything questionable is found.
3. Examine car for structural features, appearance, need for attention, surface condition, condition of paint, etc.
4. On hand powered units, examine cable pulls for loose strands, sharp edges, rough surfaces, or other potential hazards.
5. Inspect power unit, motor controls, and all accessories.
6. Check all indicators, lights, bull's-eyes, controls, safety devices, etc.
7. Check door gate and interlocks.
8. Comply with lubrication schedule.
9. Check operating voltage and running amps of hoist motor.
10. Verify that safety devices are operating properly before putting dumbwaiter back in service.
11. Clean up and remove all debris.

Tools

LUBRICANT-07 TOOL_GROUP_B SAFETY_SIGNS

This standard applies to manually operated lifts and hoists.

Special Instructions

- 1. Review manufacturer's instructions.

Checkpoint

- 1. Test operate and note condition, including all controls.
- 2. Check all hydraulic hoses, cylinders, and connections for leaks and condition. Tighten connections or replace hoses to correct leaks.
- 3. Check mechanical and hydraulic brake systems, including master cylinder fluid levels, and correct as required.
- 4. Lubricate all zerk fittings and any other points specified by the manufacturer.
- 5. Check all lift arms, stabilizers, locks, chains, cables, channels, and structure for condition and wear.
- 6. Where equipped with a battery, check charge, electrolyte level, and condition of cables and terminals.

Tools

MAT_CLEANING-03 HYDROMETER-01 HYDRAULIC_HOSE GREASE_GUN-03 BRAKE_FLUID TOOL_BASIC

This standard card applies to electric hoists that are installed in a fixed position or mounted on an overhead rail system. It does not apply to the larger motorized electric cranes.

Special Instructions

1. Review manufacturer's instructions.
2. Review standard operating procedure for controlling hazardous energy sources.
3. De-energize, tag, and lock out circuit.
4. Take appropriate safety precautions for working at heights.

Checkpoint

1. Inspect cables and pulleys.
2. Lubricate as recommended by the manufacturer.
3. Check oil in gear box and add as required.
4. Restore power, test and operate.
5. Check and adjust brake if required.
6. Remove tags from power source.

Tools

TOOL_BASIC PLATFORM GREASE_GUN-03

This standard card applies to electric hoists used to raise, lower, and support large decorative lighting fixtures in auditoriums and conference rooms.

Special Instructions

- 1. Schedule all hoists to be maintained at one time.
- 2. Schedule hoists to coincide with PM on chandelier.

Checkpoint

- 1. Check supply voltage.
- 2. Check connections at disconnect, starter, and reversing switch.
- 3. Check motor mounting frame and bolts, tighten if necessary.
- 4. Check hoist support, bracket lubricant, and pivot point.
- 5. Check sprockets and roller chain, lubricate as needed.
- 6. Check oil in gear motor, change if required.
- 7. Check and clean motor, lubricate as needed.
- 8. Check oil in hoist, change if required.
- 9. Remove safety bar and inspect for defects. Replace if required.
- 10. Inspect chain for wear, rust, and weak spots. Coat with oil or grease.

Tools

MULTIMETER LUBRICANT-07 TOOL_CLEANING-04 TOOL_GROUP_A

This standard card applies to hoisting equipment such as is used for the understage movie screen located in the Departmental Auditorium in Washington, DC This compares to the hoisting equipment used for an electric elevator, but is smaller.

Special Instructions

1. Review manufacturer's instructions.

Checkpoint

1. Controller:
 - a. Clean with blower or vacuum.
 - b. Check switches, relays, timers, resistors, contacts, connections, fuses, and overload settings.
 - c. Replace worn parts and adjust for proper operation.
2. Brakes:
 - a. Observe operation.
 - b. Inspect drum and shoe clearance, adjust if needed.
 - c. Lubricate pivot points and clean as necessary.
3. Hoist Motor:
 - a. Inspect oil level, oil pickup, and belts.
 - b. Check for excessive heat, noise, and leaks.
 - c. Lubricate in accordance with specifications.
 - d. Keep end bells clean.
4. Cables:
 - a. Inspect, lubricate, and adjust hoist cables and cable drums.
 - b. Inspect traveling cables.
 - c. Check all cable fastenings.
 - d. Inspect guide rails and guide shoes.
5. Gear Machines:
 - a. Inspect worm and gear for bottoming and back lash, and thrust end play.
 - b. Check oil level and packing.
 - c. Check for proper oil pickup.
6. Governor:
 - a. Observe operation.
 - b. Check electrical switches and seals.
 - c. Lubricate pivot points and clean as necessary.
7. Machine Room:
 - a. Dust machines, control cabinets, etc.
 - b. Sweep floors and remove all trash. Relamp as necessary.

Tools

EQUIP_CLEANING-03 LUBRICANT-07 TOOL_BASIC BARRICADES

This standard applies to electric lift trucks with 2,000 and 6,000 pound lifting capacities and operating 2,000 hours per year. Contact the lift truck manufacturer in cases when lift trucks are used more than 2,000 hours per year for guidance on adjusting the frequency of maintenance.

Special Instructions

1. Review manufacturer's instructions for specific guidelines, as lift trucks vary.
2. Review standard operating procedure for powered industrial trucks (Forklifts).
3. Review standard operating procedure for controlling hazardous energy sources.
4. Proper battery maintenance is critical, as poor battery care can lead to expensive component failures.
 - a. Maintain proper water levels.
 - b. Keep battery and terminals clean.
 - c. Keep battery properly charged.
5. Never smoke or carry an open flame in or near the battery.
6. Use caution in handling the electrolyte, it is harmful to the skin and clothing.
7. Never remove any connecting cables or straps while charger is on or there is a possibility of a load being on the batteries, (this can cause a spark that may ignite ever present hydrogen gas).
8. Wear acid resistant apron, gloves, and plastic face shield when handling electrolyte.
9. Review standard operating procedure for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
10. Take appropriate action to correct deficiencies.
11. Document all findings on equipment history record.

Checkpoint

1. Motor:
 - a. Visually inspect the motor and connections.
 - b. Measure the winding resistance.
2. Battery:
 - a. Inspect the battery condition and cleanliness.
 - b. Check and restore the electrolyte level
 - c. Check and restore the specific gravity.
3. Inspect the contactors.
4. Inspect the direction lever and direction switch.
5. Check the operation of the controller.
6. Tighten all fuses.
7. Inspect the wiring:
 - a. Tighten terminal connections.
 - b. Tighten battery connections.
8. Steering:
 - a. Check the operation of the steering wheel.
 - b. Inspect the steering links.
 - c. Inspect the steering box.
9. Brake system:
 - a. Inspect the brake lines.
 - b. Adjust and check the operation of the brake assembly. Measure the pedal height. Replace cups at 1,000 hours.
 - c. Adjust and check the operation of the parking brake.
 - d. Check and restore the brake fluid level.
 - e. Tighten the seat.
10. Lift and tilt cylinders:
 - a. Measure the lift and tilt cylinders.

- b. Measure the lift speed.
 - c. Inspect for oil leakage.
 - d. Inspect mounting and control levers.
11. Hydraulic system:
- a. Check the oil pump for leaks.
 - b. Check and restore the hydraulic oil level.
 - c. Check the operation of the oil control valve and the tilt lock valve.
 - d. Inspect the hydraulic pipes.
 - e. Check and restore the transmission fluid.
 - f. Check the differential for leaks.
12. Inspect the wheels and tires.
13. Tighten all lug nuts.
14. Inspect the wheel bearings, frame, and O.H. guard.
15. Inspect the mast/carriage and mast strips.
16. Tighten the back rest.
17. Inspect the forks and stopper pins.
18. Adjust the lift chains.
19. Inspect the anchor bolts and chain wheels.
20. Inspect the axle beam king pins and knuckles.
21. Inspect the lighting system, horn, and dash gauges.

Tools

TOOL_GROUP_B TESTER_SPEC_GRAVITY SPARE_PARTS MAT_CLEANING-05 LUBRICANT-06

This standard applies to electric lift trucks with 2,000 and 6,000 pound lifting capacities and operating 2,000 hours per year. Contact the lift truck manufacturer in cases when lift trucks are used more than 2,000 hours per year for guidance on adjusting the frequency of maintenance.

Special Instructions

1. Review manufacturer's instructions for specific guidelines, as lift trucks vary.
2. Review standard operating procedure for powered industrial trucks (Forklifts).
3. Review standard operating procedure for controlling hazardous energy sources.
4. Never smoke or carry an open flame in or near the battery.
5. Review standard operating procedure for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
6. Take appropriate action to correct deficiencies.
7. Document all findings on equipment history record.

Checkpoint

1. Motor:
 - a. Measure the brush wear.
 - b. Measure the tension.
2. Battery - measure the volts per cell.
3. Measure the controller's overcurrent limit.
4. Inspect steering box, brake line, and hydraulic pipe hoses and replace at least once every two years or if defective.
5. Inspect brake master cylinder cups and replace at least once every two years or if defective.
6. Lubricate the brake assembly.
7. Perform a natural drop test of the lift and tilt cylinders.
8. Measure the hydraulic oil pressure.
9. Replace the differential fluid.
10. Grease the wheel bearings.
11. Inspect the lift chains and replace at least once every three years or if defective.
12. Inspect the axle beam and mast supports.

Tools

SPARE_PARTS TOOL_GROUP_B LUBRICANT-08

This standard applies to adjustable loading ramps.

Special Instructions

1. Review manufacturer's instructions.
2. Review standard operating procedure for controlling hazardous energy sources.
3. Disconnect, lock and tag switch.

Checkpoint

1. Inspect structural features, framework, support members, anchor bolts, pit, platform, etc. Examine condition of bumper. Does it protect ramp properly?
2. Remove dirt and trash from pit and determine if pit drain is open.
3. Inspect motor, controls, starter, push buttons, solenoids, etc. Clean, adjust and lubricate as necessary. Be sure disconnect switch can be locked.
4. For hydraulic units:
 - a. Inspect coupling, pump, control valves, piping, relief valve reservoir, fill pipe, cap, vents, etc. clean, adjust, and lubricate as needed.
 - b. Inspect cylinder, ram, packing glands, etc. Add or renew packing as required.
 - c. Change oil as required. Review the Safety Data Sheet for disposal of used oil. If appropriate, recycle oil at an authorized station.
5. Electro-Mechanical Units:
 - a. Clean and inspect coupling, reduction gear, sprockets and chain, gear trains, screw and lever, and/or other mechanical features. Look for misalignment, loose bolts, evidence of binding or wear, excessive clearance, etc., tighten as necessary.
 - b. Examine lubrication devices. Service if required.
 - c. Test operation of ramp in all directions using a load if possible. Note if ramp holds and does not creep when load is applied or removed. Adjust if necessary.
 - d. Check manual operation, power disengagement, etc.
 - e. Lubricate as required.
 - f. Clean up work area.
6. Manual units:
 - a. Clean and inspect all mechanical features. Look for misalignment, loose bolts, evidence of binding or wear, excessive clearance, etc., tighten as necessary.
 - b. Test operation of ramp in all configurations and directions using a load if possible. Note if ramp holds and does not creep when load is applied or removed. Adjust if necessary.
 - c. If ramp is foldable, test folding in all directions and check that it completes it's full range of motion. Adjust if necessary.
 - d. Lubricate as required.
 - e. Clean up work area.

Tools

TOOL_GROUP_C LUBRICANT-07 HYDRAULIC_FLUID MAT_CLEANING-02

This standard applies to automotive lifts primarily used for vehicle maintenance.

Special Instructions

- 1. Review manufacturer's instructions.
- 2. Review standard operating procedure for controlling hazardous energy sources and confined space entry.
- 3. Disconnect, lock and tag switch.

Checkpoint

- 1. Clean the drive assembly, lift platform, travel area, etc. Remove debris.
- 2. Visually inspect moving parts and cables for signs of excessive wear.
- 3. Check conditions of drive chain, belts, cables, etc. and adjust and replace as needed.
- 4. Check hydraulic seals, hoses and fittings for leaks and or damage.
- 5. Check hydraulic fluid levels. Replace if low or dirty.
- 6. Lubricate moving parts as required. Lubricate columns and ensure they are plumb.
- 7. Operate lift through its full range of motion in both directions, using a load if possible. Note operation of controls, safety devices, leveling, and other devices and features.

Tools

TOOL_GROUP_C MAT_CLEANING-02 LUBRICANT-07 HYDRAULIC_FLUID

This standard applies to automotive lifts primarily used for vehicle maintenance.

Special Instructions

- 1. Review manufacturer's instructions.
- 2. Review standard operating procedure for controlling hazardous energy sources and confined space entry.
- 3. Disconnect, lock and tag switch.

Checkpoint

- 1. Complete the monthly maintenance for this equipment concurrent with this PM card.
- 2. Lubricate moving parts as required. Lubricate columns and ensure they are plumb.
- 3. Replace hydraulic fluid. It is crucial to keep it clean.

Tools

TOOL_GROUP_C MAT_CLEANING-02 LUBRICANT-07 HYDRAULIC_FLUID

This standard card applies to balers used to compress recyclable and waste materials into small, manageable bales. Materials such as cardboard, textiles, foam, plastics, cans and bottles are compressed in balers and then baled for transport to a garbage or recycling facility.

Special Instructions

1. Review manufacturer's instructions.
2. Review the Standard Operating Procedure for controlling hazardous energy sources.
3. Open and tag electric switches.

Checkpoint

1. Dust or wipe clean all parts of machine. Examine structural features.
2. Blow out electric motor. Inspect starter, controls, push button, upper and lower limit switch, etc. Clean and adjust as required.
3. Check drive unit, mechanical features, and all moving parts.
4. Comply with lubrication schedule.
5. Adjust operating mechanism.
6. Clean up work area.

Tools

TOOL_GROUP_B LUBRICANT-07

This standard applies to gasoline, LP gas, and diesel powered forklifts, tractors, front-end loaders, and light duty trucks. Review manufacturer's instructions before proceeding.

Special Instructions

- 1. Review manufacturer's instructions.
- 2. Review the Standard Operating Procedure for powered industrial trucks (Forklifts).
- 3. Handle flammable materials safely.
- 4. Have the proper fire extinguisher on hand. Ensure that the fire extinguisher is properly serviced and that it is in proper working order.
- 5. Use gasoline and LPG forklifts in well ventilated areas only.

Checkpoint

- 1. Change engine oil. Review the Safety Data Sheet for proper disposal of used oil. If appropriate, recycle oil at an authorized station.
- 2. Remove crankcase breather cap and wash in solvent.
- 3. Inspect air cleaner element and clean or replace as required.
- 4. Tune up engine.
- 5. Change hydraulic oil filter.
- 6. Check all belts for proper tension or wear and hoses for deterioration. Adjust or replace as necessary.
- 7. Inspect all gauges and indicators.
- 8. Test horn, backup warning and safety lights.
- 9. Test operate all tilt, lift, and lowering controls.
- 10. Test steering and adjust if necessary.
- 11. Check all hydraulic hoses, cylinders, and connections for leaks and condition. Tighten connections and replace hoses to correct leaks.
- 12. Check mechanical and hydraulic brake systems, including master cylinder fluid levels, and correct as required.
- 13. Lubricate all Zerk fittings and any other points specified by the manufacturer.
- 14. Check all lift arms, stabilizers, locks, chains, cables, channels, and structure for condition and wear.
- 15. Where equipped with a battery, check charge, electrolyte level, and condition of cables and terminals.
- 16. Check all other fluid levels and add or replace as required.

Tools

BRAKE_FLUID	HYDRAULIC_OIL_FILTER	GREASE_GUN-03	BELTS-02	AIR_CLEANER	HYDRAULIC_HOSE	TOOL_BASIC
TUNE_UP_KIT	MAT_CLEANING-03	HYDROMETER-01				

This standard card applies to propane, gasoline, and battery powered carts and scooters used to transport personnel and materials in large buildings and on compounds or complex facilities.

Special Instructions

1. Review manufacturer's maintenance instructions.
2. Review standard operating procedure for emergency eyewash and shower equipment. Note the location of emergency eyewash and/or shower equipment.
3. Work area must be properly ventilated to protect against battery gases and fumes from fuels.
4. Adequate personal protection against acid spills must be provided, i.e., face mask, rubber gloves, etc.
5. Have on hand an approved fire extinguisher. Ensure that the fire extinguisher is properly serviced and that it is in proper working order.

Checkpoint

6. Battery powered:
 - a. Remove and clean battery terminals. Clean posts and reconnect terminals.
 - b. Check electrolyte specific gravity with a hydrometer.
 - c. Add water if required.
 - d. Inspect lights, buzzers, controls, and wiring for tight connections and proper operation.
 - e. Lubricate according to manufacturer's recommendations.
 - f. Check brakes, clutch, wheels, tires, steering, and frame.
 - g. Vacuum or blow out motor air passages and check mounting.
 - h. Check and adjust as needed belts and/or chain drive.
 - i. Test run (road test).
 - j. Wire brush, treat with rust inhibitor, and paint rusted areas.
2. Engine powered:
 - a. Inspect fuel tank, connections, and lines for leaks and deterioration. Tighten or replace as required.
 - b. Change oil. Change oil, air, and fuel filters.
 - c. Change points, condenser, plugs, and rotor button if applicable.
 - d. Lubricate according to manufacturer's recommendation.
 - e. Check brakes, clutch, wheels, tires, steering, and frame.
 - f. Inspect lights, buzzers, controls, flashers, and wiring for proper operation and tighteners.
 - g. Check and adjust, as needed, belts and/or chain drive.
 - h. Start engine and perform any manufacturer's recommended tune-up procedures.
 - i. Test run (road test).
 - j. Wire brush, treat with rust inhibitor, and paint all rusted areas.

Tools

WRENCH_SPARK_PLUG	GAUGES_FEELER	BATTERY_BRUSH	ENGINE_OIL_CONTAINER	EYE_WASH	FACE_SHIELD_APRON_GLOV
GAUGE_TIRE-02	HYDROMETER-01	LEAK_DECTOR	RUST_INHIBITOR-03	TOOL_GROUP_B	WATER
					FILTER_PLUGS

PLMB-DRN-01-01Y**Roof Drains, Downspout, and Gutter Inspection**Rev-1 | 2021-11-03

This standard applies to roof drains, downspouts, and gutters.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Use care when working in high places.
3. Use safety line with harness if necessary.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check gutters, drains, and downspouts to insure that they are properly attached to the building, connections sealed, and free of debris.
2. Check drain grates/strainers/screens for condition and proper installation. Replace missing or damaged drain grates/strainers/screens.
3. If downspouts have heaters, test, operate and correct deficiencies.
4. Remove all trash, debris or unsecured material from roof area.
5. Where downspouts discharge onto lower roofs, check if there has been any scouring of the surfacing.
6. Check for missing or damaged splash blocks.

Tools

TOOL_GROUP_B LADDER-05

This standard applies to areaway, driveway, and storm drains.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Perform work in autumn after leaves have fallen.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Remove grate if it exists.
2. Clean drain and area leading to drain.
3. Remove sediment, debris and trash and dispose of properly.
4. Test drain for free water flow by flushing with a hose.
5. Reinstall grate if removed in step 1. Ensure drain grates that abut each other are installed in a linear pattern (Creating a uniform appearance).
6. Replace damaged grates.

Tools

TOOL_BASIC GLOVES_WORK HOSE-01

This standard applies to gas fired domestic hot water heaters.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. If the insulation is known or suspected to contain asbestos, check the building's asbestos management plan to see it has been tested for asbestos. If it is suspect but has not been tested, have it tested. Manage asbestos in accordance with the asbestos management plan.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Use caution when working with natural gas fired equipment. Be aware of any smells (rotten egg) that could be a natural gas leak.
6. Do not allow any open flames around equipment.
7. Perform pressure vessel inspection in accordance with requirements of American Society of Mechanical Engineers (ASME), Section VIII, Pressure Vessel and National Board Inspection Code Part 2, NB-263 RCI-1 Rules for Commissioned Inspectors.
8. Hot water heater shall be inspected and tested annually by RCI-1 commissioned inspector in accordance with NBIC Part 2 Inspection code.

Checkpoint

1. Attach drain hose. Drain several gallons from tank to remove sediment.
2. Manually check operation of safety valve. Check for corrosion around valve. Verify the safety valve inspection tag is in place. Ensure that no personnel are in area of relief piping discharge.
3. Check all connections - electric, gas and water. Tighten as necessary.
4. Check operation and setting of aquastat. Check hot water temperature with dial thermometer, and set aquastat at minimum value required for all uses.
5. Drain storage and expansion tanks, and flush to remove sediment, scale, and solid at bottom of tank.
6. Clean sight glasses on tanks.
7. Clean strainer, check condition of traps. Report and repair leaks.
8. Clean pump, controls, switches, and starters. Check operation of pump and condition of pump seal or packing, and replace as required.
9. If applicable, remove and inspect anode, replace if necessary.
10. Clean up work area and remove trash.
11. If the insulation contains asbestos, it is considered hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
12. Inspect safety valve discharge pipe to determine if discharge pipe is free and in accordance with International Plumbing Code Chapter 5 requirements.
13. Inspect for galvanic corrosion (Galvanic corrosion is caused by self-induced current created by electrical potential of two dissimilar metals in contact with an electrolyte. It can occur when two dissimilar metals (such as copper tube and steel pipe) are connected in the presence of an electrolyte).

Tools

THERMOMETER AMPMETER EQUIP_CLEANING-03 HOSE_DRAIN TOOL_GROUP_C GAUGE_GAP-01

This standard applies to electric domestic hot water heaters.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. If the insulation is known or suspected to contain asbestos, check the building's asbestos management plan to see it has been tested for asbestos. If it is suspect but has not been tested, have it tested. Manage asbestos in accordance with the asbestos management plan.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Attach drain hose. Drain several gallons from tank to remove sediment.
2. Manually check operation of safety valve. Ensure that no personnel are in area of relief piping discharge. Check for corrosion around valve.
3. Check all connections - electric and water. Tighten as necessary. Ensure power is disconnected to electric heaters prior to checking connections.
4. Check operation and setting of aquastat. Check hot water temperature with dial thermometer, and set aquastat at minimum value required for all uses.
5. Check amperage draw of upper and lower elements and compare to name plate data.
6. Clean element contacts, and check for proper closing under load.
7. Clean pump, controls, switches, and starters. Check condition of pump seal or packing, and replace as required.
8. If applicable, remove and inspect anode, replace if necessary
9. Clean up work area and remove trash.
10. If the insulation contains asbestos, it is considered hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Tools

HOSE_DRAIN AMPMETER THERMOMETER TOOL_GROUP_C EQUIP_CLEANING-03

This standard applies to independent domestic hot water heaters, which consists of a hot water storage tank with coils for hot water or steam submerged into the water to be heated. The storage capacity is from 100 to 1000 gallons, with a temperature rise from 90 deg. F to 140 deg. F. The tank will require manholes or handhole inspection plates.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Obtain operating logs.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Review operating logs to check loss of efficiency of heat exchange surfaces, indicating scale and corrosion buildup.
6. Check inspection certificate.
7. If the insulation is known or suspected to contain asbestos, check the building's asbestos management plan to see it has been tested for asbestos. If it is suspect but has not been tested, have it tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Inspect element header and exterior of tank including fittings, manholes, and handholes for leaks and signs of corrosion.
2. Hand operate and test pressure relief valve.
3. Drain and flush tank.
4. Remove tank inspection plate and inspect tank interior. Record the size and depth of pits, presence of cracks, and condition of openings, fittings, welds, rivets, and joints.
5. Check condition of heat exchanger element. Remove and clean as necessary.
6. Inspect condition of epoxy tank lining.
7. Replace all gaskets and manhole inspection plates, and tighten all bolts as required.
8. Fill tank and check for leaks.
9. Clean strainer, check condition of traps. Report leaks to the COR.
10. Clean pump, controls, switches, and starters. Check condition of pump seal or packing, replace as required.
11. Clean, test, and inspect sight glasses, valves, fittings, drains.
12. Inspect structural supports, and repair or replace damaged insulation or covering.
13. Schedule hydrostatic testing according to established procedures.
14. Return tank to service and observe temperature control operation. Adjust as required.
15. If the insulation contains asbestos, it is considered hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Tools

PAINT-03 TOOL_GROUP_C MAT_CLEAN_PATCH HOSE-02 GOOGLES WRENCHES-03

This standard applies to the three most common types of serviceable water softeners: Lime, zeolite, and anthracite.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Drain the tank:
 - a. Examine the exterior of the tank including fittings, gauges, manholes, and handholes for signs of leaks or corrosion. Repair as necessary.
 - b. Inspect structural supports and insulation or coverings for defects or deterioration.
 - c. Open the tank and remove rust or chemical deposits from interior tank surfaces.
 - d. Remove and clean all spray nozzles.
 - e. Inspect the interior of the tank for pitting, cracks, and other defects.
2. Lime Water Softener:
 - a. Dismantle vacuum breakers. Inspect stem, valve seat and spring. Repair as required.
 - b. Inspect, clean, and flush the nozzle ring.
 - c. Remove vent condenser heads and clean the tubes.
 - d. Inspect and clean the sight glass, level indicators, and level controllers.
3. Zeolite Water Softener:
 - a. Check the filter bed for proper level
 - b. Take samples of the resin according to manufacturer's instructions and send to a lab for analyses.
4. Anthracite Water Softener:
 - a. Check the filter bed for proper level

Tools

GRAVEL_SAND_CHAR TOOL_GROUP_C RESPIRATOR-01

This standard card applies to water filters installed in secondary water systems for heating and cooling systems. These filters may be either a disposable cartridge type filter or a washable and reusable filter media.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review and follow manufacturer's instructions. Specialized filters like those used for sediment, rust, copper or lead may require more frequent changing.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Drain water from filter housing.
2. Remove cover from filter housing.
3. Remove spring, "O" rings, etc., and old filter.
4. Clean and flush filter housing.
5. Clean filter media, if applicable.
6. Install filter media or cartridge.
7. Replace springs and "O" rings.
8. Replace housing cover.
9. Open inlet and outlet valves.
10. Check housing cover for leaks.

Tools

TOOL_BASIC GASKET-06 FILTER_CARTRIDGE-01

This standard applies to pneumatic tank type sewage ejectors. Components consist of sewage tank and air lines from a tank or compressor with various valves and controls. When the tank fills the valves open to eject sewage with compressed air.

Special Instructions

1. Review standard operating procedures for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule outage with operating personnel.
3. Review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Observe personal safety requirements when working with the sewage ejector and related piping.

Checkpoint

1. Inspect check valves in compressor or air storage tank inlet lines to tank, and suction and discharge lines of sewage pot.
2. Check freedom of motion, and wear on check valve clapper and clapper seat.
3. Remove sewage pot inspection plate. Inspect and clean float ball, bucket and rod or level sensor. There may be a few floats for different levels of ejection and a high level alarm.
4. Inspect float assembly linkage, shaft, keys, and keyways. Look for wear, binding, etc.
5. Change oil in immersed float switch. Check packing. Review the Safety Data Sheet for proper disposal of used oil. If appropriate, recycle oil at an authorized station.
6. Remove any obstructions from water line. Clean strainer.
7. Check solenoid valve on air inlet for operation and freedom of movement.
8. Check that vent line is clear of any obstructions.
9. Clean up work area and remove all debris.

Tools

TOOL_GROUP_C PPE_CLOTHING-02 LUBRICANT-04 EQUIP_CLEANING-03

This standard applies to sump pump type sewage ejectors that operate by means of an electric motor, pump with controls for pump start/stop and staging.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Review standard operating procedures for selection, care, and use of respiratory protection.
6. Wear rubber apron, gloves, boots, full face shield, and respirator when performing this work.
7. If a person must enter the pit, test for oxygen deficiency and supply proper ventilating equipment as needed.
8. No open flames or smoking.
9. If strainer and check valve cleaning requires removal of pump unit which should be considered a repair and not general maintenance.

Checkpoint

1. Remove cover plates, flush pit, and pump out.
2. Check operation of level sensors, floats, rods, switches, or other types of level sensors and alarms.
3. Check and inspect wiring, electrical connections and conduit for corrosion and tightness.
4. Clean pump and lubricate as required. Some pumps have sealed factory lubricated bearings that require no further service until replacement is needed.
5. Inspect check valve.
6. Inspect interior of pit for cracks.
7. Clean motor with vacuum or low pressure air (less than 40 psi). Check for obstructions in motor cooling and air flow.
8. Check for corrosion. Clean and treat with rust inhibitor as needed.
9. Inspect cover plate gaskets and replace if necessary.
10. Clean up work area and remove all debris.

Tools

METER_TOXIC_GAS EQUIP_CLEANING-03 LUBRICANT-01 PPE_CLOTHING-01 TOOL_GROUP_C VENTILATION_EQUIP

This standard applies to sump pump that operates by means of an electric motor, pump with controls for pump start/stop and staging.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Schedule outage with operating personnel.
3. Review manufacturer's instructions.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If the material removed from the pump is hazardous, dispose of properly and document in accordance with applicable waste management and disposal regulations. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
6. If strainer cleaning requires removal of pump unit which should be considered a repair and not general maintenance.
7. Excessive sediment and debris, not removed by flushing the pit should be handled on a project basis, and not considered under this standard.

Checkpoint

1. Remove cover plates, flush pit, and pump out.
2. Check operation of level sensors, floats, rods, switches, or other types of level sensors and alarms.
3. Clean pump and motor and lubricate as required. Some pumps have sealed factory lubricated bearings that require no further service until replacement is needed.
4. Inspect check valve.
5. Inspect interior of pit for cracks.
6. Inspect cover plate gaskets and replace if necessary.
7. Clean up work area and remove all debris.

Tools

TOOL_GROUP_C PPE_CLOTHING-01 LUBRICANT-01 EQUIP_CLEANING-03

This equipment includes emergency showers, eye-wash equipment, eye/face wash equipment, hand held drench hoses, and combination shower and eye-wash equipment, installed for the emergency treatment of the eyes and body of a person who has been exposed to injurious material. This includes water treatment chemicals, battery acid, cleaning solvents and compounds, etc.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. On non-plumbed units (units containing their own water supply), use manufacturer's instruction to perform maintenance.
6. Where units have been sealed, be certain seals are available to re-seal the unit after PM is accomplished.
7. Where flow alarms have been installed, check with responding unit before activating the unit.

Checkpoint

1. Activate the unit to flush the line and verify proper operation.
2. Assure that area is free of obstructions, that activation mechanisms are accessible to personnel in a distressed condition.
3. Operate valve in full open and close position. Loss of ability to close tightly will require inspection of valve seals and discs for wear and contaminate build-up.
4. If applicable, check tempering feature, verify temperature and flow is correct.
5. Check systems for cleanliness and clean if necessary.

Tools

TOOL_BASIC EQUIP_CLEANING-03

This equipment includes emergency showers, eye-wash equipment, eye/face wash equipment, hand held drench hoses, and combination shower and eye-wash equipment, installed for the emergency treatment of the eyes and body of a person who has been exposed to injurious material. This includes water treatment chemicals, battery acid, cleaning solvents and compounds, etc.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check operation of self-closing valve.
2. Check fittings.
3. Check pull lever for proper function.
4. Check shower head for signs of corrosion, encrustation, or foreign particles. Clean as necessary.

Tools

EQUIP_CLEANING-03 TOOL_BASIC

This procedure covers all septic tanks and drain fields.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Wear suitable protective clothing.
6. No open flames or smoking.
7. Use barricades around tank opening.
8. Review the available Hazard Assessment of the space. Follow OSHA guidelines and company Standard Operating Procedures for entering permit-required confined spaces. NEVER enter a permit-required confined space without a safety attendant.

Checkpoint

1. Remove cover.
2. Pump out contents of tank into container truck that meets State health requirements.
3. After contacts are removed, inspect to make sure that baffle plates are in place and tank is in good condition.
4. Run water from inside of building to ensure that there are no blockages in the waste pipe from the building to the septic tank.
5. Install cover.
6. Contents of tank should be disposed of in accordance with State health requirements.

Tools

TOOL_BASIC SCRAPER EQUIP_CLEANING-03 BARRICADES

This PM Guide is for Grease Trap, Grease Inceptor and Grease Abatement Devices.

Special Instructions

Indoor grease interceptors shall be inspected and cleaned quarterly or as allowed by local ordinance. Outdoor units may be increased if allowed by local ordinance.

Use appropriate protective clothing, especially safety glasses.

Locate Grease trap, grease trap and associated work can be located inside the building or outside.

Checkpoint

Check for any leaks

Remove baffle(s) if possible

Remove all water in the trap or interceptor to facilitate cleaning. The water shall be disposed of in accordance with local code and NOT placed in the sanitary system.

Inspect for clogging, scale, and improper positioned or missing baffles.

Scrape the sides, the lid, and the baffle(s) to remove as much of the accumulated grease as possible.

Inspect the entire system for signs of trouble. Verify that all components appear to be operating effectively.

Replace the baffle(s) and the lid. Tighten loose parts as necessary.

Verify proper flow by allowing the trap to fill with water.

On the maintenance log, record the volume of grease removed as a percentage of trap volume.

Clean up the work area and remove all trash.

Properly dispose of fat, oil, grease removed from grease trap in accordance with local ordinance.

Tools

None

This standard card applies to hot water converters using a heat exchanger to transfer the heat from a steam system to the hot water system.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Review operating logs to check loss of efficiency of heat exchange surfaces, indicating scale and/or corrosion build-up.
6. If the insulation is known or suspected to contain asbestos, check the building's asbestos management plan to see it has been tested for asbestos. If it is suspect but has not been tested, have it tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Inspect element header for signs of leaking or corrosion. Remove corrosion and encrustations.
2. Remove heat exchanger element and clean only if a loss of efficiency is indicated, or signs of leaking around header are evident.
3. Tighten all bolts around header.
4. Renew paint/protective coating/insulation as required.
5. Drain storage and expansion tanks, and flush to remove sediment, scale, etc.
6. Clean sight glasses on tanks.
7. Clean strainer, check condition of traps. Report leaks to COR.
8. Clean pump, controls, switches, and starters. Check condition of pump seal or packing, and replace as required.
9. Clean up work area and remove trash.
10. If the insulation contains asbestos, it is considered hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Tools

HOSE-01 GOOGLES MAT_CLEANING-02 PAINT-07 TOOL_GROUP_C WRENCHES-03

This standard applies to all solar collector panels.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check collector glazing for cracks or deterioration. Clean as required.
2. Check collector for damage and report same to supervisor.
3. If conditions warrant, clean collector glass with warm water or, if badly soiled, with a mild detergent and water. Never use a chemical glass cleaner, abrasives, steel wool, or any substance that could scratch the surface. Cold water should never be sprayed on hot collector surfaces.
4. Renew paint and/or protective coating.
5. Check collector framework and lines; ensure that unit is firmly bolted and anchored.

Tools

WRENCHES-02 TOOL_BASIC PAINT-07 MAT_CLEANING-02

This standard applies to all solar collector panels for domestic hot water.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Check system pressure on closed loop for loss of fluid; pressure should be 40 psig.
2. Check glycol strength. Run pH test on glycol-water solution. If required, drain entire solution and replace with a 50-50 mixture of glycol and water.
3. Check for leaks and condition of insulation on lines.
4. Remove heat exchanger element and clean only if a loss of efficiency is indicated or signs of leaking are evident around header.
5. Inspect element header for signs of leaking or corrosion. Remove corrosion and encrustations.
6. Tighten all bolts around header.
7. Clean strainer and check condition of traps. Report leaks.
8. Check and secure all pipe hangers, expansion joints, and associated items.
9. Drain storage and expansion tanks; flush to remove sediment, scale, etc.
10. Clean sight glasses and flow indicators on tanks.
11. Check valves for proper positioning.
12. Clean pump, controls, switches, and starters. Check condition of pump seal or packing and replace as required. Lubricate motor if required.

Tools

HOSE-01 TOOL_BASIC MAT_CLEANING-02 LUBRICANT-07 GOOGLES ANTIFREEZE_TESTER

This standard applies to slip type joints only. Bellows type joints are to be replaced when a leak occurs.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions for all system components.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Perform visual check for leaks
2. Replace if leaks are present
3. Clean up work area.

Tools

TOOL_GROUP_C MAT_CLEANING-05 LUBRICANT-07

This standard applies to Y-type strainers that are at remote locations or that are not included under the guide card of equipment they are paired with. Typically guide cards for various pumps include strainers as children rather than list maintenance for strainers on pump guide.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If the material removed from the pump is hazardous, dispose of properly and document in accordance with applicable waste management and disposal regulations. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
6. The maintenance of Y-type strainers should be scheduled to coincide with the maintenance of the equipment that they are associated with unless strainer cleaning requires removal of the pump unit which should then be considered a repair and not general maintenance.

Checkpoint

1. Secure strainer isolation valves.
2. Drain strainer housing.
3. Back flush if possible or remove and clean strainer cage, if applicable.
4. Replace cartridge type and clean out strainer housing, if applicable.
5. Reassemble unit or replace drain plug and open isolation valve.
6. Check unit for leaks.
7. Clean up work area and remove all debris.

Tools

TOOL_BASIC HOSE_BUCKET GASKET-07 FILTER_CARTRIDGE-02

This standard applies to the maintenance of bolted flange type strainers that are not included under the guide card of equipment they are paired with. Typically guide cards for various pumps include strainers as children rather than list maintenance for strainers on pump guide.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If the material removed from the pump is hazardous, dispose of properly and document in accordance with applicable waste management and disposal regulations. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
6. The maintenance of strainers should be scheduled to coincide with the maintenance of the equipment that they are associated with unless strainer cleaning requires removal of the pump unit which should then be considered a repair and not general maintenance.

Checkpoint

1. Remove flange cover bolts.
2. If required, use hoist or crane to remove cover plate.
3. Remove device or devices; clean and inspect for damage.
4. Clean strainer housing cover plate and any interior apparatus. Check for cracks and deterioration.
5. Install new cover gasket and reassemble.
6. Remove tags and open valves; check for leaks.
7. Clean up work area and remove all debris.

Tools

TOOL_BASIC HOSE_BUCKET GASKET-07 CRANE

This standard applies to those strainers used to remove the larger particles of debris in water that may be drawn from rivers, reflecting pools, or non-potable sources. Many are motorized self cleaning type. This water is sometimes used as condenser water in refrigeration systems. These strainers are installed just ahead of backwash strainers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If system is used as condenser water for a refrigeration unit, ensure the affected refrigeration unit is shutdown.
6. If the material removed from the pump is hazardous, dispose of properly and document in accordance with applicable waste management and disposal regulations. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Checkpoint

1. Close off main water inlet.
2. Close off condenser valves.
3. Open all drains and start drain pump if necessary.
4. Clean both strainer lid covers and remove baskets.
5. Clean inner housing, baskets, and screens.
6. Replace lid hatch seals.
7. Check sliding door seals and replace if necessary.
8. Check for pitting and corrosion and paint with rust inhibitor wherever necessary.
9. Check strainer baskets; repair or replace as necessary.
10. Remove and replace gland packing.
11. Remove and replace lubricant in gear box with manufacturer's recommended type.
12. Lubricate motor.
13. Check starter, relays, contactor, points, and wire connections.
14. Run drive mechanism and check for proper operation.
15. Install baskets, close lids, open valves, and check for leaks.
16. Replace hoist ropes if necessary and inspect pulleys.
17. Clean outside casing and touch up pitted or corroded areas with rust inhibitor.
18. Clean up work area and remove all debris.

Tools

CRANE GASKET-07 LUBRICANT-07 MULTIMETER RUST_INHIBITOR-06 TOOL_BASIC EQUIP_CLEANING-01
HOSE_BUCKET

This standard applies to those strainers used to remove the small solids found in water drawn from rivers, reflecting pools, etc. Many are motorized self cleaning type. This water is sometimes used as condenser water in refrigeration systems. These strainers are installed just after the dual strainers.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule outage with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If the material removed from the pump is hazardous, dispose of properly and document in accordance with applicable waste management and disposal regulations. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.

Checkpoint

1. Close inlet and outlet valves.
2. Drain backwash.
3. Install hoist and remove lid bolts.
4. Remove wiring harness and shaft hex nuts.
5. Lift cover flange.
6. Remove tube support rod nuts and tie bar.
7. Individually remove tubes and acid/detergent clean each tube.
8. Clean inner casing and tube sheet. Inspect for pits and cracks.
9. Remove side inspection plate and clean bottom sump.
10. Inspect arm assembly and bolt flange.
11. Inspect cutless bearing (rubber lined cylindrical bearing), bearing housing, set screw, shaft coupler bolt and nut, shaft, and distributor barrel seat.
12. Install tubes, rod nuts, and tie bars.
13. Replace shaft gasket.
14. Inspect main shaft, bearing housing, set screw, cutless bearing, drive shaft bearing, and the stud and nut (upper unit).
15. Replace gasket for main cover.
16. Install cover lid and bolts.
17. Replace packing and inspect gland bolt.
18. Replace two "O" rings on upper shaft.
19. Inspect thrust bearings and replace if necessary.
20. Replace shear key and install hex nuts.
21. Drain oil from gear case and replace.
22. Install wiring harness and lubricate motor if necessary.
23. Check all starting relay contacts and wire connections.
24. Start backwash; observe rotation and arm movement through inspection plate.
25. Clean tubing and flange for pressure gauge.
26. Secure unit and install inspection plate.
27. Close drain valve and open inlet and outlet valves.
28. Start backwash and adjust packing.
29. Inspect unit for leaks.
30. Touch up rusted and corroded areas with rust inhibitor and paint.
31. Clean up work site.

Tools

HOSE_BUCKET TOOL_BASIC	LUBRICANT-07	MULTIMETER	CRANE	EQUIP_CLEANING-01	GASKET-07	RUST_INHIBITOR-06
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This standard applies to all types of high pressure steam traps.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Check trap operation under steam pressure.
6. Remove and replace faulty traps or trap elements

Checkpoint

1. Thermostatic traps. (Bellows or diaphragm type):
 - a. Remove cap or bonnet.
 - b. Clean interior of trap, valve, and seat.
 - c. Inspect bellows and diaphragm and note by sound whether it contains liquid charge.
 - d. Replace bellows or diaphragm as necessary.
 - e. If valve seat is cut, replace seat.
2. Float and/or thermostatic traps:
 - a. Remove bonnet. Inspect linkage and float operation for leakage, defective operation or deterioration.
 - b. Examine, clean, and check operation of bellows as in above.
3. Inverted bucket trap:
 - a. Remove bonnet.
 - b. Clean interior of trap.
 - c. Inspect valve linkage mechanism and seating of valve.
 - d. Examine condition of bucket.
 - e. Examine vent or face, inlet and outlet for evidence of corrosion.
4. Impulse trap:
 - a. Remove bonnet.
 - b. Inspect valve disc, inlet valve, outlet outface.
 - c. See that fulcrum point is free of dirt.
 - d. Clean body of trap.

Tools

TOOL_GROUP_C

This standard applies to all types of low pressure steam traps.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Check trap operation under steam pressure.
6. Remove and replace faulty traps or trap elements

Checkpoint

1. Thermostatic traps. (Bellows or diaphragm type):
 - a. Remove cap or bonnet.
 - b. Clean interior of trap, valve, and seat.
 - c. Inspect bellows and diaphragm and note by sound whether it contains liquid charge.
 - d. Replace bellows or diaphragm as necessary.
 - e. If valve seat is cut, replace seat.
2. Float and/or thermostatic traps:
 - a. Remove bonnet. Inspect linkage and float operation for leakage, defective operation or deterioration.
 - b. Examine, clean, and check operation of bellows as in above.
3. Inverted bucket trap:
 - a. Remove bonnet.
 - b. Clean interior of trap.
 - c. Inspect valve linkage mechanism and seating of valve.
 - d. Examine condition of bucket.
 - e. Examine vent or face, inlet and outlet for evidence of corrosion.
4. Impulse trap:
 - a. Remove bonnet.
 - b. Inspect valve disc, inlet valve, outlet outface.
 - c. See that fulcrum point is free of dirt.
 - d. Clean body of trap.

Tools

TOOL_GROUP_C

This standard applies to water distillers used in laboratories only.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's operation and maintenance instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Secure and tag water supply.

Checkpoint

1. Drain the unit.
2. Remove cover and clean interior of unit, including heating element electrode.
3. Inspect control cabinet. Clean by using vacuum or dry compressed air to remove dust and tighten all connections.
4. Check for leaks at fittings, gaskets and gauge glass.
5. Restore to service and check operation.
6. Collect sample and perform water hardness test.

Tools

WATER_TEST_KIT TOOL_GROUP_B CONTAINER

This standard card applies to safety relief valves installed on boilers, steam lines, and other equipment. The safety valves are designed to safely relieve excessive pressure, thus preventing rupture or explosion of the pressure parts. Safety valves differ from relief valves in that the safety relief valve opens fully when the applied pressure exceeds its lifting set-point and remains open until the applied pressure drops below its reset point. Relief valve start to open when pressure overcomes the spring pressure and remains open to the degree that the applied pressure pushes it open. When pressure drops, the spring closes the valve.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. The safety relief valves are designed to be operated by steam and should only be tested when sufficient pressure exists to clear the seating area of any debris.
6. Check with foreman and operating personnel before performing this test.
7. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Inspect condition of spring, flanges, and threaded connections.
2. Inspect the manual lifting lever for obstruction and damage, Note follow manufacturers procedures for manually lifting valve via valve lever. Check for binding of the stem or seat. Make sure that the valve returns to proper position when the lever is released.
3. Inspect support brackets and tighten as required
4. Check that the discharge piping support is tight and not causing stress on the valve.
5. Clean the valve body.
6. Lubricate the stem and lever pivot.

Tools

TOOL_GROUP_C

This applies to single or double seated, diaphragm or spring loaded pilot operated valves.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Checkpoint

1. Inspect for dirt collected at bleed port and restriction elbow. Clean if necessary.
2. Inspect joints for leakage. Tighten all bolts.
3. Check for dust or other material on the upper face of the pilot pressure plate. Clean if needed.
4. Remove and clean line strainer (back-flush where possible).
5. Inspect valve head and seats for nicks or abrasions. Notify supervisor if valve requires regrinding.
6. Inspect pressure reading against set point.
7. Check for free operation of valve stem.
8. Inspect condition of diaphragm.
9. Inspect pilot line for leaks.
10. Clean up work area and remove all debris.

Tools

TOOL_GROUP_C

This applies to valves other than those used on Fire Protection systems. Maintenance for valves used on fire protection systems are described under the appropriate guide for the specific item of fire protection equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Operate valve in full open/closed position. Loss of ability to close tightly will require inspection of valve seals and discs for wear and contaminant build-ups.
2. Check for sticking valve stems and lubricate stems and fittings sparingly.
3. Replace packing; dress, re-bush, or replace packing gland assembly, if required.
4. Check for freedom of motion on valves equipped with wheel and chain for remote operation.
5. Clean up work site.

Tools

LUBRICANT-07 TOOL_GROUP_C

This standard card applies to check valves in primary fluid systems that improve operating efficiency or prevent damage to system components. It applies only to check valves not serviced in conjunction with their associated equipment.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Schedule shutdown with operating personnel.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. If materials to be worked on are known or suspected to contain asbestos, check the building's asbestos management plan to see if they have been tested for asbestos. If they are suspect but have not been tested, have them tested. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.

Checkpoint

1. Remove the cover and clean the valve seat and disc.
2. Examine the hanger, disc, and seat for cracks or wear.
3. Check seals, packing, and gaskets for deterioration; replace if necessary.
4. Reassemble valve using a new cover gasket or seal.
5. Test under operating conditions if possible.

Tools

LUBRICANT-07 TOOL_GROUP_C PACKING

This standard card applies to approved backflow prevention devices installed in water lines to prevent the backflow of contaminated water into the potable water system. Standard check valves or double check valves are not approved backflow preventers and are not included for servicing under this standard. Backflow preventers are installed in main supply lines and water lines that supply boilers, central chilled water units, cooling towers, deep sinks used for cleaning, lawn sprinklers, lavatories, kitchen equipment, fire sprinkler systems, hose bibs, laboratories, or wherever contaminated water could backflow into the potable water system.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Schedule outage of equipment with operating personnel. For cafeteria kitchens, coordinate the maintenance activity with the cafeteria operator.
5. Schedule maintenance on manually operated valves in conjunction with this activity.
6. Typically, backflow prevention devices MUST be tested by a licensed journeyman tradesperson certified by the authority having jurisdiction. Comply with this authority as required.
7. Review manufacturer's instructions for maintenance and testing procedures and required equipment. These procedures may vary depending on the model and size of the backflow preventer.
8. Study the test procedures to be used.

Checkpoint

1. On reduced pressure zone backflow preventers, perform the following tests in accordance with the manufacturer's specifications, using the appropriate test kit:
 - a. Test check valve number for tightness against reverse flow.
 - b. Test gate valve number for tightness.
 - c. Test check valve number for tightness.
 - d. Test operation of pressure differential relief valve.
2. Service the first and second checks. CAUTION: If the check valve is spring loaded, do not remove the spring retainers. Consult the manufacturer's instructions for proper servicing.
 - a. Carefully remove screws, cover, and check.
 - b. Disengage the disc and spring assembly into individual components in accordance with manufacturer's instructions. Remove any embedded foreign objects, and inspect for corrosion, worn seals, etc. Clean or replace the assembly as required.
 - c. Clean or replace seals as necessary. Apply a light coating of manufacturer's specified and FDA approved lubrication prior to installation of seals.
 - d. Reassemble the check valve module in reverse order.
 - e. Repeat for second check.
3. Service the relief valve. CAUTION: Springs may be loaded. Strictly comply with manufacturer's instructions.
 - a. Remove bolts, cover, diaphragm, and relief valve piston assembly per manufacturer's instructions.
 - "b. Clean or replace wiper seal, piston ""O"" ring, and relief valve disc as required. Apply appropriate lubricant to ""O"" ring per manufacturer's specifications prior to reinstallation."
 - c. Inspect bottom spring assembly. If defective, replace entire unit. Do not attempt to remove the spring.
4. Following the manufacturer's procedures, test and calibrate the device.
5. Following the manufacturer's procedures, vent both chambers and return the system to normal operation. Verify that there is no dripping or periodic spitting, and that the water flows properly and pressure drop is normal.
6. For cafeterias and kitchens only, inspect food cookers, post-mix carbonated beverage machines, dishwashers, hose bibs, and service sinks and determine whether a backflow preventer is installed. If missing, ensure a preventer isn't

located upstream, then initiate a work order to install one if required.

Tools

LUBRICANT-07 MANUF_TESTING TOOL_GROUP_C

PRESPOTRAIT-01-04Y

Rev-0 | 2025-01-01

Presidential Portrait Removal Certification (WORK ORDER LOG NOTE CERTIFICATION REQUIRED.)

This Job plan is for buildings to schedule, conduct and certify completion of the required prortrait removal of the outgoing President, Vice President and GSA Administrator using this work order.

Special Instructions

- 4. Remove, collect and properly dispose of all displayed and stored portraits of the outgoing President, Vice President and GSA Administrator by January 17th of the inaugural year.
- 3. To the extent practicable, picture frames should be preserved for future use.
- 2. After the portrait removals and subsequent destruction is performed, certify the completion by logging into NCMMS and copying the following message to the Work LOG tab of this work order: "I certify that the previous administration portraits have been removed and disposed of in accordance with subsection 5(e) of GSA Order ADM 1080.2 (February 28, 2023). The portraits were disposed of properly, including any remaining portraits in stock."
- 1. This work order is assigned to you as the ServiceNow Primary Point of Contact (POC) for the building listed on this work order and its purpose is to document the required outgoing President, Vice president and GSA Administrator removal and the proper destruction of portratits removed or in storage at the facility.

Checkpoint

Tools

None

SCL-C-A

Cooling Season Checklist
Rev-2 | undefined

null

Special Instructions

- 1. Perform checklist from the Work Order Tracking (HSE) application.
- 2. After opening the Work Order, go to the Details tab and under Standard Actions fill out each Review Action with an entry to complete the checklist.
- 3. When the checklist is complete, change the Work Order status to Complete.
- 4. The Attached Cooling Season Readiness Checklist can be filled out and attached to the workorder instead of using the Standard Action Checklist in the Work Order (HSE) application.

Checkpoint

Tools

None

null

Special Instructions

- 1. Perform checklist from the Work Order Tracking (HSE) application.
- 2. After opening the Work Order, go to the Details tab and under Standard Actions fill out each Review Action with an entry to complete the checklist.
- 3. When the checklist is complete, change the Work Order status to Complete.

Checkpoint

Tools

None

TANK-AGS-01-01Y

Tanks, Air, Refrigerant, LP Gas
Rev-4 | 2021-11-08

This standard card applies to three types of fixed storage tanks; air, refrigerant, and LP gas. These are large permanently mounted supply or reserve tanks. Not included are main supply tanks serviced by the supplier, or small returnable tanks used on mobile equipment or for maintenance.

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Review manufacturer's instructions.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. Maintenance shall be coordinated with required inspections.
- 5. Review standard operating procedures for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
- 6. Review standard operating procedures for selection, care, and use of respiratory protection.
- 7. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

- 1. Examine exterior of tank, including fittings, gauges, manholes, and handholes for leaks, rust, and corrosion. Wire brush and touch up paint.
- 2. Open tank and remove rust, scale and sludge by scraping, wire brushing, or shot cleaning. If the material removed from the tank is hazardous waste, refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
- 3. Prior to entering the tank, ensure that it is a safe environment for entry by testing the air. Follow appropriate safety precautions.
- 4. Inspect interior of tank carefully, recording the size and depth of pits, presence of cracks and condition of openings, fittings, weld, rivets, and joints.
- 5. Touch up interior coat with an approved protective coating.
- 6. Inspect structural supports and condition of insulation, if any. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
- 7. Perform hydrostatic test, if required. Note: Hydrostatic test of unfired pressure vessels is required of any vessel that has a capacity of greater than 15 gallons and operates at a pressure in excess of 60 psig.

Tools

PAINT-06 GAS_METER PROTECTIVE_COAT-01 RESPIRATOR-01 TOOL_GROUP_C FIRE_EXTINGUISHER

This maintenance standard applies to chemical tanks other than water softener tanks or plastic expendable vats or tanks.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review standard operating procedures for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
5. Review standard operating procedures for selection, care, and use of respiratory protection.
6. Schedule annual valve maintenance in conjunction with this activity.

Checkpoint

1. Examine exterior of tank, including fittings, gauges, structural supports, manholes, and handholes for leaks, signs of corrosion, or other defects. Correct as indicated.
2. Drain and flush the tank. If the material removed from the tank is hazardous waste, refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
3. Prior to entering the tank, ensure that it is a safe environment for entry by testing the air. Follow appropriate safety precautions.
4. Open tank and remove rust or chemical deposits from interior surfaces.
5. Inspect thoroughly the interior of the tank; record the size and depth of pits, presence of cracks, and the condition of openings, fittings, welds, rivets, and joints.
6. Check condition of agitators and/or float assemblies.
7. Clean strainer(s).
8. Touch up cost with approved protective coating.

Tools

RESPIRATOR-03 PAINT-06 TOOL_BASIC PROTECTIVE_COAT-01 FACE_SHIELD_APRON

This guide applies to all Aboveground Storage Tank (AST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress

fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

Tools

None

This guide applies to all Aboveground Storage Tank (AST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine IF any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress

fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).

Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.

7. Review/Update FST data In NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.

4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.

5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.

6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.

8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.

9. Review/Update FST in NCMMS - Tank size (GL capacity), location, substance stored, tank construction material, system piping (secondary containment, material, monitoring, feed, type monitoring). Check for completeness, accuracy and currency. Update as Needed.

10. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.

11. Review/Update FST data in NCMMS - Copies of tank notifications and reports to state agencies (e.g., installation, closure, leaks, spills, EPCRA, Tier II Reports, etc.). Check for completeness, accuracy and currency. Update as Needed. Check with your GSA Regional Environmental, Health and Safety (EHS) Specialist, Contracting Officer Representative of assistance is needed in determining the proper agency to contact.

12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.

13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.

14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.

15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.

16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.

Tools

None

This guide applies to all Aboveground Storage Tank (AST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine IF any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress

- fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).
2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).
 3. Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.
 4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.
 5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.
 6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.
 7. Review/Update FST data In NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.
 8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.
 9. Review/Update FST in NCMMS - Tank size (GL capacity), location, substance stored, tank construction material, system piping (secondary containment, material, monitoring, feed, type monitoring). Check for completeness, accuracy and currency. Update as Needed.
 10. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
 14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy. Update as Needed.
 11. Review/Update FST data in NCMMS - Copies of tank notifications and reports to state agencies (e.g., installation, closure, leaks, spills, EPCRA, Tier II Reports, etc.). Check for completeness, accuracy and currency. Update as Needed. Check with your GSA Regional Environmental, Health and Safety (EHS) Specialist, Contracting Officer Representative of assistance is needed in determining the proper agency to contact.
 12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.
 13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.
 15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.
 16. Review Posting of Filling/transfer procedures. Check for currency, readability, placement, deterioration, etc. Replace as needed.
 17. Triennially - Coordinate third-party inspection of AST by person certified by Steel Tank Institute (STI) or American Petroleum Institute (API) or State approved equivalent. Inspections must be documented and maintained for the life of the facility. Inspection results report copied to this work order and in Certificates.

Tools

None

This standard Above and Below Ground fuel oil storage tanks.

Special Instructions

1. In addition to the procedure(s) outlined in this standard (Appendix I), the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
5. Review the standard operating procedure for selection, care, and use of respiratory protection.
6. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.

Checkpoint

1. Refer to Appendix B (URL/Link provided as attachment to job plan) for requirements and procedures.

Tools

GOOGLES TOOL_GROUP_C SAFETY_HARNESS RESPIRATOR-01

This guide applies to all Aboveground Storage Tank (AST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine IF any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress

- fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).
2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).
 3. Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.
 4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.
 5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.
 6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.
 7. Review/Update FST data In NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.
 8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.
 9. Review/Update FST in NCMMS - Tank size (GL capacity), location, substance stored, tank construction material, system piping (secondary containment, material, monitoring, feed, type monitoring). Check for completeness, accuracy and currency. Update as Needed.
 10. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
 11. Review/Update FST data in NCMMS - Copies of tank notifications and reports to state agencies (e.g., installation, closure, leaks, spills, EPCRA, Tier II Reports, etc.). Check for completeness, accuracy and currency. Update as Needed. Check with your GSA Regional Environmental, Health and Safety (EHS) Specialist, Contracting Officer Representative of assistance is needed in determining the proper agency to contact.
 12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.
 13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.
 14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
 15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.
 16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.
 17. Every 5 years - Review SPCC and update if changes to the facility were made. Incorporate any pens-and-ink annotations made to the plan head of the 5-year review. Changes made to an SPCC Plan that was authored by a PE must be recertified by a PE if the 10,000 gal capacity still exists.

Tools

None

This guide applies to all Underground Storage Tank (UST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
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11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the UST system check for items such as but not limited to signs of leaks, or spills and stressed vegetation. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, etc.).
2. If UST receives deliveries at intervals greater than 30 days, check your spill prevention equipment prior to each delivery.

3. To the extent practicable, inspection must be done while transferring fuel (pipes in use) to determine leakage.
4. Check spill prevention equipment for damage and remove liquid or debris.
5. Remove any liquid or debris from the spill bucket and area surrounding the spill bucket.
6. For double-walled spill prevention equipment with interstitial monitoring, check for a leak in the interstitial area.
7. Check for and remove obstructions in the fill pipe.
8. Check the fill connection, cap, and seal to ensure it is securely on the fill pipe.
9. Check the tank vent, flow meter and gauge for serviceability and functionality.
10. Check release detection equipment to ensure it is operating with no alarms or unusual operating conditions present (for example ATG consoles or pressure or vacuum gauges).
11. Review release detection records and ensure they are current.
12. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
13. For USTS with continuous monitoring, print the monthly report and compare with previous readings to assess possible product loss.
14. For USTs without continuous monitoring, perform manual inventory reconciliation calculations, and document.
15. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.

Tools

None

This guide applies to all Underground Storage Tank (UST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

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7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
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12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
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Checkpoint

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2. If UST receives deliveries at intervals greater than 30 days, check your spill prevention equipment prior to each delivery.
3. To the extent practicable, inspection must be done while transferring fuel (pipes in use) to determine leakage.

4. Check spill prevention equipment for damage and remove liquid or debris.
5. Remove any liquid or debris from the spill bucket and area surrounding the spill bucket.
6. For double-walled spill prevention equipment with interstitial monitoring, check for a leak in the interstitial area.
7. Check for and remove obstructions in the fill pipe.
8. Check the fill connection, cap, and seal to ensure it is securely on the fill pipe.
9. Check the tank vent, flow meter and gauge for serviceability and functionality.
10. Check release detection equipment to ensure it is operating with no alarms or unusual operating conditions present (for example ATG consoles or pressure or vacuum gauges).
11. Review release detection records and ensure they are current.
12. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
13. For USTS with continuous monitoring, print the monthly report and compare with previous readings to assess possible product loss.
14. For USTs without continuous monitoring, perform manual inventory reconciliation calculations, and document.
15. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.
17. Check Hand held release detection equipment (for example tank gauge sticks or groundwater bailers) for functionality and serviceability. Replace as Needed.
18. Test mechanical components of Automatic tank gauge (ATG) and other controllers. Test the alarm, Verify the system configuration. Document results.
19. Test mechanical components of Automatic tank gauge (ATG) and other controllers. Test the battery backup. Document Results.
20. Test mechanical Probes and Sensors for proper operation. Inspect for residual buildup Ensure any floats move freely Document Results.
21. Test mechanical Probes and Sensors for proper operation. Ensure any shafts are not damaged Ensure the cables are free of kinks and breaks Test the alarm operability and communication with the controller. Document Results.
22. Test Automatic line leak detector (ALLD). Ensure the device activates (alarms, restricts flow, or shuts off flow) within an hour when simulating a release equivalent to 3 gallons per hour at 10 pounds per square inch. Document Results.

Tools

None

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8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
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10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Every 60 Days - If UST system is equipped with a rectifier for corrosion protection, The cathodic protection system must be tested by a qualified cathodic protection tester. The Amps, Watts and hours must be documented and maintained for the life of the facility.
2. Conduct visual inspection of the UST system check for items such as but not limited to signs of leaks, or spills and stressed vegetation. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, etc.).

3. If UST receives deliveries at intervals greater than 30 days, check your spill prevention equipment prior to each delivery.
4. To the extent practicable, inspection must be done while transferring fuel (pipes in use) to determine leakage.
5. Check spill prevention equipment for damage and remove liquid or debris.
6. Remove any liquid or debris from the spill bucket and area surrounding the spill bucket.
7. For double-walled spill prevention equipment with interstitial monitoring, check for a leak in the interstitial area.
8. Check for and remove obstructions in the fill pipe.
9. Check the fill connection, cap, and seal to ensure it is securely on the fill pipe.
10. Check the tank vent, flow meter and gauge for serviceability and functionality.
11. Check release detection equipment to ensure it is operating with no alarms or unusual operating conditions present (for example ATG consoles or pressure or vacuum gauges).
12. Review release detection records and ensure they are current.
13. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
14. For USTS with continuous monitoring, print the monthly report and compare with previous readings to assess possible product loss.
15. For USTs without continuous monitoring, perform manual inventory reconciliation calculations, and document.
16. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
17. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.

Tools

None

This guide applies to all Underground Storage Tank (UST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

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4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
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8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the UST system check for items such as but not limited to signs of leaks, or spills and stressed vegetation. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, etc.).
2. If UST receives deliveries at intervals greater than 30 days, check your spill prevention equipment prior to each delivery.
3. To the extent practicable, inspection must be done while transferring fuel (pipes in use) to determine leakage.

4. Check spill prevention equipment for damage and remove liquid or debris.
5. Remove any liquid or debris from the spill bucket and area surrounding the spill bucket.
6. For double-walled spill prevention equipment with interstitial monitoring, check for a leak in the interstitial area.
7. Check for and remove obstructions in the fill pipe.
8. Check the fill connection, cap, and seal to ensure it is securely on the fill pipe.
9. Check the tank vent, flow meter and gauge for serviceability and functionality.
10. Check release detection equipment to ensure it is operating with no alarms or unusual operating conditions present (for example ATG consoles or pressure or vacuum gauges).
11. Review release detection records and ensure they are current.
12. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
13. For USTs with continuous monitoring, print the monthly report and compare with previous readings to assess possible product loss.
14. For USTs without continuous monitoring, perform manual inventory reconciliation calculations, and document.
15. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.
17. Check Hand held release detection equipment (for example tank gauge sticks or groundwater bailers) for functionality and serviceability. Replace as Needed.
18. Test and certify the functionality of the Automated Leak Level Detection System (ALLD).
19. Test and Certify UST piping. Must be tested for leaks by certified third party with documentation maintained for the life of the facility (for pressurized systems).
20. Test the spill bucket to ensure tightness by third party and maintain documentation of the test for the life of the facility
21. Test functionality of the overfill prevention equipment. Test for alarm activate at the proper levels. Maintain documentation of the test for the life of the facility.
22. For steel tanks and/or piping in contact with soil test the functionality of the cathodic protection system must be tested by a certified cathodic protection tester. Upload documentation into NCMMS and of the test for the life of the facility.
23. UST piping must be tested for leaks by certified third party with documentation maintained for the life of the facility (for American Suction Systems).
24. Every 3 years - For UST systems equipped with a corrosion protection other than impressed current, have cathodic protection system tested. Test must be performed by a qualified cathodic protection tester. Upload test report to NCMMS.

Tools

None

This guide applies to all Aboveground Storage Tank (AST) and Underground Storage Tank (UST) systems used to store regulated substances (i.e., petroleum products and hazardous substances), except where the specific task is not applicable (e.g., the regulatory agency does not require registration, exempt based on content or use, etc.). Check with regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.

Special Instructions

15. Above or Underground containers used to store petroleum. Does Not include day tanks or belly tanks. Applies to aboveground Used Oil storage containers 5 gallons or greater and underground Used Oil tanks less than 110 gals, and elevator Hydraulic Oil, and Cooking Oil tanks. Tanks located in a basement or garage are considered ASTs. Check with regional Environmental, Health and Safety Specialist and local regulations to determine if other requirements impact this location and Job Plan Tasks.
14. After completing all items in the Job Plan change the work order status to "Complete"
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
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9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks are required to be registered or permitted by state or local laws.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to. If Applicable
1. Maintain ready access to the state requirements for managing storage of regulated substances. If Applicable

Checkpoint

7. Check the fill connection, cap, and seal to ensure it is securely on the fill pipe.
6. Review Posted Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.
5. Check that container is marked with the contents (i.e. Diesel, Used Oil, etc.).
4. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified)
3. Check for water in the secondary containment area.
2. Check for stressed vegetation near outside containers.
1. Conduct a visual inspection of exposed area of storage container or tank. Check for items such as but not limited to signs of leaks, corrosion, and stress fractures.

Tools

None

Fuel tanks that supply fuel to the building emergency generators (Belly Tank or Day Tank). Usually, an AST but can be an UST. Tanks in a basement or garage are considered ASTs. Otherwise regardless of the physical location (above or below ground) tanks are considered USTs if 10% or more of the combined volume of the tanks and piping is underground. Federal regulations apply to AST tanks 55 gallons or greater capacity and USTs with a capacity of 110 gallons or greater.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
2. In addition to the procedure(s) outlined in this job plan, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
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7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
9. Before beginning work, based on the tasks to be performed select appropriate PPE and safety procedures. This includes but not limited to use of protective clothing and gloves, hearing and eye protection, breathing protection, and following proper procedures for lock-out-Tag-out, de-energizing equipment, and use of the buddy system.
10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

Tools

None

Fuel tanks that supply fuel to the building emergency generators (Belly Tank or Day Tank). Usually, an AST but can be an UST. Tanks in a basement or garage are considered ASTs. Otherwise regardless of the physical location (above or below ground) tanks are considered USTs if 10% or more of the combined volume of the tanks and piping is underground. Federal regulations apply to AST tanks 55 gallons or greater capacity and USTs with a capacity of 110 gallons or greater.

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3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
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8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
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10. If the tank is in a confined space – Review the standard operating procedure for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete"

Checkpoint

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).
3. Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.
4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.
5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.
6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.
7. Review/Update FST data In NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.
8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.
9. Review/Update FST in NCMMS - Tank size (GL capacity), location, substance stored, tank construction material, system piping (secondary containment, material, monitoring, feed, type monitoring). Check for completeness, accuracy and currency. Update as Needed.
10. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.
11. Review/Update FST data in NCMMS - Copies of tank notifications and reports to state agencies (e.g., installation, closure, leaks, spills, EPCRA, Tier II Reports, etc.). Check for completeness, accuracy and currency. Update as Needed. Check with your GSA Regional Environmental, Health and Safety (EHS) Specialist, Contracting Officer Representative of assistance is needed in determining the proper agency to contact.
12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.
13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.
14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.
15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.
16. Review Posting of Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.

Tools

None

Fuel tanks that supply fuel to the building emergency generators (Belly Tank or Day Tank). Usually, an AST but can be an UST. Tanks in a basement or garage are considered ASTs. Otherwise regardless of the physical location (above or below ground) tanks are considered USTs if 10% or more of the combined volume of the tanks and piping is underground. Federal regulations apply to AST tanks 55 gallons or greater capacity and USTs with a capacity of 110 gallons or greater.

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7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
8. Tests, inspections, repairs and other maintenance activities must be performed in accordance with the manufacturer's instructions; a code of practice developed by a nationally recognized association or independent testing laboratory; or requirements determined by the implementing agency. Vendor shall be responsible for ensuring that all appropriate tasks are completed accordingly. Refer to federal and state regulations for additional details and acceptable codes of practice.
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12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to "Complete".

Checkpoint

7. Review/Update FST data In NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.
1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other

hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).
3. Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.
4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.
5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.
6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.
8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.
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12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.
13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.
14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy. Update as Needed.
15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.
16. Review Posted Filling/transfer procedures. Check for currency, readability, placement, deterioration, etc. Replace as needed.
17. Triennially - Coordinate third-party inspection of AST by person certified by Steel Tank Institute (STI) or American Petroleum Institute (API) or State approved equivalent. Inspections must be documented and maintained for the life of the facility. Inspection results report copied to this work order and in Certificates.

Tools

None

Fuel tanks that supply fuel to the building emergency generators (Belly Tank or Day Tank). Usually, an AST but can be an UST. Tanks in a basement or garage are considered ASTs. Otherwise regardless of the physical location (above or below ground) tanks are considered USTs if 10% or more of the combined volume of the tanks and piping is underground. Federal regulations apply to AST tanks 55 gallons or greater capacity and USTs with a capacity of 110 gallons or greater.

Special Instructions

1. Maintain ready access to the state requirements for managing storage of regulated substances.
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3. In addition to the procedure(s) outlined in this job plan, state, local, and GSA regional tank management requirements may apply and shall be strictly adhered to. The more stringent requirement shall prevail.
4. Check with your regional Environmental, Health and Safety (EHS) Specialist and local regulations to determine if any tank regulatory changes have occurred that impact this location and Job Plan Tasks.
5. Contact your regional Environmental Protection Specialist and Contracting Officer Representative to determine if any of the facility's tanks (ASTs or USTs) are required to be registered or permitted by state or local laws.
6. Prepare and submit completed required forms to the proper outside regulatory contact when reporting to another Federal, State or Local agency is required. In addition, submit copies of completed forms and reports to the GSA EHS Specialist, and Contracting Officer Representative (COR) as specified by local procedures and COR instructions.
7. Upload completed documents into NCMMS and the associated work order. This includes but not limited to test results, walk through/visual inspections, monthly test reports or calculations completed forms and reports to other agencies, and other supporting documentation. Retain for the life of the facility.
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11. Check the building's asbestos management plan or with the GSA Regional EHS Specialist, or Contracting Officer Representative to determine if asbestos containing material is known to be present in the area where work will be performed. Manage asbestos in accordance with the asbestos management plan. Request copy of latest asbestos management plan from COR if necessary.
12. Report spills in accordance with GSA reporting requirements. Contact the Regional GSA Regional Environmental, Health and Safety (EHS) Specialist, or Contracting Officer Representative if assistance is needed to determine reporting requirements. If no spills occurred during this period, no action is required for this task.
13. Complete the NCMMS digital inspection form (when provided) as a part of this PM before completing the work order.
14. After completing all items in the Job Plan change the work order status to ""Complete"

Checkpoint

17. Every 5 years - Review SPCC and update if changes to the facility were made. Incorporate any pens-and-ink annotations made to the plan head of the 5-year review. Changes made to an SPCC Plan that was authored by a PE must be recertified by a PE if the 10,000 gal capacity still exists.
11. Review/Update FST data in NCMMS - Copies of tank notifications and reports to state agencies (e.g., installation, closure, leaks, spills, EPCRA, Tier II Reports, etc.). Check for completeness, accuracy and currency. Update as Needed.

Check with your GSA Regional Environmental, Health and Safety (EHS) Specialist, Contracting Officer Representative of assistance is needed in determining the proper agency to contact.

3. Annually prepare and submit Tier II Report if facility stores 10,000 pounds or 1,440 gallons or more of any type of oil (diesel, gasoline, oil, etc.) or if any of the chemicals listed in 40 CFR part 355 Appendix A and Appendix B are stored in quantities at or above the set threshold limits during the year. Report must be submitted to the designated agency (EPA, or State) by 1 March of each year, or by the deadline set by the state. One report per facility. Check with your regional EHS for guidance.

7. Review/Update FST data in NCMMS - installation year, permit and registration, test results, and inspection. Check for completeness, accuracy and currency. Update as Needed.

8. Review/Update FST data in NCMMS - Tank Regulatory Operational Status (i.e., currently in use, temporarily out of service, closed in place, removed) and out of service date/date closed. Check for completeness, accuracy and currency. Update as Needed.

9. Review/Update FST in NCMMS - Tank size (GL capacity), location, substance stored, tank construction material, system piping (secondary containment, material, monitoring, feed, type monitoring). Check for completeness, accuracy and currency. Update as Needed.

10. Review/Update FST data in NCMMS - Internal and external protections including cathodic protection, spill and overfill, and release detection methods. Check for completeness, accuracy and currency. Update as Needed.

4. Review site tank management procedures and update to align with applicable current GSA, Federal and state requirements. Check for completeness, accuracy and currency. Update as Needed.

5. Review point of contact information to include Name, Address, Emergency Phone Numbers, Email, Contractor/Vendors, State, and Local authorities. Check for completeness, accuracy and currency. Update as Needed.

6. Review/Update FST data in NCMMS - All operator and owner information including training, Licenses, contact information. Records should be created and maintained as in Qualifications or Certificate module of NCMMS. Check for completeness, accuracy and currency. Update as Needed.

12. Review/Update FST data in NCMMS - Spill response plans, SPCC Plans or other release mitigation and response plans for facilities not subject to SPCC requirements). Changes shall be made to reflect current conditions and are temporary pen-and-ink entries until such time that the formal 5-year update and revision. Check entries for completeness, accuracy and currency. Update as Needed.

13. Conduct and document SPCC Plan training for oil-handling personnel. Training shall cover the O&M of equipment, discharge procedure protocols, applicable pollution control laws, rules and regulations, general facility operations and the contents of the facility SPCC Plan. Oil-handling personnel includes employees engaged in the operation and maintenance of oil storage containers or the operation of equipment related to storage containers and emergency response personnel. Document in NCMMS.

1. Conduct visual inspection of the AST using the most current version of the attached Steel Tank Institute Monthly Inspection Checklist (STI SP001) or equivalent. Check for items such as but not limited to signs of leaks, stress fractures, stressed vegetation, water in the secondary containment area. Check for NFPA 704 signage and other hazard warnings. Clearly document status of hazard warnings (i.e., NFPA Diamond present and hazards properly classified, tank is marked with the contents, etc.).

2. Prepare/submit hazardous chemical report IF hazardous chemicals are on-site in quantities at or above the reporting threshold. Submit to a materials inventory or Safety Data Sheets to the designated emergency response authorities (SERC, LEPC, Fire Department) Rules vary by location, follow local procedures and document, Update Only when needed. See 40 CFR part 355 Appendix A and Appendix B, for the list of chemicals and threshold limits (Fuel/oil the reporting quantity is 1,440 gallons).

14. Review/Update tank data in NCMMS - Filling and transfer procedures. Check for currency, completeness and accuracy.

15. Review/Update FST data in NCMMS to identify low-performing assets that may be potential candidates for upgrades, replacements, closure, based on professional judgement. Document observations and bring adverse findings to the attention of the COR and EHS POC.

16. Review Post Filling/transfer procedures. Where procedures are posted, check for currency, readability, placement, deterioration, etc. Replace as needed.

Tools

None

This standard applies to all storage tanks including hot water, cold water, and expansion tanks which water is stored.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Maintenance shall be coordinated with required inspections. This applies to hot water, cold water, expansion tanks, etc., but not to hot water converter (steam) tanks.
5. Review standard operating procedures for confined space entry. Follow OSHA guidelines for permitting of entry and atmospheric testing of confined spaces. NEVER enter a confined space without a safety watch.
6. Review standard operating procedures for selection, care, and use of respiratory protection.
7. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.

Checkpoint

1. Examine exterior of tank including fittings, manholes, and handholes for leaks, signs of corrosion, and correct as indicated.
2. Drain and flush tank.
3. Prior to entering the tank, ensure that it is a safe environment for entry by testing the air. Follow appropriate safety precautions.
4. Open tank and remove rust, scale and buildup by scraping, wire brushing or shot cleaning.
5. Inspect thoroughly the interior of tank; record the size and depth of pits, presence of cracks and condition of openings, fittings, welds, rivets, and joints.
6. Coat with epoxy or other approved protective coatings.
7. Inspect structural supports and repair or replace damaged insulation or covering. If insulation contains asbestos and is damaged or eroded, it is considered a hazardous waste. Refer to Appendix A - 1095.8 PBS Regulated Waste Management (Link/URL attached to Job Plan for reference) and state and local requirements.
8. Clean, test and inspect sight glasses, valves, fittings, drains, and controls.
9. Perform hydrostatic test if required.
10. Fill and return to service.
11. Add an EPA approved biocide.
12. Clean up work site.

Tools

GOOGLES EQUIP_CLEANING-03 SAFETY_SIGNS BIOCIDES_EPA_APPROVED RESPIRATOR-01 TOOL_GROUP_C

VTRN-ELV-01-01M

Elevator Maintenance - Hydraulic
Rev-3 | 2022-01-01

This standard applies to hydraulically operated passenger and freight elevators. Normally, these elevators are limited to 4 floors or less.

Special Instructions

- 1. Complete all maintenance requirements included in the applicable ASME A17 Elevator Safety Codes.
- 2. Utilize the Maintenance Control Program (MCP) that is required to be developed and provided by the contractor for each piece of equipment as required in section 8.6 of ASME A17.1. MCP shall include all maintenance tasks and their frequencies with month dependent upon each unit's characteristics (equipment age, condition, accumulated wear, usage, improved technology, manufacturer's recommendations, etc.)
- 3. Follow lock out/tag out procedures at all times.

Checkpoint

- 1. Emergency Phone: Verify proper operation with clear communication including adequate volume.
- 2. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Tools

None

This standard applies to hydraulically operated passenger and freight elevators. Normally, these elevators are limited to 4 floors or less.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A17.1 and A17.2, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, ...
(Continued) or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
6. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
7. Be sure that safety devices are operational before performing any inspection work.
8. The annual inspection of hydraulic elevators includes those designated as periodic in the code.
9. Use energy efficient lighting to the fullest practical extent.
10. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
11. Comply with state and local codes as applicable.

Checkpoint

1. Brakes: Completely dismantle brake assembly, clean, and inspect for wear. Replace defective parts required for proper operation. Where brake shoes are asbestos-containing, check for dust, and practice appropriate cleanup and maintenance precautions. Lubricate bearing, pins, and pivot points.
2. Selector: Inspect, clean, lubricate, replace parts, and make repairs or adjustments required for proper operation of selector unit components, including cables, chains, clutches, cams gears, fuses, motor brushes, wiring, connections, contacts, relays, tape, tape tension sheave, broken tape switch and tape wipers.
3. Controller: Thoroughly clean controller with blower or vacuum. Inspect and check the operation of switches, relays, timers, capacitors, resistors, contacts, overloads, wiring, connections, fuses, oil level of overloads and overload settings. Check for MG shutdown, high call reversal, zone control, and load by-pass door failure time. Check programming up peak, down peak, off peak, off hours. Replace worn or defective parts and adjust controller for proper operation.
4. Hoist way Doors: Clean, inspect, and lubricate all door operating mechanisms; including but not limited to rollers, upthrusts, interlocks, clutches, self closing, gibbs and sills. Replace worn or defective parts, repair and adjust door mechanisms as required for proper operation.
5. Hoist ways: Clean rails, beams, and all related iron work in hoist way. Dust hoist way walls. Clean top, bottom, and sides of car. Clean counterweight and pit area.

6. Hoist Machine and Motor-Generator: Clean with blower or vacuum. Clean end bells, and brush riggings and commutator. Inspect connections, armature and rotor clearances.
7. Buffers: Check oil level and operation of switches. Add oil or adjust switches as necessary for proper operation. Manually compress buffer; test the proper return in accordance with ASME/ANSI A17.1 Safety Code for Elevators and Escalators, Rule 201.4e(1).
8. Scheduling, Dispatch, and Signal Boards: Clean with blower or vacuum. Inspect and check the operation of all switches, relays, timers, fuses, capacitors, resistors, contacts, overloads, wiring and connections. Replace worn or defective parts and adjust for proper operation.
9. Motors: Change oil in hoist motor, motor generator set, geared machines, and gear boxes with lubricants as specified by the equipment manufacturer(s).
10. "10. Inside of Car: Emergency stop switch; car emergency signal; rated load, platform area capacity and data plate; signs in freight elevator; car enclosure; ventilation of passenger elevators; side emergency exits; car door or gate; car door or gate electric contacts; closed position of car door or gate; power opening of doors or gates; power closing of doors or gates; door reopening device; car floor and landing sill; operating control device; emergency signal device; and car lighting ."
Including emergency lighting.
11. Outside Hoist way: Car platform guard; power closing of hoist way doors; door closing force; sequence operation; hoist way enclosure; hoist way doors; vision panels; hoist way door locking device; elevator parking device; access to hoist way; and emergency doors.
12. Top of Car: Top car clearance and refuge space; stop switch top of car; top car operating device; top car light and outlet; traveling cables and junction; door gate contacts, cams, etc., hangars and connections; hoist way clearances; normal terminal stopping device; top emergency exit; crosshead data plate; construction of hoist way; floor over hoist way; hoist way smoke control; guide rails, fastenings, and alignment; pipe, wiring, and ducts; floor numbers; landing sill guards, (continued)
and recesses; car frame and stiles; and ant creep leveling device.
13. Machine Room and Machinery Space: Access to machine space; pressure tanks; lighting machine space; ventilation of machine and control space; guards for exposed equipment; control valve; terminal stopping device; pumps; relief and check valve; drives; flexible hose and fittings; tank and oil level; controller wiring fuses, etc.; piping supply line and shutoff.
14. Pit: Pit light and stop switch; pit access; car clearance and runby; construction of oil buffer; oil buffer plunger return; oil buffer oil level gage; oil buffer data plate; spring buffer and data plate; solid bumper; cylinder oil collection; pipes, valves, fittings, and supports; plunger; plunger connection to car; normal stopping device; guard between pits; pit access; illumination; traveling; car frame and platform; and pit construction.
15. Test: Flexible hose and fittings; oil buffer (perform the five year test as necessary); terminal stopping devices; firefighters service; emergency power operation; power door closing force; pressure tank (three year); pressure switch; relief valve setting; static load test; emergency terminal speed limiting device.
16. Safeties: Inspect, clean, lubricate and manually operate safety mechanisms. Replace parts or adjust as necessary.
17. Clean up and remove all debris from work area.

Tools

HAMMER	WEIGHTS_TEST	CALIPER	CHALK	ELEVATOR_HANDBOOK	ELEVATOR_MANUAL	ELEVATOR_SAFETY	
FLASHLIGHT_NON_CONDUCTIVE		GAGE_THICKNESS		LEVEL_SPIRIT	LIGHT_METER-02	LUBRICANT-07	MAT_CLEANING-05
TEST_GAGE_DOOR	WATCH_STOP	TOOL_BASIC		GAGE_SHEAVE_GROOVE	BARRICADES	MATERIAL_01	TAPE
TACHOMETER-02	MIRROR	METER	SIGN_OUT_OF_SERVICE-01		PRESSURE_GAUGE-01	SAFETY_HAT	

This standard applies to hydraulically operated passenger and freight elevators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A17.1 and A17.2, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, ...

(Continued) or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.

4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Be sure that safety devices are operational before performing any inspection work.
8. The 5 year inspections of hydraulic elevators are described in the periodic section of the code.
9. Perform the semiannual and annual inspection simultaneously.
10. Comply with state and local codes as applicable.

Checkpoint

1. Inside of Car: Power opening of doors or gates; standby power operation.
2. Machine Room: Power unit assemblies; terminal stopping devices;
3. Top of Car: Counterweight safeties; guide shoes; roller guides;
4. Inspections Made in the Pit: Hydraulic cylinders; bulkhead and Overspeed valves.

Tools

SAFETY_HAT	METER	MIRROR	TAPE	MATERIAL_01	GAGE_SHEAVE_GROOVE	TOOL_BASIC	WATCH_STOP
TEST_GAGE_DOOR	LIGHT_METER-02	LEVEL_SPIRIT		GAGE_THICKNESS	FLASHLIGHT_NON_CONDUCTIVE	ELEVATOR_SAFETY	
ELEVATOR_MANUAL	ELEVATOR_HANDBOOK	CALIPER		WEIGHTS_TEST	HAMMER	CHALK	

This standard applies to hydraulically operated passenger and freight elevators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A1 and A2, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
6. Be sure that safety devices are operational before performing any inspection work.
7. Inspection checklists adapted from the code are included for convenience.
8. The semiannual inspection of hydraulic elevators includes those designated as routine in the code.
9. Items requiring attention should be reported to the elevator shop supervisor or elevator contractor.
10. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
11. Use energy efficient lighting to the fullest practical extent.
12. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
13. Comply with state and local codes as applicable.

Checkpoint

1. Inside of Car: Emergency stop switch; car emergency signal; rated load, platform area capacity and data plate; signs in freight elevator; car enclosure; ventilation of passenger elevators; side emergency exits; car door or gate; car door or gate electric contacts; closed position of car door or gate; power opening of doors or gates; power closing of doors or gates; door reopening device; car floor and landing sill; operating control device; (Continued)
emergency signal device; and car lighting (including emergency).
2. Outside Hoist way: Car platform guard; power closing of hoist way doors; sequence operation; hoist way enclosure; hoist way doors; vision panels; hoist way door locking device; elevator parking device; access to hoist way; and emergency doors.
3. Machine Room: Lighting and receptacles; housekeeping; ventilation; fire extinguisher; pipes, wiring, and ducts; guarding of exposed auxiliary equipment; numbering of machines and disconnect switches; disconnecting means and control; controller wiring, fuses, grounding, etc.; static control; drive machine brake; traction drive machines; gears and bearings; winding drum machine; belt- or chain-drive machine; motor generator; absorption of regenerated power; AC drives from a DC source; (Continued)
traction sheaves; secondary and deflector sheaves; rope fastenings; terminal stopping devices; slack rope devices; governor, over speed switch, and seal; and car and counterweight safeties.

4. Machine Room and Machinery Space: Access to machine space; pressure tanks; lighting machine space; ventilation of machine and control space; guards for exposed equipment; control valve; terminal stopping device; pumps; relief and check valve; drives; flexible hose and fittings; tank and oil level; controller wiring fuses, etc.; piping supply line and shutoff.
5. Pit: Pit light and stop switch; pit access; car clearance and runby; construction of oil buffer; oil buffer oil level gage; oil buffer data plate; spring buffer and data plate; solid bumper; cylinder oil collection; pipes, valves, fittings, and supports; plunger; plunger connection to car; normal stopping device; guard between pits; pit access; illumination; traveling; car frame and platform; and pit construction.
6. Test: Oil buffer; terminal stopping devices; and emergency power operation.
7. Motors: Inspect connections, armature and rotor clearances of the hoist motor and motor generator set: Clean and adjust as necessary to obtain proper operation.
8. Cables: Inspect, lubricate and properly adjust hoist cables, compensating cables, governor cables, and traveling cables to their manufacturer's specifications. Check all cable fastenings. Inspect guide rails and counter weights. Check and adjust the slow down and limit switches. Adjust all other items as necessary to obtain proper operation.
9. Sheaves: Inspect, clean, and lubricate in accordance with manufacturer's specifications all deflector, compensating, and top of car sheaves.
10. Clean up and remove all debris from work areas.

Tools

WEIGHTS_TEST	MATERIAL_01	GAGE_SHEAVE_GROOVE	CALIPER	CHALK	ELEVATOR_HANDBOOK	ELEVATOR_MANUAL
WATCH_STOP	GAGE_THICKNESS	HAMMER	LEVEL_SPIRIT	LIGHT_METER-02	LUBRICANT-07	TACHOMETER-02
ELEVATOR_SAFETY	MAT_CLEANING-05	TEST_GAGE_DOOR	TAPE	TOOL_BASIC	SIGN_OUT_OF_SERVICE-01	SAFETY_HAT
PRESSURE_GAUGE-01	BARRICADES	METER	MIRROR	FLASHLIGHT_NON_CONDUCTIVE		

null

Special Instructions

1. Complete all maintenance requirements included in the applicable ASME A17 Elevator Safety Codes.
2. Utilize the Maintenance Control Program (MCP) that is required to be developed and provided by the contractor for each piece of equipment as required in section 8.6 of ASME A17.1. MCP shall include all maintenance tasks and their frequencies with month dependent upon each unit's characteristics (equipment age, condition, accumulated wear, usage, improved technology, manufacturer's recommendations, etc.)
3. Ensure MCP is available for each unit. Ensure documentation of tasks performed under the MCP requirements are available and complete for review.
4. Emergency Phone: Verify proper operation with clear communication including adequate volume.
5. Conduct standard Fire Service Test per ASME A17.
6. Emergency Alarm Bell: Ensure the elevator emergency alarm bell is operational with adequate volume.

Checkpoint

1. Lubricants: Consult the Material Safety Data Sheet for hazardous ingredients and proper Personal Protective Equipment.
2. Cleaning Supplies and Materials: Consult the Material Safety Data Sheet for hazardous ingredients and proper Personal Protective Equipment.
3. Follow lock out/tag out procedures at all times.
4. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.

Tools

None

This standard applies to electrically operated passenger and freight elevators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A11, Safety Code for Elevators and Escalators, and A2, Inspectors' Manual for Electric Elevators, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
6. Be sure that safety devices are operational before performing any inspection work.
7. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
8. The annual inspection of electric elevators are described in the periodic section of the code.
9. Perform the semiannual inspection simultaneously.
10. Comply with state and local codes as applicable.
11. Use energy efficient lighting to the fullest practical extent.
12. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
13. Comply with state and local codes as applicable.

Checkpoint

1. Inside of Car: Car lighting; standby power operation.
2. Machine Room: Static control (check and torque all connections to correct specifications, and check and lubricate cooling fan); check primary and secondary voltages phase to phase and phase to ground; winding drum machine; AC drives from a DC source; traction sheaves; terminal stopping devices; slack rope devices; governor, over speed switch, and seal; and car and counterweight safeties.
3. Top of Car: Final terminal stopping devices; and counterweight safeties.
4. Outside Hoist way: Standby power selection switch.
5. Inspections Made in the Pit: Car and counterweight buffer; final terminal stopping devices; normal terminal stopping devices; and car safeties and guiding members.
6. Firefighter's Service: Operation of elevators under fire and other emergency conditions, A11b-1973 through A11b-1980; operation of elevators under fire and other emergency conditions, A17.1-1983 through A17.1b-1983; firefighter's service A17.1-1984 th
7. Machine Room: Static control (check and torque all connections to correct specifications, and check and lubricate cooling fan); check primary and secondary voltages phase to phase and phase to ground; winding drum machine; AC drives from a DC source; traction sheaves; terminal stopping devices; slack rope devices; governor, over speed switch, and seal; and car and counterweight safeties.

8. Selector: Inspect, clean, lubricate, replace parts, and make repairs or adjustments required for proper operation of selector unit components, including cables, chains, clutches, cams gears, fuses, motor brushes, wiring, connections, contacts, relays, tape, tape tension sheave, broken tape switch and tape wipers.
9. Controller: Thoroughly clean controller with blower or vacuum. Inspect and check the operation of switches, relays, timers, capacitors, resistors, contacts, overloads, wiring, connections, fuses, oil level of overloads and overload settings. Check for MG shutdown, high call reversal, zone control, and load by-pass door failure time. Check programming up peak, down peak, off peak, off hours. Replace worn or defective parts and adjust controller for proper operation.
10. Hoist way Doors: Clean, inspect, and lubricate all door operating mechanisms; including but not limited to rollers, upthrusts, interlocks, clutches, self closing, gibs and sills. Replace worn or defective parts, repair and adjust door mechanisms as required for proper operation.
11. Hoist ways: Clean rails, beams, and all related iron work in hoist way. Dust hoist way walls. Clean top, bottom, and sides of car. Clean counterweight and pit area.
12. Hoist Machine and Motor-Generator: Clean with blower or vacuum. Clean end bells, and brush riggings and commutator. Inspect connections, armature and rotor clearances.
13. Buffers: Check oil level and operation of switches. Add oil or adjust switches as necessary for proper operation. Manually compress buffer; test the proper return in accordance with ASME/ANSI A1Safety Code for Elevators and Escalators, Rule 204e(1).
14. Scheduling, Dispatch, and Signal Boards: Clean with blower or vacuum. Inspect and check the operation of all switches, relays, timers, fuses, capacitors, resistors, contacts, overloads, wiring and connections. Replace worn or defective parts and adjust for proper operation.
15. Motors: Change oil in hoist motor, motor generator set, geared machines, and gear boxes with lubricants as specified by the equipment manufacturer(s).
16. Safeties: Inspect, clean, lubricate and manually operate safety mechanisms. Replace parts or adjust as necessary.
17. Clean up and remove all debris from work area.

Tools

LEVEL_SPIRIT	CALIPER	LUBRICANT-07	BARRICADES	CHALK	ELEVATOR_HANDBOOK	ELEVATOR_MANUAL
ELEVATOR_SAFETY	FLASHLIGHT_NON_CONDUCTIVE		GAGE_SHEAVE_GROOVE	GAGE_THICKNESS	HAMMER	LIGHT_METER-02
MAT_CLEANING-05	MATERIAL_01	WEIGHTS_TEST	WATCH_STOP	TOOL_BASIC	TEST_GAGE_DOOR	METER
TACHOMETER-02	SIGN_OUT_OF_SERVICE-01	SAFETY_HAT	TAPE	MIRROR		

This standard applies to electrically operated passenger and freight elevators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A11, Safety Code for Elevators and Escalators, and A2, Inspectors' Manual for Electric Elevators, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Be sure that safety devices are operational before performing any inspection work.
8. The 5 year inspection of electric elevators are described in the periodic section of the code.
9. Perform the semiannual and annual inspection simultaneously.
10. Comply with state and local codes as applicable.

Checkpoint

1. Inside of Car: Power opening of doors or gates; standby power operation.
2. Machine Room: Traction sheaves; terminal stopping devices;
3. Top of Car: Counterweight safeties.
4. Inspections Made in the Pit: Car and counterweight buffer.

Tools

TEST_GAGE_DOOR	WEIGHTS_TEST	TAPE	WATCH_STOP	ELEVATOR_MANUAL	CALIPER	TACHOMETER-02	
ELEVATOR_HANDBOOK	ELEVATOR_SAFETY		FLASHLIGHT_NON_CONDUCTIVE		GAGE_SHEAVE_GROOVE	GAGE_THICKNESS	HAMMER
LIGHT_METER-02	MATERIAL_01	METER	MIRROR	LEVEL_SPIRIT	CHALK	SAFETY_HAT	

This standard applies to electrically operated passenger and freight elevators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This elevator inspection guide is adapted from the ANSI/ASME Codes A17.1, Safety Code for Elevators and Escalators, and A17.2, Inspectors' Manual for Electric Elevators, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections. Adhere to local codes when applicable.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on top of the car or in the pit. Do not enter any pit containing standing water.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Be sure that safety devices are operational before performing any inspection work.
8. The semiannual inspection of electric elevators includes those designated as routine in the code.
9. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
10. Comply with state and local codes as applicable.

Checkpoint

1. Inside of Car: Door reopening device; stop switches; operating control devices; car floor and landing sill; car lighting; car emergency signal; car door or gate; door closing force; power closing of doors or gates; power opening of doors or gates; car vision panels and glass car doors; car enclosure; emergency exit; ventilation; signs and operating device symbols; rated load, platform area, and data plate; standby power operation; restricted opening of car or hoist way doors; and car ride.
2. Machine Room: Lighting and receptacles; housekeeping; ventilation; fire extinguisher; pipes, wiring, and ducts; guarding of exposed auxiliary equipment; numbering of machines and disconnect switches; disconnecting means and control; controller wiring, fuses, grounding, etc.; static control; drive machine brake; traction drive machines; gears and bearings; winding drum machine; belt- or chain-drive machine; motor generator; absorption of regenerated power; AC drives from a DC source; (continued)
traction drive machines; gears and bearings; winding drum machine; belt- or chain-drive machine; motor generator; absorption of regenerated power; AC drives from a DC source; traction sheaves; secondary and deflector sheaves; rope fastenings; terminal stopping devices; slack rope devices; governor, over speed switch, and seal; and car and counterweight safeties.
3. Top-of-car stop switch; car top light and outlet; top-of-car operating device and working platforms; top-of-car clearance and refuge space; top counterweight clearance; car, overhead, and deflector sheaves; normal terminal stopping devices; final terminal stopping devices; broken rope, chain, or tape switch; car leveling devices; top emergency exit; counterweight and counterweight buffer; counterweight safeties; hoist way smoke control; pipes, wiring, and ducts; hoist way clearances; (Continued)

- multiple hoist ways; traveling cables and junction boxes; door and gate equipment; car frame and stiles; guide rails fastening and equipment; governor rope; governor releasing carrier; wire rope fastening and hitch plate; suspension rope; and compensating ropes and chains.
4. Outside Hoist way: Car platform guard; hoist way doors; vision panels; hoist way door locking devices; access to hoist way; power closing of hoist way doors; sequence operations; elevator parking devices; emergency doors blind hoist ways; and standby power selection switch.
5. Inspections Made in the Pit: Pit access, lighting, and stop switch, and condition; car and counterweight buffer; final terminal stopping devices; normal terminal stopping devices; traveling cables; governor-rope tension devices; compensating chains, ropes, and sheaves; car frame and platform; and car safeties and guiding members.
6. Motors: Inspect connections, armature and rotor clearances of the hoist motor and motor generator set. Clean and adjust as necessary to obtain proper operation.
7. Cables: Inspect, lubricate and properly adjust hoist cables, compensating cables, governor cables, and traveling cables to their manufacturer's specifications. Check all cable fastenings. Inspect guide rails and counter weights. Check and adjust the slow down and limit switches. Adjust all other items as necessary to obtain proper operation.
8. Sheaves: Inspect, clean, and lubricate in accordance with manufacturer's specifications all deflector, compensating, and top of car sheaves.
9. Clean up and remove all debris from work areas.

Tools

WEIGHTS_TEST	TOOL_BASIC	TEST_GAGE_DOOR	WATCH_STOP	BARRICADES	TACHOMETER-02	CALIPER	CHALK
ELEVATOR_HANDBOOK	ELEVATOR_MANUAL	ELEVATOR_SAFETY	ELEVATOR-04	FLASHLIGHT_NON_CONDUCTIVE			
GAGE_SHEAVE_GROOVE	GAGE_THICKNESS	HAMMER	PRESSURE_GAUGE-01	LEVEL_SPIRIT	SIGN_OUT_OF_SERVICE-01		
SAFETY_HAT	MIRROR	METER	MATERIAL_01	MAT_CLEANING-05	LUBRICANT-07	LIGHT_METER-01	TAPE

This standard applies to all elevators, electric or hydraulic types.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Items requiring attention should be reported to the elevator shop supervisor or elevator contractor.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
5. Comply with state and local codes as applicable.

Checkpoint

1. Machine Room: Dust drive machines, motor generator sets, and control cabinets. Sweep floor, re-lamp inoperative machine room lights and remove trash. Inspect controllers, signal, scheduling and dispatch panels, selectors and selector tapes. Make maintenance adjustments necessary for proper operation.
2. Pits: Dust ironwork, sweep floor, empty drip pans, re-lamp inoperative lights, remove trash and check for leaks. Check run-by, buffer switch, compensating switch and emergency stop switch. Make maintenance adjustments necessary for proper operation.
3. Motors; hoist motor, motor generator and regulators: Inspect oil level, oil pick up and belts. Make maintenance adjustments necessary for proper operation. Check for excessive heat, noise and leaks. Lubricate in accordance with manufacturer's specification. Renew brushes as necessary for proper operation. Keep end bells, commutators and brush riggings clean.
4. Geared Machines: Inspect worm and gear for bottoming and backlash, thrust end play, bearing wear, oil pick up, oil level and packing. Check for leaks and empty drip pan. Renew brushes as necessary. Keep end bells, commutators and brush riggings clean.
5. Brakes: Observe operation of brake. Inspect drum and shoe clearance and adjust as necessary for proper operation. Clean and lubricate pivot points. Where brake shoes are asbestos-containing, check for dust, and practice appropriate cleanup and maintenance precautions.
6. Car Gate: Clean, lubricate and inspect hangers and all related gate operating mechanisms. Check rollers, up-thrust, interlock, gear box, motor brushes, door control box, cables, safety edge, light rays, gibbs, sills, proper operating speed and force close speed. Make maintenance adjustments necessary for proper operation.
7. Speed Governor: Observe operation, including tension sheave. Check electrical switches for proper operation; check that inspection seals are in place and not broken. Clean and lubricate pivot points.
8. Lighting: Re-lamp all inoperative lamps located in lamp hatchway, hall landings, position indicators, car stations, dome, and wherever else required. Clean light diffusers and car stations.
9. Hydraulic Machines: Observe operation of motor and pump, oil lines, tank, controls, plunger and packing. Adjust as necessary for proper operation. Correct excessive creeping. Test manual and emergency control. Clean and lubricate as necessary. Check for and repair leaks on oil lines, tank, and packing. Test manual and emergency control. Clean and lubricate all equipment as necessary for proper operation.
10. Emergency Items: Check that emergency procedure sign is in place and inspection certificate is current and visible (or filed in buildings manager's office). Check top and side exits and related switches for proper operation. Test alarm bell, emergency stop switch, communications system, fire recall service (key capture, minimum one floor run on Phase II), and any other emergency recall features. (Continued)
Repair, replace, and adjust parts and equipment as necessary to insure operation in accordance with the manufacturer's specifications. Document each test performed, including the date and results of each test.
11. Clean up and remove all debris from work areas.

Tools

SIGN_OUT_OF_SERVICE-01 LUBRICANT-07 MAT_CLEANING-05 BARRICADES TOOL_BASIC

This standard applies to all elevators, electric or hydraulic types (4 floors or less).

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Review manufacturer's instructions.
3. Comply with state and local codes as applicable.
Items requiring attention should be reported to the elevator shop supervisor or elevator contractor.

Checkpoint

1. Inspected the following equipment looking for proper operation.
 - a. motor-generator unit
 - b. hoist machine
 - c. controls and governor
 - d. Doors, door hangers, closers, interlocks and door operators
2. Emergency Items: Check that emergency procedure sign is in place and inspection certificate is current and visible (or filed in buildings manager's office).
3. Clean up and remove all debris from work areas.

Tools

MAT_CLEANING-05 TOOL_BASIC

This standard applies to all elevators, electric or hydraulic types (4 floors or less).

Special Instructions

- 1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
- 2. Items requiring attention should be reported to the elevator shop supervisor or elevator contractor.
- 3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
- 4. Use energy efficient lighting to the fullest practical extent.
- 5. If asbestos containing materials are present, check the building's asbestos management plan as described in the Environmental Management portion of the Safety and Environmental Management Program. Manage asbestos in accordance with the asbestos management plan.
- 6. Comply with state and local codes as applicable.

Checkpoint

- 1. Guides: Inspect, clean, lubricate, and properly adjust all roller guides, guide shoes, and rail oilers.
- 2. Car Tops: Remove all dirt, dust, and oil. Clean and lubricate fans.
- 3. Emergency Lighting: Check the operation of the emergency lights; replace as necessary.
- 4. Solid State Components and Circuit Boards: Inspect printed circuit board and other solid state devices for cleanliness, condensation spots, evidence of heating and deterioration. Check and replace defective solid state devices.
- 5. Leveling: Inspect leveling operation. Clean and lubricate switches, vanes, and all other related parts. Adjust to obtain the proper leveling at all landings in both the UP and DOWN directions within 1/4 inch above or below the landing sill.
- 6. Clean up and remove all debris from work area.

Tools

BARRICADES TOOL_BASIC SIGN_OUT_OF_SERVICE-01 MAT_CLEANING-05 LUBRICANT-07

This standard applies to escalators and moving walkways.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This escalator and moving walk inspection guide is adapted from the ANSI/ASME Codes A1 and A2, hereinafter referred to as the code. Refer to these documents for comprehensive instructions on elevator inspections.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
4. or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof
5. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
6. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on around this equipment.
7. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
8. Be sure that safety devices are operational before performing any inspection work.
9. Set up a barricade at the escalator entry point. Use signs to direct passengers to stairs. Once escalator is vacant, set up another barricade at the exit point, and stop the escalator.
10. Comply with state and local codes as applicable.

Checkpoint

1. Moving Walk Broken Treadway Devices. Test by manually operating the actuating device.
2. Stop Switches. Test all stop switches, including those located in machine spaces, and all mainline disconnect switches for proper operation.
3. Escalator Skirt-Obstruction Devices. These devices are to be manually operated by simulating an actual obstruction.
4. Brakes. Test the operation of the electrically released brake by manually operating the other safety devices and observing.
5. Starting Switches. Test for proper operation and type.
6. Escalator Reversal Stop Devices. Inspect and test operation.
7. Thoroughly clean escalator, working from top to bottom.
8. Check operation of all safety devices, including skirt switches, handrail switches, broken chain switches, lower unit tension devices, etc.
9. Follow manufacturer's recommendations for lubrication.
10. Perform semi-annual work as prescribed by the manufacturer.
11. After work has been completed, perform periodic inspection and complete the Certificate of Inspection.
12. Remove barricades and place escalator back into service.

Tools

SAFETY_HAT	PRESSURE_GAUGE-01	METER	MIRROR	TACHOMETER-02	TAPE	BARRICADES	GAGE_SHEAVE_GROOVE
TOOL_BASIC	WATCH_STOP	TEST_GAGE_DOOR	LIGHT_METER-02	LEVEL_SPIRIT		GAGE_THICKNESS	
FLASHLIGHT_NON_CONDUCTIVE		ELEVATOR_SAFETY	ELEVATOR_MANUAL	ELEVATOR_HANDBOOK	CHALK	CALIPER	
WEIGHTS_TEST	HAMMER	MATERIAL_01	PRESSURE_GAUGE				

This standard applies to escalators and moving walkways.

Special Instructions

1. Utilize the Maintenance Control Program (MCP) that is required to be developed and provided by the contractor for each piece of equipment as required in section 8.6.8 of ASME A17.1. MCP shall include all maintenance tasks and their frequencies with month dependent upon each unit's characteristics (equipment age, condition, accumulated wear, usage, improved technology, manufacturer's recommendations, etc.)
2. Ensure MCP is available for each unit. Ensure documentation of tasks performed under the MCP requirements are available and complete for review. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
3. Set up a barricade at the escalator entry point. Use signs to direct passengers to stairs. Once the escalator is vacant, set up another barricade at the exit point, and stop the escalator. Upon completion of work remove barricades and place escalator back into service.
4. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
5. "Comply with state and local codes as applicable. This escalator and moving walk inspection guide is adapted from the ANSI/ASME Codes A1and A2, hereinafter referred to as the ""code."" Refer to these documents for comprehensive instructions on elevator"

Checkpoint

1. Ride escalator. Check operation for smoothness, unusual vibration or noise, condition of handrails, etc.
2. Inspect comb plates at both ends of escalator for broken teeth and check for proper clearance between combs and step teeth and check for broken step treads.
3. Operate each emergency stop button and note that the escalator stops and alarm is operational. If the escalator has the capabilities of running in both directions, stop buttons should function properly for each direction of travel. Observe the stopping distance.
4. Lubricants: Consult the Material Safety Data Sheet for hazardous ingredients and proper Personal Protective Equipment.
5. Cleaning Supplies and Materials: Consult the Material Safety Data Sheet for hazardous ingredients and proper Personal Protective Equipment.

Tools

LEVEL_SPIRIT	SAFETY_HAT	WEIGHTS_TEST	WATCH_STOP	TEST_GAGE_DOOR	TACHOMETER-02	PRESSURE_GAUGE-01	
MIRROR	TAPE	METER	CALIPER	BARRICADES_ROPES	MATERIAL_01	CHALK	ELEVATOR_HANDBOOK
ELEVATOR_MANUAL	ELEVATOR_SAFETY	FLASHLIGHT_NON_CONDUCTIVE	GAGE_SHEAVE_GROOVE	GAGE_THICKNESS	HAMMER	LIGHT_METER-02	

This weekly maintenance standard applies to passenger escalators.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. Set up a barricade at the escalator entry point. Use signs to direct passengers to stairs. Once escalator is vacant, set up another barricade at the exit point, and stop the escalator.
3. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
4. Comply with state and local codes as applicable. This escalator and moving walk inspection guide is adapted from the ANSI/ASME Codes A1 and A2, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator.

Checkpoint

1. Ride escalator. Check operation for smoothness, unusual vibration or noise, condition of handrails, etc.
2. De-energize, tag and lockout the electrical circuit.
3. Inspect comb plates at both ends of escalator for broken teeth and check for proper clearance between combs and step teeth and check for broken step treads.
4. Check clearance between steps and skirt panel. Look for anything (loose trim, screws or bolts) that could snag or damage clothing or cause injury. Check operation of handrail brushes.
5. Clean escalator machine space.
6. Clean and lubricate step rollers, step chain, drive gears or chains, handrail drive chains, etc., according to manufacturer's instructions. Observe gears and chains for signs of wear, misalignment, etc. Adjust as required.
7. Check motor for signs of overheating; clean excess grease.
8. Inspect controller for loose leads, burned contacts, etc. Repair as required. Clean excess grease and interior of controller. Check for loose connections.
9. Clean handrails as required.
10. Check escalator lighting. Replace bulbs as required.
11. Operate each emergency stop button and note that the escalator stops. If the escalator has the capabilities of running in both directions, stop buttons should function properly for each direction of travel. Observe the stopping distance.
12. Clean up and remove all debris from work area.
13. Remove barricades and place escalator back into service.

Tools

BARRICADES_ROPES LUBRICANT-07 MAT_CLEANING-05 TOOL_BASIC

This standard applies to escalators and moving walkways.

Special Instructions

1. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to.
2. This escalator and moving walk inspection guide is adapted from the ANSI/ASME Codes A1 and A2, hereinafter referred to as the "code". Refer to these documents for comprehensive instructions on elevator inspections.
3. Inspectors shall meet the requirements of the ANSI/ASME QEI-1 Standard for the Qualifications of Elevator Inspectors and shall be recognized by GSA. Inspectors and inspection supervisors shall be certified by an ASME accredited organization in accordance with the requirements of ASME QEI-1. Inspections and tests shall be performed by an inspector employed by GSA or by an outside inspector authorized by GSA to perform them. They shall be witnessed by an inspector employed by GSA, (Continued)
or by a qualified person authorized by GSA to witness the tests and inspections on its behalf. Immediately following these inspections and tests, the inspector shall submit to GSA a statement certifying that the inspections and tests have been performed and a report on the results thereof.
4. Avoid the wearing of loose clothing and neckties during the performance of this inspection.
5. Be at all times vigilant of the location and movement of cars, counterweights, projections, rotating machinery, etc. Note the clearance available when working on around this equipment.
6. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
7. Be sure that safety devices are operational before performing any inspection work.
8. Set up a barricade at the escalator entry point. Use signs to direct passengers to stairs. Once escalator is vacant, set up another barricade at the exit point, and stop the escalator.
9. Comply with state and local codes as applicable.

Checkpoint

1. Speed Governors. Where a speed governor is required, test by manually operating the trip mechanism.
2. Escalator Broken Step-Chain Device. Test the broken step-chain device by operating it manually.
3. Broken Drive-Chain Devices. Test the broken drive-chain device by operating the actuating mechanism manually.
4. Moving Walk Broken Treadway Devices. Test by manually operating the actuating device.
5. Stop Switches. Test all stop switches, including those located in machine spaces, and all mainline disconnect switches for proper operation.
6. Fire Shutters. Test the fire shutters to verify that their operation will stop the escalator or moving walk. If power operated, the force and reversing requirements must be met.
7. Escalator Skirt-Obstruction Devices. These devices are to be manually operated by simulating an actual obstruction.
8. Brakes. Test the operation of the electrically released brake by manually operating the other safety devices and observing.
9. Starting Switches. Test for proper operation and type.
10. Escalator Reversal Stop Devices. Inspect and test operation.
11. Escalator Step Upthrust Devices. Test by manually displacing the step, which should cause the device to operate.
12. Remove steps as required to provide clear access to escalator pan and place steps on tarp.
13. Thoroughly clean escalator, working from top to bottom.
14. Clean all tracks and check for wear or rippling. File tracks as required. Check all step and chain rollers. Adjust step and chain roller upthrust as required. Adjust transfer bars or guides or replace as necessary.
15. Remove upper panel on each side of escalator, taking care not to scratch or gouge panel. Check handrail tension device and handrail drive assembly. Clean, adjust and lubricate as required.
16. Check operation of all safety devices, including skirt switches, handrail switches, broken chain switches, lower unit tension devices, etc.
17. Follow manufacturer's recommendations for lubrication.

- 18. Perform annual work as prescribed by the manufacturer.
- 19. Re-assemble entire unit, thoroughly cleaning steps and check for broken treads as they are replaced. Check entire unit for proper running clearances. Re-shim steps as required.
- 20. After work has been completed, perform periodic inspection and complete the Certificate of Inspection.
- 21. Remove barricades and place escalator back into service.

Tools

LIGHT_METER-02	HAMMER	GAGE_THICKNESS	GAGE_SHEAVE_GROOVE	FLASHLIGHT_NON_CONDUCTIVE	ELEVATOR_SAFETY		
ELEVATOR_MANUAL	ELEVATOR_HANDBOOK	CHALK	MATERIAL_01	BARRICADES_ROPES	CALIPER	METER	TAPE
MIRROR	PRESSURE_GAUGE-01	TACHOMETER-02	TEST_GAGE_DOOR	WATCH_STOP	WEIGHTS_TEST	SAFETY_HAT	
LEVEL_SPIRIT							

This standard card applies to wheelchair lifts, normally found at entrances and in entrance lobbies of various public buildings.

Special Instructions

1. Utilize the Written Maintenance Program (WMP/MCP) that is required to be developed and provided by the contractor for each piece of equipment as required in section 11.1 of ASME A18.1. 2017. MCP shall include all maintenance tasks and their frequencies with month dependent upon each unit's characteristics (equipment age, condition, accumulated wear, usage, improved technology, manufacturer's recommendations, etc.)
2. Ensure WMP/MCP is available for each unit. Ensure documentation of tasks performed under the MCP requirements are available and complete for review. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered to
3. In addition to the procedure(s) outlined in this standard, the equipment manufacturer's recommended maintenance procedure(s) and/or instruction(s) shall be strictly adhered.
4. Review manufacturer's instructions for all system components.
5. Follow lock out/tag out procedures at all times. De-energize or discharge all hydraulic, electrical, mechanical, or thermal energy prior to beginning work.
6. Set up a barricade at the lift entry point.

Checkpoint

1. Thoroughly clean the drive assembly, lift platform, travel area, etc. Remove trash and debris.
2. Inspect structural features, framework, support members, anchor bolts, lift platform, etc., for condition and structural integrity. Tighten any loose bolts or screws.
3. Lubricate screw mechanisms, gears, sprockets and chains, pillow blocks, bearings, etc., as required.
4. Check condition of drive chains or belts and adjust or replace as required.
5. Operate lift through its full range of travel, in both directions. Note operation of controls, safety devices, leveling, and other devices or features.

Tools

EQUIP_CLEANING-03 TOOL_BASIC